

Factor Investing | Handbook



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Contents

Preface.....	5
Factor Investing – A New Dimension In The Asset Allocation Puzzle.....	6
Preface From MSCI	10
 1. Introduction.....	 11
The Rise Of Factor Investment	12
Deploying Multi-Factor Index Allocations In Institutional Portfolios.....	28
List Of Authors	59
 2. Choosing Right Factors.....	 63
Index Performance In Changing Economic Environments	
A Macro-Economic Framework	64
Incorporating Risk Premia Mandates In A Strategic Allocation.	
A Client Case Study: Wyoming Retirement System.....	119
Foundations Of Factor Investing	129
Multi-Factor Indexes Made Simple	
A Review Of Static And Dynamic Approaches.....	176
List Of Authors	199
 3. Loading Factor Exposures To Portfolios.....	 205
Tactical Tilts And Multi Asset Class Risk Premia.....	206
Building Smarter Equity Value Strategies	223
Minimum Expected Tail LossEquity Portfolios	241
Equity Factors In Action: The Need For Multi-Regional-Currency-	
Hedged Exposures.....	250
Using Passive Factor Exposures To Build A Robust Multi-Factor	
Portfolio	268
List Of Authors	284

4. Indexing A Factor Portfolio287
Efficient Implementation Of Alternative Beta Index Portfolios.....288
Challenges When Hedging Currency Risks In ETFs299
List Of Authors318

5. Factor Portfolios Put In Practice.....321
A Swiss Pension Fund Case Study.....322
An Alternative And More Efficient Approach For Investment
Managers To Access Broad Commodities327
Italian Portfolio Managers Using Passive Factor Tilts.....336
A Need For Factor Allocations: A Showcase Of A German
Insurance Company349
Combining Core Satellite Investing With Factor Allocation:
A Swiss Private Bank Model357
Exploring The Liquidity Of A Risk Weighted ETF.....365
List Of Authors370

About The Contributors373

Preface

Factor Investing – A New Dimension In The Asset Allocation Puzzle

Thomas Merz

UBS Global Asset Management

Conventional investment portfolios are typically characterized by a static asset allocation and by diversification across a range of asset classes. In search of higher returns and greater risk diversification, institutional as well as private investors have implemented a wide range of investment strategies which aim to find the best possible solution to the afore-mentioned objective.

In the past fourteen years, however, markets have uncovered various shortcomings in the paradigm that asset classes are independent, with little to no overlap, and which, in aggregate, are able to cover the investment universe. Many of the conventionally constructed portfolios using the traditional asset class and sub-asset class approach, have failed to deliver proper diversified portfolio characteristics. Observations of rising correlations between asset classes during the depths of the Financial and Eurozone Sovereign Debt Crisis caused many investors to question basic assumptions about traditional portfolio models. They also became aware that such high correlations were due to the significant overlap in their underlying common risk factor exposures. Ideally, investors could construct portfolios using various components with independent risk drivers that were individually rewarded by the market for their level of risk. These portfolio building blocks (factors) would explain the majority of their return and risk characteristics. While it is true that an asset-based investing approach provides an indirect way to invest in factor returns, the main advantage of a direct factor investment approach is that such factors can be used in portfolios that are more diversified and more efficiently harvest the factor premia offered by the market than traditional methods.

Factor investing has become a more widely deployed tool in the universe of investment strategies, as exemplified by recent academic literature¹. For example, Bhansali (2011) argues that investors would be better off diversifying their portfolios across risk factors than asset classes. This perspective is based on the observation that macro-economic forecasts can be easily mapped to risk factors, whereas mapping to asset classes² is obscured. Recent research indicates that four out of five underlying risk factors can explain approximately 70 % of the variation in most liquid assets. Ang (2013) considers that in large portfolios it is very hard to find excess returns that are not related to factors. Given that many mispricing opportunities are not scalable, active manager decisions tend to be correlated, as it is hard to find truly independent portfolio strategies. Security-specific bets are swamped at the portfolio level by macro-economic and factor risks. Furthermore, Podkaminer (2013) advocates that factors are basic building blocks of asset classes and a source of common risk exposures across asset classes. In his view, factors make up the smallest systematic (or non-idiosyncratic) units that influence investment return and risk characteristics. Although the construction of ex ante optimized portfolios using factor premia inputs is possible, there are still significant challenges to overcome in order to address practical portfolio constraints. This includes the need for active, frequent rebalancing, creation of forward looking assumptions and the consensus of the relevant factor premiums utilized. How risk factors are to be included in the portfolio construction process, and potentially combined with conventional portfolio assets, remains a lively debate amongst practitioners.

Our *Factor Investing Handbook* is intended to serve as a guide to better understanding as well as facilitate a dialogue between asset owners and solution providers such as UBS Global Asset Management. The overall aim of this publication is to provide our existing and future clients with useful information on specific factor investing know-how and analysis, leading to better investment and risk management decisions in the future. This publication presents both fundamental

1 See for example Ang (2013), Ang, Goetzmann and Schaefer (2009), Asness, Moskowitz and Pedersen (2013), Bender et al. (2013, 2010), Bender, Hammond and Mok (2014), Bhansali (2011), Fama and French (2014), Jostova et al. (2013), Huij et al. (2014), Ilmanen (2011), Ilmanen and Kizer (2012), Page and Taborsky (2011), Pappas et al. (2012), Podkaminer (2013), Rosen and Saunders (2012), Witte, Ples and Corominas (2013).

2 Which in his view can be seen as complex baskets of risk factors.

thought as well as practical applications pertaining to factor investing. The first two chapters provide basic insights regarding the factor investment approach. This information targets investment professionals and academics who have yet to dive deeper into the material of factor premia. Chapter 3 and 4 offer balanced and practical views on the application of factor based investing. These essays strive to develop applicable investment solutions for practitioners who are asked to find efficient ways to include factor premia in their portfolios. The final chapter is devoted to real-life examples of portfolio construction processes which have already embraced factor investing. It provides a sounding board for academics and practitioners to compare their ideas regarding factor portfolios with the returns of an array of existing invested client portfolios.

We would like to thank all of the authors for their valuable contributions. By including articles from both academics and practitioners, the *Factor Investing Handbook* will serve as a non-exhaustive source of fruitful, as well as controversial discussions on the topic of factor premia focused investment processes.

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Preface From MSCI

Dimitris Melas

MSCI

The last ten years have seen a substantial growth in the development and adoption of factor indexes which represent the returns of systematic market factors.

As part of this evolution, MSCI has developed a family of factor indexes that capture the performance of six well-known, researched and documented factors: volatility, size, quality, value, yield and momentum.

Factor indexes have changed the way institutional investors think about Beta and Alpha, for 50 years the cornerstone of equity investing. In the past, the prevailing paradigm was indexes for Beta and active management for Alpha. Factor indexes create the possibility of seeking factor returns by means of passive vehicles. Factor indexes are not a replacement for benchmark indexes or active managers. Active mandates will continue to play a central role: skillful active managers can add value through market timing, sector rotation and stock selection. The advent of factor indexes might help true active managers differentiate their unique alpha generating skills and demonstrate clearly how they add value beyond what is possible through factor indexation.

MSCI has been uniquely positioned as an industry leader for over 40 years in both Equity Factor Models and Equity Indexes. Combining this expertise to create Factor Indexes was a natural step. We are pleased to offer a sample of our published research in this Factor Handbook.

1. Introduction

The Rise Of Factor Investment

Ignacio Munoz-Alonso

IE Business School

The Beginnings: Harry Markowitz, William Sharpe And The CAPM

Factor models relate specific factors assembled in a joint structure denominated *Return Generating Process*, able to generate random asset returns. Factor investing is the systematic use of factor models to invest in market segments which achieve better returns than those in other sectors. The notion that returns could be associated to quantifiable and predetermined factors surged in the post-war period as an attempt to overcome what many considered to be an arbitrary approach for the determination of financial prices and expected returns. Value investing was the prevailing school of thought in those days and its practice was a combination of intuition, experience and a fair amount of patience.

It all started some decades ago when a group of young mathematicians and economists realized the potential of applying operations research and linear programming techniques to the selection and pricing of financial assets³. Many of those techniques were developed during the war as a way of improving military decision-making in risky and uncertain conditions. For example, the fastest and safest route for a convoy of ships crossing the Atlantic, knowing that enemy submarines could be there with some probability. Or the number of fragments a bomb should produce after blowing up, considering that smaller fragments had a higher probability of hitting a target, yet would have much less impact.

Harry Markowitz, a graduate student in Chicago and one of those in the group, proposed an answer to a similar kind of question posed by Milton Friedman,

3 A good discussion on the development of modern finance can be found in Fox, J., 2009.

a member of Markowitz's dissertation committee: »How much return are you willing to sacrifice to increase the probability that you will achieve your goal?« Markowitz called his approach *portfolio selection* and was first published in *The Journal of Finance* in October 1952.

His method had strong theoretical foundations underpinned by solid statistical and optimization techniques, along with a rigorous characterization of the agent's behaviour in conditions of uncertainty. For the first time, it was possible to make decisions without recourse to perfect foresight. To do this, Markowitz applied the Von Neumann–Morgenstern preference theory, which in essence allows individuals to think probabilistically, thus setting the groundwork for a systematic treatment of risk in modelling an individual's behaviour. The portfolio selection method considered one period preference functions and probability distributions that could be fully characterised only by their first two moments, the mean and the variance, pointing towards what later was to be the first factor model in finance, the Capital Asset Pricing Model (CAPM).

In this one period framework individuals were only concerned with maximizing a vector of returns given a measure of risk described by a matrix of variances and covariances between the different assets returns, or equivalently, minimizing that risk given a required set of returns. And diversification did, of course, its magic: the larger the number of assets in a portfolio, the lower its variance given a constant aggregated return. In the absence of correlations between asset returns in the portfolio, $\sigma_{i,j}=0$, the portfolio variance could be reduced to almost zero as additional assets were included, though, in practice, the minimum variance achievable was determined by the correlation between all the assets contained in the most populated portfolio, the market portfolio. The portion of risk that could be diversified away by means of diversification was denominated idiosyncratic, specific or diversifiable risk, and the limit to risk diversification was that inherent to the market portfolio, denominated systematic or market risk. The programme of the portfolio selection method was, then, to form mean-variance »efficient portfolios«, those resulting from the minimization of its aggregated risk, given a set of returns by means of a proper selection of assets weights, dictated by the correlations of assets against each other.

One practical problem of this approach was that, as an increasing number of assets was added to the portfolio, the calculation of the variance covariance matrixes started to appear fairly unmanageable, especially given the limited computational capacity in those days. Estimation of the variance covariance matrix for n securities requires n variances and $n(n-1)/2$ covariances for a total of $2n + n(n-1)/2$ parameters. If, for example, $n=100$, we would need to estimate 5150 parameters in order to determine an optimal allocation.

Calculating a single efficient portfolio could become an unbearable task in terms of time and computational cost. In order to find any practical use for this new portfolio method, it would seem imperative to derive simpler models that are not so data intensive and which capture enough reality.

A solution to this problem was provided by William Sharpe in his doctoral thesis, later published in *The Journal of Finance* in September 1964⁴. The idea was to define an alternative metric of risk that could yield the same result but with simpler mathematics. Restricting the asset class to securities, Sharpe thought that, rather than comparing the covariance of each separate security to every other one, a similar result could be obtained with a measure of the covariance between an individual security and the rest of the portfolio as a whole, the market portfolio, a value-weight aggregation of all existing risky securities in the market. He represented that measure in his dissertation with the capital letter »B«, later transmuted into the Greek letter »beta« that was defined as:

$$\beta = \frac{\sigma_{m,i}}{\sigma_m^2}$$

Where:

$\sigma_{m,i}$ = Covariance between the market portfolio m and security i

σ_m^2 = Variance of the market portfolio m

4 Lintner (1965) and Mossin (1966) independently arrived at similar conclusions, which is why the CAPM model is often referred as the Sharpe–Lintner–Mossin model.

But Sharpe went beyond the mere statistical problem and investigated the consequences on asset prices in a one-period environment where all agents were rational, return maximizers in the mean-variance space and shared the same beliefs about the probability distributions of returns. What resulted was an equilibrium model for securities pricing in which the expected return of a security was directly linked to the volatility of the market, measured by the beta: rational investors would be willing to pay more, and, thus, accept a lower return for a security with a lower beta, that is, for lower volatilities relative to the market.

However, Sharpe was missing a piece in order to progress. A few years before Sharpe wrote his dissertation, Tobin (1958) showed in his Separation Theorem that the mean-variance preferences could be derived by restricting the probability distributions over which choices are made to a two asset family. The major result of Tobin's analysis focused on the choice between a single risky asset and a risk-free or cash asset. The consequence of this result was that if all individuals hold the same beliefs they will all invest their total wealth in two portfolios formed by a combination of the risky efficient portfolio reflected in the market portfolio, and the risk-free asset in different proportions, accommodated to their subjective risk preferences.

Supply and demand of risk premia should then adjust to produce a situation where all securities keep an equilibrium relationship θ where excess returns are compensated by their corresponding higher risks in a stable situation. If the price of risk is the same for every security in the market, then the market for risky securities is cleared:

$$\theta = \frac{E(R_i) - R_f}{\sigma_{m,i}} = \frac{E(R_j) - R_f}{\sigma_{m,j}} = \dots = \frac{E(R_m) - R_f}{\sigma_m^2}$$

Where:

$E(R_i)$ = expected return of security i

R_f = the risk-free return

$E(R_m)$ = expected market return

$\sigma_{m,i}$ = Covariance between the market portfolio m and security i

σ_m^2 = Variance of the market portfolio m

A straight consequence from this equilibrium situation was that every individual security excess return could be characterized by the well-known CAPM expression, where security excess returns could be explained by one single factor, the beta that linearly relates them with observed market premia:

$$E(R_i - R_f) = \beta[E(R_m) - R_f]$$

The Empirical Evidence Of CAPM And The Development Of Multifactor Models

The CAPM was formulated as an equilibrium model that explains where, under certain assumptions, returns should be, rather than where they are. Though it is regarded as the central paradigm of factor modelling, it was soon evident from early tests that the Sharpe version of the CAPM was rather distant from reality. There was a positive correlation between beta and average returns, but it was too »flat«⁵.

Jensen (1968) produced the first times series regression test for the Sharpe CAPM under the assumption that, if excess returns $(R_i - R_f)$ could be fully explained by its expected CAPM premium $\beta(R_m - R_f)$, then the intercept term in the time series regression, the »Jensen's Alpha«, is zero for each asset. The evidence that the positive relation was too »flat«, indicated that the intercept was larger than the average risk-free rate and that the beta was less than the average market return, or in other words, that there were other factors beyond the beta, not included in the formulation, that were driving returns. This was evidenced in other early tests, such as Douglas (1968), Black, Jensen and Scholes (1972) as well in later cross section regression tests, like Fama and French (1992).

As well as the empirical inconsistencies of the model, some theoretical criticisms also made an impact on the perceived reliability of the CAPM. Among them was the Roll's Critique (Richard Roll, 1977) which questioned the possibility of unanimous strategies followed by every investor, as there was not

5 Fama, French, 2004.

a unique and unquestionable definition of what the market is. The market portfolio, as the risky component of risk-free-risky portfolio partition followed by every investor, was difficult if not impossible to characterize, as it was not, directly, observable. It is not only that many other assets classes beyond securities and other financial assets could be included in that definition, such as real estate holdings, it is also that some other drivers of investor's attitudes towards investing, such as consumption preferences or human capital, which account for a large portion of total wealth, were not included in the definition. The corollary of all this was that, under such an indeterminacy, the CAPM was never tested and probably never will be.

The lack of the explanatory power of the CAPM together with its theoretical questioning, stimulated the investigation of other alternative routes beyond the market beta as the single explanatory factor.

In one camp, there were those who advocated for a *relative pricing* approach, in which asset prices could be explained given the price of some other comparable assets. It is no longer relevant where those prices come from and if they are where they should be. What matters is the identification of the factors common to the co-movement of the returns of substitute assets, as well, along with the discovery of those with a higher probability of producing risk premia. Each individual asset or portfolio, the target portfolio, the must-be-seen-as-a portfolio that can be replicated by forming an identical risk-return profile by combining other existing market assets, the index portfolio equivalent to the market portfolio in the CAPM. Returns should reflect the possibility of substitution between these two replicable asset portfolios.

There is no recourse to any equilibrium condition to explain returns with its frequently restrictive and unrealistic assumptions. Returns, and prices, are not absolute anymore, they are relative. Under this approach, many of the problematic assumptions of the CAPM, such as unanimous beliefs on the distribution of returns or complete rationality, can be ignored. Instead there are only a few central and intuitive sets of assumptions: the *law of one price* holds, precluding the possibility of arbitrage between assets or portfolios; there are at least a few well-informed investors, arbitrageurs, able to profit from arbitrage opportunities and clearing the market; and the possibility of free formation of portfolios, that is, there are enough liquid assets in the market to replicate

comparable portfolios. In addition, the index portfolio does not need to be the market portfolio, it suffices to be well diversified, thus circumventing the problematic Roll's Critique.

It was Stephen Ross (1976), in his Arbitrage Pricing Theory (APT), who first presented the idea that investors can diversify away any idiosyncratic risk and, being exposed, only to factor risk. His idea was that many portfolios of stocks could be reasonably approximated as linear combinations of the return of a few basic common »factors« in such a way that:

$$R_i = a + \beta_1 f_1 + \beta_2 f_2 + \dots + \varepsilon$$

Where:

a = risk-free asset return

R_i = return of security i

$\beta_n f_n$ = the factors f and their respective loads, β

ε = error term, the idiosyncratic component in each security

Ross APT, the first multifactor risk model, states that if ex-post returns of a certain security or portfolio can be approximated by a proper selection of factors, say, macro-economic factors and a few industry groupings, then the error term in the above regression should be small, as the structure of the return would be, largely, captured by the return generating process. If that is not the case, then one can buy the focus portfolio R_i and short the right-hand side combination of factors and earn a high return with little risk, that is, an arbitrage opportunity would arise. Equally, factors can be properly discriminated in order to separate those containing more premia.

The choice of factors depends on the type of assets being considered, but apart from the market portfolio which is frequently included, those tend to relate to exogenous macro-economic variables influencing the return performance of each specific security, such as inflation, interest rates, or business cycles, as well as commonly used lagging and leading macro indicators. For example, to price airline stocks, we may need to look at the price of fuel, interest rates, if the company finance the fleet with loans, and perhaps GDP will have a relatively high load factor as the airline industry is highly cyclical. Under the

logic of the APT, other comparable securities of companies operating under a similar set of influences should reflect returns aligned with the loadings of the same factors of their model to explain comparable returns. Observed departures from the model factor loadings would offer arbitrage opportunities, or more likely, near arbitrage opportunities. Interestingly, the method requires, in theory, only a few, even one, in the limit, well-informed active investor, thus easing the need for a strict market efficiency assumption.

Perhaps the problem with multifactor models is the loss of generality, one of the advantages of the CAPM. The more factors that are included, the more specific it becomes to certain configurations. In an attempt to generalize APT factors, Chen, Ross and Roll (1986) later developed a pioneer model including a set of four pre-specified macro-economic factors based on unexpected changes in four factors: industrial production, the slope of the term structure of interest rates, risk premia (default spreads) and inflation. They tested the model for the US economy in the period 1953–1983 and found statistical significance in the first three factors and weaker evidence of the significance of the inflation factor.

This type of model flourished throughout the decades of the 70s and 80s, but, unfortunately, later tests demonstrated the difficulties for the correct identification and measurement of all the true systematic risk factors together with some serious statistical weaknesses that limited the explanatory power of the APT. Difficulties in implementing macro-economic factor models also arise from the problems in making comparisons across different countries and markets. Macro-economic, historical data often present inadequacies and inconsistencies. And finally, the role of expectation, as the true driving force behind the pricing of securities, can make it impossible to construct a measure of systematic macro-economic factors.

A more extreme application of this class of non-equilibrium multifactor pricing models, in terms of its distance from absolute pricing models formulations, is the *statistical* factor modeling methods. Under this approach there are no *a priori* assumptions made about the factor specifications, nor their loadings. These are treated *ex ante* as unknown parameters and factor values and loadings are identified, instead, from the covariances of asset returns, alone, taken from cross-sectional or historical data. It is a statistical search

for the highest sources of covariation from the raw data, irrespective of its economic interpretation. The intuition behind this method is that algorithms can be created to identify and extract the most relevant, or systematic, information from the raw data represented in a sample of variances and covariances of returns from a set of arbitrary securities. Such algorithms rank statistical factors according to their explanatory capacity of returns and aggregate them as linear combinations of the existing securities returns, in such a way that a relatively limited number of linear factors can explain a large proportion of returns variation and risk premia; additional factors displaying little explanatory capacity. The primary methods used to identify factors and sensitivities are principal components and factor analysis.

Statistical factor models can derive satisfactory results with a smaller number of factors than fundamental and macro-economic models. The results of statistical factor models continue to improve as computational capacity and data quality improve. However, one needs to treat statistical factors as »mathematical artifacts«, which are often difficult to interpret in economic terms, thus limiting its application for valuation, portfolio construction or risk management purposes.

The third school of thought, post CAPM, is represented by those who questioned some of the assumptions and theoretical departures of the CAPM model but still continued to adhere to *absolute asset pricing* methods. This may be called the camp of the *fundamentalist*. They expanded the notion of equilibrium by incorporating additional fundamental information common to every security, such as book-to-market (B/M), leverage or size, that could be introduced in the CAPM formulation together with the market portfolio to improve its explanatory capacity under the recognition that beta was not the only single factor able to capture all asset returns.

Perhaps the most popular of these models is the Fama–French (1993, 1996) three-factor model. Using cross sectional and time series data they find that the size, and book-to-market (B/M) ratios add to the explanation of expected returns provided by market beta when assembled in a linear model. Based on this, they propose a three-factor model for expected returns:

$$E(R_i) - R_f = \beta_{i,M} [E(R_M) - R_f] + \beta_{i,S} E(SMB) + \beta_{i,h} E(HML)$$

Where:

$E(R_i) - R_f$ = expected, excess returns

$\beta_{i,M} [E(R_M) - R_f]$ = excess return on the market, which is the value-weighted market portfolio return minus the risk-free rate, similar to CAPM

$\beta_{i,S} E(SMB)$ = small minus big

$\beta_{i,H} E(HML)$ = high minus low

In this equation, SMB (small minus big) is the difference between return on diversified portfolios of small and big stocks, HML (high minus low) is the difference between return on diversified portfolios of high and low book-to-market values of stocks and the betas are the slopes in the regressions $R_i - R_f$ on $R_m - R_f$ SMB and HML. All these factors should be counted as risk premium factors and, therefore, good candidates for long-term factor investing.

Despite the relative success and early adoption of the model by practitioners, it soon became clear, from the early days of its life, that some relevant fundamental factors were missing from the explanation. In particular, an important and disturbing observation was that returns appeared to display some form of inertia, independent to the three variables included in the original formulation. Stocks that do well, relative to the market in the last three to twelve months, tend to do well for the next few months, and stock that does poorly continues to do poorly. One response to this was provided by Carhart (1997) by adding a momentum factor to the three-factor model, described as the difference between the returns on diversified portfolios of short-term winners and losers.

Though the spirit of the Fama–French model is still alive and well, and has been the source of inspiration for many managers and academics that have improved and refined the fundamental modelization over the last 20 years, it is also true that it is not free from controversy. Specific problems arise, in particular, with the highly debated SMB factor. This has not been observed in the United States since the early 1980s and does not exist outside the U.S. Recent research concluded that its presence in early tests was likely driven by a mistake in how researchers treated missing data for delisted stocks (Shumway and Warther, 1999 and Hsu and Kalesnik, 2014). Other factors appear to produce different results in international markets and different phases of the business

cycle, but in general, it seems fair to say that the B/M factor is the one that has proven a more evident, significant explanatory capacity.

A more recent departure from the Fama–French original three-factor model is their five-factor model (Fama, French, 2014) that is based on the evidence provided by Novy–Marx (2012) and Titman et al. (2003). In this new departure, Fama and French propose the inclusion of two more factors in the original model, one accounting for the difference in profitability between strong and weak companies and another that captures the investment effort made by the firm. Though the intuition on the importance of these new factors is clear, there are already analytical, as well as empirical, problems that obscure the interpretation of the inclusion of these two new variables. In general, the larger the number of factors, the higher the possibility of correlations between them, making it difficult to isolate the true contribution of each individual factor, as is the case with the correlation exhibited between investment and size or investment and returns. In addition, it is not easy to interpret some results, which makes it difficult to correctly identify what we are looking for. There is evidence that firms that significantly increase their capital investments tend to underperform relative to comparables, as if managers were destroying value when offered an abundance of resources (Titman et al., 2003). Whether it is a temporary effect related to the maturity phase of new investments, or a permanent behavioural factor, remains elusive.

And, of course, there is the question about the assumption of rationality of investors, not specific to the Fama–French models, but generic to any quantitative formulation departing from strong behavioural assumptions. Whether it is rationality, consistent return maximization, or arbitrage seeking, there is a set of predetermined, consistent behaviour underpinning the results of all factor models, perhaps with the exception of the statistical ones. Another school of thought in the argumentation of asset pricing, the *behaviouralist*, is not impressed at all by the results of any of these models. It is irrational behaviour that explains why factors earn excess returns, they claim. Investors follow herds, feel fear when markets overreact and often like to chase winners without looking too much at any model. If enough investors exhibit these biases, it can lead to observable factor anomalies, difficult to capture by any factor model, that will be short-lived and unstable. And yes, we have to agree that if not for the entirety of the return variations, unquantified behavioural factors may be playing a role. The problem is that if measuring macro-economic

factors or market returns – what market? – is challenging, behavioural factors may be more problematic, if not impossible to quantify.

As Fama and French concede, *»When one observes a positive relation between expected cash flows and expected returns that are left unexplained by the CAPM or the three-factor model, one can't tell whether it is the result of irrational pricing or a misspecified asset pricing model«* (Fama, French, 2004).

The Current State Of Affairs And The Contemporary Models: A Zoo Of New Factors

Five decades have passed since the identification of the first single-factor model, and during this time the proliferation of research, investment managers, young PhDs, improved data quality and cheaper and increased computing capacity have created an environment where almost every imaginable factor has been tested. From the good old betas to the Google hits, a stock received to find risk premia, managers now run portfolios with 30, 40, even 80, factors, perhaps dangerously approaching the trivial conclusion that returns depend on everything but eluding a formulation of true premia drivers. It was John Cochrane (2011) who referred to this situation as the »zoo« of new factors. Factor inflation erodes the quality of the proposition, but, the fact is, factor investment is far from being a hype and has become a dominant paradigm in portfolio management as a serious stream of action, despite the questionable capacity of many of the myriad of factors commonly used in the practice of portfolio allocation to deliver true premia.

Harvey, Liu and Zhu (2014) investigated the progression of factor discoveries from 1965 to 2013 and found that the number of factors had grown exponentially, and since 2004 at a rate significantly larger than the rate of published papers. Over that last decade, the 164 factors published in papers, double the 84 factors discovered in all previous years. They tested 316 factors, proposed over this decade, including those in working papers, and found that 70% of them had a Sharpe Ratio that was less than 0.5. Of the 296 factors published, somewhere between 160 and 130 would be considered false discoveries according to their testing methodology, roughly 40 – 50%.

Further analysis of factor inflation is provided by Levi and Welch (2014). They analysed 12 factors in 49 industries for a total of almost 600 factors and found that only 51% of them produced positive premia.

One would think that perhaps the proliferation of new factors is reaching the point of marginal decreasing returns and that the tide should start to retreat. But the truth is that the cost of running back tests by the thousands, declines equally exponentially with cheaper and more powerful technology and more indepth knowledge. So the cost of running portfolios with an unmanageable number of factors, many of them often yielding poor results, making investment strategies and their results difficult to interpret. But that may collide with the investors' desire to believe that analytical and computational power can discover even the more subtle driver of premia; no matter how hidden it is, it will be found. And managers don't do much to contradict that view.

A good example of the popularization of factor investing is the smart beta wave. Though it is hard to find an investing expert who likes the term smart beta – it makes William Sharpe »definitionally sick« – its popularity has grown rapidly and there were \$350 million of such denominated smart beta assets at the end of 2014, with an increase of 30% during the year according to Bloomberg. Factors proliferate under the label smart beta, which is not problematic as long as one makes intelligent decisions regarding the separation of factors that are for real from those who add nothing to capture premia.

Factor creation is not the main problem, anymore, and new discoveries will probably not be a problem in the future. For, contrary to what we expect, this trend will continue, it is perhaps time to call for a quantum leap in discipline and a revisitation of the factor selection criteria. It will not do any harm to move for a while from quantity to quality, perhaps focusing the efforts in formulating more compact lists of candidates able to generate return premia.

Hsu and Kalesnik (2014) reviewed the literature and proposed a list of the characteristics that a factor should present to be accepted as truly containing risk premia:

1. The factor was discovered many decades ago; it has survived numerous database revisions as well as extensive out-of-sample data.
2. The factor has been vetted, replicated, and debated in top academic journals over decades.
3. The factor works in non-U.S. countries and regions.
4. The factor premium does not change, materially, due to minor variations in the factor definition/construction.
5. The factor has a credible reason to offer a persistent premium:
 - it is related to a macro risk exposure, or
 - it is related to a deeply rooted behavioural bias that is present in a meaningful fraction of investors, or
 - it is related to an institutional feature that cannot be easily changed.
6. The factor exceeds a more stringent *t*-stat threshold of 3.5 (preferably 4.0) instead of 2.0 to adjust for data-snooping and other biases evidenced by the recent explosion in factor proliferation.

In the remarkable journey factor investing has made during its five decades of existence from the labs of the universities to becoming a dominant stream in investment practice, a phenomenal amount of talent and effort has been deployed in the quest for factors. Its success is unquestionable, if measured by the size of assets managed with the technology and the metrics of the industry structure. The increasing number of managers, academics and investors that are attracted every year, guarantee that we will continue to see new challenges and creativity. Some fundamental questions remain open, a few of them will be answered, for some others, perhaps, we will never reach a definitive conclusion. Models, even the most successful, will not last forever. But there is no doubt, factor investing will continue to offer extraordinary opportunities for an intelligent practice in the times to come.

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Deploying Multi-Factor Index Allocations In Institutional Portfolios

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Executive Summary

This paper is the second in a three-paper series focusing on factor investing. In the first paper, »Foundations of Factor Investing«, we discussed six factors – Value, Low Size, Low Volatility, High Yield, Quality and Momentum – that have, historically, earned a premium over long periods, represent exposure to systematic sources of risk and have strong theoretical foundations. We also discussed how they could be captured through indexation. In this paper, we turn to the question of how institutional investors interested in factor investing may allocate to, and across, factors.

In particular, we introduce a new framework for how institutional investors might consider implementing factor allocations through a passive mandate, replicating a single multi-factor index. We call this type of allocation »multi-factor index allocation«. Multi-factor indexes combine select factor indexes into single mixes, created and controlled by the investor. Multi-factor indexes have, historically, demonstrated four key benefits: diversification, transparency, flexibility and cost-efficiency via reduced turnover.

Most importantly, with regard to diversification, combining factors could have helped offset the cyclicity in single-factor performance, historically. When

multiple factor indexes are combined into a single multi-factor index, diversification across factors has led to:

- Lower volatility and a higher Sharpe Ratio
- Higher information ratios and lower tracking errors
- Less regime dependency over business cycles

Next, we look at how factor allocations fit in the traditional institutional portfolio setting. Factor allocations have the potential to change the landscape of mandate structures by offering a new way to achieve exposure to systematic factors that formerly could only be captured through active mandates. Factor index-based investing can be viewed as active decision-making implemented through passive replication. As such, factor allocations should be tailored to each institution.

The first step is to assess the role of factor investing in the institution's portfolio. The two main dimensions that drive factor investing are the institution's objectives and constraints (i.e. governance structure, horizon, risk budget). For example, those seeking to enhance risk-adjusted returns may be looking for a dynamic allocation (higher return and higher risk), a defensive allocation (moderate return and lower risk), or a balanced allocation (something in between).

Once the institution has established its investment objectives and identified factors that might contribute to these objectives, it must also decide how to structure and implement the factor allocation. The main criteria for deciding which combination of indexes to deploy, depend on the institution's assessment of the tradeoff between investability and factor exposure (which is tied to performance). Indexes with greater investability generally have lower factor exposure and vice versa. In this implementation phase, there can also be significant turnover reduction benefits to combining multiple factors in a multi-factor index. In particular, »natural crossing« effects may reduce turnover, provided that the allocation is structured around a single passive mandate (or multiple mandates structured to replicate the same index, passively) with synchronized rebalancing dates. Since there are different index alternatives with varying levels of exposure versus investability, the appropriate index implementation depends on the institution's objectives and constraints.

Implementing Factors Through Multi-Factor Index Allocations – Multi-Factor Indexes: A New Approach For Institutional Mandates

In a previous paper, »Foundations of Factor Investing«, we discussed why some institutional investors seek exposure to systematic factors and introduced the notion of factor indexes that represent factor returns. We focused on six factors (Value, Low Size, Low Volatility, High Yield, Quality and Momentum) that, historically, have earned a premium over long periods and have strong theoretical foundations. As stated previously, in this paper, we discuss a new framework for how institutional investors might consider implementing factor allocations through a passive mandate, replicating a single multi-factor index, so-called »multi-factor index allocation«. Multi-factor indexes combine select factor indexes into single mixes, created and controlled by the investor.

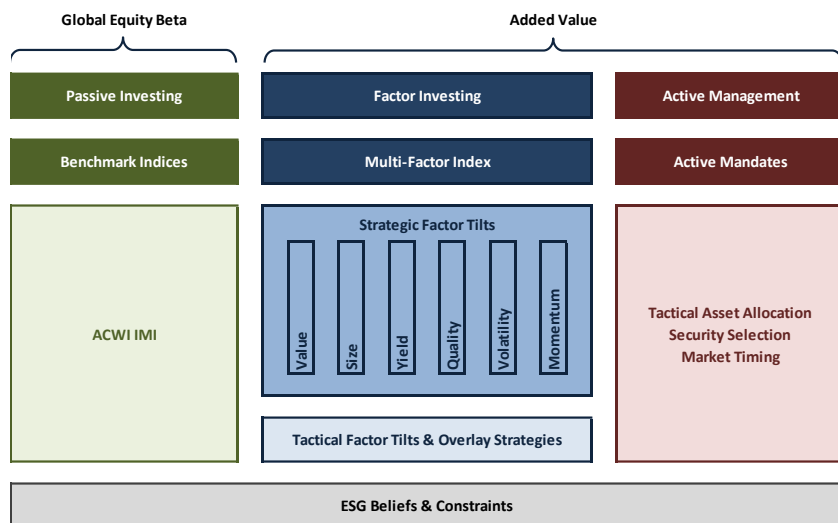
Traditionally, institutional investors structured their allocations around two main sources of return:

- (*Passive*) *Beta*: Based on modern portfolio theory, beta is the return the institution gets from broad exposure to the market, or the full equity investment opportunity set. It is achievable by means of a portfolio that passively tracks the market, represented, typically, by a market capitalization-weighted index. For instance, in a global portfolio, global equity beta is represented by a broad market capitalization-weighted index such as the MSCI ACWI Investable Market Index (IMI).
- (*Active*) *Alpha*: Alpha is the additional return that active management can provide. It is excess return (or value-added) over the market capitalization-weighted index. Traditionally, active managers have sought to identify and capture two types of alpha: market inefficiencies and systematic factors associated with excess risk-adjusted returns.

Factor allocations have the potential to change the landscape of mandate structures by offering a new way to achieve exposure to systematic factors that formerly could only be captured through active mandates. Exhibit 1 shows how we can view these allocations as part of a new category in between traditional

passive mandates, which replicate market cap-weighted portfolios, and active mandates. Factor index-based investing can be viewed as active decision-making implemented through passive replication.⁶

Exhibit 1: Factor Allocations within Institutional Mandates



Multi-factor index allocations offer a new approach for institutional investors to seek factor returns⁷. Their four key potential benefits are:⁸

- *Flexibility*: Institutions have full control over the selection and the weights of individual factor indexes within a multi-factor index and can adjust the strategic factor allocation, dynamically, through time. The most appropriate

⁶ Note that in Exhibit 1, and throughout the paper, we generally refer to factor index allocations through a multi-factor index but a factor index allocation could also consist of only one single-factor index. In this case, the benefits of indexation (transparency and simplicity) would apply, but not the diversification and natural crossing effects.

⁷ Historically, active managers would have provided institutions with exposure to multiple factors. For instance, quantitative active funds can use optimizers to create portfolios with targeted factor exposures. But there are significant potential benefits to an index-based approach (transparency, cost-effectiveness and flexibility).

⁸ Note that the benefits of »Transparency« and »Cost Efficiency« would be potentially applicable for single-factor index allocations as well.

combination of individual factor indexes can be customized to account for institutional constraints (e.g., ESG policies, plan rules, etc.). Operationally, the multi-factor approach provides flexibility as it can be created and managed easily within the passive mandate and without having to change the structure or the terms of the mandate. Because the multi-factor allocation relies on standardized indexes, it allows for the flexibility of employing existing passive instruments such as ETFs for tactical overlays. We view this as a »building block« approach.

- *Transparency*: Multi-factor index allocations provide full transparency regarding the strategy's underlying building blocks. They allow for easy and consistent analysis of the aggregate positions, exposures, and risks of the portfolio along with the individual indexes, all with the same level of granularity.
- *Cost efficiency*: Because multi-factor indexes can be replicated passively, multi-factor index allocations can provide a potentially cost-effective alternative to active funds. Moreover, blending multiple factor indexes in a multi-factor index may create natural crossing opportunities which can reduce turnover and potentially reduce the transaction costs of rebalancing.
- *Diversification*: Factor returns have been highly cyclical, historically, with sensitivity to macro-economic and market forces. They also have underperformed the overall market for long periods of time. However, they do not all react to the same drivers and, hence, can have low correlations between each other. Consequently, multi-factor index allocations have, historically, demonstrated similar premiums over the long run to the individual factors but with milder fluctuations.

In Exhibit 1, the category »Factor Investing« contains both »Strategic Factor Tilts« and »Tactical Factor Tilts and Overlay Strategies«. The former refer to strategic static tilts deployed as a long-term strategy while the latter refers to dynamic allocations in which investors overweight/underweight factor allocations based on their forward looking expectations.

Also in Exhibit 1, »Pure Alpha« can still be provided by active management which comprises value-adding activities that are not captured by passive factor

allocations to indexes. »Pure Alpha« includes stock selection and sector rotation strategies, as well as top-down asset allocation strategies where factor tilts are not driving excess return.

Deploying Factor Allocations

Many institutions have struggled to determine the appropriateness of factors for their own plan, including what role these allocations might play, which factors should be adopted, and how factor indexes can be used.⁹

There are, generally, three main parts to the process for an institution deploying factor allocations:

- Assess the institution's objectives and constraints
- Select candidate factors
- Decide how to structure the implementation

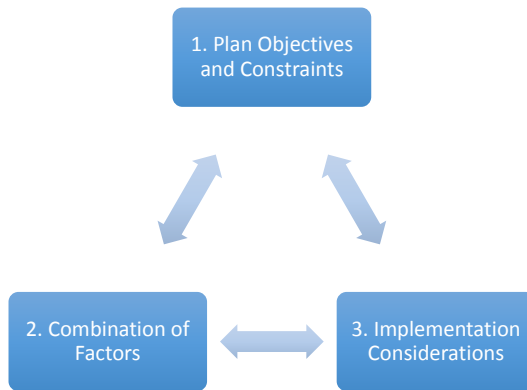
In this framework, the institution must first assess the role of factor investing and what it hopes to achieve. This includes setting the investment objectives, assessing the internal governance structure, and establishing key constraints such as risk tolerance. Once the role of factor investing has been established, the institution can then evaluate candidate factors. As discussed in »Foundations of Factor Investing«, certain factors have strong theoretical foundations and have earned a persistent premium over long periods. The institution must form a belief about whether a factor's long-term historical premium will persist as part of this step.

The third part of the deployment decision process in this framework is implementation. Among the available options for implementation (including via active managers) we focus, here, on passive implementation based on indexes.

9 To add to the difficulty, there has been a rapid proliferation of factor indexes and investment products. Even the breadth of names alone – factor indexes, strategy indexes, smart beta, alternative beta, to name just a few – have challenged even the most sophisticated investors.

Exhibit 2 illustrates the process for an institution to evaluate its objectives, the relevant candidate factors, and the implementation structure. The plan's objectives and constraints inform the combination of the factors chosen and the degree of investability required in the factor allocation. For instance, very large allocations may not be capable of implementation for certain highly concentrated or long-short strategies.

Exhibit 2: Dimensions for Implementing Multi-Factor Index Allocations



Note that the institution's objectives and constraints drive the factor allocation decision (not the indexes themselves); a point that is often lost in the arguments about why one index might be superior to another. Simply focusing on a particular index's rules and construction process, leads to the slippery slope of data-mining. There are thousands of options for generating indexes by varying the weights or criteria for selecting stocks. Any given set of index construction rules can lead to outperformance of the market through statistical sampling alone.¹⁰

Before evaluating any factor indexes, the institution should identify its goals for factor investing and evaluate potential candidate factors based on criteria

¹⁰ In fact, a recent paper by Arnott, Hsu, Kalesnik, and Tindall (2013) argues that any non-price-weighted portfolio will outperform a cap-weighted portfolio because of size and value effects. In our framework, we start with the pure factors first – Value and Size – and choose the most appropriate index based on key metrics such as factor exposure, investability, tracking error, and concentration.

that follow from their objectives and constraints. Choosing a factor index is an implementation decision that transforms the objectives, goals, and factor beliefs into actual allocations. Next, in Section II, we discuss how an institution's objectives and constraints motivate the appropriate choice and blend of factors and factor indexes. In Section III, we look at investability, which is key to determining how to structure the factor index allocation. There are different index alternatives with varying levels of exposure versus investability. Therefore, the appropriate index implementation depends on how the institution prioritizes exposure versus investability which, in turn, is based on the institution's objectives and constraints.

Selecting The Right Blend Of Factors

As an institution seeks the right blend of factors, the starting point is the institution's own profile. Factor allocations should be driven, first and foremost, by the institution's investment objectives and constraints (governance structure, horizon, risk budget, etc.).

- *Objectives:* Different investors have different objectives for factor investing. To put it another way: different investors have different problems which factor investing is meant to address. One institution may seek to enhance risk-adjusted returns, limit downside risk, or improve returns by holding the current level (or market level) of risk (or beta) constant. Another institution might be trying to replicate the performance of certain style managers, for instance, existing value and small cap managers. Different investors will also have different beliefs regarding the persistence of factors.
- *Constraints:* Constraints can also vary among investors. Key constraints are associated with the institution's governance structure which is tied to its investment horizon and risk tolerance. Often, the stronger the governance structure an institution has, the longer the horizon and the higher the risk tolerance it has. Institutions with very strong governance structures and long horizons are better able to withstand long periods of underperformance, and perhaps be compensated for bearing this risk. Funding ratios and the size of managed assets can also affect investability constraints.

1. Introduction

Before selecting factors, the institution should begin by screening out any candidate factors which it does not expect to persist in the future. In other words, ***all candidate factors should be those the institution believe will persist in the future.*** Thus, the ***institution's objectives and constraints, together, drive the choice of factors*** among these candidates. For example, an institution seeking to enhance risk-adjusted returns may be looking for a somewhat more aggressive allocation (higher return and higher risk), a defensive allocation (moderate return and lower risk) or a balanced allocation (something in between).

Exhibit 3: Factors have Historically Exhibited Different Performance Characteristics

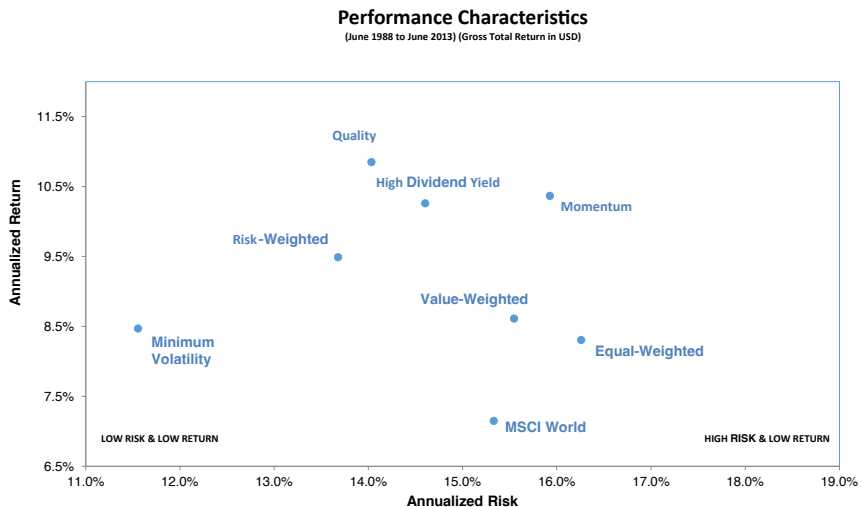


Exhibit 3 shows the historical return and risk characteristics (June 1988 to June 2013) of seven MSCI Factor Indexes capturing »risk premia« factors introduced in »Foundations Of Factor Investing«. These are factors that have earned a premium over long periods and which have solid theoretical foundations (factor indexes based on the MSCI World Index are shown). The Low Volatility factor, represented by the MSCI World Minimum Volatility and World Risk Weighted Indexes, and the Quality factor, represented by the MSCI World Quality Index, both have

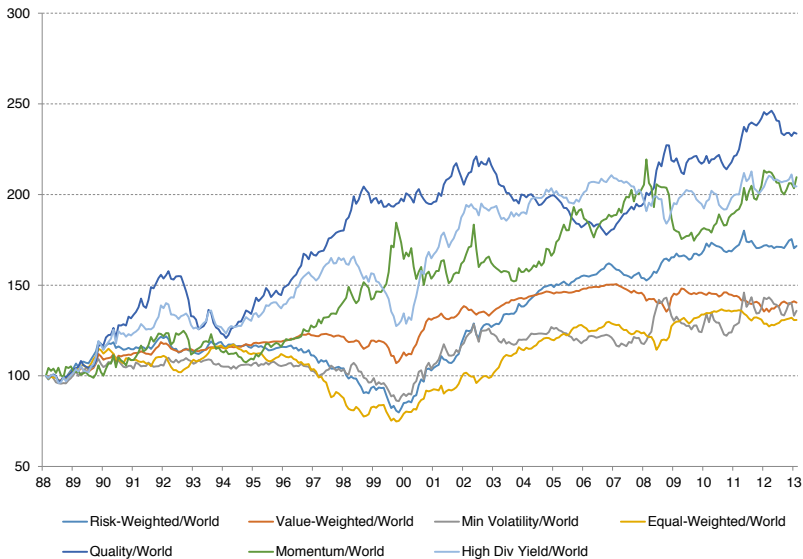
lower risk than the MSCI World Index. The Value and Yield factors represented by the MSCI World Value Weighted and MSCI High Dividend Yield Indexes had risk levels close to the market. The Low Size factor and Momentum factors, represented by the MSCI World Equal-Weighted, and MSCI Momentum Indexes, respectively, have had higher returns. All seven indexes have historically shown higher Sharpe Ratios than the MSCI World Index. Determining the appropriate factors to allocate to, might depend on the institution's return, risk, or Sharpe Ratio objectives.

Correlations Matter When Selecting Factors: The Diversification Effects Of Multi-Factor Index Allocations

Factor selection should also take into account the correlations between factors which affect portfolio-level risk. Factor returns have historically been highly cyclical. Exhibit 4 shows the cumulative returns relative to the market cap-weighted index (MSCI World Index). Each of the factor indexes shown has undergone, as a minimum, two-to-three consecutive year periods of underperformance. Some factors underwent even longer periods; the Small Cap or Low Size factor (captured by the MSCI World Equal-Weighted Index in the Exhibit) went through a six-year period of underperformance in the 1990s.

But while individual factor returns have all been cyclical, their periods of underperformance have not been identical. Systematic factors have historically been sensitive to macro-economic and market forces but not in the same way. For instance, during the period between 2001 and 2007, the Momentum, Value, Low Volatility, and Low Size factors experienced positive excess returns over the market, but the Quality factor experienced negative returns. In contrast, from 2007-on, Quality fared well while Momentum and Value did not. Combining Quality with Momentum and Value, for instance, achieved smoother returns over time, historically, and diversified across multi-year cycles.

Exhibit 4: All Systematic Factors are Cyclical (Cumulative Relative Returns, June 1988 to June 2013)



There is also strong empirical evidence that factors performed differently over various parts of the business cycle. Some factors, such as Value, Momentum, and Size, have, historically, been pro-cyclical, performing well when economy growth, inflation and interest rates are rising. Other factors, such as Quality and Low Volatility, have, historically, been defensive, performing well when the macroenvironment was falling or weak. Similar to macro business cycles, investors may seek factors that perform well under different types of market cycles, such as high/low market volatility. Measuring the sensitivity of factors to macro-economic cycles is an area that still requires further research. For recent research in this area, see Winkelmann et al. (2013).

The historical diversification effects can further be seen in the correlations between the monthly active returns shown in Exhibit 5. Notably, the active returns of the MSCI World Quality and MSCI Momentum Indexes have been very low or negatively correlated with the other factor indexes shown. However, the majority of the correlations range from about 0.30 to -0.30.

Exhibit 5: Correlations Of Relative Monthly Returns (June 1988 to June 2013, USD Gross Returns)

	MSCI World Risk- Weighted	MSCI World Value- Weighted	MSCI World Mini- mum Volatility	MSCI World Equal- Weighted	MSCI World Quality	MSCI World Mo- men- tum	MSCI World High Div. Yield
MSCI World Risk- Weighted	1.00						
MSCI World Value- Weighted	0.61	1.00					
MSCI World Minimum Volatility	0.65	0.14	1.00				
MSCI World Equal- Weighted	0.75	0.63	0.12	1.00			
MSCI World Quality	0.07	0.00	0.24	-0.26	1.00		
MSCI World Momentum	0.04	-0.26	0.16	-0.20	0.38	1.00	
MSCI World High Div. Yield	0.62	0.71	0.51	0.26	0.35	0.04	1.00

When multiple factor indexes are combined into a single multi-factor index, diversification across factors has, historically, led to:

- Lower volatility and higher Sharpe Ratio
- Higher information ratios and lower tracking errors
- Less regime dependency over business cycles

To illustrate, Exhibit 6 shows a multi-factor index where four individual indexes are combined: the MSCI World Quality Index, MSCI World Value-Weighted Index, and MSCI World Momentum Index, and MSCI World Risk-Weighted Index.¹¹ While the returns are a linear combination of the individual indexes, the risk metrics are not. The high Information Ratio of 0.83, substantially higher than the four individual indexes, reflects how well they diversified each other during this period.

Exhibit 6: Combining Multiple Factors Offers Substantial Diversification Effects (May 1999 to September 2013)

	World Standard	MSCI World Quality Index	MSCI World Risk-Weighted Index	MSCI World Value-Weighted Index	MSCI World Momentum Index	Multi-Factor Index
Total Return* (%)	4.2	5.3	8.6	5.5	6.9	6.7
Total Risk* (%)	16.3	14.3	14.6	17.2	16.7	14.9
Sharpe Ratio	0.18	0.26	0.47	0.25	0.33	0.34
Annualized Active Return (bps)		110	440	120	270	250
Tracking Error* (%)		4.5	5.6	3.6	9.0	3.0
Information Ratio		0.25	0.79	0.35	0.30	0.83
Max Rel. Draw-down (Active Returns) (%)		20.5	16.0	10.7	21.6	5.7
Max Rel. Draw-down Period (Active Returns) (in Months)		52	10	9	19	2

* Annualized in USD for the 05/31/1999 to 09/30/2013 period.

** Annualized one-way index turnover for the 05/31/1999 to 09/30/2013 period.

¹¹ The multiple-index combination is rebalanced semi-annually at the same time as the underlying indexes in May and November.

	World Standard	MSCI World Quality Index	MSCI World Risk-Weighted Index	MSCI World Value-Weighted Index	MSCI World Momentum Index	Multi-Factor Index
Max Relative Drawdown (%)		44.5	50.9	57.5	52.5	51.4
Max Relative Drawdown Period (in Months)		16	16	16	16	16

Exhibit 6 also includes maximum drawdown, relative to the MSCI World Index, and the maximum relative drawdown period in months. Both measures capture prolonged periods of underperformance. This measure of risk is equally, if not more, important than traditional measures of risk, such as standard deviation of returns, because it arguably captures »career risk«. Even for institutions with long stated horizons, the investment staff will often be forced to reassess allocations if the portfolio underperforms for too long. The multi-factor index has significantly lower drawdown measures than the individual indexes, historically.

In sum, there are important diversification effects in combining multiple factors, historically. Multi-factor indexes achieved the same historical premium, over the long run, as the individual factors but with milder fluctuations. Actual use cases include a Canadian pension plan, which adopted a combination of MSCI Risk-Weighted, MSCI Quality, and MSCI Value-Weighted Indexes, and a US pension plan which chose a combination of MSCI High Dividend Yield, MSCI Quality, and MSCI Value-Weighted Indexes. These, and other use cases presented later in Section III, further illustrate the benefits of multi-factor indexes.

Considerations For Combining Factor Indexes

Tying all this together, we arrive at the main considerations for selecting the right blend of factors. It starts with the institution's objectives and constraints, its beliefs regarding which factors are likely to persist, and in some cases,

return expectations for the factors. When choosing an appropriate factor combination, the key criteria are risk, correlations with other factors, and performance in different business cycles, as shown in Exhibit 7.

Exhibit 7: Considerations for Combining Factor Indexes

Factor	Historical Risk	Historical Correlation	Historical Business Cycle
Value	Comparable to market	Low with Momentum and Quality	Pro-cyclical
Momentum	Comparable to market	Low with Value, Yield, and Quality	Pro-cyclical
Low Size	Higher than market	Low with Min Volatility, Yield, and Quality	Pro-cyclical
Quality	Lower than market	Low with Value, Size, Yield and Momentum	Defensive
Low Volatility	Lower than market	Low with Value and Momentum	Defensive
Yield	Lower than market	Low with Size, Quality and Momentum	Defensive

Other criteria that can affect factor selection include sources of return as well as return patterns. For example, an institution may prefer income to capital appreciation or prefer factors which imply higher yields. In addition, an institution may be particularly sensitive to the possibility of a prolonged draw-down and seek factors that are less likely to go through multi-year periods of underperformance or, as illustrated earlier in Exhibit 6, blends of factors that minimize prolonged underperformance. Thus, the criteria for choosing factors and combinations of factors could include a variety of characteristics such as *return (including forward looking expectations)*, *risk*, *Sharpe Ratio*, *diversification effects*, *yield levels*, *beta*, *general liquidity characteristics*, *downside risk*, and *risk of prolonged periods of underperformance*.

Exhibit 8 provides examples of how factor allocations can be tailored by the institution to its objectives.¹²

¹² Some institutions may not have explicit performance goals. Rather, they may be seeking ways to make explicit, the tilts that the plan's active managers already take.

Exhibit 8: Factor Allocations Are Based On the Institution's Objectives and Constraints

Sample Objective	Example Allocation (Pure Factors)	Example Index Allocation
Diversified Balanced Mix	Value, Low Volatility, Momentum, Quality	MSCI Multi-Factor Index: Value-Weighted, Risk-Weighted, Momentum, and Quality
Diversified Dynamic Mix	Low Size, Momentum, and Value	MSCI Multi-Factor Index: Equal-Weighted, Momentum, and Value-Weighted
Diversified Defensive Mix	Low Volatility, Value, and Quality	MSCI Multi-Factor Index (MSCI Quality Mix): Minimum Volatility, Value, and Quality
De-Risking with Yield-Enhancement	Low Volatility and High Dividend Yield	MSCI Multi-Factor Index: Minimum Volatility, and High Dividend Yield

In sum, there is no universal factor solution, either in the form of a single factor, or a combination of factors, that is right for all institutions. Actual use cases are helpful in understanding different types of allocations. Several examples are shown in Section III.

Implementation Considerations

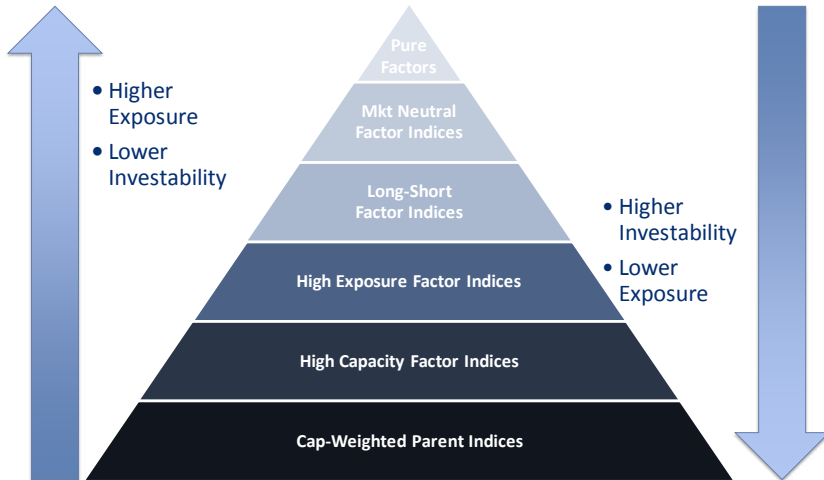
In this section, we discuss in more detail the critical aspects of implementation. We focus, in particular, on a potential framework for how to incorporate the investability dimension in the selection of the individual factor indexes. We also discuss how factor indexes can be combined in a multi-factor index to reduce trading cost by benefiting from potential natural crossing. This last element requires the allocation to be structured around a single passive mandate (or multiple mandates structured to replicate, passively, the same index) with synchronized rebalancing dates.

Understanding The Exposure Vs. Investability Tradeoff

In selecting the individual factor indexes that make up a multi-factor index or in selecting a single-factor index, the most critical point we can stress here is that there is a tradeoff between the strength of the exposure to a factor and the

investability of the strategy that reflects it. There is a range of index alternatives that have varying levels of investability and exposure to a specific factor. Exhibit 9 shows a general framework which visually displays the different index options.

Exhibit 9: Capturing Factors Through Indexation



The most investable index, by definition, is the one whose weights are proportional to free float adjusted market capitalization, at the bottom part of the pyramid. The factors at the top (i.e. the Fama–French or Barra factors) are the theoretical or pure factors that the institution may wish to capture, but that are research, rather than investability, oriented. The closest factor indexes to market capitalization-weighted indexes are High Capacity Factor Indexes. These are indexes that hold all the stocks in the parent index but tilt the market cap weights toward the desired factor. As we move up, High Exposure Indexes hold a subset of names in the parent index and can employ more aggressive weighting mechanisms. The investor who seeks to control active country or industry weights or exposures to other style factors, or who desires to limit turnover, tracking error, or concentration, can use High Exposure Indexes that employ optimization or systematic stock screening. Next, Long/Short Factor Indexes add leverage (e.g., 150/50, 130/30) primarily to hedge out residual exposure to other factors, and lastly Market-Neutral Factor Indexes

are pure long/short Indexes that have zero market exposure.¹³ The leveraged index categories typically employ optimization.

Moving up the pyramid yields lower investability and greater exposure to the pure factor.

Factor Exposure

What do we mean by »factor exposure«? Factor exposure captures the degree to which the index captures the pure non-investable factor. To assess the strength of the factor exposure of a particular index, one can use a factor model (which can be used to calculate any portfolio or index's exposure to the factors in that model). Factor exposure is typically expressed as standard deviations away from the cap-weighted average of the market.¹⁴ Note that for most factor models, which typically employ linear exposures and regressions, the exposure of an index to an underlying factor is just the weighted exposure of the individual stocks' exposures to the factor in question. (Factor exposure is also often called »signal strength« in the language of quantitative equity managers.)

As one moves up the pyramid, typically higher levels of factor exposure are achieved which translates into higher returns if factor returns scale with exposure, and as long as incidental bets are controlled.¹⁵ This last point is important because more concentrated portfolios often have larger sector and country active weights, or even unintended exposures to factors other than the factor of interest. If these are not controlled, they can incidentally negatively affect returns, detracting from the intended factor return.

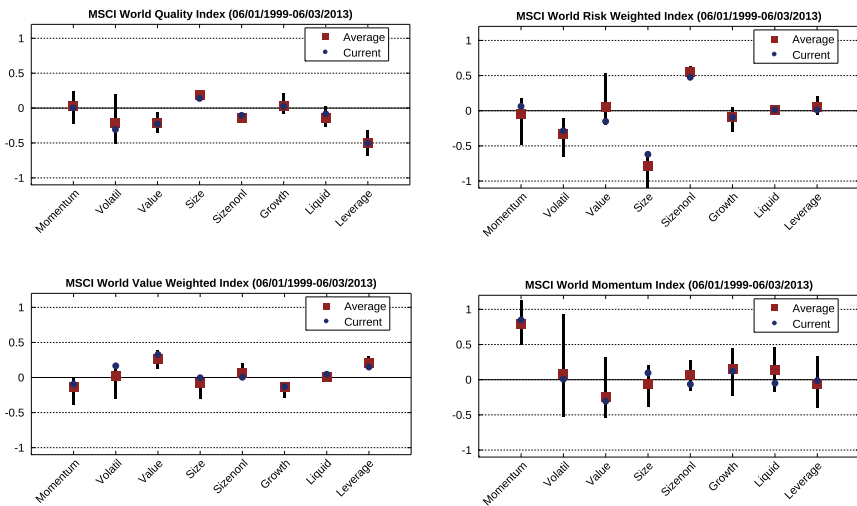
13 Active country and sector weights will be zero and exposures to all other style factors will be zero.

14 An active/relative exposure of 0.25 to the Barra Value factor can be interpreted as the portfolio or stock's Value characteristics being 0.25 standard deviations higher than the market cap-weighted benchmark.

15 One can have higher exposure to the desired factor but the positive impact on returns may be negated by other exposures (either to other factors or countries or sectors). Controlling exposures to other factors is possible through optimization. For example, in a Value factor index, one might want to neutralize exposures to other factors such as Low Size and Momentum.

In Exhibit 10, we illustrate factor exposures using the Barra multi-factor models which estimate factor portfolios using multivariate regressions and have the advantage of specifying factors with little co-linearity. As an example, Exhibit 10 shows the active exposures (relative to the MSCI World Index) of four of the factor indexes. In all cases, the indexes have significant exposure, with the expected sign, to the most relevant pure factors. The MSCI World Value-Weighted Index has an exposure of 0.28 to the Barra GEM2 Value Factor, which is above the usual 0.20–0.25 rule of thumb for statistically significant exposures. In some cases, an index may have significant exposure to factors other than the intended factor. For example, the MSCI World Risk-Weighted Index has a significant small cap bias as seen by the large negative exposure to the GEM2 Size factor. In this case, the small cap bias contributes to the excess return of the World Risk-Weighted Index. Institutional investors should be aware of these potential secondary exposures and understand/manage them appropriately.

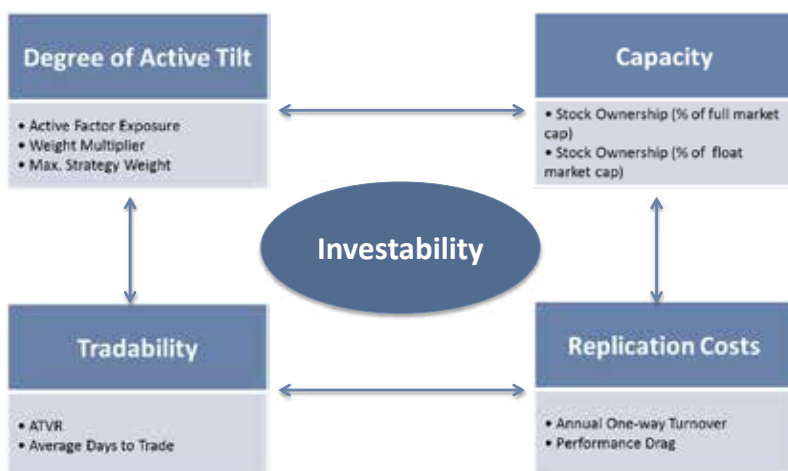
Exhibit 10: Factor Exposures (Factor Exposures For Select World Factor Indexes Using The Barra GEM2 Model, Average And Current Exposures, June 1999 To June 2013)



Investability

What do we mean by »investability«? Investability refers to how liquid and tradable the index is. It also refers to how scalable the allocation to an index replicating vehicle might be. There are multiple dimensions to investability. As shown in Exhibit 11, they include Tradability/Liquidity, Turnover/Cost of Replication and Capacity – for a given degree of active tilt.¹⁶ Tradability/Liquidity quantifies how liquid the stocks are in the index replicating portfolio and how tradable the portfolio is. Metrics include days to trade individual stocks at rebalancings and during the initial construction, along with days to trade a certain portion of the portfolio (given a certain portfolio size and a set limit to the amount traded on a single day). Turnover/Cost of Replication measures the turnover of the index at rebalancing, which scales with trading costs. The higher the turnover, in general, the higher the cost of trading. Capacity quantifies (for a given portfolio size) the percentage of a stock's free float or full market capitalization that the portfolio would own. The degree to which a portfolio is »active«, relative to the index, has been traditionally used by many active asset managers to characterize their active strategies' performance. Metrics, such as active share and maximum strategy weight, capture this.

Exhibit 11: Dimensions for Investability



¹⁶ These dimensions were first discussed in Bambaci et al. (2013).

Note that some indexes may score well on all four dimensions; the MSCI Value-Weighted Indexes, for instance, have, historically, exhibited low turnover, high capacity, and good tradability. Others may have good capacity and tradability but incur high turnover (i.e. Momentum).

Investability Vs. Exposure

Since, as we have seen, indexes nearer the top of the pyramid are less investable and less liquid but have greater factor exposure, there is an *unavoidable tradeoff* between the purity or exposure of a factor index and the investability of a factor index. One can, generally, only achieve purer factor exposure by sacrificing investability and being willing to take on greater amounts of active risk and complexity. The appropriate index thus depends on the institutional investor's own preferences for factor exposure and investability. Institutions must make a self-assessment of where they desire to be on the pyramid.

It is also important to note that institutions, typically, care about tracking error, or risk relative to the market cap-weighted parent index. In particular, many plans have active risk budgets at the plan level.¹⁷ As we move up the pyramid, tracking error, generally, increases. Plans with low tracking error targets may want to limit the discussion to the lower end of the pyramid while those with higher tracking error limits may consider options further up the pyramid.

Exhibit 12 shows characteristics of the MSCI Factor Indexes over the period June 1988 to June 2013. Higher capacity indexes typically hold a broad set of names (i.e. all the names in the broad market parent index) and are weighted with investability in mind. As previously discussed in Bambaci et al. (2013), the MSCI Value-Weighted Indexes effectively employ a weighting scheme that combines a score based on value characteristics and market capitalization, and are an example of a high capacity index.

As illustrated in Exhibit 12, the MSCI World Value-Weighted Index has the lowest active risk (tracking error) and very low turnover among the indexes

¹⁷ Many institutional investors have a maximum (or target) level of desired risk, usually in the form of return standard deviation, but sometimes gauged by downside measures such as maximum drawdown or expected shortfall.

shown. Other »Weighted« indexes (all of which hold all the names in the parent index) also exhibit relatively low tracking errors and turnover. The other indexes (the MSCI Momentum Indexes, MSCI Quality Indexes, and MSCI Minimum Volatility Indexes) are more concentrated indexes, holding only a subset of the names in the parent index. These indexes exhibited higher tracking errors and lower levels of investability. (The MSCI Minimum Volatility Indexes are turnover constrained to 20% but other measures of investability are more similar to the MSCI Momentum and MSCI Quality Indexes.)

Exhibit 12: MSCI World Factor Indexes (Main Characteristics, June 1988 to June 2013)

Index	Factor Exposures*	Total Return	Total Risk	Active Return	Active Risk	Annual Turnover	Pair-wise Correlation
MSCI World	--	7.1	15.4	0.0	0.0	3.9	NA
MSCI World Equal-Weighted	Size	8.3	16.3	1.2	5.2	31.8	0.22
MSCI World Minimum Volatility	Volatility	8.5	11.6	1.4	6.7	20.0	0.30
MSCI World Value-Weighted	Value	8.6	15.6	1.5	3.6	20.3	0.30
MSCI World Risk-Weighted	Size, Volatility	9.5	13.7	2.4	5.3	27.2	0.46
MSCI World Quality	Growth, Leverage	10.9	14.0	3.8	5.9	27.6	0.13
MSCI World Momentum	Momentum	10.4	15.9	3.3	8.5	127.5	0.03
MSCI World HDY	--	10.3	14.6	3.2	6.5	22.0	0.41

* In the column »Factor Exposures« we show the Barra Global Equity Model (GEM2) factors which are statistically significant on average ($>+/- 0.20$), with the expected sign, since December 1997. Note that there is no »Yield factor« in the GEM2 Model. Instead, Yield is a component (with a weight of 10%) in the GEM2 Value factor. Turnover reported is the average annual one-way turnover based on history from June 1988 to June 2013.

Reducing Trading Costs By Leveraging The Benefits Of Natural Crossing

In addition to the investability dimensions we have discussed so far, investors should also consider the potential to reduce trading costs at each rebalancing through operational efficiency. As we discussed in Section II, historically, there have been significant diversification effects when combining multiple factors. In the implementation phase, there can also be significant investability benefits to combining multiple factors in a multi-factor index.

Combining factor indexes may reduce turnover from »natural crossing« effects. On the index rebalancing dates, the composite index would be rebalanced back to its target weights (e.g., 50/50) and turnover may be reduced as a company deleted from one factor index might be added as a constituent of another factor index. Take, for example, a stock whose price is falling over time. As the price falls, it may drop out of a Momentum Index but the lower price could push the stock into a Value Index. Those shares which overlap the two indexes would be internally crossed. This »natural crossing« leads to lower index turnover and, by implication, lower transaction costs in a portfolio replicating the index.

The historical effects of natural crossing are shown in Exhibit 13. In this example, we show a blend of the MSCI World Quality Index, MSCI World Risk-Weighted Index, MSCI World Value-Weighted Index, and MSCI World Momentum Index. The four indexes are equally weighted and rebalanced, semi-annually, at the same time the underlying indexes are reconstituted (note that the rebalancing for the individual indexes and the rebalancing across indexes needs to be synchronized for the natural crossing to take place). The annual turnover for the individual indexes is 22.98%, 22.04%, 18.30%, and 89.62%, respectively. If these four indexes were replicated separately, their combined turnover would be 40.81%. When they are replicated as a single portfolio in a single mandate, the combined turnover is significantly lower, at 31.91%. The turnover declines by 8.9 percentage points. What does this mean in terms of trading costs in index replicating portfolios? If trading costs are 50 basis points (a relatively conservative assumption for globally developed market equities), the round-trip trading costs would be 41 basis points for the separately managed portfolio, and 32 basis points for the combined multi-factor

index-based portfolio. The latter option saves the investor close to 9 bps in transaction costs.

Exhibit 13: Crossing Benefits Resulted in Lower Turnover and Lower Trading Costs (Simulated Turnover of Separate and Combined Equally Weighted Allocations to Select MSCI Factor Indexes)

	MSCI World Quality Index	MSCI World Risk-Weighted Index	MSCI World Value Weighted Index	MSCI World Momentum Index	Separate Mandates (A)	Combined Mandates (B)	Reduction (A) - (B)
Turnover (%)	22.98	22.04	18.30	89.62	40.81	31.91	8.90
Performance Drag in bps (at 25 bps)*	11.49	11.02	9.15	44.81	20.40	15.95	4.45
Performance Drag in bps (at 50 bps)*	22.98	22.04	18.30	89.62	40.81	31.91	8.90
Performance Drag in bps (at 75 bps)*	34.47	33.06	27.45	134.42	61.21	47.86	13.35

Annualized for the 05/31/1999 to 9/30/2013 period.

** Performance drag aims to represent the total two-way annualized index level transaction cost assuming various levels of security level transaction cost.*

Our conclusion here is that these natural crossing effects may often be overlooked and deserve consideration given the potential additional savings.

Multi-Factor Index Allocations: Examples

The right blend of factors will depend on the institution's preferences for various aspects of performance (return, risk, correlations, etc.), investability, and factor exposure, which in turn reflects the institution's objectives and constraints.

Actual use cases can be helpful in understanding how institutions have actually addressed these issues in adopting multi-factor index combinations.

Based on real use cases, in the first example, we show a strategic or long-term static allocation that is designed to be well diversified. Factor indexes in this example are the MSCI Value, MSCI Momentum, MSCI Risk-Weighted and MSCI Quality Indexes. The four factors are implemented as a single composite multi-factor index that is rebalanced, semi-annually. The index allocation is executed as a passive internal mandate.

The second example focuses on an allocation that provides lower absolute volatility with higher yield. The desire to »de-risk« is driven by the institution's projections of a bearish, low growth market. At the same time, the institution seeks to achieve higher yields while de-risking. This allocation is implemented as a passive external mandate on a multi-factor index combining Low Volatility via the MSCI Minimum Volatility Indexes and Yield via the MSCI High Dividend Yield Indexes.

One additional use case is an extension of the second example. The »core portfolio« in the second use case (MSCI Minimum Volatility and MSCI High Dividend Yield Indexes) can also be augmented by tactical factor allocations to factor indexes such as MSCI Momentum, MSCI Quality, MSCI Value-Weighted and MSCI Equal-Weighted Indexes. These exposures could be adjusted over time based on forward looking views and deployed via four exchange-traded funds tracking the MSCI Indexes. An external consultant or active manager could play a role in advising on the tactical overlay decision.

How do institutions in practice arrive at allocations like these? Institutions must evaluate a number of key dimensions that we have discussed already in this paper – Performance (Risk, Returns, etc.), Factor Exposure, Investability, and the Effects of Combining Multiple Indexes. Exhibit 14 summarizes a few of the key dimensions that might help the institution form a view on different combinations.

In order to help institutions understand whether their objectives are being met, given various combinations of factor indexes, MSCI has developed IndexMetrics, a structured framework for the analysis of multi-factor blends. The next paper in this series »Introducing MSCI IndexMetrics« describes this framework in greater detail.

Exhibit 14: Key Metrics for Evaluating Different Combinations of Factor Indexes in Structuring a Multi-Factor Index Allocation

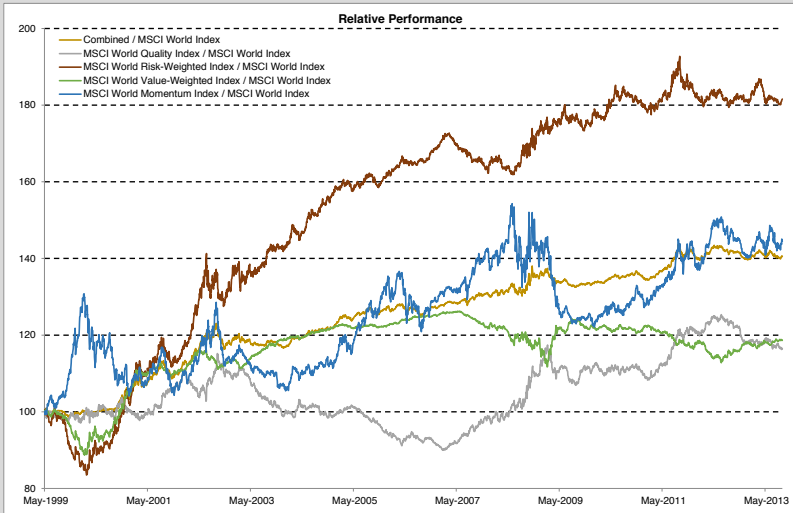
Performance	Exposure	Investability	Combination
– Total Returns, Total Risk, Sharpe Ratio – Active Returns, Tracking Error, Information Ratio – VaR, Expected Shortfall – Maximum Drawdown (percentage, length) – Relative Maximum Drawdown – Years of Consecutive Underperformance	– Active Factor, Sector, Region Exposures – Relative Valuation and Fundamental Ratios	– Liquidity – Cost of Replication (Turnover) – Capacity – Concentration	– Diversification of Returns – Turnover Reduction

Example #1: Strategic Long-Term Risk-Adjusted Return

Strategic Allocation:

- MSCI World Value-Weighted Index 25%
- MSCI World Risk-Weighted Index 25%
- MSCI World Quality Index 25%
- MSCI World Momentum Index 25%

As shown in Exhibit 15, the four indexes exhibited significantly different returns over various sub periods. They have, historically, provided high levels of diversification.

Exhibit 15: Performance Using Historical Returns (May 1999 to September 2013)

The result of combining the four indexes is a balanced portfolio which exhibited return enhancement at lower risk levels than the market, historically.

Exhibit 16: Performance Using Historical Returns (May 1999 to September 2013)

Historical Gross Total Return, USD	MSCI World	MSCI World Quality Index	MSCI World Risk-Weighted Index	MSCI World Value-Weighted Index	MSCI World Momentum Index	Combined
Total Return Performance						
Total Return* (%)	4.2	5.3	8.6	5.5	6.9	6.7
Total Risk* (%)	16.3	14.3	14.6	17.2	16.7	14.9
Return/Risk	0.26	0.37	0.59	0.32	0.41	0.45
Sharpe Ratio	0.18	0.26	0.47	0.25	0.33	0.34
Active Return Performance						

Historical Gross Total Return, USD	MSCI World	MSCI World Quality Index	MSCI World Risk-Weighted Index	MSCI World Value-Weighted Index	MSCI World Momentum Index	Combined
Active Return* (%)	0.0	1.1	4.4	1.2	2.7	2.5
Tracking error* (%)	0.0	4.5	5.6	3.6	9.0	3.0
Information Ratio	N/A	0.25	0.79	0.35	0.30	0.83
Trading Costs / Investability						
Weighted Average Days to Trade***	0.01	0.04	0.04	0.01	0.18	0.03
Turnover** (%)	3.1	23.0	22.0	18.3	89.6	32.0
Performance Drag in bps (at 50 bps)	3.1	23.0	22.0	18.3	89.6	32.0

* Annualized in USD for the 05/31/1999 to 09/30/2013 period.

** Annualized one-way index turnover for the 05/31/1999 to 09/30/2013 period.

*** Average of last four index reviews ending 09/30/2013. Assuming a fund size of USD 10 bn and a maximum daily trading limit of 20%.

Example #2: De-Risking With Yield Enhancement

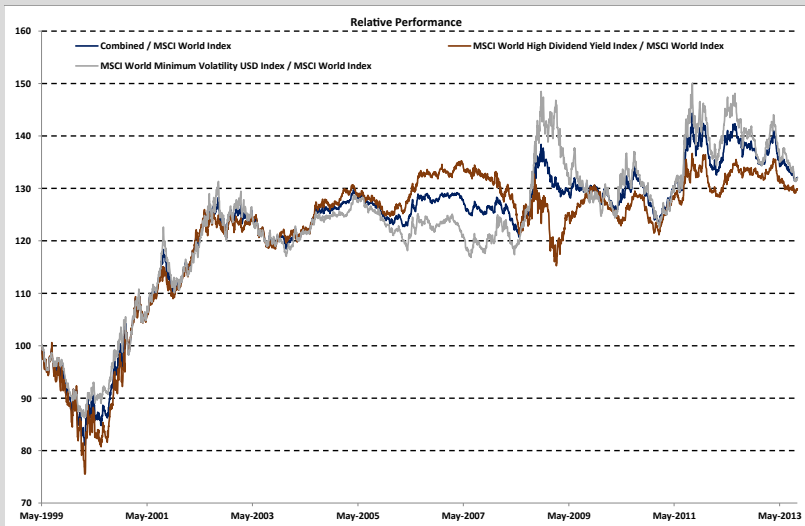
Strategic Allocation:

- MSCI World Minimum Volatility Index 50%
- MSCI World High Dividend Yield Index 50%

Some institutional investors have sought to enhance yield in recent years and at the same time reduce overall volatility. As shown in Exhibit 17, combining the MSCI World High Dividend Yield and MSCI World Minimum Volatility Indexes would have improved the historical performance of the portfolio over

the May 1999 to June 2013 period, significantly, with overall lower volatility. Meanwhile, the average dividend yield for this period was 3.3% for the multi-factor index compared to 2.2% for the market cap-weighted parent MSCI World Index.

Exhibit 17: Performance Using Historical Returns (May 1999 to September 2013)



The result of combining a high yield factor index with a low volatility index is a portfolio which has, historically, exhibited substantial risk-adjusted return enhancement, as seen below:

Exhibit 18: Performance Using Historical Returns (May 1999 to September 2013)

Historical Gross Total Return, USD	MSCI World Index	MSCI World High Dividend Yield Index	MSCI World Minimum Volatility (USD) Index	Combined
Total Return Performance				
Total Return* (%)	4.2	6.1	6.3	6.3
Total Risk* (%)	16.3	16.1	11.4	13.5
Return/Risk	0.26	0.38	0.55	0.46
Sharpe Ratio	0.18	0.30	0.37	0.33
Active Return Performance				
Active Return* (%)	0.0	1.9	2.0	2.0
Tracking error* (%)	0.0	6.1	7.9	6.2
Information Ratio	N/A	0.31	0.26	0.33
Trading Costs / Instability				
Weighted Average Days to Trade***	0.01	0.4	1.2	0.5
Turnover** (%)	3.1	20.4	27.1	23.1
Performance Drag in bps (at 50 bps)	3.1	20.4	27.1	23.1
Yield				
Dividend Yield (%)****	2.3	4.0	2.6	3.3

* Annualized in USD for the 05/31/1999 to 09/30/2013 period.

** Annualized one-way index turnover for the 05/31/1999 to 09/30/2013 period.

*** Average of last four index reviews ending 09/30/2013. Assuming a fund size of USD 10 bn and a maximum daily trading limit of 20%.

**** Monthly averages for the 05/31/1999 to 09/30/2013 period.

Conclusion

In this paper, we have discussed a framework for how institutional investors might consider deploying factor allocations based on factor indexes. The framework comprises three key steps. In the first step, the institution assesses the role of factor investing in its portfolio. The second step identifies which factor(s) are appropriate for the institution's portfolio. Finally, the third step implements the factor index allocation. This includes structuring the portfolio to take into account potential diversification effects between factors and the institution's preferences for investability and factor exposure. Factor allocations can play a variety of roles in the investment process, depending on the objectives and constraints of the investor.

Because they reflect systematic factors that respond to macro-economic and macro market forces, factor indexes can underperform the overall market for periods of time that may exceed an investment committee's patience. However, many of these factors respond differently to macro-economic and macro market forces, so they have, historically, low correlations which may yield strong diversification effects for combining multiple factors in an allocation. We demonstrated how combining factor indexes in a »Multi-Factor Index« captured these diversification effects as well as additional benefits such as lower turnover as a result of internal crossing.

In the next paper in this series, *»Introducing MSCI IndexMetrics: An Analytical Tool for Factor Investing«*, we describe the concepts discussed here in this second paper, more precisely. We define the metrics used in this paper as part of the overall criteria for identifying factors, selecting factor indexes and selecting combinations of factor indexes. These metrics are all part of MSCI's new analytics engine – »MSCI IndexMetrics« – which transforms the concepts into a set of actionable and concrete quantitative metrics.

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Remy Briand holds an MSc in Computer Sciences from INSA (Lyon) and an MBA from HEC (Paris).

As Global Head of Research at MSCI, Remy Briand is responsible for research across all MSCI products, including indexes, risk models and ESG (environmental, social and governance) ratings. He leads a global team of 150 researchers designing products and services that help institutional investors make more informed investment decisions.

In 2010, Remy led the restructuring of the various ESG businesses acquired through MSCI's purchase of the RiskMetrics Group. MSCI ESG Research Inc. was created with a mission to help investors integrate ESG factors into their investment processes. As Head of Index Research from 2007 to 2010, Remy initiated an innovation programme which led to the creation of the MSCI Frontier Markets Indexes, MSCI Minimum Volatility Indexes and MSCI Factor Indexes.

Prior to joining MSCI in 2001, Remy was an equity portfolio manager and head of research at Credit Lyonnais Asset Management in Paris. He began his career as a private equity analyst at Credit Lyonnais.

Over his 25-year career as an investor, independent thinker and business leader, Remy has gained unique insights on topics such as global investing, emerging markets, sustainable investment and financial innovation. He regularly shares his views at industry conferences and with financial news media.

Dimitris Melas is a Chartered Financial Analyst (CFA) and holds an MSc in Electrical Engineering, an MBA in Finance, and a PhD in Financial Mathematics from the London School of Economics.

As Global Head of New Product Research, Dimitris Melas and his team are responsible for all research efforts to enhance existing MSCI indexes and to develop new index methodologies. The team is also responsible for conducting research, publishing articles, and giving client and conference presentations to promote the use of MSCI Indexes in the institutional investment process.

Prior to joining MSCI in 2006, Dimitris worked at HSBC Asset Management where he was Head of Research and Head of Quantitative Strategies, leading a team of analysts and fund managers that provided analytical research to the global asset management business and managed several quantitative equity portfolios. At HSBC, he was also a member of the Global Investment Strategy Group, a committee of CIOs and other senior investment professionals responsible for setting investment strategy and asset allocation policy for all balanced and multi-asset class portfolios managed by HSBC Asset Management.

Dimitris has published several research papers in leading academic and industry journals. His paper »Efficient Replication of Factor Returns« published in the Journal of Portfolio Management was voted »Best Index-Related Research Paper« at the 6th Annual William F. Sharpe Indexing Achievement Awards. Dimitris is frequently invited to speak at investment conferences and has contributed several articles and interviews in leading financial media including the Financial Times, Bloomberg and CNBC.

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As Head of Equity Applied Research for the Americas, Raman Aylur Subramanian conducts research on MSCI product applications, and presents the results in interactive sessions with clients. Raman joined MSCI in 1999, and has been working in a variety of research roles. Raman has established close relationships with clients and gained an in-depth understanding of their needs with respect to the MSCI Indexes.

Prior to joining MSCI, Raman worked in the Indian petroleum industry where he worked on sales and product management.

Madhu Subramanian is Executive Director at MSCI where he co-leads New Product Development initiatives within Index Research. Madhu and his research team are responsible for developing new index methodologies and enhancing existing MSCI Indexes. The team is also responsible for conducting and publishing leading edge research on topics covering global investing, Risk Premia Indexes and Economic Exposure based products to promote the use of MSCI products in the institutional investment process.

Mr. Subramanian is a CFA charter holder and holds a bachelor's degree in Mechanical Engineering in addition to a Master's degree in Management. He brings nearly 16 years of experience spanning roles in corporate finance, risk and investment management. Madhu has published several papers on Emerging Markets investing and Risk Premia Indexes in leading industry journals such as Journal of Indexes, Journal of Alternative Investments and Journal of Index Investing.

2. Choosing Right Factors

Index Performance In Changing Economic Environments A Macro-Economic Framework

Abhishek Gupta
Altaf Kassam
Raghu Suryanarayanan

MSCI

Executive Summary

In this paper, we use our MSCI factor and sector index data and MSCI Macro-economic and MSCI Asset Pricing models to explore the impact of macro-economic scenarios on relative index performance and allocation to these indexes in a structured way. We build on:

1. Historical analysis using our 40+ years' historyⁱ⁸ of MSCI factor and sector indexes, and
2. Long-term analysis based on forecasts from MSCI Macro-Economic and MSCI Asset Pricing Models.

Our results show Industrials, Consumer Discretionary and Information Technology have been the three most cyclical sectors (highest correlations on average to the business cycle measured by the OECD's Composite Leading Indicator, CLI); and Health Care, Consumer Staples, and Utilities the most defensive (most negative correlations). In factor indexes, Equal Weighted and Value Weighted have been the most cyclical, and Minimum Volatility and Quality the most defensive.

ⁱ⁸ Please refer to Appendix I for details of the start dates and simulation dates of the index data used.

The models forecast that the MSCI Momentum, Value-Weighted, and Equal-Weighted Factor Indexes could exhibit the highest active returns following a large positive shock to trend growth in Developed Market (DM) economies (most »growth sensitive«). They also model that Minimum Volatility, High Dividend Yield and Quality could outperform following a large negative shock (»growth hedging«). Materials, Consumer Discretionary and Financials are the most growth sensitive sectors in the model, while Consumer Staples, Health Care, and Utilities are the most growth hedging. Here, short-term historical performance and long-term, model-based performance concur.

In our historical analysis we add a measure of inflation (CPI) and divide history into periods where CLI/CPI are rising/falling separately. The results are intuitive for the CLI, especially for the Minimum Volatility Factor Index and defensive sectors, although less conclusive for CPI. Our model-based analysis looks at the long-term sensitivity of the MSCI factor and sector indexes to inflation. Here, results differ from our historical analysis as inflation impacts equities in different ways over the short and long-term. First, higher inflation, generally, has a negative impact on future long-term real GDP growth. Thus, Small Caps may see short-term outperformance as inflation rises, but be relatively more impacted by inflation in the long-term, as they are more sensitive to real GDP growth. Secondly, factors such as High Dividend Yield, and sectors such as Utilities, are generally long duration with stable nominal cash-flows. Higher inflation could have a more negative impact on their real cash-flows over time.

We classify regimes, historically, through a bivariate framework of CLI/CPI rising/falling, and examine the performance of our indexes across these regimes. We found Momentum and Equal-Weighted Factor Indexes performed best in high growth regimes, and Minimum Volatility showed the largest outperformance in the presence of low growth and high inflation. Defensive sectors have outperformed best during periods of low economic growth and low inflation. Finally, we show the historical performance over the last 40 years of the MSCI factor and sector indexes through regime transitions. We found that when growth has remained positive or strengthened then the Equal-Weighted Factor Index has historically outperformed and Minimum Volatility underperformed. On the other hand, in an environment of slowing growth, an investment in Quality has generally outperformed, and when coupled with low inflation, Minimum Volatility has outperformed. In sectors, Utilities have generally

underperformed in environments of strong growth, but, when growth has fallen, a Consumer Staples overweight has historically outperformed. If this drop in growth comes with increasing inflation, Energy has been the standout performer.

Our frameworks and models have important implications for asset allocation: deviations away from the market cap portfolio could depend on an investor's macro-economic views and tolerance for macro-economic uncertainty. For an uncertainty-tolerant investor, assuming DM economic growth returns quickly to its trend, the models indicate allocating towards indexes that are the most sensitive to economic growth. In contrast, if slow growth and low inflation are believed to persist, the models indicate allocating towards the least growth sensitive indexes¹⁹.

Introduction

From »Abenomics« to »taper tantrums«, soaring home and energy prices igniting inflation to a lack of consumer spending and driving deflation, a recurrent theme of the last few years among investors has been concern over the changing state of the economy and its impact on their investments.

As a result, institutional investors have started explicitly accounting for macro-economic conditions in their asset allocation decisions and, as part of that shift, have tried to create a process for designing macro-economic-sensitive portfolios. Our paper attempts to give this process a framework, building on past (Kouzmanenko and Nagy, 2009) and recent (Winkelmann et al., 2013) MSCI research.

The two pillars we have built this framework on are:

1. Historical analysis using our 40+ years' history²⁰ of MSCI factor and sector indexes to show to what degree economic intuition has been borne out historically in the short-term, and

19 The results presented in this paper are based on the MSCI Macro-Economic, Asset Pricing and Asset Allocation Models. The models are statistical, data driven models, and do not contain investment choices or recommend any specific investments.

20 Please refer to Appendix I for details of the exact start dates and simulation dates of the index data used.

2. long-term analysis based on forecasts from the MSCI Macro-Economic and MSCI Asset Pricing Models.

We base our historical analysis on the observation made in our December 2013 paper, »Deploying Multi-Factor Index Allocations In Institutional Portfolios«, that factor returns have historically been highly cyclical and that their periods of underperformance have not been identical. Systematic factors have historically been sensitive to macro-economic and market forces but not in the same way. So, there seems to be strong empirical evidence that factors have performed differently over different parts of the business cycle. Some factors such as Value, Momentum, and Size have historically been pro-cyclical, outperforming when economic growth and volatility are rising. Quality and Low Volatility have been more defensive, outperforming in a weak macro environment.

We recognize that asset allocation and investment policy should reflect forward-looking views. An emerging consensus in the Macro-Finance academic literature is that macro-economic risks are persistent shocks to trend growth and inflation. This literature suggests that investors should care about quarter-to-quarter shocks to macro variables (such as real GDP growth or inflation) only because they may carry signals about persistently higher (or lower) growth in the future. Very large shocks (either positive or negative) may signal the potential for meaningful changes in a trend more than relatively muted shocks. Although recently reported growth and inflation figures have been muted, many institutional investors are concerned about a very significant positive/negative shock sometime in the future that is indicative of significantly higher or lower average growth and inflation rates in the future.

Our analytic framework and models suggest that portfolios differ in their sensitivity to trend growth and inflation shocks. Nominal bonds are driven, predominantly, by inflation risk, although real growth risk also matters. In contrast, equities are mostly driven by real economic growth risk. The impact of inflation on equities could depend on both the type of equity portfolio and the length of investment horizon. Within bonds, high duration bonds could be more impacted by macro shocks relative to short duration bonds. Equity strategies could also exhibit varied sensitivity to real growth shocks. For example,

the models suggest that Value, Small Cap, and cyclical sectors could be more sensitive than Growth, Large Cap, and defensive sectors.

Designing macro-sensitive portfolio strategies relies on the definition of macro-economic risk and measuring the link between asset prices and this risk. To this end, we have developed a framework and models to provide institutional investors with a structure to:

- Classify investment strategies along their short-term and long-term exposures to macro-economic shocks
- Generate scenarios of macro-economic factors, such as GDP and inflation
- Evaluate the impact of macro-economic scenarios on strategy returns and asset allocation.

The rest of the paper proceeds as follows:

We start by classifying the MSCI factor and sector indexes based on their historical short-term and model-based long-term responses to economic growth and inflation. Next, using historical data, we classify economic regimes through a bivariate framework which groups economic regimes into four quadrants, depending on whether the CLI and the CPI are rising or falling. We, then, examine the differential performance of the MSCI Index universe across these regimes and the relative performances of the indexes through transitions from one regime to another.

Finally, we apply the MSCI Asset Pricing and Asset Allocation Models to assess the likely performance of these indexes under statistically plausible macro-economic scenarios and to derive combinations of MSCI factor and sector indexes that could potentially benefit from, or offer protection in, changing economic regimes²¹.

21 The results presented in this paper are based on the MSCI Macro-Economic, Asset Pricing and Asset Allocation Models. The models are statistical and data driven models, and do not contain investment choices or recommend any specific investments.

Equity Factors And Sectors Capture Premia Or Hedge Risk

Both our short-term historical analysis and the long-run analysis using the MSCI Macro-Economic and Asset Pricing Models suggest that MSCI factor and sector indexes differ in their sensitivity to real economic growth and inflation. In this section, we classify the MSCI factor and GICS 10 sector indexes based on their historical short-term and model projected long-term responses to economic growth and inflation.

We focus on MSCI World factor and sector indexes, although our approach can be applied across other universes for which the requisite macro-economic data are available. For factor index data we use the current family of seven MSCI Factor Indexes: Minimum Volatility (optimized in USD), Risk-Weighted, Value-Weighted, Momentum, Quality, High Dividend Yield and Equal-Weighted. The earliest start dates for these indexes are shown in Exhibit 19 of the Appendix. For the Minimum Volatility Factor Index we generated a proxy to simulate data before the official start of May 1988. A description of this proxy is given in Exhibit 20 of the Appendix. Sectors are defined in our analysis by the top-level layer of the Global Industry Classification Standard (GICS®) that assigns companies to one of ten economic sectors. Before 1994, these definitions are extended by mapping the Barra model industry classification to the GICS sectors to create historical notional sector indexes. The mapping table is shown in Appendix III. All our analyses use USD Gross Total Return Indexes, and month-end returns.

Our historical studies use two different data series to classify economic regimes: the OECD CLI, to measure the overall state of the economy or a point in the business cycle, and the OECD All Items CPI, to assess the rate of inflation. Further details on the data chosen and their construction are given in the Appendix.

Classification Based On Short-Term Historical Analysis – Equity factor and sector indexes can be classified as cyclical or defensive

Exhibit 1: Correlation coefficient between the year-on-year (YoY) relative performance of regional factor and sector indexes versus parent index and YoY change of the corresponding OECD CLI index. Based on returns from 1975 through December 2013 including simulated data. Please refer to Appendix I for details of exact start dates and simulation dates of the index data used.

MSCI World Universe – Correlation with OECD-Total CLI

Sectors	Correlation	Factors	Correlation
Energy	-0.24	Equal-Weighted	0.14
Materials	0.23	High Dividend Yield	-0.25
Industrials	0.36	Minimum Volatility*	-0.50
Consumer Discretionary	0.27	Momentum	-0.11
Consumer Staples	-0.51	Quality	-0.43
Health Care	-0.57	Risk-Weighted	-0.19
Financials	0.12	Value-Weighted	0.05
IT	0.25		
Telecom	-0.19		
Utilities	-0.47		

* see Exhibit 3

As an extension to our 2009 paper »Sector Performance Across Business Cycles«, we updated data through end-2013 and added factor indexes to the sector indexes previously studied. We have restricted the results in our paper to the MSCI World Index universe although our analysis could be extended to other regions and countries (e.g. we have already performed the same analysis for the USA) given the availability of data. As per the previous study, we expect the »cyclical« factors' relative performance to the market to be positively correlated with the rate of change of the CLI and the »defensive« factors' relative

performance to the market to be negatively correlated. In other words, cyclical factor/sector indexes outperform the market in a growing economy, while defensives outperform in a slowdown.

We evaluate this relationship by calculating correlation coefficients between the year-on-year (YoY) relative performance of each factor and sector index with its parent index and the YoY change of the CLI. Exhibit 1, above, shows these results for MSCI World factor and sector indexes.

Concentrating on sector indexes (we highlight the top/bottom three in terms of correlation level), we found that Health Care, Consumer Staples and Utilities (in increasing order of correlation) were the most »defensive« (most negative correlation of annual active return to YoY CLI changes) and Industrials, Consumer Discretionary and IT were the most »cyclical«. These results are intuitive.

For the factor indexes, the defensive nature of four: Minimum Volatility, Quality, High Dividend Yield and Risk-Weighted, is very clear. However, the supposedly cyclical factor indexes (Equal-Weighted, Value-Weighted and Momentum) are more of a puzzle, with only the Equal-Weighted Factor Index showing a high positive correlation to the CLI, and the sensitivity of the Momentum Factor Index to economic conditions actually appearing to be negative. We see, later, that splitting the economic regime into periods of economic growth/decline and then adding a second variable, such as CPI, helps to discriminate performances in these cyclical indexes.

Examining Differential Short-Term Index Performance In Periods Of Decreasing/Increasing Economic Activity And Inflation

Our previous analysis described the response of MSCI World factor and sector indexes to movements in the CLI *as a whole* by looking at the correlations of annual changes. We did not divide periods of rising or falling CLI, nor show the actual magnitude of any differential return.

2. Choosing Right Factors

Institutional investors are increasingly questioning the effect of inflation on their portfolios as well as growth (perhaps splitting economic outcomes, albeit crudely, into »good growth« and »bad growth« scenarios). This leads us to add CPI to the explanatory variables. We now partition economic regimes into rising and falling levels of CLI and CPI, *separately*, as defined in Exhibit 2 below.

Exhibit 2: Definitions of »rising« and »falling« variable used for CLI and CPI in univariate tests

Univariate Analysis	CLI	CPI
Test for whether variable is Rising/Falling	MoM Change	3m-36m Moving Average

Results for these univariate tests are shown in Exhibit 3 below for the MSCI World factor and sector indexes. The average monthly gross active returns for each index in the rising/falling variable states is shown and the higher return by index for each variable is highlighted in green, and the lower return in yellow. Where the average active return differential is less than 10bps we use a black border.

Looking first at CLI (first two columns), results are similar to the previous section, except for Momentum, which is now »cyclical« (higher active return in times of increasing CLI than decreasing).

For the sector indexes, results are more clear with half of the sectors »defensive« (higher active return in times of decreasing CLI), and half of them »cyclical«, with the defensive sectors – Utilities, Consumer Staples and Health Care, showing large differentials with respect to changes up or down in CLI.

Exhibit 3: Univariate analysis of differential performance of factor and sector indexes in regimes of rising/falling economic variables – MSCI World Index

Univariate Analysis

	CLI		CPI	
Index - MSCI World	Decreasing	Increasing	Decreasing	Increasing
Equal Weighted	0.0%	0.3%	0.2%	0.0%
High Dividend Yield	0.2%	0.1%	0.2%	0.2%
Minimum Volatility*	0.3%	-0.3%	0.0%	0.1%
Momentum	0.2%	0.3%	0.2%	0.3%
Quality	0.3%	-0.1%	0.0%	0.2%
Risk-Weighted	0.2%	0.2%	0.2%	0.1%
Value-Weighted	0.0%	0.2%	0.1%	0.0%
Energy	0.1%	0.3%	0.1%	0.4%
Materials	-0.3%	0.3%	0.0%	0.0%
Industrials	-0.3%	0.2%	0.0%	-0.1%
Consumer Discretionary	-0.2%	0.1%	0.0%	-0.2%
Consumer Staples	0.8%	-0.4%	0.1%	0.4%
Health Care	0.8%	-0.4%	0.0%	0.5%
Financials	0.1%	0.0%	0.1%	-0.1%
Information Technology	-0.4%	0.5%	0.1%	-0.1%
Telecom Services	0.2%	-0.2%	0.2%	-0.5%
Utilities	0.8%	-0.7%	0.1%	-0.1%

Average Monthly Gross Active Returns relative to MSCI World from Dec 1975 to Dec 2013.

* Based on official Index Levels from May 1988; Low Volatility Tilt Index prior to that includes simulated data. Please refer to Appendix I for details of exact start dates and simulation dates of the index data used. Indexes where the return differential is less than 10 basis points are boxed.

Looking next at performances in decreasing/increasing CPI states (middle two columns), there is no large differentiation between the factor indexes' responses. For example, there is no change in sign for periods of rising versus

falling CPI. Later results show that using CLI and CPI together, and focusing on transitions, gives more differentiated responses between factor indexes.

For sectors, performances are more varied than factors, with the split of indexes similar to that for CLI (comparing periods of increasing CLI to periods of increasing CPI) except Health Care and Consumer Staples now show higher active return in times of increasing CPI than decreasing, and Industrials, Consumer Discretionary and IT »swap sides« with higher returns in decreasing CPI.

Exhibit 4 below summarizes the results of the MSCI World Index univariate analyses by Top 2 factor and Top 3 sector indexes in terms of absolute return differential across decreasing and increasing states of the independent macro-economic variable.

Having started our historical analysis by looking at the correlation of factor and sector indexes to a single variable (CLI) across time, we next showed how dividing responses by rising and falling CLI and CPI separately can give a more granular sensitivity to these indexes. While this analysis focused on short-term performance and sensitivity to changes in economic growth and inflation, the next section applies our analytic framework and models to study the long-term sensitivity of MSCI World factor and sector indexes to real GDP growth and inflation.

Exhibit 4: Summary of results for univariate analysis by top factor and sector indexes – MSCI World Index

Most Responsive*	CLI	CPI
Factors	Minimum Volatility** (0.6%) Equal-Weighted (-0.4%)	Equal-Weighted (0.2%) Quality (-0.2%)
Sectors	Utilities (1.4%) Health Care (1.2%) Consumer Staples (1.2%)	Telecom Services (0.6%) Health Care (-0.5%) Energy (-0.3%)

*Shown as average active monthly gross returns in periods of decreasing variable minus periods of increasing variable.

** Based on official Index Levels from May 1988; Low Volatility Tilt Index prior to that includes simulated data. Please refer to Appendix I for details of exact start dates and simulation dates of the index data used.

Classification Based On The Long-Term MSCI Asset Pricing Model

The MSCI Asset Pricing Model²² follows the basic principle of modern asset pricing that the competitive equilibrium value of an asset equals the expected discounted value of current and future asset cash flows. The application of this fundamental principle of asset valuation leads to the conclusion that macro risk has an impact on valuation and risk via two channels: cash flows and discount factors.

Our principal finding is that the cash flows earned by different equity portfolios can respond differently to persistent shocks to real output, and that these differences can emerge over longer time horizons. The discount factor that the model uses is sensitive to macro risks. As the model's discount factor is estimated from bond market data, this introduces inflation risk as a source of macro risk.

We now apply the model to show how factor and sector indexes differ in their long-term sensitivity to real GDP growth and inflation. Exhibit 5 summarizes the results of the long-term analysis. The long-run response of equity factor and sector cash flows to pervasive macro shocks suggests that macro shocks are associated with undiversifiable financial risk. Because asset cash flows respond only over a long horizon, this risk is a long-term risk. As with all undiversifiable risk, this risk must be priced. Because portfolios respond differently to macro shocks, they will have different exposures to long-run risk and, consequently, have different prices.

As expected from our previous studies²³, the model shows Momentum, Value-Weighted, Equal-Weighted, Risk-Weighted, and Small Cap exhibiting high sensitivity to real GDP growth risk, relative to the capitalization-weighted index, in the long-run. Among the sectors, the model shows Cyclicals, such as Materials, Consumer Discretionary and Financials, as carrying higher

22 See »Pricing and Analyzing Macro Risk«, MSCI Market Insight, April 2013, for more details about the MSCI Asset Pricing Model.

23 See »Macroeconomic Risk and Asset Cash-Flows«, MSCI Research Insight, March 2013, and »Pricing and Analyzing Macro Risk«, MSCI Market Insight, April 2013.

exposure to real economic growth relative to the market portfolio, while defensives exhibit lower exposure.

Exhibit 24 in the Appendix shows that, according to the model, large positive shocks to economic growth could have a greater positive long-term impact on the return to these indexes relative to the market. This suggests these indexes could potentially command higher long-run average returns or premia. We label these portfolios as »growth sensitive« portfolios.

The model indicates that Minimum Volatility, Quality and High Dividend Yield, and defensive sectors such as Consumer Staples, Health Care and Utilities are less sensitive to real GDP growth relative to the market. This suggests these portfolios could potentially help hedge large negative shocks to real economic growth relative to the market portfolio. We label these portfolios as »growth hedging« portfolios.

Thus, in terms of sensitivity to economic growth, the long-term model-based analysis broadly agrees with the historical short-term analysis presented in the previous section.

Exhibit 5: Equity strategies exhibit varied long-run sensitivity to macro-economic risk

Macro Risk	Growth/Inflation Sensitive Indexes	Growth/Inflation Hedging Indexes
	Growth Sensitive	Growth Hedging
Real GDP Growth Risk	Equal-Weighted Momentum Risk-Weighted Value-Weighted Small Cap Materials Consumer Discretionary Financials	High Dividend Yield Quality Minimum Volatility Energy Industrials Consumer Staples Health Care Information Tech Telecom Services Utilities

Macro Risk	Growth/Inflation Sensitive Indexes	Growth/Inflation Hedging Indexes
	Inflation Sensitive	Inflation Hedging
Inflation Risk	Equal-Weighted Momentum High Dividend Yield Quality Risk-Weighted Small Cap Materials Industrials Consumer Discretionary Consumer Staples Healthcare Information Tech	Energy Financials Telecom Services Utilities Minimum Volatility

The Exhibit classifies the MSCI World factor and sector indexes along their positive or negative sensitivity to real GDP growth and inflation over long horizons, relative to the MSCI World Index.

However, the results suggest that inflation impacts equities in different ways over the short and long-term. Exhibit 5 and Exhibit 25 in the Appendix show that equity factor and sector indexes may also be differentiated according to their long-term sensitivity to inflation. Sectors such as Energy, Utilities and Telecom Services may benefit from higher inflation relative to the market, as profits to these sectors are either hedged against potential upside risks to inflation, or increase with inflation. Overall, though, inflation could have a negative impact on equities. There could be two reasons for this. The first explanation relies on the observed negative impact of higher inflation on future long-term real GDP growth. Thus, Small Cap and the Consumer Discretionary sector may be more adversely impacted by inflation relative to the market as they are also more sensitive to real GDP growth risk in the long-term. The second explanation is related to equity duration. Factor indexes such as High Dividend Yield and Quality, and sector indexes such as Consumer Staples, are characterized as high equity duration portfolios with stable nominal cash-flows. In turn, higher inflation could have a negative impact on these portfolios' real cash-flows and returns.

This classification analysis carries important implications for strategic and medium term macro-sensitive asset allocation. Indeed, as we will see in the final section of this paper, our framework and models indicate that an investor's investment horizon, macro-economic views and willingness to tolerate macro-economic uncertainty could inform asset allocation decisions. Observed historical premia on strategies such as Value and Size could be compensation for persistent shocks to trend growth. These premia are also consistent with the hypothetical investor holding the market portfolio. However, specific investors could reasonably deviate from holding market capitalization weights. In particular, long-horizon investors that have greater tolerance for macro-economic uncertainty might consider tilting towards high growth sensitive portfolios relative to the capitalization-weighted portfolio, especially, if they perceive the outlook for real economic growth to be positive.

Before turning to the implications for macro-sensitive asset allocation, in the following section, we explore the historical performance of factor and sector indexes during historically plausible economic regimes. In particular, using historical data, we classify economic regimes through a bivariate framework that groups them into four quadrants, depending on whether the CLI and the CPI are rising or falling, and examine the differential performance of the index universe across these states, and the relative performances of the indexes through transitions from one regime to another.

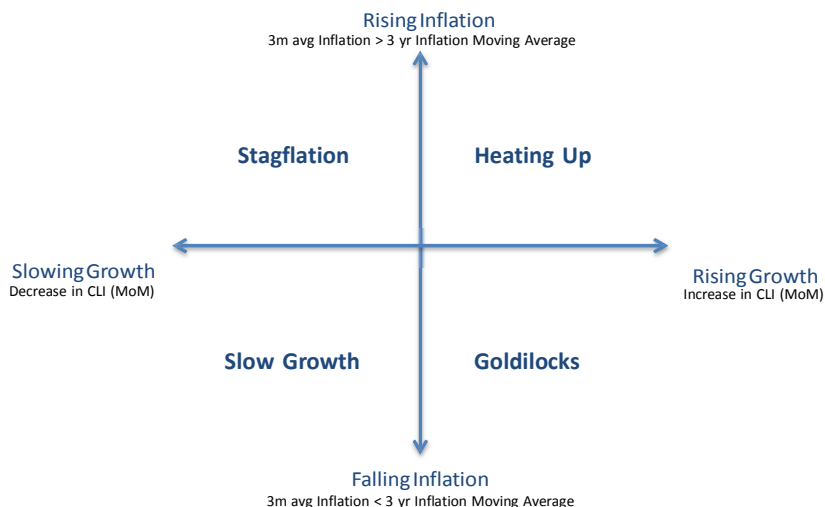
Equity Factors And Sector Performance Differed During Regimes And Through Regime Transitions – Index Performance During Economic Regimes Classified Using CLI And CPI

The motivation for this research came from requests from institutional investors to better understand how their portfolios of factor and sector indexes behave in different economic regimes. So far, we have sought to answer that question using single macro-economic variables in isolation and have shown how our indexes respond when these single variables are rising or falling. In other words, we tried to answer the question: »I think economic activity/

inflation is going to increase/decrease over the near term – how has this historically affected my portfolio?»

While we hope this is both interesting and useful, a more realistic analysis that fits better with institutional investors' views of the world, is to look at variables in conjunction. However, there is a panoply of economic variables for investors to choose from and, so, a huge number of possible combinations exist. In this section, we err once again on the side of parsimony, and focus our analysis on a pair of variables that institutional investors naturally think of in conjunction: the Business Cycle and Inflation. When assessing possible future economic conditions, institutional investors look not only at economic growth (e.g. using the CLI), but also divide growth into »good« and »bad« regimes depending on accompanying inflation. A more realistic question is: »I think economic growth is going to dip *and* inflation is going to pick up – what has happened to portfolios like mine under stagflation in the past?»

Exhibit 6: Schematic of economic regime classification using CLI and CPI



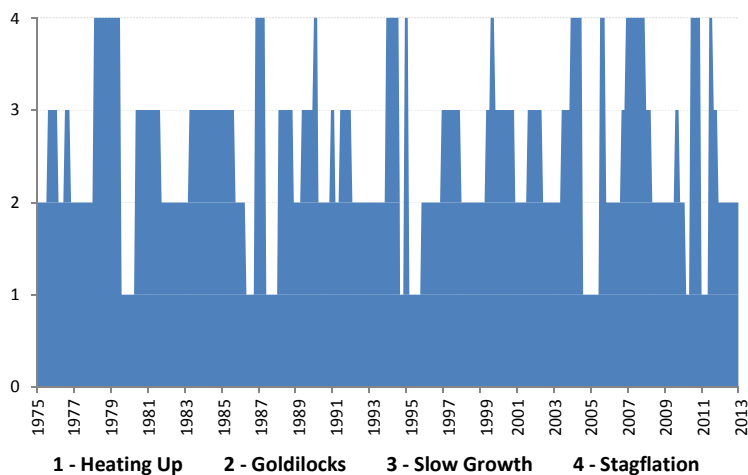
We try to answer this question by dividing economic regimes into four outcomes, depending on whether economic growth is strong or weak and whether inflation is rising or falling, as shown in Exhibit 6, above. We name these four regimes:

1. »Goldilocks« – Rising Growth and Falling Inflation
2. »Slow Growth« – Slowing Growth and Falling Inflation
3. »Stagflation« – Slowing Growth and Rising Inflation
4. »Heating Up« – Rising Growth and Rising Inflation

Our definitions of »Slowing/Rising« Growth and »Falling/Rising« Inflation correspond exactly to the definitions of Decreasing/Increasing CLI/CPI that we used in the univariate analysis in the previous section (see Exhibit 2). Similarly, for »Falling/Rising Inflation« we use the exact same definitions as for our univariate analyses. When classifying regimes we require that each regime must persist for a minimum of three months, before being classified as a new state of the economy.

Using these definitions, we arrive at the graph in Exhibit 7 which shows the persistence and frequency of these changing regimes over time:

Exhibit 7: Economic regimes classified by using OECD-Total CLI and CPI



1 Heating Up 2 Goldilocks 3 Slow Growth 4 Stagflation

Exhibit 8, below, shows the relative frequency of these four regimes over approximately 38 years of analysis. We see that »Goldilocks« (Rising Growth and Falling Inflation) has been the most common regime, followed by »Slow Growth«, »Stagflation« and, finally, »Heating Up«:

Exhibit 8: Relative frequency of economic regimes using bivariate classification: OECD-Total

Macro-Economic States	State Likelihood
Heating Up	11%
Goldilocks	40%
Slow Growth	32%
Stagflation	17%

Exhibit 9 below shows the historical results of the bivariate analysis for the MSCI World factor and sector indexes. We work our way through the results, from the least frequently observed economic regimes (»Heating Up«) to the most historically likely (»Goldilocks«).

In »Heating Up«, for factor indexes, Momentum and Minimum Volatility are the standouts in terms of relative performance, with Momentum delivering its largest average active monthly return in this regime. Both these results are intuitive, as are the relative intransigence of Risk-Weighted, High Dividend Yield and Quality, and the outperformance of Equal-Weighted and Value Weighted.

On the sector front for »Heating Up«, the results are also rather intuitive, with Materials and IT leading the way, and Utilities, followed by Telecom Services, firmly bringing up the rear.

Moving next to »Stagflation«, Minimum Volatility is again the standout, showing the mirror of its active return in »Heating Up«, with the more cyclical Equal-Weighted and Value-Weighted Factor Indexes performing the worst. On the sector front there is a very clear differentiation between Energy (with the highest active return in this regime through all scenarios over all sector and factor indexes) and the defensive Consumer Staples, Health Care and Utilities sectors

showing strong outperformance. The more cyclical sectors all underperform, with Consumer Discretionary having the worst relative performance (again intuitive in an environment of rising prices but slowing economic growth).

Exhibit 9: Bivariate analysis of performance of factor and sector indexes in economic regimes classified by rising/falling OECD-Total CLI and All Items CPI – MSCI World Index

Bivariate Analysis

Index – MSCI World	Heating Up	Goldilocks	Slow Growth	Stagflation
Equal-Weighted	0.2%	0.4%	0.0%	-0.1%
High Dividend Yield	0.0%	0.1%	0.2%	0.2%
Minimum Volatility**	-0.4%	-0.3%	0.3%	0.4%
Momentum	0.3%	0.3%	0.2%	0.3%
Quality	0.1%	-0.1%	0.3%	0.3%
Risk-Weighted	0.0%	0.2%	0.2%	0.1%
Value-Weighted	0.1%	0.2%	0.1%	-0.1%
Energy	-0.1%	0.3%	-0.2%	0.9%
Materials	0.5%	0.3%	-0.4%	-0.4%
Industrials	0.2%	0.2%	-0.3%	-0.2%
Consumer Discr	0.2%	0.1%	0.0%	-0.5%
Consumer Staples	0.0%	-0.5%	0.8%	0.6%
Health Care	0.2%	-0.5%	0.8%	0.6%
Financials	-0.1%	0.0%	0.3%	-0.2%
Information Tec	0.3%	0.6%	-0.4%	-0.4%
Telecom Svc	-0.5%	-0.2%	0.6%	-0.4%
Utilities	-1.0%	-0.5%	0.8%	0.6%

Average Monthly Gross Active Returns relative to MSCI World from Dec 1975 to Dec 2013.

*** Based on official Index Levels from May 1988; Low Volatility Tilt Index prior to that includes simulated data. Please refer to Appendix I for details of exact start dates and simulation dates of the index data used.*

»Slow Growth« sees a positive outperformance for all the MSCI World factor indexes, and the differentiation in this regime comes through the sector indexes

with the classically defensive Utilities, Health Care and Consumer Staples outperforming at the expense of the cyclical IT, Materials and Industrials.

Exhibit 10: Summary of results for bivariate analysis by top factor and sector indexes – MSCI World

Most Responsive	Heating Up	Goldilocks	Slow Growth	Stagflation
Factors	Minimum Volatility** (-0.4%)	Equal-Weighted (0.4%)	Minimum Volatility** (0.3%)	Minimum Volatility** (0.4%)
	Momentum (0.3%)	Momentum (0.3%)	Quality (0.3%)	Momentum (0.3%)
Sectors	Utilities (-1.0%)	Information Tech (0.6%)	Consumer Staples (0.8%)	Energy (0.9%)
	Materials (0.5%)	Utilities (-0.5%)	Utilities (0.8%)	Health Care (0.6%)
	Telecom Svc (-0.5%)	Health Care (-0.5%)	Health Care (0.8%)	Consumer Staples (0.6%)

*Average Monthly Gross Active Returns relative to MSCI World from Dec 1975 to Dec 2013.
 ** Based on official Index Levels from May 1988; Low Volatility Tilt Index prior to that includes simulated data. Please refer to Appendix I for details of exact start dates and simulation dates of the index data used.*

Finally, the most common »Goldilocks« macro-economic scenario sees Equal-Weighted as the top performer among factor indexes, closely followed by Momentum, whereas Minimum Volatility and Quality underperform. From the sector indexes, IT, Energy and Materials are the clear winners, with the defensive trifecta of Utilities, Health Care and Consumer Staples once again underperforming. Exhibit 10, above, summarizes the results of the MSCI World Index bivariate analyses by Top 2 factor and Top 3 sector indexes in terms of absolute active return per quadrant.

Looking at correlation between indexes through different economic regimes

Our December 2013 paper, »Deploying Multi-Factor Index Allocations in Institutional Portfolios« observed the cyclical nature of factor returns, historically,

and suggested that institutional investors could potentially smooth out some of this cyclicity through diversification by intelligently combining factor indexes. In that paper, we looked at the long-run average correlations of active returns between factors, but given our observations above on the variability of factor returns through different economic regime classifications, we wanted to show, as well, how factor return correlations behaved in different economic regimes.

Exhibit 11: Active factor return volatilities and correlations in four economic regimes

Annualized Tracking Errors

Index - MSCI World	Benchmark (Volatility)	Equal-Weighted	High Dividend Yield	Minimum Volatility**	Momentum	Quality	Risk-Weighted	Value-Weighted	Average Pair-wise Correlation
Full History	15.0%	5.4%	6.1%	6.0%	8.8%	5.9%	5.7%	3.8%	0.27
Heating Up	10.1%	4.5%	5.8%	4.1%	9.0%	6.0%	4.7%	3.3%	0.22
Goldilocks	12.7%	5.1%	6.0%	5.1%	7.5%	6.1%	5.5%	3.3%	0.23
Slow Growth	16.3%	5.8%	6.8%	6.4%	8.7%	5.7%	6.2%	3.7%	0.29
Stagflation	18.9%	5.6%	5.4%	7.6%	11.1%	5.2%	5.4%	4.6	0.32

**Based on official Index Levels from May 1988.

Results are shown in Exhibit 11 above. In more volatile economic regimes (i.e. into »Slow Growth« and »Stagflation«), average tracking errors tended to increase for the more cyclical indexes (e.g. Equal-Weighted, Momentum and Value-Weighted), whereas there was no clear pattern for the more defensive indexes. Average pairwise factor index active return correlations tended to increase with market volatility, a commonly observed phenomenon.

Looking At Index Performance Through Historical Regime Transitions

Our previous set of analyses classified the current state of the economy into one of four economic regimes, based on the levels and co-movement of the OECD CLI and CPI indicators, and showed the average monthly active return in each regime.

For the final part of our historical analysis we look not just at relative factor and sector index performances within a single regime, but at how indexes perform in relative terms when we move from one regime to another (regime transitions). An example of the type of question we are seeking to answer is: »I believe the global economy is in a state of ›Slow Growth‹ now, and I'm worried that growth will remain low but inflation will pick up. How has my index portfolio performed, historically, through such transitions into a ›Stagflation‹ scenario?«

Exhibit 12: Regime transition historical frequency matrix for OECD-Total CLI and CPI indicators

Likelihood of transitioning between regimes over 3 months		TO			
FROM		Heating Up	Goldilocks	Slow Growth	Stagflation
	Heating Up	55%	6%	11%	28%
	Goldilocks	3%	73%	20%	3%
	Slow Growth	0%	27%	65%	8%
	Stagflation	23%	8%	12%	58%

Exhibit 12, above, shows the historical frequencies of transitions from 1976 to 2013 between each of our four economic regimes to each other, over a three-month period.

The first point that is clear from these results is that the most likely state to be in after three months is the same state you are in now (the diagonal elements of the table). Also, this regime persistence (an embodiment of the

statistical phenomenon of autocorrelation) is higher for the more common (»Goldilocks« and »Slow Growth«) than the less common (»Stagflation« and »Heating Up«) regimes.

Secondly, the chances of radical regime shifts where both the CLI and CPI indicators change sign (so moving more than two states up or down in Exhibit 7 above), e.g. from »Slow Growth« to »Heating Up« or from »Goldilocks« to »Stagflation«, are relatively rare. In fact, the particular example of moving from »Slow Growth« (low growth and low inflation) to »Heating Up« (high growth and high inflation) over three months has historically never happened in our sample from 1976 to 2013. Again, this is intuitive, if one imagines the macro-economic environment following a continuous path around the quadrants of Exhibit 6 above, then diagonal jumps should be less likely, but not necessarily impossible.

Looking back at our initial results for the bivariate analysis (Exhibit 9) we can think of these single-regime results as being similar to a frequency-weighted combination of the results as we move from one regime to the other. In other words, if we focus on the »Heating Up« column of results from Exhibit 9 they could be seen as a combination of returns observed when we start in the »Heating Up« state, and then either remain in »Heating Up« (the most common ›transition‹) or move to »Stagflation« (next most common), »Goldilocks«, or »Slow Growth« (least common). In essence, each column in Exhibit 9 will now generate a transition table in its own right, the columns of which will be the resulting regimes over three months, and the values of which will be the active returns to the factor and sector indexes over these transitions. In total, Exhibit 9 will be represented by four such tables. These four tables are shown for the MSCI World Index in Exhibit 13 below.

Exhibit 13: Active returns for MSCI World factor and sector indexes over regime transitions

From Heating Up To				
	Heating Up	Goldilocks	Slow Growth	Stagflation
Equal-Weighted	0.8%	-2.2%	0.6%	-1.9%
High Dividend Yield	-0.2%	5.5%	0.5%	1.4%
Minimum Volatility**	-1.4%	-0.6%	0.2%	2.3%
Momentum	0.6%	1.0%	-1.9%	2.2%
Quality	0.0%	4.1%	-1.0%	-0.3%
Risk-Weighted	0.3%	-1.2%	0.3%	-0.4%
Value-Weighted				
	0.4%	1.6%	0.9%	-1.0%
Energy	-0.1%	5.0%	-5.6%	-1.6%
Materials	1.3%	-4.6%	1.8%	-2.5%
Industrials	1.0%	-3.9%	1.6%	-1.2%
Consumer Discr	0.7%	-2.5%	5.5%	-1.1%
Consumer Staples	-0.2%	-0.1%	0.4%	4.1%
Health Care	0.0%	3.9%	1.5%	4.2%
Financials	0.4%	0.8%	-0.7%	-0.2%
Information Tech	-0.3%	6.0%	1.7%	-4.2%
Telecom Svc	-2.4%	1.2%	5.5%	3.4%
Utilities	-2.2%	0.3%	-2.0%	4.7%

From Goldilocks to				
	Heating Up	Goldilocks	Slow Growth	Stagflation
Equal-Weighted	-0.4%	1.1%	-0.5%	-0.9%
High Dividend Yield	-1.2%	-0.3%	1.3%	1.4%
Minimum Volatility**	-2.4%	-0.8%	1.9%	0.5%
Momentum	-0.4%	1.0%	0.7%	-2.8%
Quality	-3.8%	-0.7%	2.0%	1.9%
Risk-Weighted	-1.4%	0.5%	0.8%	-0.8%
Value-Weighted	-0.7%	0.3%	0.1%	0.6%
Energy	6.0%	0.7%	1.8%	3.4%
Materials	2.6%	0.8%	-2.0%	0.6%
Industrials	-0.5%	0.7%	-0.3%	-2.7%
Consumer Discr	-1.8%	0.1%	-1.3%	-2.3%
Consumer Staples	-1.9%	-1.3%	3.3%	0.1%
Health Care	-2.4%	-1.3%	3.3%	3.2%
Financials	2.7%	0.0%	0.0%	-1.3%
Information Tech	-0.6%	1.7%	-1.5%	4.1%
Telecom Svc	-2.5%	-0.9%	1.0%	-1.3%
Utilities	-8.9%	-1.8%	2.9%	-1.6%

From Slow Growth To				
	Heating Up	Goldilocks	Slow Growth	Stagflation
Equal-Weighted		1.8%	-0.1%	0.5%
High Dividend Yield		0.8%	0.9%	0.3%
Minimum Volatility**		-2.0%	1.1%	0.2%
Momentum		-0.7%	1.0%	0.1%
Quality		-0.8%	0.6%	0.6%
Risk-Weighted		0.1%	0.9%	0.8%
Value-Weighted		1.1%	0.2%	0.5%
Energy		-1.6%	-0.4%	-0.7%
Materials		0.9%	-1.8%	0.0%
Industrials		0.8%	-1.2%	1.0%
Consumer Discr		2.1%	0.1%	-2.8%
Consumer Staples		-2.7%	2.4%	2.9%
Health Care		-3.8%	2.2%	1.0%
Financials		1.0%	0.9%	2.7%
Information Tech		3.7%	-1.9%	-0.8%
Telecom Svc		-0.8%	2.4%	-3.2%
Utilities		-2.0%	2.8%	4.7%

From Stagflation To				
	Heating Up	Goldilocks	Slow Growth	Stagflation
Equal-Weighted	1.0%	1.0%	2.0%	-0.1%
High Dividend Yield	-0.2%	0.2%	2.6%	0.5%
Minimum Volatility**	-1.5%	-0.4%	3.2%	0.7%
Momentum	2.0%	1.8%	-1.8%	1.6%
Quality	0.9%	1.6%	1.6%	1.5%
Risk-Weighted	0.1%	0.8%	3.4%	0.2%
Value-Weighted	0.4%	0.5%	1.8%	-0.2%
Energy	1.7%	-2.5%	3.2%	4.0%
Materials	2.3%	0.8%	3.6%	-0.8%
Industrials	0.6%	0.0%	0.7%	-1.2%
Consumer Discr	0.6%	2.2%	-0.8%	-1.6%
Consumer Staples	-1.5%	-0.1%	8.1%	1.1%
Health Care	-0.6%	-0.3%	8.2%	1.6%
Financials	-1.3%	0.3%	-0.4%	-1.4%
Information Tech	3.2%	0.0%	-6.9%	-0.6%
Telecom Svc	-2.0%	1.9%	1.1%	-1.9%
Utilities	-3.2%	-1.4%	5.4%	0.3%

Average Quarterly Gross Active Returns relative to MSCI World from Feb 1976 to Dec 2013.

*** Based on official Index Levels from May 1988; Low Volatility Tilt Index prior to that includes simulated data. Please refer to Appendix I for details of exact start dates and simulation dates of the index data used.*

Looking at the regime transition results and starting this time with the most common historical regime, »Goldilocks« (bottom-left table in Exhibit 13), and focusing first on the most common »transition« – remaining in »Goldilocks« – we see that the factor indexes that outperform the most are Equal-Weighted and Momentum, and those that underperform the most are Minimum Volatility and Quality. Within the sector indexes, IT is the clear outperformer, and the now familiar defensive triumvirate of Utilities, Health Care and Consumer Staples strongly underperforms. Comparing these transition results to the overall »Goldilocks« results (second column of Exhibit 9) we see the results are similar in rank and sign, but, for the transition, are greater in magnitude.

2. Choosing Right Factors

This fits with our explanation that the results in Exhibit 9 are an average over all transitions, and staying in the same state is the most likely »transition«.

When in a »Goldilocks« state, the second most common transition is to move into »Slow Growth«. When we look at the results for this transition we see that it largely features the same indexes as for »Goldilocks« but with greater magnitude and opposite sign. For example, the returns to the Minimum Volatility and Quality factor indexes flip from negative to positive and for the Equal-Weighted factor index vice versa. Similarly, Consumer Staples and Health Care move from underperforming to outperforming, and IT now underperforms.

For the least common transitions, to »Heating Up« and »Stagflation« (3% historical frequency each) results become more dispersed by index and larger in absolute terms. Summary results are shown for the MSCI World Index in Exhibit 14.

Exhibit 14: Summary of highest absolute active returns by index split by transitions – MSCI World

		Ending Regime								
		Heating Up		Goldilocks		Slow Growth		Stagflation		
		Index	Response*	Index	Response*	Index	Response*	Index	Response*	
Starting Regime	Most responsive	Factors	1 Momentum Minimum	1.97%	Momentum	1.77%	Risk-Weighted	3.41%	Momentum	1.64%
			2 Volatility**	-1.53%	Quality	1.63%	Minimum Volatility**	3.15%	Quality	1.55%
	Sectors	1 Utilities	-3.24%	Energy	-2.48%	Health Care	8.16%	Energy	3.96%	
		2 Information Tech	3.22%	Consumer Discr	2.16%	Consumer Staples	8.09%	Telecom Svc	-1.87%	
		3 Materials	2.79%	Telecom Svc	1.87%	Information Tech	-6.94%	Consumer Discr	-1.63%	
	Slow Growth	Factors	1		Minimum Volatility**	-2.00%	Minimum Volatility**	1.10%	Risk-Weighted	0.83%
			2		Equal-Weighted	1.75%	Momentum	0.97%	Quality	0.55%
		Sectors	1		Health Care	-3.84%	Utilities	2.79%	Utilities	4.70%
			2		Information Tech	3.66%	Consumer Staples	2.40%	Telecom Svc	-3.18%
			3		Consumer Staples	-2.72%	Telecom Svc	2.30%	Consumer Staples	2.92%
		Goldilocks	Factors	1 Quality	-3.80%	Equal-Weighted	1.06%	Quality	2.00%	Momentum
	2 Minimum Volatility**			-2.36%	Momentum	0.99%	Minimum Volatility**	1.92%	Quality	1.91%
	Sectors		1 Utilities	-8.92%	Utilities	-1.82%	Consumer Staples	3.30%	Information Tech	4.12%
			2 Energy	5.96%	Information Tech	1.68%	Health Care	3.30%	Energy	3.45%
			3 Financials	2.75%	Consumer Staples	-1.32%	Utilities	2.93%	Health Care	3.19%
	Heating Up		Factors	1 Minimum Volatility**	-1.43%	High Dividend Yield	5.52%	Momentum	-1.95%	Minimum Volatility**
		2 Equal-Weighted		0.77%	Quality	4.12%	Quality	-0.98%	Momentum	2.15%
		Sectors	1 Telecom Svc	-2.38%	Information Tech	6.01%	Energy	-5.62%	Utilities	4.67%
			2 Utilities	-2.22%	Energy	5.04%	Telecom Svc	5.51%	Health Care	4.21%
			3 Materials	1.28%	Materials	-4.58%	Consumer Discr	5.49%	Information Tech	-4.21%

Average Quarterly Gross Active Returns relative to MSCI World from Feb 1976 to Dec 2013.

*** Based on official Index Levels from May 1988; Low Volatility Tilt Index prior to that includes simulated data. Please refer to Appendix I for details of exact start dates and simulation dates of the index data used.*

To summarize the key results for the most common »Goldilocks« as the starting scenario, and working our way from »Heating Up«, through »Goldilocks« and »Slow Growth« to »Stagflation« as the ending scenarios, we see the following commonalities:

1. If you think the economic regime is remaining benign or strengthening, then, historically, the Equal-Weighted Factor Index has outperformed and Minimum Volatility has underperformed in such transitions in the past.
2. On the other hand, if you think growth is slowing then an investment in the Quality Factor Index has generally outperformed, and if you think inflation is also staying low (moving to »Slow Growth«) then Minimum Volatility has also outperformed historically.
3. In terms of sector indexes, an underweight to Utilities has generally outperformed in environments where growth has remained strong, but when growth has fallen then a Consumer Staples overweight has historically outperformed. However, if this drop in growth is accompanied by a pickup in inflation (i.e. move to »Stagflation«) then Energy has been the standout performer historically.

Differences become more marked as we move into the less frequent economic regimes, which potentially argue for a more local approach to asset allocation in unusual or rapidly changing economic environments. This local approach is taken up in the next section, where we show how to identify plausible scenarios for GDP growth and inflation with the help of the MSCI Macro-Economic Model, and explore the impact of these macro-economic forecast scenarios on selected equity index returns and allocations to these indexes.

Macro Views Inform Equity Strategy Returns And Asset Allocation Decisions

So far, we have shown how factor and sector indexes have differed, historically, in their short-term performance, under different economic growth and inflation regimes, and during regime transitions. We have also shown, using the MSCI Asset Pricing Model, how to classify factor and sector indexes along their exposure to long-term economic growth and inflation risk. Together, the analyses of the previous sections are a first step towards establishing a structured framework to help institutional investors to explicitly incorporate macro-economic views into their asset allocation decisions.

We now take a forward-looking and longer-term perspective. Institutional investors are also interested in assessing their portfolio response to unprecedented macro-economic shocks that potentially carry an uncertain, long-term impact on trend growth and inflation. For example, since 2008, investors remain uncertain about the time required for DM economies to revert back to their long-term trend, and about the impact of a slower or faster recovery on factor and sector returns, and asset allocation. Another example is the uncertain consequence of unprecedented »Quantitative Easing« monetary policy and its reversal.

This section shows how our framework and models could help institutional investors address these questions in a structured way. We, first, apply the MSCI Macro-Economic Model to identify plausible macro-economic forecast scenarios and assess their likelihood relative to a baseline. We, then, apply the MSCI Asset Pricing and Dynamic Asset Allocation Models to assess the likely performance of growth sensitive and growth hedging indexes under these scenarios and to derive combinations of MSCI factor and sector indexes that could potentially benefit from, or offer protection in, the same scenarios.

Identifying Plausible Macro-Economic Forecast Scenarios

Macro-economic forecast scenarios can be evaluated in a structured way. The first step in this process is to define a baseline and the investment horizon. The aim of the baseline scenario is to set bands against which stress scenarios

can be measured. In this paper, we use the three-years out baseline (median) forecasts from the MSCI Macro-Economic Model shown in Exhibit 15. The MSCI Macro-Economic Model is an innovative statistical model used to generate forecasts and scenarios over different horizons of selected macro-economic variables, such as real GDP growth and inflation²⁴.

According to the model's baseline forecasts, both real economic growth and inflation for DM economies are projected to remain persistently well below their long-term average, at about 1.5%, even three-years out.

While declines in inflation may usually be attributable to declines in energy prices, other factors may be at work behind the recent declines and persistence of low rates. An important question is whether the low realized and projected inflation is a consequence of a prolonged period of sub-par growth since 2008, and will eventually recover as economic growth returns to trend, or whether it signals a slower recovery.

One of the benefits of the MSCI Macro-Economic Model is that it also provides confidence bands for the forecasts. The confidence bands for the baseline forecasts, also from Exhibit 15, suggest that a faster return to the long-term average trend growth of 3.2%, together with benign inflation, is plausible. More precisely, the model assigns about a 30% chance for real GDP growth for DM economies overall to exceed the long-term trend three-years out. In particular, the recently observed and continued improvements in industrial production,

24 The MSCI Macro-Economic Model is an innovative Bayesian Vector Autoregressive model (BVAR). BVARs are a standard statistical tool for multivariate time series analysis. They have widespread application in macro-economic analysis. A benefit of BVARs is that they produce forecasts over many periods, and calculate confidence bands over those same periods. The MSCI Macro-Economic Model builds on the long-run risk model for real GDP growth described in our previous paper (»Macro-Economic Risk and Asset Cash Flows«). In addition to real GDP growth and corporate profits, only observed on a quarterly basis, typically with a lag, the model also applies the latest advances in academic research to combine these quarterly variables with variables available on a timely, monthly basis. The monthly variables are: CPI inflation, term spread (10-Year minus 3 month), money growth (M2), Fed base rate, the SPGS Commodity Index, unemployment rate, exports, capacity utilization, and labor unit cost. Lastly, we impose restrictions (Bayesian priors) based on empirical regularities found in the observed time series of these variables. Further details on the MSCI Macro-Economic Model can be found in Appendix IX.

retail sales, and unemployment in the US (see Appendix VIII) could help quicken the pace of recovery in DM (and EM) economies, should they persist.

Exhibit 15: World (DM) real GDP Growth and CPI inflation forecasts from the MSCI Macro-Economic Model

	Long-Term Trend through 2007	Last 4 Quarters	MSCI Year-Ahead Forecasts			MSCI Three-Years Out Forecasts		
			Baseline	Low	High	Baseline	Low	High
World (DM) Real GDP Growth	3.2	1.5	1.6	0.2	3.1	1.5	-0.2	3.1
World (DM) Consumer Price Inflation	3.3	1.4	1.3	0.5	2.1	1.5	0.6	2.5

The Exhibit shows the long-term trend growth and inflation rates through 2007, and compares these rates to the current rates, and to the MSCI Macro-Economic Model's Baseline, Low, and High growth and inflation forecasts for the year-ahead and three-years out. The Low and High forecasts provide the 30-70 confidence bands around the Baseline. All rates are annualized percentage rates.

Turning to inflation, the model confidence bands suggest that high inflation seems an unlikely outcome. In fact, the model suggests that there is even a 30% chance of a significant further decline in inflation three-years out, down from 1.5% to 0.6%. In turn, a widespread deflation may also deteriorate real growth²⁵.

Following these observations, we choose to focus on the following three plausible medium term (three years) macro-economic scenarios for DM economies:

- MSCI Model Baseline scenario (persistence of below long-term trend growth and inflation)
- Return to Long-term Trend Growth, coupled with moderate inflation
- Slow Growth (slower growth and lower inflation)

²⁵ The realized inflation figure over the last four quarters, from Exhibit 28 in the Appendix, suggests that deflation has already gained ground in Italy and Spain.

Exhibit 16: Three plausible macro-economic scenarios based on the MSCI Macro-economic Model

		MSCI Macro-Economic Model Forecasts	
		(Three-Year Horizon, Annualized)	
Three-Year Horizon Macro-Economic Scenario	MSCI Macro-Economic Model Implied Probability	DM Real GDP Growth	DM Inflation
Current		1.5%	1.4%
MSCI Model Baseline	50%	1.5%	1.5%
Return To Long-Term Trend Growth	10%	3.1%	2.5%
Slow Growth	10%	-0.2%	0.6%

The Exhibit shows, in the second column, the probability implied by the MSCI Macro-Economic Model for the MSCI Model Baseline, Return to Long-term Trend Growth, and Slow Growth scenarios. The last two columns show the three-year horizon MSCI Macro-Economic Model forecasts for real GDP growth and inflation for DM economies, conditional on the current economic conditions, and each of the three scenarios. All rates are annualized percentage rates.

The likelihood for these scenarios and the model-implied real GDP growth and inflation forecasts under these scenarios are summarized in Exhibit 16. The following sections apply the MSCI Asset Pricing and Dynamic Asset Allocation Models²⁶ to assess the impact of these three plausible scenarios on the three-year horizon performance of the MSCI World Index (the »market«), a growth sensitive and a growth hedging combination of MSCI factor and sector indexes, and macro-sensitive asset allocations to these indexes.

Forecast Macro Scenarios Could Impact Strategy Returns And Asset Allocation

As we saw in the first section, factor and sector indexes can be grouped into growth sensitive and growth hedging categories based on both historical and model-based analyses. In this section, we focus on two equally weighted

²⁶ See the Appendix for a detailed description.

portfolios of these factor and sector indexes. The first portfolio blends all the growth sensitive indexes. We label it as »Growth Sensitive Strategy«. The second portfolio blends all the growth hedging indexes. We label this portfolio as »Growth Hedging Strategy«.

Exhibit 17 shows that macro-economic scenarios could have varied implications for equity market and strategy returns. Under the MSCI Model Baseline scenario, real GDP growth and inflation forecasts are projected to remain persistently low through 2014. In turn, the implied returns from the MSCI Asset Pricing Model are projected to remain broadly stable under this scenario.

At the same time, the MSCI Asset Pricing Model suggests that a faster return to trend growth in DM economies is positive for equities. As shown in Table 4, the model-implied annualized return on the MSCI World Index could increase by 1.4% per year over three years. Exhibit 17 also shows how equity strategies could differ in their response to a positive shock to trend growth. While the growth sensitive strategy could experience a larger increase in returns of 1.9% per year over three years, the return to the growth hedging strategy could only increase by the much lower amount of 0.7% per year over the same horizon.

Exhibit 17: Three-year horizon MSCI Asset Pricing Model projected real returns, conditional on macro-economic scenarios

Three-Year Horizon Macro-Economic Scenario	MSCI Asset Pricing Model Projected Real Return (Three-Year Horizon, Annualized)		
	MSCI World Index	Growth Sensitive Strategy	Growth Hedging Strategy
Current	5.3%	6.3%	5.3%
MSCI Model Baseline	5.2%	6.2%	5.2%
Return To Long-Term Trend Growth	6.7%	8.2%	6.0%
Slow Growth	3.8%	4.0%	4.5%

The Exhibit shows the annualized, three-year horizon projected real returns from the MSCI Asset Pricing Model conditional on the current economic conditions, the MSCI Model Baseline, the Return to Long-term Trend Growth, and Slow Growth scenarios.

In contrast, the Slow Growth scenario could be negative for equities, with annualized returns down from 5.3% to 3.8%, and could reverse the relative performance of growth sensitive and growth hedging strategies.

Macro-economic scenarios could also impact asset allocation decisions. Indeed, the MSCI Dynamic Asset Allocation Model²⁷ suggests that deviations away from the cap-weighted market portfolio are dictated by an investor's tolerance for macro-economic uncertainty, investment horizon, and macro views. In our analysis, investors care about both the risk inherent in their current investment opportunity set, and also the uncertainty about the evolution of investment opportunities over time. It is the evolution of investment opportunities over time that is most sensitive to persistent shocks to current economic conditions. Investors could be uncertainty-tolerant or uncertainty averse. Uncertainty-tolerant investors could be long-term institutional investors with the willingness or capacity to withstand the short-term effects of pervasive shocks to the economy.

At the same time, as discussed in the first section, observed premiums on growth sensitive indexes are compensation for potential and persistent shocks to trend growth. Thus, in the long-term, assuming that economic growth could eventually revert to its long-term average, the model indicates that a strategic tilt towards the growth sensitive indexes could allow uncertainty-tolerant institutional investors to benefit from growth sensitive premiums²⁸. However, if negative shocks to trend growth are believed to persist for a longer time than expected, the model indicates that uncertainty-tolerant investors could benefit from a tactical tilt towards growth hedging indexes²⁹.

27 See Exhibit 29 in Appendix X for a description of the equilibrium asset allocation concept underpinning the MSCI Dynamic Asset Allocation model.

28 For more details on this point and the implications of our framework and models for long-term strategic asset allocation, see »Macro Risk and Strategic Asset Allocation: Deconstructing Risk Parity Portfolios«, MSCI Market Insight, June 2013.

29 The results presented in this paper and in Exhibit 18 are based on MSCI Macro-Economic, Asset Pricing and Asset Allocation Models. The models are statistical and data driven models, and do not contain investment choices or recommend any specific investments.

Exhibit 18: Three-year horizon macro-sensitive allocations to growth sensitive and growth hedging strategies, conditional on macro-economic scenarios

Three-Year Horizon Macro-Economic Scenario	Three-Year Horizon Macro-Sensitive Asset Allocations		
	MSCI World Standard Index	Growth Sensitive Strategy	Growth Hedging Strategy
Current	48%	0%	52%
MSCI Model			
Baseline	46%	0%	54%
Return To Long-Term Trend Growth	53%	47%	0%
Slow Growth	0%	0%	100%

The Exhibit shows, for an uncertainty-tolerant institutional investor, the optimal three-year horizon optimal asset allocations from the MSCI Dynamic Asset Allocation Model, conditional on the current economic conditions, the MSCI Model Baseline, Return to Long-term Trend Growth, and Slow Growth scenarios.

Exhibit 18 illustrates this point, for a particular uncertainty-tolerant institutional investor. If this investor believes in the MSCI Model Baseline scenario, where growth and inflation could remain below trend, the model calculates an optimal 54% tilt towards the growth hedging strategy. In contrast, if this same investor rather believes that economic growth in DM economies could quickly return to its trend, the model calculates an optimal 47% tilt towards the growth sensitive strategy over the growth hedging strategy tilt of the baseline scenario. Last, if this investor assumes that growth and inflation could decline further relative to the Baseline scenario (»slow growth«), the model indicates allocating fully to the growth hedging strategy.

Taken together, the historical analysis of the previous section and the model-based analysis of this section illustrate how the impact of macro-economic scenarios on factor and sector index performance, and optimal allocations to these indexes could be assessed in a structured way. The results outlined in this paper open up a number of important questions. In particular, one question is how to evaluate and back-test strategies that are conditioned on macro-economic views. Another question is how these strategies can be implemented

in a cost effective fashion. These issues will be explored in subsequent papers in the series.

Conclusion

A recurrent theme of the last few years among investors has been concern over the changing state of the economy and its impact on their investments. As a result, many institutional investors have started explicitly accounting for macro-economic conditions in their asset allocation decisions and, as part of that shift, are trying to create a process for identifying macro-economic-sensitive portfolios.

In this paper, the first in a series, we attempt to give this process a framework. We build both on a historical analysis using our 40+ years' history of MSCI factor and sector indexes and a long-term analysis based on forecasts from the MSCI Macro-Economic and MSCI Asset Pricing Models.

We start by classifying the MSCI factor and GICS 10 sector indexes based on their historical short- term and model-based long-term responses to economic growth and inflation. Next, using historical data, we classify economic regimes through a bivariate framework that groups economic regimes into four quadrants, depending on whether the CLI and CPI are rising or falling, and examine the differential performance of our index universe across these states, and the relative performances of the indexes through regime transitions. Finally, we apply the MSCI Asset Pricing Model to assess the likely performance of these indexes under statistically plausible forecasts and macro-economic scenarios, and to derive combinations of MSCI World factor and sector indexes that could potentially benefit from or offer protection under changing economic regimes.

Our results suggest that factor and sector indexes exhibit varied sensitivity to the economy. The historical analysis show Industrials, Consumer Discretionary and Information Technology as the three most cyclical sectors (highest correlations on average to the business cycle); and Health Care, Consumer

Staples, and Utilities as the most defensive (most negative correlations). In factor indexes Equal-Weighted and Value-Weighted have been the most cyclical, and Minimum Volatility, and Quality the most defensive.

The long-term models forecast that the Momentum, Value-Weighted, and Equal-Weighted Factor Indexes could exhibit the highest active returns following a large positive shock to trend growth in Developed Market (DM) economies (most »growth sensitive«). They also model that Minimum Volatility, High Dividend Yield and Quality could outperform following a large negative shock (»growth hedging«). Materials, Consumer Discretionary and Financials are the most growth sensitive sectors in the model, while Consumer Staples, Health Care, and Utilities are the most growth hedging. Here, short-term historical performance and long-term, model-based performance concur.

Our historical analysis adds a measure of inflation (CPI) and divides history into periods where CLI/CPI are rising/falling separately. The results are intuitive for the CLI, especially for the Minimum Volatility Index and defensive sectors, although CPI results are less conclusive.

However, the results suggest inflation impacts equities in different ways over the short and long-term. The model-based analysis looks at the long-term sensitivity of equity factor and sector indexes to inflation. Inflation could impact equities in two ways. Firstly, higher inflation generally has a negative impact on future long-term real GDP growth. Thus, Small Caps may see short-term out-performance as inflation rises, but be relatively more impacted by inflation in the long-term, as they are more sensitive to real GDP growth. Secondly, factors such as High Dividend Yield and Quality and sectors such as Consumer Staples, are generally long duration with stable nominal cash-flows. Higher inflation will have a more negative impact on their real cash-flows over time.

The behaviour of factor and sector indexes may also be sensitive to economic regimes and regime transitions. We classify economic regimes through a bivariate («Quadrant») framework of CLI/CPI rising/falling, and find that the Momentum and Equal-Weighted Factor Indexes have performed best in »Heating Up« and »Goldilocks« scenarios, while the Minimum Volatility Index showed the largest positive relative performance in a »Stagflation« environment. Our results also suggest that defensive sectors have outperformed most in »Slow Growth«. Turning to regime transitions, in an economic regime where growth is remaining positive or strengthening then the Equal-Weighted Factor Index has historically outperformed and Minimum Volatility underperformed. On the other hand, in an environment of slowing growth an investment in Quality has generally outperformed, and when coupled with low inflation then Minimum Volatility has outperformed also. In sectors, Utilities have generally underperformed in environments of strong growth, but when growth has fallen, then a Consumer Staples overweight has historically outperformed. If this drop in growth comes with increasing inflation (»Stagflation«) then Energy has been the standout performer.

Our framework and models have important implications for asset allocation: deviations away from the market cap portfolio depend on an investor's macro-economic views and tolerance for macro-economic uncertainty. For institutions that have their own views on short-term changes in macro-economic conditions that could have persistent effects on trend growth and inflation, the analysis above could help lead to tactical asset allocation between factor and sector indexes. For an uncertainty-tolerant investor assuming DM economic growth returns quickly to its trend, the models indicate allocating towards growth sensitive indexes. In contrast, if slow growth and low inflation are believed to persist, the models indicate allocating towards growth hedging indexes³⁰. Subsequent papers in the series will further explore cost effective implementations of macro-sensitive strategies based on MSCI Factor, Sector, and Thematic Indexes.

30 The results presented in this paper are based on the MSCI Macro-Economic, Asset Pricing and Asset Allocation Models. The models are statistical, data driven models, and do not contain investment choices or recommend any specific investments.

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Appendix

Start Dates For Equity Data

Exhibit 19: Start dates of factor and sector index data

Index Name	Gross Index Start Dates	Live Date (If Different)
<i>Cap-Weighted Indexes</i>		
MSCI World Index (Large+Mid Cap)	Dec 31, 1969	
<i>Factor Indexes</i>		
MSCI World Equal-Weighted Index	May 31, 1973	Jan 22, 2008
MSCI World High Dividend Yield Index	Nov 28, 1975	Oct 31, 2006
MSCI World Minimum Volatility Index (USD)	May 31, 1973	Apr 14, 2008 *
MSCI World Momentum Index	May 31, 1973	Dec 11, 2013
MSCI World Quality Index	Nov 28, 1975	Dec 18, 2012
MSCI World Risk-Weighted Index	May 31, 1973	Apr 06, 2011
MSCI World Value-Weighted Index	May 31, 1973	Dec 07, 2010
<i>Sector Indexes</i>		
MSCI World Sector Indexes	May 31, 1975	**

* Official history from May 31, 1988: Simulated using top-300 risk-weighted stocks prior.

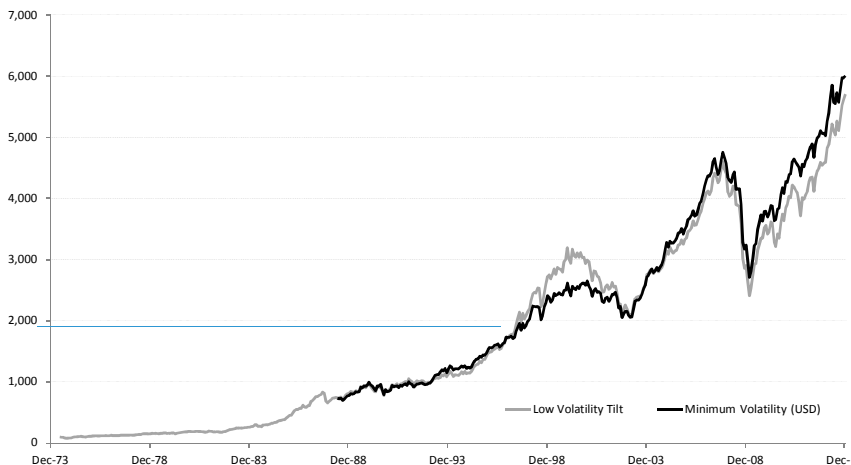
** Official history from December 31, 1994: Simulated using sector mapping (Exhibit 21).

Generation Of Minimum Volatility Index Proxy

Simulation Of Low Volatility Index Data Prior To 1988

For the MSCI Minimum Volatility (USD) Index, MSCI has official data available publicly back to the end of May 1988, which is when the first instance of the Barra risk model used to perform the optimization for the index was created. For the purposes of our study we have simulated a proxy for the MSCI Minimum Volatility Indexes for World and USA prior to May 31, 1988. The simulated proxy selects the top 300 stocks by lowest three-year weekly variance from the MSCI World Index, constructs a »score« which is the inverse of this variance, and then weights stocks in proportion to this score $\times$ market capitalization. Exhibit 20 compares the performance of our simulated Low Volatility Tilt Index with the official MSCI Minimum Volatility (USD) Index on the MSCI World universe. The historical tracking error between the two has been 4.40%.

Exhibit 20: Historical Performance of MSCI World Minimum Volatility (USD) Index and simulated World Low Volatility Tilt



Barra To GICS® Sector Mapping

(To Simulate GICS Sectors Pre-1994)

Exhibit 21: Barra Industry to GICS Sector mapping

GICS Sector	Barra Industry Classification
ENERGY	ENERGY SOURCES ENERGY EQUIPMENT & SERVICES
MATERIALS	BUILDING MATERIALS & COMPONENTS CHEMICALS FOREST PRODUCTS & PAPER METALS - NON FERROUS METALS - STEEL MISC. MATERIALS & COMMODITIES CONSTRUCTION & HOUSING GOLD MINES
INDUSTRIALS	AEROSPACE & MILITARY TECHNOLOGY INDUSTRIAL COMPONENTS MACHINERY & ENGINEERING TRANSPORTATION - AIRLINES TRANSPORTATION - ROAD & RAIL TRANSPORTATION - SHIPPING WHOLESALE & INTERNATIONAL TRADE
CONSUMER DISCR	APPLIANCES & HOUSEHOLD DURABLES AUTO-MOBILES RECREATION, OTHER CONSUMER GOODS TEXTILES & APPAREL BROADCASTING & PUBLISHING LEISURE & TOURISM MERCHANDISING
CONSUMER STAPLES	BEVERAGES & TOBACCO FOOD & HOUSEHOLD PRODUCTS
HEALTH CARE	HEALTH & PERSONAL CARE
FINANCIALS	BANKING FINANCIAL SERVICES INSURANCE REAL ESTATE
INFORMATION TECH	DATA PROCESSING & REPRODUCTION ELECTRICAL & ELECTRONICS ELECTRONIC COMPONENTS, INSTRUMENTS BUSINESS & PUBLIC SERVICES
TELECOM SVC	TELECOMMUNICATIONS
UTILITIES	UTILITIES - ELECTRICAL & GAS

Further Details On Economic Data Series – OECD CLI

The starting point of our study was to use some measure of the overall state of the economy to classify the current economic regime. Some of the existing literature on this topic defines economic expansions and contractions according to changes in gross domestic product (GDP) or references the National Bureau of Economic Research (NBER) business cycle definitions. This literature finds that the excess returns of a sector rotation strategy are not significant (for example, Stangl et al. (2009)).

We chose to start our historical analysis by extending earlier MSCI work³¹ which classified sectors as defensive or cyclical based on the strength of their co-movement with the business cycle. In that study, we used the Organization for Economic Cooperation and Development's (OECD) Composite Leading Indicator (CLI) series to define expansions and contractions (as stock prices usually anticipate economic up-and downturns). When defining business cycles in this way, we find that performance differentials between cyclical and defensive sectors are important.

As described by the OECD, the system of CLIs is designed to provide early signals of turning points in business cycles, i.e. fluctuation of the economic activity around its long-term potential. Details on the data series chosen by country, the weighting schemes used, and the data filtering employed are available in the OECD's »System of Composite Leading Indicators« methodology document.

We compare the MSCI World factor and sector indexes to the OECD – Total CLI series using monthly observations.

31 Kouzmenko, R. and Z. Nagy, 2009, »Sector Performance Across Business Cycles,« MSCI Research Bulletin.

CPI

Having looked at factor and sector performance based on overall response to changes in the OECD CLI, we wanted to add an extra dimension of discrimination using inflation, firstly, as a separate univariate variable and, then, also as a second independent variable in conjunction with CLI.

Once again we turned to the OECD to source our measure of inflation. The measure of inflation we chose was the annual inflation rate, i.e. the movement of the Consumer Price Index (CPI) from one month to the same month of the previous year expressed as percentage over time.

Within the family of six CPIs calculated by the OECD we use »CPI All items«. In a similar fashion to CLI, further details on the OECD's CPI dataset and calculations are available in the OECD's Methodological Notes. As with CLI, we compare MSCI World sector and factor Indexes to OECD – Total CPI using monthly observations.

Stress Tests

Using Actual Historical Economic Regime Changes

In this paper we have tried to help investors understand how their equity factor and sector investments have performed, historically, in the presence of macro-economic uncertainty by characterizing their responses as various macro-economic variables have changed. One frequent question we receive from investors is: »How would my portfolio have performed through (for example) the 1990–91 economic slowdown?« Such scenario analyses are commonly called stress tests, and we have taken a subset of those used in our BarraOne risk management software and shown how the MSCI factor and sector indexes have performed over these in Exhibit 22.

2. Choosing Right Factors

Exhibit 22: (a) MSCI World factor indexes, and (b) MSCI World sector indexes active returns and (c) scenario dates for historical market stress periods

(a)

Event	MSCI World Return	Equal-Weighted	High Dividend Yield	Minimum Volatility ^{***}	Momentum	Quality	Risk-Weighted	Value-Weighted	Change in Adjusted CLI	Change in Rolling Inflation	Change in VIX
1987 Market Crash (Aug. to Nov.)	-16%	-4%	-7%	-2%	8%	-9%	-4%	-5%	0.02	0.9%	28.28
1990 – 1991 Economic Slow Down	-13%	0%	15%	1%	7%	19%	4%	7%	-1.83	1.1%	4.01
1994 US Rate Hike	-1%	-1%	-4%	-1%	-7%	1%	-1%	0%	0.88	1.0%	2.55
1997 – 1998 Asian Financial Crisis	3%	-14%	6%	3%	6%	7%	-6%	0%	-0.01	0.0%	0.47
2001 Dot-com Slowdown	-37%	4%	11%	15%	11%	12%	15%	2%	-0.48	-1.1%	1.22
2007 – 2008 Oil Price Rise	-1%	-5%	-10%	-4%	17%	9%	-5%	-6%	-0.88	2.1%	12.09
2007 – 2008 Subprime Mortgage Meltdown	-6%	-3%	-3%	3%	5%	5%	-1%	-3%	-0.67	1.5%	-0.99
2010 Peripheral European Debt Crisis	-10%	2%	3%	9%	1%	2%	5%	0%	0.17	-0.5%	6.42
2011 US Debt Ceiling Act	-18%	0%	6%	13%	7%	7%	6%	0%	-0.91	0.5%	28.21

(b)

Event	MSCI World Return	World Energy	World Material	World Indust.	World Cons. Discr.	World Cons. Staples	World Health Care	World Fins.	World Info. Tech.	World Telco	World Utils.	Change in Adjusted CLI	Change in Rolling Inflation	Change in VIX
1987 Market Crash (Aug. to Nov.)	-16%	-11%	2%	1%	-7%	0%	-3%	5%	-4%	3%	17%	0.02	0.9%	28.28
1990–1991 Economic Slow Down	-13%	9%	-8%	-8%	-3%	18%	23%	-7%	11%	5%	4%	-1.83	1.1%	4.01
1994 US Rate Hike	-1%	2%	7%	0%	0%	5%	7%	-6%	16%	-5%	-9%	0.88	1.0%	2.55
1997–1998 Asian Financial Crisis	3%	-2%	-18%	-11%	-3%	2%	11%	1%	3%	16%	9%	-0.01	0.0%	0.47
2001 Dot-com Slowdown	-37%	16%	14%	-2%	-1%	30%	14%	1%	-20%	-16%	-1%	-0.48	-1.1%	1.22
2007–2008 Oil Price Rise	-1%	50%	46%	2%	-19%	7%	-8%	-27%	3%	0%	19%	-0.88	2.1%	12.09
2007–2008 Subprime Mortgage Meltdown	-6%	13%	16%	-2%	-7%	11%	3%	-11%	-5%	2%	12%	-0.67	1.5%	-0.99
2010 Peripheral European Debt Crisis	-10%	-2%	-1%	1%	4%	7%	1%	-2%	-2%	12%	8%	0.17	-0.5%	6.42
2011 US Debt Ceiling Act	-18%	-2%	-6%	-4%	2%	10%	7%	-6%	7%	8%	12%	-0.91	0.5%	28.21

(c)

Event	Start Date Used	End Date Used
1987 Market Crash (Aug. To Nov.)	31-Jul-87	30-Nov-87
1990 – 1991 Economic Slow Down	29-Dec-89	31-Jan-91
1994 US Rate Hike	31-Jan-94	30-Dec-94
1997 – 1998 Asian Financial Crisis	30-Jun-97	30-Jan-98
2001 Dot-com Slowdown	28-Feb-01	31-Oct-02
2007 – 2008 Oil Price Rise	29-Dec-06	30-Jun-08
2007 – 2008 Subprime Mortgage M	31-Jul-07	31-Mar-08
2010 Peripheral European Debt Cr	31-Mar-10	31-Aug-10
2011 US Debt Ceiling Act	29-Apr-11	30-Sep-11

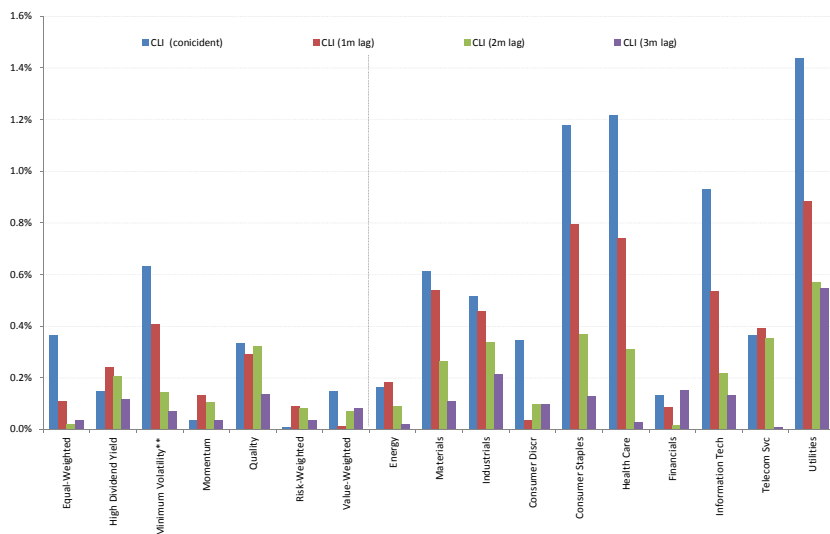
A Note About Our Lack Of Perfect Foresight

Our historical analysis has been built by looking back in time and classifying economic regimes via the levels of, and changes in, two variables, the CLI and CPI, and then seeing how equity factor and sector index returns have historically moved. Our historical analysis ended in December 2013, and it is March 2014 at time of writing, so all the historical data we have needed has been available to us.

However, economic data is rarely, if ever, available at the same time as the period which it attempts to describe, and can also be subject to revision. For example, OECD CLI and CPI data is generally produced with a two-month lag (so January but not February 2014 data is available now). In short, we only have enough data now to tell us which state we were in two months ago.

One solution to avoid this look-ahead bias would be to lag our OECD data by two months, but as Exhibit 23 shows, the index responses decay as we lag the leading indicators. The equity markets respond to leading indicators without a lag.

Exhibit 23: Absolute active return differential between periods of increasing and decreasing CLI as data is lagged



Another idea would be to assume that the market can anticipate some of the components of these leading indicators, even though they aren't officially observed or recorded until later and use the various forecasts which exist, but the predictive capability of these forecasts is hard to prove.

So, while we don't find lagging our data or using economic forecasts useful, we do feel that we can give investors the comfort that this is not simply an exercise in historical data crunching, but can be useful in making forward looking investment decisions, for the reasons explained below:

Firstly, as shown in Exhibit 12 above, knowing which regime we were in two months ago can give us reasonable certainty of the regime we are in now, as transitions happen gradually and smoothly.

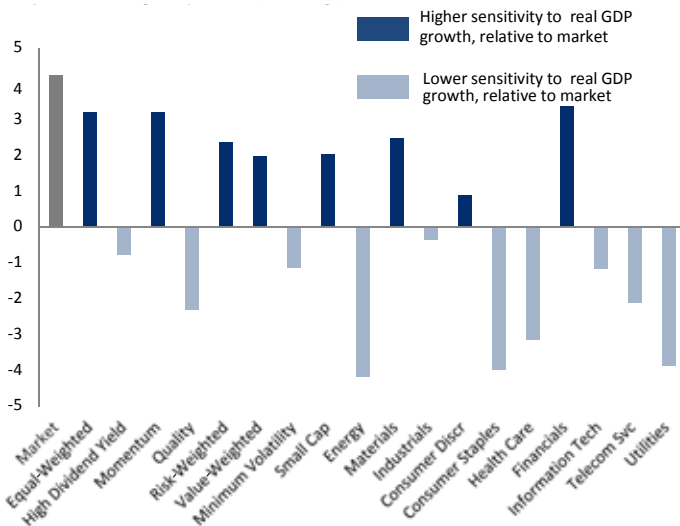
Secondly, we believe investors can use equity returns over the periods between economic variable observations to anticipate regime transitions (including the »transition« of staying in the same state). It is clear that there is a relationship between stock returns and leading indicators although we don't claim that there is

predictive value. Examining actual returns and comparing these to those observed historically over regime transitions can give us an indication on the likelihood and direction of an upcoming state change, although we will never be able to say with certainty which regime we are in now or where we will be in the future.

Finally, as shown in the long-term model-based analysis section of this paper, in the long run stocks tend to react slowly to GDP shocks. It can take up to five years or longer for such a shock to be incorporated into returns. Whereas, the aim of our historical analysis was to show the relationship between economic variables and short-term equity index returns, investors thinking strategically about long-term equity allocations under economic regime shifts should focus on this framework.

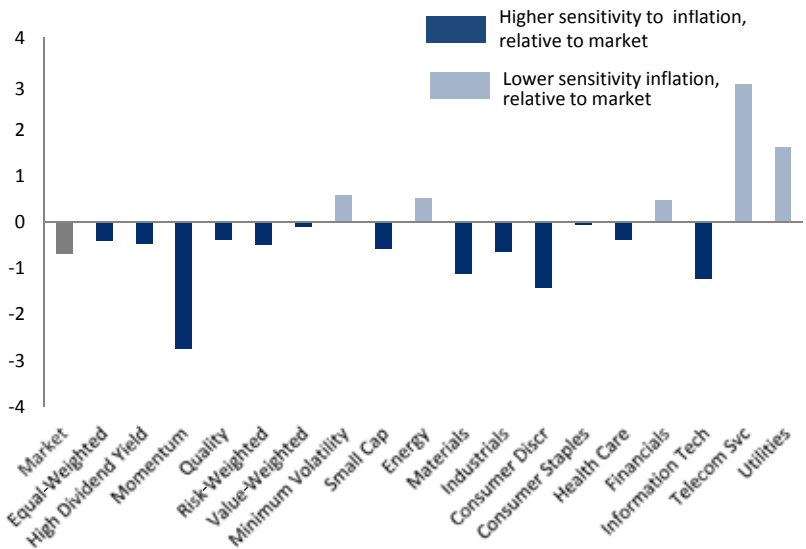
Equity Portfolios Returns Differ In Their Long-Run Sensitivity To Macro-Economic Shocks

Exhibit 24: A return to trend growth in DM economies is positive for equity real returns
Three-year cumulative impact on portfolio value, relative to the capitalization-weighted portfolio (Percentage)



The Exhibit shows the three-year cumulative impact of a 1.5% positive shock to real GDP growth on the MSCI World Index («Market»), MSCI World factor and sector indexes real returns. Return responses are relative to the baseline three-years out growth forecast for DM economies of 1.5%.

Exhibit 25: High inflation is negative for equity factor and sector returns
Three-year cumulative impact on real portfolio value, relative to the capitalization-weighted portfolio (Percentage)



The Exhibit shows the three-year cumulative impact of a 1.5% positive shock to inflation on the MSCI World Index («Market»), MSCI World factor and sector indexes real returns. Return responses are relative to the baseline three-year out inflation forecast for DM economies of 1.5%.

Monthly Macro-Economic Indicator Heat-Map

Exhibit 26: Monthly macro-economic indicator heat-map

As of 03/02/2014	Industrial Produc- tion	Retail Sales	Unem- ploy- ment	CPI Inflation	Unit Labor Cost	Trade Balance	Ex- ports
US	0.40	0.24	-1.08	0.67	0.23	-0.06	-0.01
Canada	0.44	-3.75	-0.36	-0.25	0.10	0.77	0.13
UK	0.44	0.18	-0.59	-0.27	0.00	0.04	-0.26
Germany	0.16	0.25	0.20	-0.65	0.11	0.48	0.01
France	0.24	0.11	0.19	-1.03	0.05	0.14	-0.11
Spain	0.23	-0.55	0.26	-0.53	-0.24	-0.01	-0.15
Italy	0.04	-0.02	0.85	-2.88	0.01	1.26	0.04
Japan	0.75	0.45	-0.63	1.10	-0.65	-1.49	0.60
China	-0.07	0.08	-0.18	0.05	0.18	1.26	0.41

The heat map shows, for each variable, the change in their realized value over previous quarter's trend, normalized by their historical standard deviation. Shaded figures indicate above one standard deviation changes (positive or negative). Red colored scores contribute to a potential decline in real GDP relative to previous quarter trend. Green colored scores contribute to a potential increase in real GDP relative to previous quarter trend. Bold figures indicate scores that remain in positive/negative territory.

The MSCI Macro-Economic Model: Measuring Macro-Economic Risk

The MSCI Macro-Economic Model is a collection of single country models that are related to one another through global growth and global inflation. Each country-specific model is a mixed- frequency Bayesian Vector Autoregressive model (BVAR) that aims to generate timely forecasts of macro-economic variables (such as real GDP growth, CPI inflation), and forecast scenarios for these macro-economic variables.

Introduced by Sims (1980a, 1980b), BVARs are commonly used, data driven statistical models that explain the joint behaviour of observable time series of multiple variables. These models are parsimonious devices to capture the complex relationships between different macro-economic variables, and their evolution over time. In a BVAR model, each variable mechanically depends on its own lagged values, and the lagged values of all the other variables. A joint probability distribution (or prior) is specified for the time series of macro-economic variables, and systematically updated based on available information using Bayes rule.

The specification of prior distributions for the time series of macro-economic and financial variables is standard in the academic literature, following the work of Doan, Litterman, and Sims (1984). These distributions impose statistical restrictions in models employing a large number of variables with few observations. These restrictions avoid over-fitting the model to the data. The prior distribution employed in the MSCI Macro-Economic Model is grounded on observed long-run empirical regularities in macro-economic variables.

An attractive feature of our model is that it produces timely forecasts. In particular, it overcomes the important problem of delays in macro-economic data releases by applying the statistical methodology developed in Schorfheide and Song (2012). This methodology efficiently combines monthly macro-economic indicators observed in a timely fashion, together with the typically lagging quarterly macro-economic variables. For example, in addition to real GDP growth and corporate profits that are only observed on a quarterly basis, typically, with a lag, the MSCI Macro-Economic Model also made efficient use of variables available on a more timely, monthly basis, such as CPI inflation, term spread (10-Year minus 3 months), money growth (M_2), Fed base rate, the SPGS Commodity Index, unemployment rate, exports, capacity utilization, and labor unit cost.

Given observed historical time series on macro-economic variables, and the data-driven prior distribution assumption, Monte Carlo simulation techniques are used to estimate the parameters of the model. Macro-economic forecasts and forecast scenarios are mechanically derived using the estimated lead-lag

dynamic structure of the model. As an example, Exhibits 27 and 28 show the model's forecasts as of March 2, 2014 for the year ahead and three years out, and confidence bands for these forecasts.

Exhibit 27: Global trend growth is projected to remain below long-term average

Country								
Global	3.8	2.5	2.4	0.7	4.0	2.3	0.4	4.2
US	3.4	2.4	2.3	0.9	3.8	2.3	0.8	3.9
Canada	3.5	2.2	1.9	0.6	3.0	1.8	0.3	3.2
UK	2.8	2.7	2.3	1.0	3.6	1.6	0.3	3.2
France	3.4	1.2	1.1	-0.1	2.2	1.0	-0.3	2.3
Germany	1.8	1.8	2.3	0.5	4.1	2.0	0.0	4.1
Spain	4.0	0.2	1.7	0.6	2.8	1.8	0.2	3.4
Italy	3.0	-1.1	-0.5	-2.2	1.2	-0.7	-2.8	1.2
Japan	3.5	2.0	1.1	-1.2	3.5	0.8	-1.8	3.1
Australia	3.5	2.3	2.5	1.4	3.6	2.6	1.4	3.7
China	7.5	7.8	8.4	6.7	10.0	8.6	6.6	10.6
South Korea	7.5	4.1	3.1	0.7	5.5	2.8	0.3	5.4
India	6.3	3.4	5.0	2.5	7.1	4.7	1.6	7.4
Brazil	2.7	0.2	-0.4	-3.8	2.7	1.5	-2.6	4.6
Russia	6.9	0.0	0.3	-4.9	5.7	1.0	-6.1	7.6

The Exhibit shows year-ahead and three-years out forecasts for real GDP growth from the MSCI Macro-Economic Model, and compares these forecasts to the historical long-term trend through 2007 and the past year's average. All growth rates are annualized percentage rates. »Low« and »High« forecasts represent the 30–70 confidence bands around the baseline forecasts.

Exhibit 28: Global inflation is projected to remain persistently low and low long-term average

Country	Long-Term Trends Through 2007	Last Four Quarters	MSCI Year Ahead Forecasts			MSCI Three-Years Out Forecast		
			Base-line	Low	High	Base-line	Low	High
Global	3.8	.8	1.7	1.0	2.4	1.8	0.9	2.7
US	4.0	.7	1.7	1.3	2.0	1.7	1.2	2.3
Canada	2.5	.3	1.4	0.8	2.0	1.4	0.8	2.1
UK	2.7	.6	2.5	1.2	3.7	2.7	1.5	4.0
France	1.8	.5	0.7	-0.2	1.5	1.0	0.0	2.1
Germany	2.0	.1	1.5	0.7	2.3	1.6	0.7	2.5
Spain	3.3	.7	0.0	-0.8	0.8	0.8	-0.4	2.2
Italy	6.4	.2	1.1	0.0	2.1	1.2	0.0	2.3
Japan	3.5	.7	0.7	-0.1	1.5	0.8	-0.1	1.6
Australia	3.4	.6	2.3	1.6	3.1	2.5	1.8	3.2
China	0.4	.3	2.5	0.8	3.9	2.6	0.5	4.6
South Korea	6.6	.9	2.0	1.1	2.9	2.0	0.9	3.1
India	7.2	.9	8.6	5.8	11.7	9.5	6.0	13.1
Brazil	7.2	.1	5.5	3.3	7.6	5.6	3.2	7.9
Russia	13.1	6.2	4.5	1.8	7.3	5.5	1.1	9.7

The Exhibit shows year-ahead and three-years out forecasts for consumer price inflation from the MSCI Macro-Economic Model, and compares these forecasts to the historical long-term trend through 2007 and the past year's average. All inflation rates are annualized percentage rates. »Low« and »High« forecasts represent the 30–70 confidence bands around the baseline forecasts.

MSCI Equilibrium Asset Allocation Model

The asset pricing model that we use to establish long-run returns is based on assumptions about a »representative« investor. In our analysis, this »stand-in« investor cares about both the risk inherent in today's investment opportunity set, and also the uncertainty about the evolution of investment opportunities over time. It is the evolution of investment opportunities over time that is most sensitive to shocks to economic conditions. Our research shows how an investor's investment horizon and willingness to tolerate macro-economic uncertainty informs asset allocation decisions.

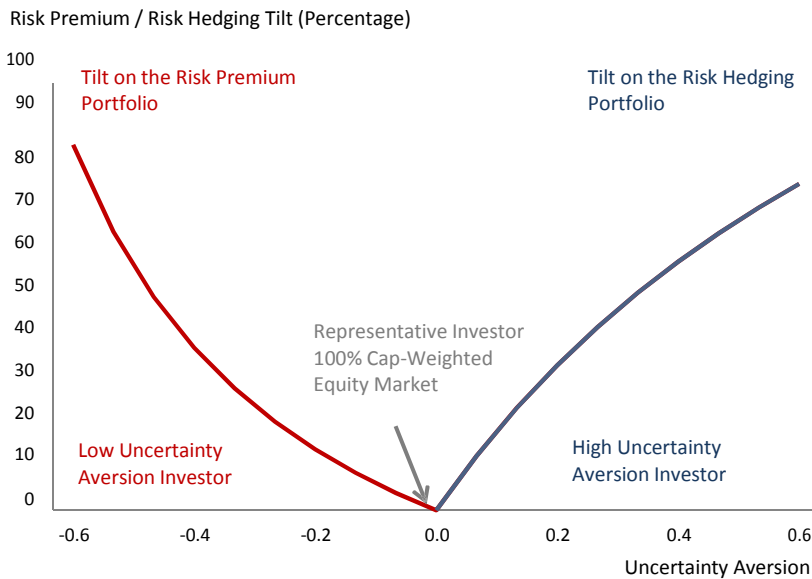
Our framework and models suggest that the observed historical premiums on strategies such as value and small cap are compensation for persistent shocks to trend growth. These premiums are also consistent with the hypothetical investor holding the market portfolio. However, specific investors can reasonably deviate from holding market cap weights. The issue is: what would cause investors to deviate from holding the market portfolio?

In the usual CAPM framework, investors care about risk aversion. Differences in risk aversion among investors lead to adjustments in the split between risky and risk-free assets, but not in the composition of the risky portfolio; in equilibrium, the risky portfolio for all investors is the market portfolio. In our analysis, investors are both risk averse and uncertainty averse. Aversion or tolerance to uncertainty is what dictates an investor's aversion to, or preference for, high cash flow beta portfolios. Preferences for high cash flow betas, in turn, drive allocations to either the risk premium portfolio or the risk hedging portfolio.

Exhibit 29 illustrates these points. In the graph, the representative investor holds the market portfolio where all strategies and assets are held in their capitalization weights. Now suppose that an investor can tilt towards either risk premium or risk hedging portfolios. For this example, the risk premium portfolio is assumed to be a value-weighted portfolio of US value, small cap, consumer discretionary, financials, real estate and materials, while the risk hedging portfolio is assumed to be a value-weighted portfolio of US growth, large cap, utilities, health care, industrials, telecoms, energy, information technology, and consumer staples.

What happens to the strategy mix as uncertainty aversion changes? As Exhibit 29 illustrates, the uncertainty averse investor (represented by a high uncertainty aversion parameter) tilts away from the market portfolio and allocates a larger proportion of the portfolio to the risk hedging portfolio. By contrast, the uncertainty seeking investor (represented by a low uncertainty aversion parameter) tilts away from the market portfolio and allocates a larger proportion of the portfolio to risk premium assets.

Exhibit 29: Portfolio holdings change with uncertainty aversion



The Exhibit shows how long-run portfolio holdings change with an investor's uncertainty aversion. The representative investor holds the 100% capitalization-weighted equity market portfolio. The uncertainty aversion (UA) scale is normalized so that the representative investor has an UA of 1. For example, an investor with an UA of 1.2 is 20% more uncertainty averse than the representative investor, and an investor with an UA of 0.8 is 20% more uncertainty-tolerant. The representative investor's uncertainty aversion was calibrated to match the observed equity market average return of 7.9% from 1950 to 2011. Investors that are more uncertainty averse tilt towards the Risk Hedging portfolio, while investors that are more uncertainty-tolerant tilt towards the Risk Premium portfolio.

The asset allocation differences in Exhibit 32 raise the question: why does uncertainty aversion vary among investors? One possibility is that there are classes of investors for whom short-run horizon effects are important. These investors may not be willing to tolerate large short-term swings in portfolio value, nor have the patience for the long-term resolution of uncertainty. An example of such an investor class could be individual investors. By contrast, other classes of investors may exist for whom short-term effects are less relevant. They may be long-lived organizations whose short-term spending needs are not driven by portfolio value. Consequently, they may have the patience required for the long-term resolution of uncertainty.

While investor attitudes about uncertainty can vary by investor class, it is important to recall that the aggregate of all investor holdings must equal the market portfolio. Looking again at Exhibit 29, the differences in asset allocation illustrate how an equilibrium, where the representative investor holds the market portfolio, can be compatible with differences in asset allocations across investor classes.

Incorporating Risk Premia Mandates In A Strategic Allocation. A Client Case Study: Wyoming Retirement System

Raman Aylur Subramanian

MSCI

The Challenge

Wyoming Retirement System (WRS), a public pension plan sponsor with a 50% target policy allocation to global equities, wished to explore options for diversifying its global equity portfolio. The investment staff had three main objectives: to lower volatility, improve risk-adjusted returns and decrease fees. To achieve these goals, WRS examined extensive academic and industry research and ultimately decided to include risk premia mandates in its strategic allocation.

The WRS Approach

The policy benchmark for the WRS equity allocation is MSCI ACWI, a broad global benchmark that includes the large and mid-capitalization segments of the global equity universe. Panel A of Exhibit 1, illustrates the WRS policy benchmark in comparison to the pension plan's final strategic allocation – with about 70% of the total equity allocation assigned to passive managers, and the other 30% mandated to active managers. The WRS investment staff decided to allocate 70% of the total passive allocation to a portfolio that tracked MSCI ACWI IMI, a broad global benchmark that encompasses the large, mid *and small* capitalization segments of the global equity universe. This

2. Choosing Right Factors

allocation provided a strategic overweight of approximately 10% to global small caps (size risk premium) in addition to capturing the core global equity risk premium. They also elected to make two strategic allocations – each of 15% – to portfolios that passively tracked the MSCI ACWI Value-Weighted and MSCI ACWI Risk-Weighted Indices. The two portfolios aimed to capture the value and low volatility risk premia, respectively. By design, these two portfolios were tilted towards lower capitalization stocks, which led to approximately equal allocations to the size, value and low volatility risk premia in the overall strategic equity allocation.

Panel B of Exhibit 1, presents the risk and return characteristics of the plan's strategic allocation to passive managers versus its policy benchmark which is assumed to be 100% invested in MSCI ACWI. All return estimates in Exhibit 1 ignore both transaction costs and management fees. It can be seen that a strategic allocation to risk premia, along with a core allocation to the market portfolio, provided better risk adjusted performance over the policy benchmark during the analysis period.

Exhibit 1: Strategic Allocation to Risk Premia

<i>Policy Benchmark Portfolio For Equities (MSCI ACWI Index)</i>		<i>Strategic Allocation to Risk Premia (market+size+value+volatility)</i>
Panel A: Portfolio Allocations		
<i>Market Beta</i>	100%	70 % MSCI ACWI IMI Index
<i>Size</i>		
<i>Value</i>		15 % MSCI ACWI Value-Weighted Index
<i>Low Volatility</i>		15 % MSCI ACWI Risk-Weighted Index
Panel B: Annualized Risk and Return Characteristics (November 1995 to March 2012)		
<i>Return (%)</i>	6.2	6.9
<i>Risk (%)</i>	16.6	16.5
<i>Return/Risk</i>	0.37	0.42
<i>Tracking Error</i>	0.00	1.82
<i>Beta</i>	1.00	0.98

Source: MSCI. Portfolio weights are rebalanced quarterly to the target weights.

Before coming to the above-mentioned final passive equity allocation, the WRS investment staff conducted what it described as an extremely thorough due diligence. In the following section we review the decision points (or due diligence questions) that the WRS investment staff reports that it considered over the course of their risk premia allocation process. These measures supplemented the existing policies and procedures regarding equity allocation, as approved under WRS investment policy.

Certain Due Diligence Considerations For Risk Premia Allocations – Cyclical Of Risk Premia Into The Future

Numerous studies have demonstrated the existence of risk premia in the past. However, the active returns of risk premia strategies are cyclical. As more (fewer) investors are willing to bear a particular kind of risk, the *ex ante* compensation for that risk decreases (increases). The first decision point for the pension plan concerned the future behaviours of risk premia. To what extent did WRS investment staff think that the past risk-adjusted outperformance of risk premia could be expected to persist in the future?

The interplay of investor flows and factor premia is complex, particularly when examined over varying time periods. It is expected that relative risk premia returns will wax and wane in the future, as in the past. (Exhibit 2, for example, displays the cyclicalities found in historical risk premia returns from May 1994 – February 2012.) The magnitude and duration of these cyclical patterns largely depends upon the sources of each risk premium. For example, the value premium can be explained by both a rational phenomenon, priced in equilibrium as compensation for systematic risk, or by the irrational or behavioural view that reflects the tendency of certain investors to overreact to good and bad news.

Exhibit 2: Risk Premia Performance over Time



Source: MSCI. Note: Small minus Large represents the performance difference of the MSCI World Small Cap Index relative to the MSCI World Large Cap Index. Value minus Growth represents the performance difference of the MSCI World Value Index relative to the MSCI World Growth Index. Low Vol minus High Vol represents the performance difference of simulated volatility equity index consists of the bottom one-third (Low Vol) and top one-third (High Vol) of the MSCI World Index market capitalization ranked by security variance. The security variance is calculated using weekly returns over 52 weeks prior to the semi-annual rebalancing date.

The WRS investment staff concluded that by diversifying across multiple uncorrelated risk premia the pension could improve the risk-adjusted returns of its equity allocation. For example, by allocating to lower volatility risk premia and assuming a similar expected return as the market portfolio, the pension decided it could improve the overall risk adjusted return of its equity allocation. Exhibit 3 displays the correlations between different risk premia and the global market beta as represented by the MSCI World Index. Most of the correlations among the risk premia shown in Exhibit 3, are low or negative and confirm that these individual risk premia did capture unique return characteristics and offered diversification over a nearly 18-year period.

Exhibit 3: Correlations among Risk Premia (Correlations over May 1994 – February 2012)

	MSCI World Index	Small minus Large	Value minus Growth	Low Vol minus high Vol
MSCI World Index	1.00			
Small minus Large	0.04	1.00		
Value minus Growth	-0.23	-0.02	1.00	
Low Vol minus High Vol	-0.73	-0.21	0.53	1.00

Source: MSCI. Note: Small minus Large represents the performance difference of the MSCI World Small Cap Index relative to the MSCI World Large Cap Index. Value minus Growth represents the performance difference of the MSCI World Value Index relative to the MSCI World Growth Index. Low Vol minus High Vol represents the performance difference of simulated volatility equity index consists of the bottom one-third (Low Vol) and top one-third (High Vol) of the MSCI World Index market capitalization ranked by security variance. The security variance is calculated using weekly returns over 52 weeks prior to the semi-annual rebalancing date. Strategic versus Tactical

A second decision point for the pension plan was to determine whether an allocation to risk premia should form a part of its strategic equity mix or whether it should be a purely tactical one.

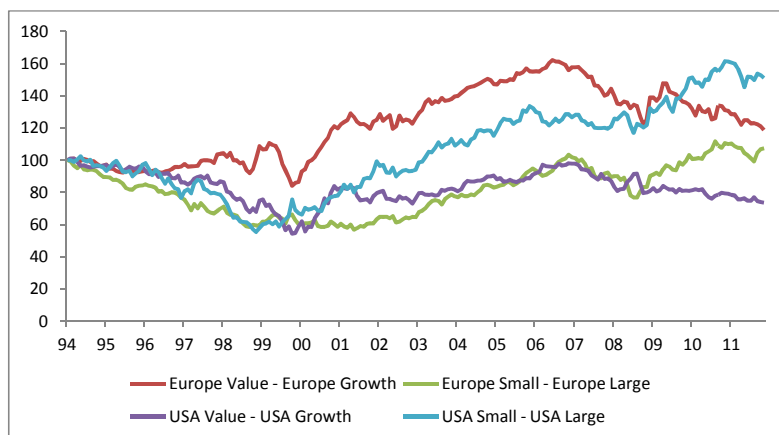
Often pension plans seek to obtain risk premia tilts in their strategic allocations by assigning mandates to managers who implicitly or explicitly aim to capitalize on these premia and add alpha. Several studies have shown that for a majority of managers, their performance results are attributable to having captured a risk premium with no persistent alpha over and above the premium itself (Grinblatt et al., 1995, Carhart, 1997). This raises the question as to whether investors should strategically allocate to risk premia rather than obtaining these exposures through a manager selection process. One key benefit to making strategic allocations to risk premia is that the plan can better control portfolio risk. Each risk premia allocation has a distinct risk-return profile relative to the policy benchmark, which is typically a market capitalization-weighted global index. By recognizing the implications of adding risk premia exposures at the strategic asset allocation level, the WRS investment staff believed that they could better control the intended deviation (or active exposure relative to the policy benchmark) from the WRS equity policy benchmark, the MSCI ACWI Index.

Which Risk Premia To Capture?

Empirical studies have documented the existence of risk premia across various asset classes. The better known risk premia within equities include, size (Banz, 1981), value (Fama and French, 1992), momentum (Jegadeesh and Titman, 1993) and low volatility (Haugen and Baker, 1991). The next question the WRS investment staff confronted was to select which of these four equity risk premia they wanted to have strategic exposure to. As WRS sought to capture the risk premia in a passive manner, they were concerned that transaction costs might eat away at any potential risk premia outperformance. The impact of transaction costs on the capture of a broad market or the equity risk premium through a market capitalization index is low, because the market portfolio is a buy-and-hold portfolio and is associated with a very low level of turnover. The turnover for size, value and low volatility risk premia indices are also modest and, therefore, transaction costs are unlikely to erode their respective premia. WRS investment staff carefully evaluated the expected transaction costs and ultimately decided to strategically allocate to the size, value and low-volatility risk premia.

Global Versus Local Risk Premia

The next decision point for WRS concerned allocations to domestic versus global risk premia. Like the risk premia observed in the USA equity markets, many academic studies have confirmed similar premia in global equity markets (Chan, Hamao, and Lakonishok (1991), Fama and French (1998), Rouwenhurst (1998), Asness, Moskowitz, and Pedersen (2009)). The WRS policy objective was to capture a global equity risk premium, therefore, the WRS investment staff wished to capture additional risk premia in a global context. Exhibit 4 displays the historical existence of size and value risk premia across equities in the USA and Europe. Interestingly, over the May 1994 – February 2012 analysis period, both markets showed similar performance patterns for these risk premia.

Exhibit 4: Risk Premia Across Markets

Optimal Allocation

A fifth decision point for WRS was to determine how much capital to allocate to each of the risk premia in order to obtain a well-diversified portfolio. It is possible to apply an optimization process for allocating among a portfolio of risk premia to maximize the portfolio Sharpe Ratio; however, this exercise requires accurate estimates of risk and return for each of the risk premia and may be prone to estimation errors. One method for overcoming the challenges of optimization is to use an equal-weighted approach. Equal weighting is a special case of mean-variance optimization that assumes all the risk premia have the same correlation coefficients, together with identical means and variances. Equally weighted portfolios are widely used in practice and they have been shown to be efficient out-of-sample (DeMiguel, Garlappi and Uppal (2009)). The WRS investment staff decided to apply an equal-weighted approach to diversify across the risk premia.

Funding Risk Premia

The next WRS decision point related to how they would fund their risk premia allocation. Should they decrease the core passive allocation or decrease the allocation to active managers? The WRS investment staff pursued a more passive investing approach overall, allocating 70% of its equity to a passive strategy and allocating 30% of its total passive equity portfolio to the passive capture of risk premia.

Implementing Risk Premia

A further decision point related to how WRS would capture the risk premia. Should they use passive long-only risk premia tilted portfolios or funds that apply proprietary algorithms to create long-short portfolios to capture the risk premia. Although these proprietary funds have the potential benefit of capturing risk premia in a purer form (versus long-only funds), they can be constrained by capacity (especially on the short side of the portfolios) and, in general, are more expensive to gain access to. The WRS objective was to access to risk premia in a passive and cost effective manner, and therefore they agreed to implement a long only risk premium tilted portfolio.

Other Considerations

Today many active managers create passive fund options that track benchmarks that they create and calculate themselves. These self-created benchmarks introduce a potential conflict of interest and a governance concern that pension boards and trustees often raise – that is, the separation of roles between index providers and managers whose performance is, generally, measured against index benchmarks. The WRS investment staff identified this as the final, but the most critical, due diligence issue that it debated before making its final decision as to which risk premia benchmarks they should choose. Ultimately, WRS opted to go with an independent index provider, without the potential conflicts of interest entailed by a money manager or investment consultant.

Additionally, the WRS investment staff recommended adopting risk premia benchmarks that were constructed with a rules-based, objective, transparent, investable and replicable methodology and with appropriate index management guidelines to keep the costs of replication reasonably low.

Conclusion

The course of inquiry which led WRS to strategically allocate to granular risk premia may be useful for other pension plans, even those operating under different constraints. This case study demonstrates how allocating to risk premia in the strategic asset allocation process enabled one sponsor to potentially lower volatility, improve risk-adjusted returns and decrease fees.

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Foundations Of Factor Investing

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Executive Summary

Factor investing has become a widely discussed part of today's investment canon. In this paper, we discuss the rationale for factor investing and how indexes can be constructed to reflect factor returns in cost-effective and transparent ways.

A factor can be thought of as any characteristic relating a group of securities that is important in explaining their return and risk. A large body of academic research highlights that long-term equity portfolio performance can be explained by factors. This research has been prevalent for over 40 years; Barra (now an MSCI company) for instance has undertaken the research of factors since the 1970s.

Certain factors have historically earned a long-term risk premium and represent exposure to systematic sources of risk. Factor investing is the investment process that aims to harvest these risk premia through exposure to factors. We currently identify six equity risk premia factors: Value, Low Size, Low Volatility, High Yield, Quality and Momentum. They are grounded in academic research and have solid explanations as to why they historically have provided a premium.

MSCI has created a family of factor indexes that are designed to reflect the performance of those six equity risk premia factors. In turn, indexation has provided a powerful way for investors to access factors in cost-effective and transparent ways. Factor allocations can be implemented passively using factor indexes, which may bring potential cost savings to institutional investors. Furthermore, factor indexes bring transparency to factor allocations, which

helps alleviate the well-known problem of manager style drift and has positive implications for risk management.

We note that factor indexes should not be viewed as replacements for market cap indexes. Market capitalization-weighted indexes represent both the opportunity set of investors as well as their aggregate holdings. Market cap-weighted indexes are also the only reference for a truly passive, macro consistent, buy and hold investment strategy. They aim to capture the long-term equity risk premium with structurally low turnover, very high trading liquidity and extremely large investment capacity. In contrast, factor indexes rebalance away from a neutral market cap starting point. As such, they represent the result of an active view or decision. Investors must form their own belief about what explains the historical premium and whether it is likely to persist.

Factor returns have also been highly cyclical. These systematic factors have been sensitive to macro-economic and market forces and have underperformed the overall market for long periods of time. However, they have not all reacted to the same drivers and, hence, any one of them can have low correlations relative to other factors. Diversification across factors has historically reduced the length of these periods of underperformance. Thus, the MSCI Factor Indexes provide building blocks that allow investors to assemble multi-factor allocations based on their preferences for performance and risk, their investment beliefs on individual factors, and their investability constraints.

Introduction

Factor investing has become a widely discussed part of today's investment canon. This paper is the first in a three-paper series focusing on factor investing. In this paper we lay out the rationale for factor investing and how indexation can capture factors in cost-effective and transparent ways.³² Specifically, institutional and individual investors around the world have asked many of the same questions:

³² The next paper in this series covers various aspects of implementation including use cases we have seen.

- What are factors? Why have they provided better risk-adjusted return, historically, and how likely is that to persist in the future?
- How does one capture factors via allocations to investable indexes?
- How should investors think of factor indexes relative to market cap weighted indexes and active management?

This paper focuses on the above questions. In Section I, we highlight that factors should be grounded in the academic literature and should be important in explaining portfolio returns in standard performance risk and attribution models. We also distinguish between generic factors and factors that reflect risk premia. The latter are factors that earn a persistent risk-adjusted premium over time and reflect exposure to sources of systematic risk.

In Section II, we discuss the proposed drivers of factors' excess returns. Understanding the potential drivers of factor returns is critical to forming a belief about their likelihood to persist in the future. Different theories have been advanced to explain why factors have historically earned a premium. One view is that factor returns are compensation for bearing systematic risk. A second view is that factor returns arise from systematic errors; either investors exhibit behavioural biases or investors are subject to different constraints (e.g., time horizons, ability to use leverage, etc.).

In Section III, we address the question of how factors should be viewed relative to market capitalization-weighted portfolios. The latter represent both the opportunity set of investors as well as their aggregate holdings. The market cap-weighted benchmark is the only reference for a truly passive, macro consistent, buy and hold investment strategy which aims to capture the long-term equity risk premium with extremely low turnover, very high trading liquidity and infinite investment capacity. Factor portfolios on the other hand rebalance away from a neutral market cap starting point. As such, they represent an active view.

In Section IV, we note the significant cyclicity of factor returns. There is, importantly, no free lunch attached to factor investing. All factors have experienced periods of underperformance and some factors have been highly

cyclical. Their cyclicality may in fact be one of the reasons they have not been arbitrated away.

In Section V, we introduce the concept of capturing factors through indexation. Until recently, capitalizing on systematic factors could only reasonably be done by active managers. Factor indexes allow institutional investors to create passive factor allocations in the transparent and cost efficient framework of indexation. Finally, in Section IV, we illustrate the use of indexation through the MSCI Factor Indexes.

What Are Factors? – Factors Have Their Roots In Academic Literature

The question of what drives stock returns has been a staple of modern finance. The oldest and most well-known model of stock returns is the Capital Asset Pricing Model (CAPM), which became a foundation of modern financial theory in the 1960s (Lintner, 1965; Mossin, 1966; Sharpe, 1964 and Treynor, 1961). In the CAPM, securities have only two main drivers: systematic risk and idiosyncratic risk. Systematic risk in the CAPM is the risk that arises from exposure to the market and is captured by beta, the sensitivity of a security's return to the market. Since systematic risk cannot be diversified away, investors are compensated with returns for bearing this risk. In other words, the expected return to any stock could be viewed as a function of its beta to the market.³³

Later, Ross (1976) proposed a different theory of what drives stock returns. »Arbitrage pricing theory« (APT) holds that the expected return of a financial asset can be modeled as a function of various macro-economic factors or theoretical market indexes.³⁴ We can credit Ross with popularizing the original

33 The expected return to a stock would just be its beta times the assumed market return. Investors would need no compensation for the idiosyncratic return which would diversify across many stocks.

34 The model-derived rate of return would then be used to price the stock correctly. The stock price should equal the expected end of period price discounted at the rate implied by the model, and if the price diverged, arbitrage would bring it back into line.

term »factors«, as the models he popularized were called »multi-factor models.« Importantly, APT, unlike the CAPM, did not explicitly state what these factors should be. Instead, the number and nature of these factors were likely to change over time and vary across markets. Thus the challenge of building factor models became, and continues to be, essentially empirical in nature.

In general, a factor can be thought of as any characteristic relating a group of securities that is important in explaining their returns and risk. As noted in the early CAPM-related literature, the market can be viewed as the first and most important equity factor. Beyond the market factor, researchers generally look for factors that are persistent over time and have strong explanatory power over a broad range of stocks.³⁵ Since, unlike stock returns, factors cannot be directly observed, there of course remains a vigorous debate about how to define and estimate them.³⁶ There are three main categories of factors today: macro-economic, statistical, and fundamental.³⁷ Macro-economic factors include measures such as surprises in inflation, surprises in GNP, surprises in the yield curve, and other measures of the macro economy (see Chen, Ross, and Roll (1986) for one of the first most well-known models). Statistical factor models identify factors using statistical techniques such as principal components analysis (PCA) where the factors are not pre-specified in advance.³⁸

35 Miller (2006) discusses three key statistical criteria for factors: persistence over time, »large enough« variability in returns, relative to individual stock volatility, and application to a »broad enough« subset of stocks within the defined universe.

36 Factor returns can be constructed by building factor portfolios that »mimic« the target factor (as in the Fama–French approach). Factor returns can alternatively be estimated through cross-sectional regression (as in the Barra approach). As far as estimation techniques go, there is a nearly infinite range of techniques that have been applied to factor estimation – principal components analysis, panel regressions, Bayesian models, latent factor models, to name a few.

37 Connor (1995) gives a comprehensive overview of these three types of factor models.

38 The question of which type of model is the best continues to be hotly debated. Typically, the appropriateness of a model comes down to the use case. And of course, within each type of model, there continue to be many debates over the best way to construct a model and specify factors.

Arguably, the mostly widely used factors today are fundamental factors. Fundamental factors capture stock characteristics such as industry membership, country membership, valuation ratios, and technical indicators, to name a few. The most popular factors today – Value, Growth, Size, Momentum – have been studied for decades as part of the academic asset pricing literature and the practitioner risk factor modeling research. Rosenberg and Marathe (1976) were among the first to describe the importance of these stock traits in explaining stock returns³⁹, leading to the creation of the multi-factor Barra risk models. Later, one of the best known efforts in this space came from Eugene Fama and Kenneth French in the early 1990s. Fama and French (1992, 1993) put forward a model explaining US equity market returns with three factors: the »market« (based on the traditional CAPM model), the size factor (large vs. small capitalization stocks) and the value factor (low vs. high book to market). The »Fama–French« model, which today includes Carhart’s (1997) momentum factor, has become a canon within the finance literature. In the past few decades, researchers have studied a host of other stock traits, from income statement and balance sheet measures like earnings revisions and accruals to technical indicators like volatility and relative strength (momentum). The latest research has even looked at non-traditional factors like the number of »Google« hits a stock receives or the number of times it is mentioned in mainstream media.

Exhibit 1 summarizes six of the most widely studied factors. More recently, Low Volatility, Yield, and Quality factors have become increasingly well-accepted in the academic literature (see Appendix A).

39 In fact, they argued that the impact of macro-economic events on individual securities was better captured through these stock characteristics.

Exhibit 1: Well-Known Systematic Factors from the Academic Research

Systematic Factors	What It is	Commonly Captured by
Value	– Captures excess returns to stocks that have low prices relative to their fundamental value	– Book to price, earnings to price, book value, sales, earnings, cash earnings, net profit, dividends, cash flow
Low Size (Small Cap)	– Captures excess returns of smaller firms (by market capitalization) relative to their larger counterparts	– Market capitalization (full or free float)
Momentum	– Reflects excess returns to stocks with stronger past performance	– Relative returns (3-mth, 6-mth, 12-mth, sometimes with last 1 mth excluded), historical alpha
Low Volatility	– Captures excess returns to stocks with lower than average volatility, beta, and/or idiosyncratic risk	– Standard deviation (1-yr, 2-yrs, 3-yrs), Downside standard deviation, standard deviation of idiosyncratic returns, Beta
Dividend Yield	– Captures excess returns to stocks that have higher-than-average dividend yields	– Dividend yield
Quality	– Captures excess returns to stocks that are characterized by low debt, stable earnings growth, and other »quality« metrics	– ROE, earnings stability, dividend growth stability, strength of balance sheet, financial leverage, accounting policies, strength of management, accruals, cash flows

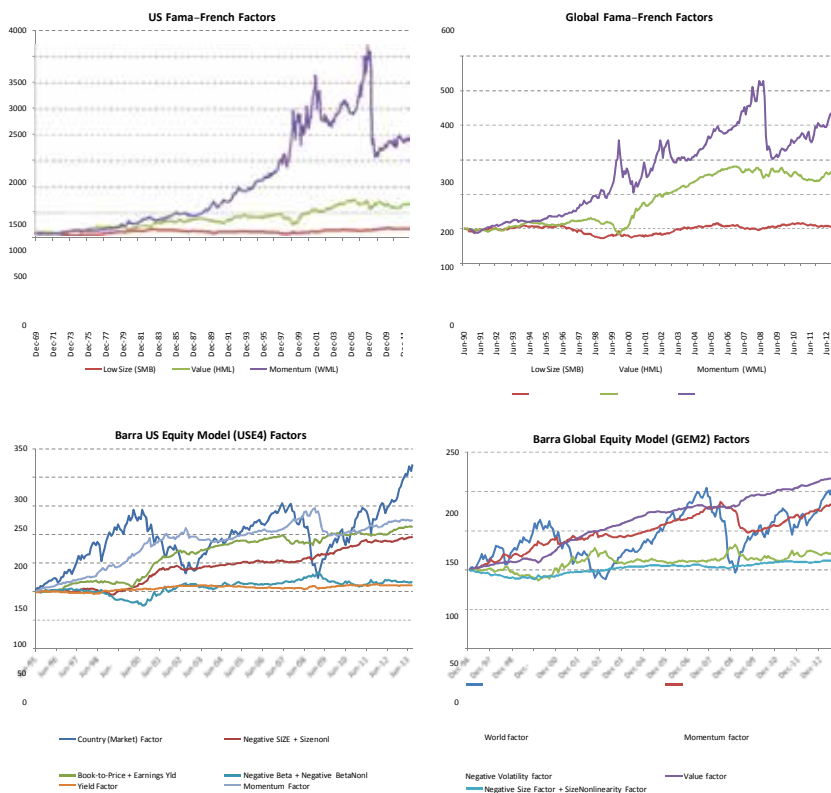
Empirical studies show that these factors have exhibited excess returns above the market.⁴⁰ For instance, the seminal Fama and French (1992) study found that the average small cap portfolio (averaged across all sorted book-to-market portfolios) earned monthly returns of 1.47% in contrast to the average large

⁴⁰ Note that the Barra Volatility factors reflect high volatility stocks relative to low volatility stocks. The premium to low volatility stocks can be assessed by taking the negative of the Volatility factors' returns.

2. Choosing Right Factors

cap portfolio's returns of 0.90% from July 1962 to December 1990. Similarly, the average high book-to-market portfolio (across all sorted size portfolios) earned 1.63% monthly returns compared to 0.64% for the average low book-to-market portfolios. Fama–French's cumulative factor returns through August 2013 are shown in Exhibit 2 (sourced from Kenneth French's website).⁴¹ The factors are long-short portfolios that do not include the market portfolio; the positive cumulative returns reflect returns excess of the market.

Exhibit 2: Well-Known Systematic Factors from the Academic Research (Cumulative Returns)



⁴¹ Note that Fama–French US factor returns are available beginning in July 1962. The chart begins in December 1969 for visual clarity.

Also shown are factor returns from the Barra US and Global Equity Risk Models (USE4 and GEM2).⁴² These factor portfolios and their accompanying returns are estimated in a different way from Fama–French, but they are all long-short portfolios without features or constraints that would make them investable in practice (e.g. limits on position sizes).⁴³ Decades of research by Barra has also found empirical evidence of several important factors beyond the language employed by multi-factor models like Barra, we refer here to style and strategy factors and not industry and country factors. The charts confirm the historical premiums observed to value, momentum, smaller cap, higher yield, and lower volatility stocks.

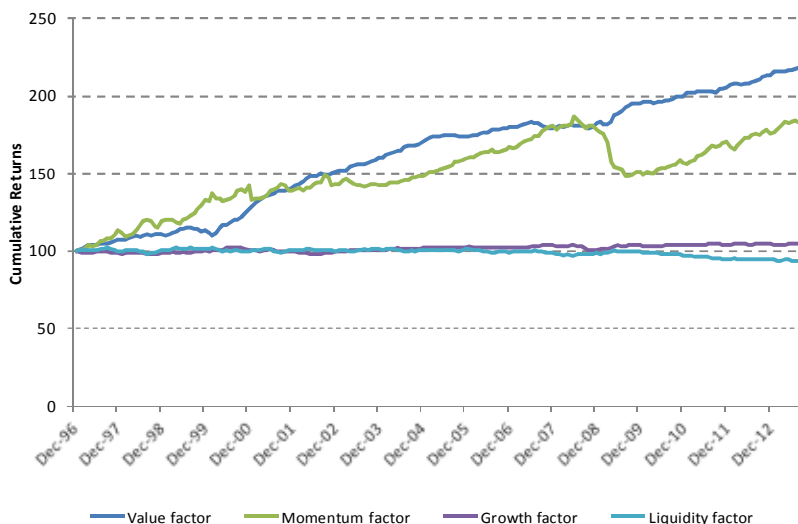
Risk Premia: Factors Which Earn Persistent Premium Over Long Periods

We must stress here that the term »factor« is often used throughout the industry liberally and may refer to one of thousands of different stock characteristics. The term »factor« is often applied to any stock characteristic that can be shown to be important, either in explaining risk or returns (typically relative to the market but not always). In the Barra Risk Models, for instance, there are factors that are significant in explaining returns but which do *not* earn a premium over reasonably long horizons. For instance, as shown in Exhibit 3, factors such as Growth and Liquidity, which capture high growth stocks and highly liquid stocks, have not outperformed the market over long periods. These types of factors are not considered good candidates for long-term factor investing.

42 In some cases, factor returns have been combined or the sign has been reversed so the magnitudes of the premia are visually comparable.

43 For factor researchers, the difference between the Fama–French and Barra approaches is not trivial. Fama–French factors are found through a process of sorting and bucketing stocks to build factor-mimicking portfolios while Barra factors are found through cross-sectional multivariate regression. Menchero (2010) provides a good discussion of the differences in factor estimation methodologies.

Exhibit 3: Differentiating Between Factors Reflecting Risk Premia and Other Factors (Cumulative Returns to Barra Global Equity Model (GEM2) Factors)



Note: Factors that can earn a premium over long periods include Value and Momentum while other factors do not, such as Growth and Liquidity.

Thus, we distinguish between generic factors and »risk premia factors« which have earned a persistent significant premium over long periods and reflect exposure to sources of systematic risk. All of the Fama–French factors count as risk premia factors since the aim of those original studies was to isolate asset pricing drivers. However, not all of the Barra Risk Model factors are candidate factors since the purpose of those models focuses on forecasting risk and explaining fund performance.

In another vein, there are well-known alpha signals such as earnings revisions or earnings momentum, which have been used over the years by active managers to generate excess returns. These stock characteristics are typically called »alpha signals« because they do not explain risk well (i.e., they are not volatile) but do earn persistent premia over time. In our nomenclature, alpha signals could potentially be candidates for risk premia factors though they

must, going back to the previous section, have strong theoretical foundations that support their existence.

Factors Are Important In Explaining Manager Returns

In addition to historically exhibiting excess returns above the market, an equally important rationale for factor investing is the wealth of evidence that they can account for a significant portion of mutual fund returns and institutional active fund returns. Empirical studies confirm that it is difficult for active managers to earn alpha. Studies by Malkiel (1995), Gruber (1996), Wermers (2000, 2003), and Jones and Wermers (2011) found that the median active manager generally does not outperform the cap-weighted benchmark net of fees and even the small subset of those who do outperform are only able to maintain that outperformance for short periods.

Studies have also found that active managers often tilt their portfolios towards well-known factors such as Value and Size and that these tilts account for a substantial portion of managers' returns over market capitalization-weighted benchmarks. For instance, Fama and French (2010) found that mutual funds in aggregate (CRSP database) experienced net returns that underperformed the Fama–French factor benchmarks by about the costs in expense ratios from 1984 to 2006. Ang, Goetzmann and Schaefer (2009) applied a similar analysis to the Norwegian Government Pension Fund's active returns and found that although added value from active management was still net positive, much of the behaviour of the Fund's small active return could be explained in terms of systematic factors. Recently, Mok, Bender, and Hammond (2013) showed that traditional Fama–French factors accounted for roughly 50% of average alpha using US institutional fund returns.⁴⁴

In fact, two of the systematic factors, Value and Size, form the basis of what has become a fund categorization standard – »style box investing«. Style box investing, sometimes referred to as style investing, was first popularized by

44 Manager returns were sourced from eVestment from March 2002 to March 2012.

Morningstar in 1992⁴⁵ and is a framework in which institutions allocate to managers within style segments (large/mid/small and value/growth). In fact, style investing was originally based on the research of Sharpe (1988, 1992) who identified the importance of these factors in driving fund returns.⁴⁶

How Likely Is It That Excess Returns To Risk Premia Factors Will Persist In The Future?

A question that comes up repeatedly from institutional investors is: »How likely are the factors' excess returns to persist in the future?« There is a vigorous debate about this question across academics and practitioners alike, but the short answer is, it depends on what forces have driven a factor historically and whether those forces will continue to persist.

What Drives Factor Returns: Systematic Risk Vs. Systematic Errors

In general, there are two main camps in the debate over what drives factor returns – one based on the view that markets are efficient and that factors reflect »systematic« sources of risk, and one based on the view that investors either exhibit behavioural biases or are subject to different constraints (e.g., time horizons, ability to use leverage, etc.).

In the first camp, the term »systematic« refers to the fact that risks to these stock traits cannot be diversified away (in the true spirit of Ross' (1976) APT Model). This argument is consistent with »efficient markets theory« which assumes markets are efficient and investors are rational. Here, factors earn excess returns because there is »systematic risk« attached to them. For example, some have argued that the small cap premium is return earned for exposure to companies that are less liquid (Liu, 2006), less transparent (Zhang,

⁴⁵ http://news.morningstar.com/pdfs/FactSheet_StyleBox_Final.pdf

⁴⁶ Arguably Sharpe's and other, similar, empirical studies became the foundation for the »Morningstar Style Box« in 1992.

2006), and more likely to be distressed (Chan and Chen, 1991; Dichev, 1998). Others have argued that factors like Value, Size, and Momentum are linked to important macro-economic factors such as growth and inflation and because of their sensitivity to shocks in the economy, must bear a return premium (Winkelmann et al., 2013).

In the second camp, factors are thought to earn excess returns because of investors' »systematic errors«. One subgroup of this camp, rooted in the »behavioural finance« literature, suggests that investors exhibit behavioural biases due to cognitive or emotional weaknesses. Examples include chasing winners, over-reacting, overconfidence, preferring »familiar« investments such as securities of the companies they work for or the country they live in (»home bias«), and myopic loss aversion. If enough investors exhibit these biases, as long as it is prohibitively costly for rational investors to arbitrage these biases away, it can lead to the factor anomalies we observe.

Within this second camp, another subgroup has tied factor performance to investor constraints and frictions/flows that arise from regulatory and industry practice. In these studies, anomalies can arise from investors behaving rationally but subject to constraints. For instance, consider different investor time horizons. Studies have shown that stocks with low liquidity earn a premium over long horizons (10 years plus) since most investors have shorter horizons (3 and 5 year horizons) and prefer stocks that are liquid over the shorter horizons. Investors with longer horizons earn a premium for bearing this horizon risk. Other examples have applied constraint-based arguments for Low Volatility and Momentum. For instance, Baker et al. (2011) cite the use of institutional benchmarks, and the subsequent preference for relative returns, as one reason why the low volatility premium exists. Dasgupta, Prat and Verardo (2011) argue that reputation concerns cause managers to herd, and this generates momentum under certain circumstances.⁴⁷

In fact, many of the well-known factors have multiple theories supporting them. Exhibit 5 summarizes main theories proposed for factor anomalies.

⁴⁷ Specifically that the market makers trading with the managers are either monopolistic or myopic.

Exhibit 5: Theories Behind the Excess Returns to Systematic Factors

Systematic Factors	Systematic Risk-based Theories ⁴⁹	Systematic Errors-based Theories ⁵⁰
Value	– Higher systematic (business cycle) risk	– Errors-in-expectations – Loss aversion – Investment-flows-based theory
Low Size (Small Cap)	– Higher systematic (business cycle) risk – Proxy for other types of systematic risk	– Errors-in-expectations
Momentum	– Higher systematic (business cycle) risk – Higher systematic tail risk	– Under-reaction and over-reaction – Investment-flows-based theory
Low Volatility	– N/A	– Lottery effect – Overconfidence effect – Leverage aversion
Dividend Yield	– Higher systematic (business cycle) risk	– Errors-in-expectations
Quality	– N/A	– Errors-in-expectations ⁵¹

For example, let's consider the theories behind the Value factor. Value investing has been widely discussed since Graham and Dodd first wrote about it 1934 («*Security Analysis*»). The Value factor captures the positive link between stocks that have low prices relative to their fundamental value and returns in excess of the capitalization-weighted benchmark. There are several explanations for the existence of this effect. In the efficient markets view, the value premium is compensation for higher real or perceived risk. Cochrane (1991, 1996) and Zhang (2005) suggest that contrary to their leaner more flexible, growth counterparts, value firms have less flexibility to adapt to unfavorable economic environments. Chen and Zhang (1998) later found that value stocks are riskier due to their high financial leverage

⁴⁸ These include all efficient markets-based theories which assume investors are fully rational.

⁴⁹ These include behavioural theories that assume investors are not fully rational as well as constraint-based theories that assume investors are rational but operate under constraints.

⁵⁰ Investors may mistakenly value stocks based on reported earnings if reported earnings are not reflective of a company's true fundamentals.

and large uncertainty in future earnings. Recently, Winkelmann et al. (2013) show value and small cap portfolios are more immediately sensitive to economic (real GDP) shocks than growth and large cap portfolios. The premium to value can consequently be viewed as compensation for macro risk.

From a behavioural perspective, the premium may exist as a result of loss aversion and mental accounting biases. According to Barberis and Huang (2001), investors become less concerned about future losses on stocks with recent good performance because any losses will be cushioned by prior gains («loss aversion» bias). This bias induces investors to perceive the stock to be less risky than before and discounts its future cash flows at a lower rate. Conversely, if a stock performs poorly in the recent past, investors will raise its discount rate. This results in a value premium in the cross-section of stocks since a stock with a high price-dividend ratio (a growth stock)⁵¹ is often one that has done well in the past, accumulating prior gains for the investor who then views it as less risky and requires a lower average return. A stock with a low price-dividend ratio (a value stock) has often had dismal prior performance; an investor may now view it as riskier, and require a higher future return. Lakonishok, Shleifer and Vishny (1994) proposed other behavioural biases that may help explain the Value premium such as investors extrapolating past growth into the future, chasing high-flying glamour stocks, or simply over-reacting to news.

More recently, using an institutional framework/constraint-based argument, Vayanos and Woolley (2011) propose that value and momentum effects arise because negative shocks to assets' fundamental values trigger outflows from funds holding those assets. Outflows cause asset sales, which amplify the shocks' negative effects. If the outflows are gradual because of investor inertia or institutional constraints, then the amplification is also gradual and momentum effects arise. Moreover, because flows push prices away from fundamental value, value effects also arise. Both effects arise despite investors and fund managers being rational.

51 Note that Barberis and Huang (2001) define value/growth using price-to-dividend ratio but the argument would apply to other definitions such as price-to-book value and price-to-earnings.

The examples discussed above represent just a small part of the wealth of research on the value premium. Additional references and discussions of other factors can be found in Appendix A.

Assessing The Likelihood Of Future Performance

For institutional investors evaluating factor allocations, arguably the most critical part of the process for adopting the approach is forming a *view or belief* about what drives the factor(s) in question. In practice, this first step often focuses on assessing the academic literature and the theories behind what drives excess returns to factors.

For institutional investors who subscribe to the »systematic risk« perspective, a factor can potentially persist indefinitely if it is actually compensation for bearing undiversifiable risk. For those who subscribe to the »behavioural bias« perspective, a factor can potentially persist as long as there are strong reasons why investors will continue to exhibit the behavioural biases in question and their actions are too costly to be arbitrated away by rational investors (or those who recognize these biases as such).⁵² Lastly, for those who subscribe to the »constraint-based« view, a factor can potentially persist only as long as those constraints remain in place.

Regardless of their drivers, Melas, Briand, and Urwin (2011) point out that as factor strategies become increasingly popular, institutional investors may over time begin to experience relatively lower factor returns. The reason is that, unlike market capitalization-weighted portfolios, factor portfolios lack macro consistency. An index or a portfolio is macro consistent if it can be held by all institutional investors without disturbing market prices. If all institutional investors tried to hold the same factor portfolio, the factor would experience diminishing active returns relative to the market. The interplay of investor

⁵² Some have argued that the return to factors observed in the data is roughly equal to the cost of arbitrage; see Shleifer and Vishny (1997). In practice, institutional investors may have different costs of arbitrage; thus their ability to exploit certain return anomalies may vary as well. Institutions with low costs of arbitrage may be able to exploit mispricings which average investors may not. These points should likely be factored into an evaluation of factor investing by large institutions.

flows and factor premia is complex particularly when viewed over time. While the idea that increasing popularity could diminish future returns is widely appreciated, investors should keep in mind that increased capital flows going into a popular strategy will produce a tailwind to the strategy's performance over the medium term, prior to any possible future reversal.

Factor Investing Versus Market Cap Investing

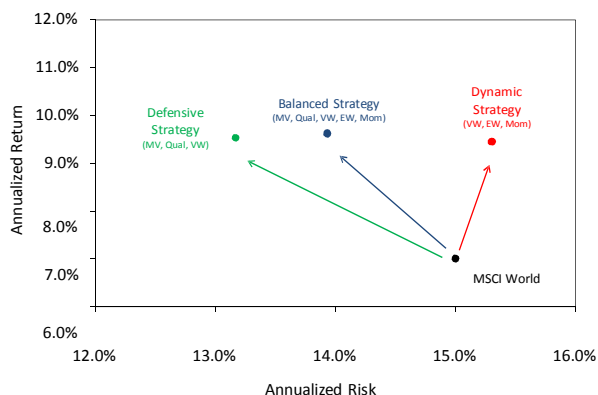
Some researchers have argued that market cap weighting is inherently flawed and have advocated replacing market cap allocations with factor allocations; see for example, Arnott, Hsu, and Moore (2005). Our position is quite the opposite. First, a market capitalization-weighted index reflects the available opportunity set of equity investments as well as the aggregate holdings of all investors. If an investor wants to understand how equities have performed, the best gauge is a market cap-weighted benchmark.⁵³ Second, market cap investing, the strategy that replicates a market cap index, is unique in being macro-consistent, meaning it is the only portfolio that all investors can hold.

Factor indexes do not reflect the full equity opportunity set and they are not macro consistent. Instead, they represent strategic tilts away from market capitalization weighted benchmarks. Exhibit 6 shows one way to view factor indexes relative to the market capitalization-weighted index using the MSCI World Index as an example. Some factors (and factor combinations) have higher returns and higher volatility while others have higher returns and lower volatility. They represent active decisions away from the market capitalization-weighted index. One intuitive way to view them is in the rubric of traditional active funds (e.g., defensive, balanced, and dynamic strategies).

53 Moreover, market capitalization-weighted can be viewed as the equilibrium portfolio if investors are rational.

2. Choosing Right Factors

Exhibit 6: Factors Represent Strategic Bets Away from the Market Capitalization-Weighted Index (Return and Risk, June 1988 to June 2013)



Note: The strategies in Exhibit 6 are equally weighted combinations of the MSCI Factor Indexes shown in parentheses. VW = MSCI World Value-Weighted Index, EW = MSCI World Equal-Weighted Index, Qual = MSCI World Quality Index, Mom = MSCI World Momentum Index, MV = MSCI World Minimum Volatility Index. All are MSCI Factor Indexes based on the MSCI World Index.

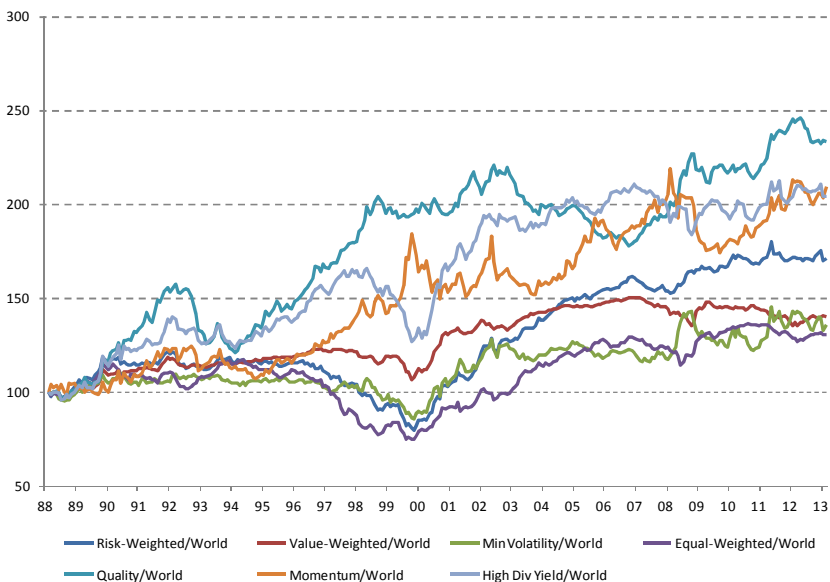
In sum, a market capitalization-weighted index is the only appropriate candidate for a truly passive, macro consistent, buy and hold investment strategy that aims to capture the long-term equity risk premium with structurally low turnover, very high trading liquidity and extremely large investment capacity. In contrast, investing in factors represents active views away from the market portfolio and investors must form their own belief about what explains the premium and whether it is likely to persist. Thus, like traditional active strategies, factor index strategies should be assessed in the long run against a market capitalization-weighted benchmark.

Is Factor Investing A Free Lunch? The Importance Of Factor Cyclicity

A key element of factor investing is factor cyclicity. While factor indexes have exhibited excess risk-adjusted returns over long time periods, over short

horizons factors exhibit significant cyclicity, including periods of underperformance. Exhibit 7 shows that each of the factor indexes has experienced at a minimum a consecutive two-to-three year period of underperformance. Some factors historically have undergone even longer periods; the Small Cap or Low Size factor (captured by the MSCI World Equal-Weighted Index in the exhibit) went through a six-year period of underperformance in the 1990s.

Exhibit 7: All Systematic Factors are Cyclical (Cumulative Relative Returns, June 1988 to June 2013)



Thus, there is no free lunch attached to factor investing. Given strong anecdotal evidence that the majority of investors have relatively shorter horizons, their cyclicity may in fact be one of the reasons they have not been arbitrated away. Investors with shorter horizons would not be able to benefit from the full cycle required for factor investing. Some may argue the premia exist to reward long-horizon investors for bearing that risk.

A key part of the decision process for institutional investors implementing factor allocations is what to do about the cyclicality. Possible approaches include the following:

- Setting an appropriately long time horizon
- Establishing an explicit timing mechanism for the initial investment
- Adopting a multiple factor approach and employing factors that diversify each other

The first is difficult, in practice, because the horizon required would typically be far longer than most institution's strategic reviews. Only institutional investors with extremely long time horizons (15 years and up for most of the factors shown in Exhibit 7) would be arguably insensitive to the timing of entry. The second option is challenging since factors, like markets, have been documented to be extremely hard to time. Most institutions would rather avoid timing decisions given their inherent difficulty.

The third is the most straightforward option. While all factors have been historically cyclical, their periods of underperformance have not coincided. For instance, during the period between 2001 and 2007, the MSCI World Momentum, MSCI World Value-Weighted, MSCI World Risk-Weighted, and MSCI World Equal-Weighted Indexes all outperformed the MSCI World Index, but the MSCI World Quality Index underperformed. From 2007-on, in contrast, the MSCI World Quality Index outperformed while the MSCI World Momentum and MSCI World Value-Weighted Indexes did not fare as well. Combining Quality with Momentum and Value-Weighted factor indexes for instance can yield a »smoother ride« and diversify across multi-year cycles. Or said another way, employing multiple factors is one way to address their cyclicality. In the second paper of the series »Deploying Multi-Factor Index Allocations in Institutional Portfolios«, we discuss in more detail the allocation of equity portfolios across multiple factors.

Capturing Factors Through Allocations To Investable Indexes

The original studies on factors were intended to identify which stock characteristics explained returns. These studies were not concerned with whether those factors were actually investable. Specifically, the factor portfolios constructed by the academics in these studies were not designed for actual implementation. For instance, the Fama–French portfolios include all listed equities (in the US, listed on NYSE and AMEX) including many small illiquid names, are pure long/short portfolios with no accommodation to the size of short positions, and are rebalanced monthly leading to turnover that is considerably higher than institutional benchmarks. In other words, because these factor studies did not assume investors were necessarily investing in these factors, the factors they described did not, nor were they meant to, take into account features key to implementation: transaction costs, liquidity, investability, and capacity.

Until recently, the ability to capitalize on factors could only reasonably be done by active managers. Value investing and small cap investing have been staples of active management for decades. But over the last decade, index providers recognized that factors could be captured in transparent rules-based ways. Investors realized that factor strategies could outperform the market similar to their theoretical factor counterparts while having strong liquidity and investability characteristics.⁵⁴ Today, these indexes go by a number of names – alternative beta, smart, beta, fundamental indexing, etc.

What has this ability to capture factors through indexation meant for institutional investors? Factor indexes could, in fact, revolutionize the investment industry by providing a new way of investing beyond traditional market capitalization-weighted portfolio and active portfolios. Factor indexes could provide:

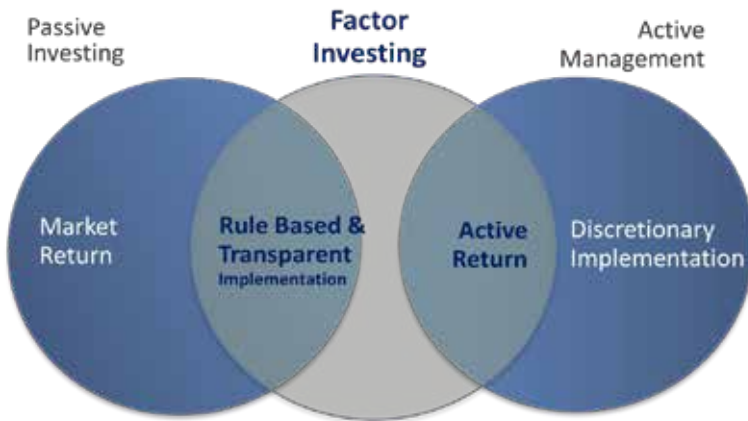
⁵⁴ Melas, Briand, and Urwin (2011) first discussed ways to capture factors through transparent rules-based long-only investable portfolios.

2. Choosing Right Factors

- *More relevant benchmarks:* Assigning an active manager a more relevant benchmark means investors can more accurately gauge what active managers are adding
- *Potentially more cost-effective implementation choices:* Funds that passively track these indexes can now potentially deliver returns previously only available through active management at potentially lower fees
- *Transparent implementation choices:* Funds that passively track these indexes can now be used in lieu of more opaque active options with positive implications for risk management

The benefits to institutional investors of factor indexes could be significant. As illustrated in Exhibit 8, factor allocations can be implemented in simple and low cost ways like traditional passive market cap-weighted allocations. We note here that factors do *not* replace active management (at least fully) because many value-added activities such as stock selection cannot be replicated by factor indexes.

Exhibit 8: Passive Factor Allocations Combine Attractive Elements of Both Traditional Passive and Active Mandates



Beyond cost implications, factor indexes offer a fully transparent way to passively invest in factors. Transparency alleviates the well-known problem of manager style drift and also has positive implications for risk management. Institutions have full »look-through« of how exposed their portfolios are to the factors.

The advent of factor indexes also heralds a change in traditional investing paradigms. As shown in Exhibit 9, index-based factor allocations could imply a significant shift in the way institutional investors view the asset allocation and active management process. Instead of focusing on diversifying across active managers in multiple alpha mandates, institutions would focus first on diversification across multiple index mandates. Risk control would then focus, principally, on managing exposure to these factors while active management would be defined more narrowly as sources of return that exclude factors.

Exhibit 9: Potential Redefinition of the Way Institutions Invest

	Current Framework	Possible New Framework
Strategy	– Diversification across managers in – multiple alpha mandates – Asset owner manages strategy through asset allocation and manager selection	– Diversification across strategies in – multiple factor index mandates – Asset owner manages strategy through factor allocation
Roles and Tools	– Alpha is defined broadly: bottom-up skill, top-down sources and timing – Risk control is principally through asset allocation and manager diversification	– Alpha is defined more narrowly – excluding factors – Risk control focuses on managing exposure to risk factors
Economics	– Active mandates dominate the portfolio's costs – Large line-up of external managers with small internal asset owner staff	– Active mandates co-exist with factor mandates producing lower costs – Larger internal staff managing more assets with fewer external managers

An Illustration With MSCI Factor Indexes

MSCI's family of factor indexes, comprising the six key factors –Value, Low Size, Low Volatility, High Yield, Quality and Momentum – are shown in Exhibit 10. These indexes are constructed for flagship global indexes like the MSCI ACWI, MSCI World, MSCI Emerging Markets, and MSCI EAFE Indexes, as well as many individual country indexes.

Exhibit 10: MSCI Family of Factor Indexes

Systematic Factors	MSCI Indexes
Value	– MSCI Value-Weighted Indexes: Capture Value factor by weighting according to four fundamental variables (Sales, Earnings, Cash Flow, Book Value)
Low Size (Small Cap)	– MSCI Equal-Weighted Indexes: Capture low size effect by equally weighting all stocks in a given parent index
Momentum	– MSCI Momentum Indexes: Reflect the performance of high momentum stocks by weighting based on 6- and 12-month momentum scaled by volatility
Low Volatility	– MSCI Minimum Volatility Indexes: Reflect empirical portfolio with lowest forecast volatility using minimum variance optimization – MSCI Risk-Weighted Indexes: Capture low volatility stocks by weighting based on the inverse of historical variance
Dividend Yield	– MSCI High Dividend Yield Indexes: Select high dividend stocks with screens for quality and potential yield traps
Quality	– MSCI Quality Indexes: Capture high quality stocks by weighting based on debt-to-equity, return-on-equity, and earnings variability

Just as »there is more than one way to skin a cat«, there is indeed more than one way to capture systematic factors through indexes. Index methodology comprises several aspects: the choice of stock universe, weighting scheme, and rebalancing frequency. All three have important implications for the traits investors care about in the resulting index-investability and liquidity, factor exposure, returns, risk, and tracking error. For instance, the index could contain all the names in the starting universe and merely reweight the names.

Alternatively, the index could contain a narrow subset of names.⁵⁵ The former would have lower tracking error, higher investability, but lower exposure to the pure factor and potentially lower returns.

More generally, there is a tradeoff between the exposure to the factor (and potential returns) and the investability of a factor index. One can generally only achieve purer factor exposure by sacrificing investability and being willing to take on greater amounts of active risk. Detailed discussion about different index choices is discussed later in the second paper of this series »Deploying Multi-Factor Index Allocations in Institutional Portfolios«.

Exhibit 11 shows key characteristics including the performance of some of the MSCI World Factor Indexes for the period June 1988 to June 2013.

Exhibit 11: MSCI World Factor Indexes (Main Characteristics, June 1988 to June 2013)

Index	Factor Exposures	Total Return	Total Risk	Active Return	Active Risk	Annual Turnover	Pairwise Correlation
MSCI World	-	7.1	15.4	0.0	0.0	3.9	NA
MSCI World Equal-Weighted	Size	8.3	16.3	1.2	5.2	31.8	0.22
MSCI World Minimum Volatility	Volatility	8.5	11.6	1.4	6.7	20.0	0.30
MSCI World Value-Weighted	Value	8.6	15.6	1.5	3.6	20.3	0.30
MSCI World Risk-Weighted	Size, Volatility	9.5	13.7	2.4	5.3	27.2	0.46

⁵⁵ MSCI Factor Indexes span both types. MSCI Factor Indexes that contain a broad universe of names (denoted by having the term »Weighted« in the index name) usually contain all the names in a broad »parent index« such as the MSCI ACWI Index and reweight the stocks (differently from market capitalization-weighted) based on the desired factor characteristic. These broad indexes are characterized by very low tracking error and high degrees of investability.

2. Choosing Right Factors

<i>Index</i>	<i>Factor Exposures</i>	<i>Total Return</i>	<i>Total Risk</i>	<i>Active Return</i>	<i>Active Risk</i>	<i>Annual Turnover</i>	<i>Pairwise Correlation</i>
MSCI World Quality	Growth, Leverage	10.9	14.0	3.8	5.9	27.6	0.13
MSCI World Momentum	Momentum	10.4	15.9	3.3	8.5	127.5	0.03
MSCI World HDY	-	10.3	14.6	3.2	6.5	22.0	0.41

In the column »Factor Exposures« we show the Barra Global Equity Model (GEM2) factors which are statistically significant on average ($>+/- 0.20$), with the expected sign, since December 1997. Note that there is no »Yield factor« in the GEM2 Model. Instead, Yield is a component (with a weight of 10%) in the GEM2 Value factor. Turnover reported is the average annual one-way turnover based on history from June 1988–June 2013.

Higher capacity indexes include the MSCI Value-Weighted Index. As illustrated in Exhibit 17 using MSCI World variants, it exhibits the lowest tracking error among the available indexes since it holds all the constituents of the parent index. The other indexes (the MSCI Momentum Indexes, MSCI Quality Indexes, and MSCI Minimum Volatility Indexes) are more concentrated indexes, holding only a subset of the names in the parent index. These indexes exhibit higher tracking errors and lower levels of investability. (The MSCI Minimum Volatility Indexes are turnover-constrained to 20% but other measures of investability are more similar to the MSCI Momentum and Quality Indexes.) The MSCI Momentum and MSCI Quality Indexes historically exhibited the strongest historical returns while the MSCI Minimum Volatility Indexes exhibited sizable risk reduction.

All the MSCI World Factor Indexes shown in Exhibit 11 outperform the MSCI World Index over the period shown, June 1988 to June 2013. Moreover, they all have turnover levels that are well within reason. For instance, assuming 50 bps in one-way trading costs, the cost of trading for the indexes would range between 20.3 basis points annually for the MSCI World Value-Weighted Index to 127.5 bps annually for the MSCI World Momentum Index. Both estimates are more than offset by their historical return premiums over this period – 150 bps and 330 bps respectively.

Conclusion

Factor investing is based on the existence of factors that have earned a premium over long periods, reflect exposure to systematic risk, and are grounded in the academic literature. Early financial theory established that for stocks, exposure to the market was a significant driver of returns (e.g., the CAPM). Later, researchers like Barr Rosenberg, Eugene Fama and Kenneth French extended the CAPM to include certain systematic factors that also were important in explaining returns. Tilts towards these factors such as Value, Low Size, and Momentum historically produced excess long-term returns and there were strong theoretical foundations behind these factors.

Until now, passive investing has focused on capturing market beta through market capitalization-weighted indexes. The only way institutional investors could get access to factors was through active management. Indexation is opening a new way for factor investing today by allowing investors to access factors through passive vehicles that replicate factor indexes. MSCI Factor Indexes provide access to six solidly grounded factors – Value, Low Size, Low Volatility, High Yield, Quality and Momentum. These indexes have historically earned excess returns over market capitalization-weighted indexes and experienced higher Sharpe Ratios.

This paper is the first in a three-paper series focusing on factor investing. In the next paper of this series, »Deploying Multi-Factor Index Allocations in Institutional Portfolios«, we address the topic of allocating across factors and what role they play in the institutional portfolio.

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Appendix A: Additional Information About The Academic Literature

There is vast literature on systematic factors. Subgenres which fall under the umbrella of systematic factors include asset pricing anomalies, stock market anomalies, multi-factor models, risk factor models, alpha signals, etc. To start, it is useful to begin with several good general references on these issues:

- »Dissecting Anomalies« by Eugene Fama and Ken French (2008)
- »Explaining Stock Returns: A Literature Review« by James L. Davis (2001)
- »Market Efficiency, Long-Term Returns, and Behavioral Finance« by Eugene Fama (1997)
- »The Efficient Market Hypothesis and Its Critics« by Burton Malkiel (2003)
- »The New Palgrave Dictionary of Economics« (2008) by Steven Durlauf and Lawrence Blume, 2nd ed.

- »Anomalies and Market Efficiency« by G. William Schwert⁵⁶ (Ch. 15 in Handbook of the Economics of Finance, by Constantinides, Harris, and Stulz, 2003)
- »Investor Psychology and Asset Pricing« by David Hirshleifer (2001)⁵⁷

While the construction of factor models occupies a somewhat segregated landscape from the market anomalies literature, this research provides guidance on how alpha signals and risk factors are identified and estimated by quantitative managers. Some good starting references include:

- »The Barra Equity Risk Model Handbook«
- »Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk« by Richard Grinold and Ronald Kahn
- »Modern Investment Management: An Equilibrium Approach« by Bob Litterman
- »Quantitative Equity Portfolio Management: Modern Techniques and Applications« by Edward Qian, Ronald Hua, and Eric Sorensen
- Additional references are provided below for the main systematic factors discussed in this paper: Value, Low Size, Momentum, Low Volatility, Quality, and Yield.

Value

The Value factor captures the positive link between stocks that have low prices relative to their fundamental value and returns in excess of the capitalization-weighted benchmark. A value strategy consists of buying stocks that have low

⁵⁶ <http://schwert.ssb.rochester.edu/hbfech15.pdf>

⁵⁷ »Investor Psychology and Asset Pricing«, David Hirshleifer, The Journal of Finance, 56(4), August, (2001), 1533–1597.

prices normalized by some indicator of company fundamentals (such as book value, sales, earnings, or dividends, etc.) and selling stocks that have high prices (also normalized). Value investing has been widely discussed since Graham and Dodd first wrote about it 1934 («Security Analysis»). It was later formalized by Basu (1977) who was the first to test the notion that value-related variables might explain violations of the CAPM. He found a significant positive relation between E/P ratios and average returns for US stocks that could not be explained by the CAPM. Later, studies documented a significant positive relation between Book-to-Price ratios and average returns as well; see Rosenberg, Reid and Lanstein (1985) and DeBondt and Thaler (1987) for example. Other measures of value have also been found to have positive excess returns including value ratios that use cash flow. The Value effect in its many forms has been reproduced by numerous researchers for many different sample periods and for most major securities markets around the world (see Hawawini and Keim (2000)). We note that critics of the value premium have argued that empirical evidence is based on data mining and point out the sample-dependency of empirical studies; see for instance Black (1993).

A discussion of theories behind the Value premium appears in Section II.

Low Size

The Size factor captures the excess returns of smaller firms (by market capitalization) relative to their larger counterparts even after adjusting for betas and other factors like Value. This result was first discovered by Banz (1981), and triggered a vast literature on the topic. Banz (1981) showed smaller companies listed on the NYSE on average delivered higher risk-adjusted returns than larger companies using data from 1936-1975. This »size effect« was later generalized by the Fama–French Three-Factor Model in the early 1990s.

The small cap premium has been found to exist even after influences are controlled for: market beta, the value effect, the momentum effect, liquidity effects, leverage, and so forth. Moreover, the phenomenon has been identified across the world in both developed and emerging markets; see Rizova (2006) for a synopsis of work applying the Fama–French framework to individual countries such as Australia, Canada, France, Germany, and the UK.

There are several theories explaining this phenomenon, and the debate continues today. In the efficient market view, Fama and French (1992, 1993) originally hypothesized that small caps have higher systematic risk which earns them a higher return premium. Subsequent researchers suggested that size may proxy for other unobservable and underlying risk factors associated with smaller firms such as liquidity (Amihud, 2002), information uncertainty (Zhang, 2006), financial distress (Chan and Chen, 1991) and default risk (Vassalou and Xing, 2004). Chan, Chen and Hsieh (1985) in particular argue the most powerful factor in explaining the size effect is the spread between low and high quality corporate bonds, which is essentially an indication of the macro environment and default risk.

From the behavioural perspective, similar behavioural arguments for the value premium are also made for the size effect, i.e. incorrectly extrapolating past into the future, chasing high-flying glamour stocks, or overreacting to news; see (Lakonishok, Shleifer, and Vishny, 1994). The explanations of the size effect remain highly debated.

Skeptics of the size premium point to the survivorship bias in the relevant research which typically does not include busted companies. This seems a legitimate claim given small companies often have sustainability issues. Others argue the size premium is difficult – if not impossible – to capture in the real world given the low trading volume of small cap stocks.

Momentum

The Momentum factor reflects future excess returns to stocks with stronger past performance. In other words, stock prices tend to exhibit trend over certain horizons; winners continue to win and losers continue to lose. Jegadeesh and Titman (1993) led one of the first seminal studies on momentum in the US stock market, which shows buying past winners and selling past losers produced significant »abnormal« returns in 1965–1989. In a study of mutual fund performance, Carhart (1997) expanded the Fama–French Three-Factor Model to a Four-Factor Model to include momentum as an additional explanatory variable. Rowenhorst (1998) found an internationally diversified portfolio of past winners outperformed a portfolio of past losers by about 1% per month,

using a sample of 2,000 European stocks from 1978 to 1995. Fama and French (2012) found strong momentum returns in North America, Europe, and Asia Pacific but not Japan in the sample period of 1989–2011. They also confirmed the robustness of the Four-Factor Model, i.e. momentum is a persistent factor not captured by either value or size. Empirically, the Four-Factor Model seems to work best with global stocks excluding micro caps.

Asness (1995, 1997) not only confirmed earlier findings on the momentum effect at the country level, but also showed that winners and losers tend to revert over the long-term, i.e. winners underperforming and losers outperforming in 3–5 years out. In fact, empirical research suggests the momentum effect is most prominent in the following 3–12 months, after which it will likely disappear. This implies the momentum strategy requires relatively high turnover in order to work. Geczy and Samonv (2013) conducted a 212-year backtest of the momentum strategy in the US market, which shows the momentum effect is statistically significant and not a product of data-mining.

The theory underlying this premium is still matter of extensive discussion. Unlike value and size, there is no satisfactory efficient markets-based theory to explain the momentum factor. The most widely cited theories are all behavioural. Investors either over-react (Barberis, Shleifer and Vishny, 1998, and Daniel, Hirshleifer and Subrahmanyam, 1998) or under-react to news (Hong, Lim and Stein, 2000), both of which may lead to the momentum effect under varying assumptions. Without diving deep into behavioural finance, these »irrational« reactions are driven by overconfidence, self-attribution, conservatism bias, aversion to realize losses, representative heuristic (tendency to identify an uncertain event by the degree to which it's similar to the parent population), or simply lack of analyst coverage. Or as another example, Dasgupta, Prat and Verardo (2011) argue that reputation concerns cause managers to herd, and this generates momentum under certain assumptions.⁵⁸ More recently, Vayanos and Woolley (2011) propose a framework based on the dynamics of institutional investing rather than individual biases. In their framework, momentum and value effects jointly arise because of flows between investment funds. Negative shocks to assets' fundamental values trigger outflows from funds holding those

⁵⁸ Specifically that the market makers trading with the managers are either monopolistic or myopic.

assets while outflows cause asset sales, which amplify the shocks' negative effects. If the outflows are gradual because of institutional constraints or inertia, then momentum effects arise. Moreover, because flows push prices away from fundamental value, value effects also arise.

Common criticisms of the momentum strategy include data mining, high turnover, crowded trading, and the risk of a sudden reversal – which is difficult to predict and manage. History shows the probability of a short-term reversal is positively correlated with volatility, and forecasting volatility is anything but easy. In addition, like other factor strategies, momentum could go through an extended period of negative performance. This suggests it's perhaps better to combine momentum with other factor strategies than using it alone. Asness, Moskowitz and Pedersen (2010) supported this argument with a study on the interaction between value and momentum. They found potential diversification benefits from combining the two strategies as value and momentum can be negatively correlated within and across asset classes.

Low Volatility

The Low Volatility factor captures excess returns to stocks with lower than average volatility, beta, and/or idiosyncratic risk. The empirical evidence for this factor is a puzzle since it is clearly at odds with one of the most basic principles in finance, that higher volatility is associated with higher returns (Blitz and Vliet, 2007). While the CAPM model asserts that riskier assets should earn higher returns, research around the Low Volatility factor shows that the opposite is true—less risky stocks outperform the market.

Haugen and Baker's (1991) critique of capitalization-weighted benchmarks was the first to document the effect. They showed that for the 1972 to 1989 period, low volatility stocks in the US performed better than the capitalization-weighted alternative. Later, Chan, Karceski and Lakonishok (1999), Schwartz (2000), Jagannathan and Ma (2003) and Clarke, de Silva and Thorley (2006) confirmed these results for the US market using a range of volatility measures. Geiger and Plagge (2007), Nielsen and Subramanian (2008) and Poul-laouec (2008) all find qualitatively similar results for global markets. Ang et al. (2006, 2009) found that the low volatility effect persists both in the US

and globally, based on extensive periods of time (for US stocks, 1963 to 2003, and for international stocks, 1980 to 2003).

Metrics used for identifying low volatility stocks range along a broad spectrum, with realized volatility on one end and forecast volatility and correlations on the other. Some operationalize low volatility as low beta. The metrics for identifying low volatility stocks also vary, with realized (historical) volatility on one end, and forecasted (implied) volatility and correlations on the other end. However, the research findings appear robust despite such differences.

The low volatility anomaly clearly contradicts the EMT and assumptions of the CAPM. The explanations here are mostly behavioural. The most common explanation is the »lottery effect«, i.e. people tend to take bets with a small expected loss but a large expected win, even though the probability of a loss is much higher than the win, and the weighted average of the outcome may be negative. This is similar to buying a lottery whereby the customer pays a small sum for potentially winning a large amount of money albeit at a very low probability. Some argue that buying a low price, volatile stock is similar to buying a lottery with similar risk-return payoffs, and thus it can be an attractive bet for investors. Therefore investors often overpay for high volatility stocks and underpay for low volatility stocks due to the »irrational« preference for volatile stocks.

Other behavioural explanations include:

- Representativeness – The success of a few, well publicized high volatility stocks make all volatile stocks seem good investments, and the speculative nature of such stocks is often ignored by investors.
- Overconfidence – Investors are overconfident in their ability to forecast the future, and the extent of their differences in opinions are higher for stocks with more uncertain outcomes (high volatility stocks). In the real world, it is easier and more practical to express a positive view (buy) than a negative view (short sell). This means the pessimists cannot effectively express their view of a volatile stock, leaving only the optimists to keep driving up its price. The result is overpricing and lower returns for high volatility stocks.

- Agency issue – There appears to be a natural tendency to avoid low volatility stocks in asset management because there is often less attention and support (in the form of research and trading) for low volatility stocks.
- Asymmetric behaviours in bull vs. bear markets – Investors behave differently in bull markets vs. bear markets. When the market is going down, there is often a huge dispersion of beta between low volatility and high volatility portfolios (i.e. the low volatility portfolio will have much lower beta). Therefore, the low volatility portfolio experiences much less drawdown than the high volatility portfolio. When the market is going up, this dispersion is not so great, and thus the low volatility portfolio underperforms only slightly. Yet, the low volatility portfolio still does better over the long term (Sefton et al., 2011)⁵⁹.

Quality

Finally, Baker, Bradley, and Wurgler (2011) propose a theory that is not based on behavioural bias but instead based on the use of equity benchmarks. Low volatility stocks often have lower betas, and overweighting them leads to higher tracking errors for institutional investors. Such tracking errors need to be justified by sufficient excess returns (alpha). In other words, institutional investors cannot buy low volatility stocks indiscriminately. To a certain extent, the benchmark issue prohibits institutional investors from fully exploiting the low volatility anomaly.

Critics of the volatility factor argue that buying a basket of low volatility stocks may lead to unintended bets on valuation and sectors, because such stocks tend to be expensive and concentrated in just a few sectors (such as financials). There is also a philosophical argument that low volatility investing is successful at reducing risk but lacks an investment thesis on returns. Therefore, technically speaking, it is a risk management, as opposed to an investment, tool.

⁵⁹ This »minimal drawdown« advantage can be easily demonstrated as follows: A high volatility portfolio may go down 50% in value in a bear market, and will need to go up 100% (double) in order to recover to its original value. However, a low volatility portfolio may go down only 30% in value, and need to go up only 43% to recover to its pre-crisis level. This puts the low volatility portfolio well ahead of the high volatility portfolio during the initial stage of a recovery.

The Quality factor aims to capture the excess return of »high quality« companies vs. the market. This has been a widely adopted concept in fundamental analysis (stock selection) but a relatively new phenomenon in quantitative investments. The main challenge is how to define the »quality« factor consistently and objectively using quantitative indicators. The quality of companies is a highly subjective matter. Though there is no standard definition, it is commonly associated with a company's competitiveness, efficiency, transparency, growth, financial and operating leverage, profit sustainability, and return-on-equity (ROE).

Sloan (1996) was one of the first to validate the excess returns to high earnings quality stocks, where accruals proxy for earnings quality. (Low accruals reflect higher earnings quality). Besides accruals, other measures of earnings quality include, earnings persistence, smoothness, and loss avoidance, and they are reviewed by Dechow, Ge, and Schrand (2010)⁶⁰. These other measures have been related to stock returns by Perotti and Wagenhofer (2011), although the general finding is that accruals have been the strongest quality-based predictor of stock returns. In recent years, Bender and Nielsen (2013) and Kozlov and Petajisto (2013) have reconfirmed the accruals effect for the 2000s. Also, Asness, Frazzini, and Pedersen (2013) recently show that quality as measured by profitability, stable growth, and high payout ratio, has significantly higher risk-adjusted returns.

More recently, some have argued that the original Dodd & Graham and Buffett investing styles include an important quality component. »[T]he secret to Buffett's success is his preference for cheap, safe, high-quality stocks combined with his consistent use of leverage to magnify returns while surviving the inevitable large absolute and relative drawdowns this entails« according to Frazzini, Kabiller and Pedersen (2012). In other words, the original »value« based style of investing pioneered by Dodd & Graham is really one that favours stocks that appear to be safe and have high quality, but are priced as bargains. Frazzini, Kabiller and Pedersen (2012) show that Berkshire Hathaway's alpha becomes insignificant when controlling for »Betting-Against-Beta« and quality factors.

60 Chen, Novy-Marx and Zhang (2011) made an ambitious attempt to include ROE in an alternative three-factor model for explaining the cross-sectional stock returns.

There is very little literature on why the Quality factor works. It's also difficult to provide a unified theory since the definition of quality varies. According to the Fama–French model, all systematic risks are ultimately economic risks. In light of this argument, it is not difficult to see the connection between stock returns and quality measures such as financial leverage and earnings growth. Campbell, Polk and Vuolteenaho (2010) offered a form of this »fundamental« explanation, arguing the primary source of systematic risks of both growth and value stocks is the cash flow fundamentals as opposed to market sentiments. From a corporate finance angle, it is easy to see how firm quality impacts stock prices. For example, a well-run company often manages its capital carefully and reduces the risk of over-leveraging or over-capitalization. The steady growth in earnings will further reduce its need for capital market financing, which will support its stock price. This will trigger a positive feedback loop making the company more competitive in the eyes of its customers and investors.

Critics of the Quality factor argue it is highly subjective and contains behaviour biases. Active managers are often strong proponents of the Quality factor but argue it can only be captured through fundamental analysis and stock picking.

Yield

The Yield factor aims to capture the outperformance of high dividend yield stocks. Dividend yield is commonly used as a valuation tool and closely related to price-to-book and price-to-earnings ratios. Therefore, assuming the payout ratio is uniformly distributed. The compounding of dividend reinvestments also contributes to the excess return of high dividend yield stocks. In addition, changes in the dividend tax regulations may also account for the dividend factor.

Blume (1980) found a positive relationship between the risk-adjusted returns and the expected dividend yield of dividend-paying stocks. Litzenberger and Ramaswamy (1979) later confirmed there is a positive but non-linear relationship between stock returns and expected dividend yields, assuming investors have no knowledge of the future ex-dividend month. Fama and French (1988) found the dividend yield has more explanatory power for longer term

returns over 2–4 years. Tweedy, Browne Company LLC (2007) offered a summary of the empirical evidence of the dividend factor in both US and UK markets for over 40 years. Various attempts have been made to integrate the dividend yield into a model to forecast stock returns using creative statistical techniques (Hodrick, 1992). Dividend yield has also been shown to have cross-sectional return predictability. Although similar in construction to the value ratios, the explanatory power of dividend yields is most often attributed to the differential taxation of capital gains and ordinary income as described in the after-tax asset pricing models developed by Brennan (1970) and Litzenberger and Ramaswamy (1979).

The Yield factor can be controversial. Black and Scholes (1974) concluded there is no significant relationship between dividend yield and stock returns after adjusting for risks. Ang and Bekaert (2007) found the excess return predictability by the dividend yield is not statistically significant or robust, which is contrary to Fama and French's findings above. The difference is mainly due to the statistical techniques used by the two research groups. Ang and Bekaert (2007) further noted the dividend is an unreliable indicator since it is »smoothed and manipulated« by the company.

Appendix B: Bibliography For Appendix A

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Multi-Factor Indexes Made Simple

A Review Of Static And Dynamic Approaches

Mehdi Alighanbari
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MSCI

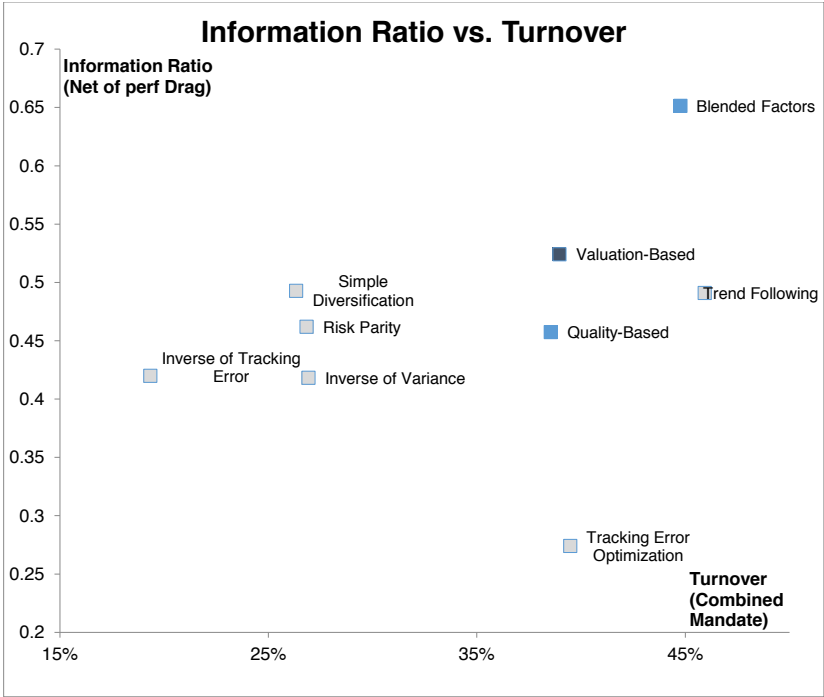
Executive Summary

Factor investing is gaining rapid acceptance by institutional and retail investors alike. Investors who were attracted by the long-term outperformance of risk premia, generally started with single-factor allocations. However, factor index returns⁶¹ have been cyclical and their active returns are weakly correlated. As a result, investors are increasingly turning their attention towards multi-factor index allocations. Not only have their returns been smoother, historically they have offered a diversification effect.

We examined nine multi-factor index strategies, which include one simple, equal-weighted strategy (Simple Diversification); five rules-based/optimization-based weighting strategies (Inverse of Variance, Inverse of Tracking Error, Trend Following, Risk Parity and Tracking Error Optimization); and three fundamental-based weighting strategies (Valuation-Based, Quality-Based and Blended Factors). Except for equal weighting, the other eight strategies involve dynamic adjustment of factor weights. The relative merit of each strategy, as measured by the information ratio versus turnover – a key element of cost – can be seen in Exhibit 1.

61 MSCI Factor Indexes provide exposure to six factors – size, value, quality, momentum, low volatility and yield – that have produced excess returns over long time periods while maintaining transparency, investability and replicability.

Exhibit 1: Information Ratio vs. Turnover of Multi-factor Indexes



Our analyses yield two key conclusions:

- The Simple Diversification strategy has been highly effective, historically.⁶² Many simple, rules-based, dynamic weighting strategies have failed to match this equal-weighted strategy’s performance, after accounting for index turnover cost.
- The fundamentals-based approaches have produced attractive results in simulation. The three strategies tested in this study have delivered higher

62 Previous research has also demonstrated that it is very difficult for an optimal portfolio to outperform one employing simple diversification. See DeMiguel, Victor, Lorenzo Garlappi and Raman Uppal (2009), »Optimal Versus Naïve Diversification: How Inefficient is the 1/N Portfolio Strategy?«, The Review of Financial Studies, Vol. 22, No. 5, 1915–1953.

active returns against the Simple Diversification strategy – pointing to the potential benefits of exploiting fundamental insights in the construction of a multi-factor index. Such strategies, however, are active in nature and typically come with the extra »costs« of higher turnover and greater complexity.

Understanding how factors can be combined, as well as the merits and disadvantages of various approaches, can help investors make more informed investment decisions. As investors explore the frontier of multi-factor investing, it is reassuring to know that a simple, equal-weighted approach has provided a compelling risk and return profile, historically. This strategy index brings simplicity, transparency and robustness to the investment process and can serve as an attractive starting point for factor allocation – especially, in the absence of active investment views and skills.

Introduction

MSCI has published several papers on factor investing, ranging from the foundation of factor investing and implementing factor indexes in equity portfolios to the behaviour of factor indexes in different macro-economic environments.⁶³ We have also discussed how combining multiple factor indexes has, historically, offered diversification and provided a smoother return stream than single-factor indexes. In this paper, we explore different multi-factor investing approaches in greater detail, based on 36 years of MSCI factor index

63 Subramanian, Aylur, Raman, Bender, Jennifer, Briand, Remy and Melas, Dimitris, »Foundations of Factor Investing« (2013). http://www.msci.com/resources/research_papers/foundations_of_factor_investing.html;
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history. We compare the return and risk characteristics of various dynamic factor allocation strategies against the Simple Diversification strategy and discuss the benefits and tradeoffs of the different approaches to factor allocation.

The main questions we address are:

1. How does a simple, equally weighted combination of factor indexes (Simple Diversification) compare to individual factor indexes?
2. Can factor indexes be combined in a dynamically managed manner that improves upon the Simple Diversification approach?
3. What are the tradeoffs between the Simple Diversification approach to factor allocation and more dynamic approaches?

We identify six risk premia – size, value, quality, momentum, low volatility and yield – as the basic building blocks for this analysis. These risk premia have been proven to earn long-term excess return over the market, historically, and are solidly grounded in academic literature.

We create the Simple Diversification multi-factor index using six corresponding MSCI World factor indexes. Using a series of rules-based and optimization algorithms to mimic dynamic approaches to index construction, we demonstrate how factor indexes can be combined beyond a simple equal-weighted strategy. We also illustrate how a rules-based interpretation of fundamental or valuation signals can be employed in the construction of multi-factor indexes.

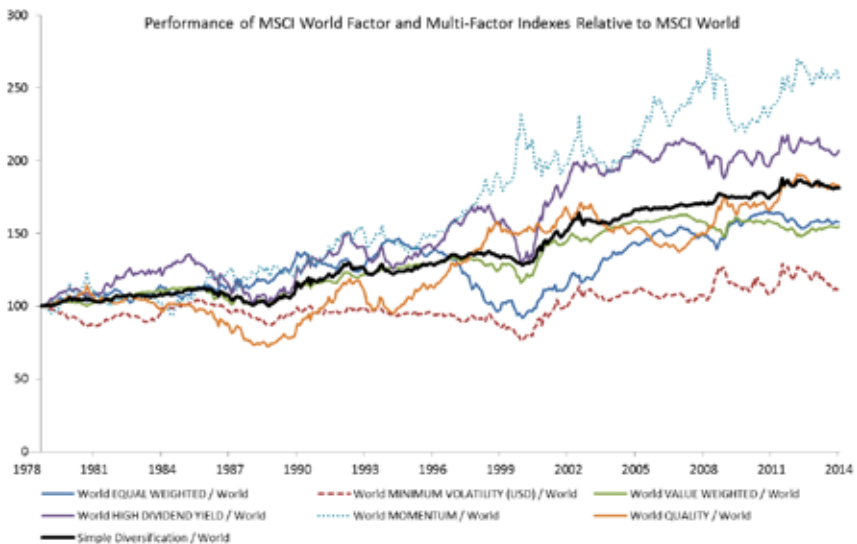
To ensure the robustness of analysis, we perform a multi-period rolling window analysis to eliminate possible biases from picking arbitrary start and end dates. We study the historical probability of factor indexes achieving excess returns or lower risk against the cap-weighted benchmark with different time horizons. The analysis, which can be found in Appendix A, also sheds light on the implications of investment horizons for factor investing.

Combining Factors: Possible Approaches

A Six-Factor Simple Diversification Index

A Simple Diversification multi-factor index arguably provides the simplest combination of factors. It is created by equally weighting factor indexes. We use six MSCI World factor indexes – Equal-Weighted, Value-Weighted, Quality, Momentum, Minimum Volatility and High Dividend Yield – to represent six well-researched risk premia. The index is rebalanced semi-annually in May and November, consistent with the rebalancing schedule of the MSCI Factor Indexes. We consider the Simple Diversification a »static« approach to factor allocation, as the weight for each factor is defined as $1/n$.

Exhibit 2: Simple Diversification has Historically Offered a Smoother Ride



The historical index performances of the Simple Diversification strategy and its underlying factor indexes can be seen in Exhibit 2. The multi-factor index captured the same long-term risk premia but offered smoother performance

than any of the underlying factor indexes. This result is not surprising: the long-term outperformance and low active correlations among the MSCI Factor Indexes help explain this phenomenon, as can be seen in Exhibit 3.

Exhibit 3: Active Return Correlations among MSCI World Factor Indexes are Low

Active Return Correlation	EQUAL-WEIGHTED	MINIMUM VOLATILITY (USD)	VALUE-WEIGHTED	HIGH DIVIDEND YIELD	MOMENTUM	QUALITY
EQUAL-WEIGHTED						
MINIMUM VOLATILITY (USD)	0.15					
VALUE-WEIGHTED	0.58	0.16				
HIGH DIVIDEND YIELD	0.26	0.45	0.73			
MOMENTUM	-0.15	0.09	-0.33	-0.03		
QUALITY	-0.15	0.18	0.15	0.44	0.25	

The Simple Diversification multi-factor index also exhibited lower volatility (13.9%) than any single-factor index except for the Minimum Volatility Index, as can be seen in Exhibit 4.

While a Simple Diversification multi-factor index may look naïve in terms of construction, it represents a reasonable starting point for an investor who wants to gain exposure to systematic risk premia but does not have specific views on the expected risk or return of the underlying factor indexes nor the skills to actively manage factor exposures.

Exhibit 4: Return/Risk Profiles of Single-Factor Indexes vs. the Six-Factor Simple Diversification Index⁶⁴

	MSCI World	MSCI Equal-Weighted	MSCI World Volatility	MSCI World Min	MSCI World Value-Weighted	MSCI High Div Yield	MSCI World Momentum	MSCI World Quality	Simple Diversification
Total Return*	10.6%	12.0%	10.9%		11.9%	12.9%	13.5%	12.5%	12.4%
Total Risk*	15.1%	15.7%	12.0%		15.2%	14.6%	16.3%	14.5%	13.9%
Return/Risk	0.70	0.77	0.91		0.79	0.88	0.83	0.86	0.90
Maximum Drawdown	-53.7%	-54.8%	-43.1%		-57.3%	-58.8%	-52.6%	-44.5%	-52.0%
Active Return*	1.4%		0.4%		1.4%	2.3%	3.0%	1.9%	1.9%
Performance Drag (bps) **	17.2		26.8		18.1	20.2	91.0	22.7	26.3
Active Return (Net of Performance Drag)	1.3%		0.1%		1.2%	2.1%	2.1%	1.7%	1.6%
Tracking Error*	5.0%		6.1%		3.5%	6.4%	8.4%	6.0%	3.3%
Information Ratio***	0.25		0.02		0.34	0.33	0.24	0.28	0.49
Maximum Active Drawdown	-37.2%		-27.4%		-13.3%	-24.3%	-23.7%	-36.6%	-10.7%
One-way Index Turnover ****	3.0	17.2	26.8		18.1	20.2	91.0	22.7	
Separate Mandates									35.4
Combined Mandate									26.3

* Annualized gross return (USD) from 11/30/1978 to 03/31/2014.

** Performance drag calculated based on annualized two-way index turnover for combined mandate assuming a transaction cost of 50bps.

*** Information Ratio is calculated using active return (net of performance drag).

**** Annualized one-way index turnover for the 05/31/1999 to 03/31/2014 period.

64 Throughout this paper, we have used data from November 30, 1978 to March 31, 2014 for our simulations. The exception is Turnover where we have used data from May 31, 1999 to March 31, 2014, due to availability and limitation of data and analyses.

Simple Rules-Based And Optimization Weighting Approaches

Going beyond Simple Diversification in a dynamic multi-factor index requires active views on factors together with the skills to manage the related exposures. A dynamic factor allocation model adjusts weights regularly; it typically involves overweighting factors that are expected to outperform and underweighting factors that are expected to underperform. The underpinning investment belief is that factors have different return streams and active factor allocation can and will add value.

There are many possible approaches to achieve a dynamic factor allocation. In this paper, we will focus only on some examples that can be replicated with a set of mechanical rules and do not seek to represent all possible dynamic factor weighting strategies.

We start with a set of simple rules-based and optimization-based weighting strategies:

- The Inverse of Variance and Risk Parity strategies can be considered as risk-based approaches to factor allocation. The underlying investment beliefs are that overweighting factors with lower volatility or balancing the risk contribution of each factor in the multi-factor index could improve risk diversification and help achieve better risk-adjusted returns.
- The Inverse of Tracking Error and Tracking Error Optimization approaches bring a risk budgeting dimension into the picture. The former aims to minimize the tracking error of the multi-factor index without optimization. The latter seeks to maximize the return outcome using mean-variance optimization subject to a tracking error constraint.
- Finally, the Trend Following strategy takes a conventional momentum strategy and applies it to factor allocation.

All dynamic strategies are rebalanced in May and November. The underlying investment belief, possible approach and weighting scheme of each strategy are outlined in Exhibit 5.

Exhibit 5: Examples of Simple Rules-Based and Optimization-Based Strategies

Multi-factor Strategy	Investment Belief	Possible Approach	Weighting Scheme
Inverse of Variance	Factors with lower volatility could produce a better risk/return profile or improved risk diversification	Weight factors by inverse of volatility	$w_i = \frac{1}{\sigma_i^2}$
	Balancing the risk contribution of factors would improve risk diversification	Weight factors so that the marginal contributions of all factors to overall risk are equal	$RC_i = RC_j \quad \forall i, j \in \{1, \dots, 6\}$ where $RC_i = w_i \frac{\partial \sigma_p}{\partial w_i}$
Inverse of Tracking Error	A multi-factor index will add value but this may be at the cost of higher tracking error risk	Weight factors by inverse of squared tracking error to parent index	$w_i = \frac{1}{TE_i^2}$
	Multi-factor indexes will add value but will be subject to a tracking error constraint	Optimize the index using mean-variance optimization	$\max_{w_i} \sum_{i=1}^6 w_i r_i$ s. t. TE_{36m} Expected alphas: Past three-year factor index returns Tracking error constraint: Realized 36-month tracking error of an equally weighted multi-factor index
Trend Following	Factors with strong past performance will deliver strong future performance	Weight by past performance; overweighting winners and underweighting losers	$w_i = \frac{Mom Z_{factor}^i}{\sum_i Mom Z_{factor}}$

See Appendix B for detailed descriptions of strategies.

Return/Risk Profiles Of Simple Rules-Based And Optimization-Based Strategies

We examine the return/risk profiles of the rules-based and optimization-based factor allocation approaches along with the Simple Diversification strategy.

As we can see in Exhibit 6, the Inverse of Variance and Risk Parity strategies produced risk and return characteristics that were fairly similar to those of the Simple Diversification strategy during the November 1978 to March 2014 period. This result can be partly explained by the fact that weights of various factor indexes are fairly stable in these two strategies and did not significantly differ from equal weighting.⁶⁵

Inverse weighting each factor index based on its tracking error would not have added much value either. In fact, the Inverse of Tracking Error approach generated a lower return but higher total risk than the Simple Diversification strategy. The result suggests that a multi-factor index that simply overweights low tracking error factor indexes may not necessary lead to better risk or return outcomes.

Exhibit 6: Performance of Rules-Based and Optimization-Based Multi-Factor Strategies

	MSCI World	Simple Diversification	Inverse of Variance	Risk Parity	Inverse of Tracking Error	Tracking Error Optimization	Trend Following
Total Return*	10.6%	12.4%	12.3%	12.4%	12.0%	11.8%	12.8%
Total Risk*	15.1%	13.9%	13.6%	13.7%	14.3%	14.6%	14.1%
Return/Risk	0.70	0.90	0.91	0.91	0.84	0.81	0.91
Maximum Drawdown	-53.7%	-52.0%	-50.8%	-51.4%	-54.9%	-53.8%	-49.6%
Active Return*		1.9%	1.8%	1.8%	1.4%	1.2%	2.2%
Performance Drag (bps) **		26.3	26.9	26.8	19.3	39.5	45.9

⁶⁵ See Appendix C for factor weight changes.

2. Choosing Right Factors

	MSCI World	Simple Diversification	Inverse of Variance	Risk Parity	Inverse of Tracking Error	Tracking Error Optimization	Trend Following
Active Return (Net of Performance Drag)		1.6%	1.5%	1.6%	1.2%	0.8%	1.7%
Tracking Error*		3.3%	3.6%	3.4%	2.9%	3.0%	3.5%
Information Ratio***		0.49	0.42	0.46	0.42	0.27	0.49
Maximum Active Drawdown		-10.7%	-11.9%	-11.0%	-13.0%	-12.9%	-10.7%
One-way Index Turnover *****	3.0						
Separate Mandates		35.4	38.9	37.1	35.1	74.0	87.0
Combined Mandate		26.3	26.9	26.8	19.3	39.5	45.9

* Annualized gross return (USD) from 11/30/1978 to 03/31/2014.

** Performance drag calculated based on annualized two-way index turnover for combined mandate assuming a transaction cost of 50bps.

*** Information Ratio is calculated using active return (net of performance drag).

***** Annualized one-way index turnover for the 05/31/1999 to 03/31/2014 period.

Optimization techniques⁶⁶ are typically employed when investors have a set of objectives and constraints they would like to incorporate into their portfolios. But optimization can be complex, often requiring accurate risk and return inputs as well as careful calibration of optimization parameters.

66 We use the Barra Open Optimizer to construct the Risk Parity and Tracking Error Optimization strategies. Given that optimization is performed at the factor index allocation level, historical MSCI factor index level returns are used to create the covariance matrix and the expected return estimation. The expected return of a strategy index is estimated using its last 36-month average performance. The covariance matrix is computed directly using the last 36-month index returns. Indexes are optimized every six months and then rebalanced.

Optimization can be performed at the individual equity level or at the factor index allocation level. For this study, optimization was employed at the factor allocation level. Based on the selected parameters, the Tracking Error Optimization multi-factor index outperformed the cap-weighted benchmark but underperformed other multi-factor strategies including Simple Diversification. The Tracking Error Optimization strategy produced an information ratio of only 0.27, the lowest in this study.

The result can be partially attributed to the fact that we used a rather simplistic covariance estimate based on only six MSCI factor indexes in the study. An alternative and more robust approach would be to optimize the multi-factor strategy from the security level with more refined risk estimation; we will explore this approach in a subsequent study.

The only rules-based strategy that outperformed the Simple Diversification strategy is the Trend Following approach. The strategy produced slightly higher return/risk and information ratios, suggesting that factor indexes exhibited some forms of momentum behaviour that could be exploited.⁶⁷ However, the Trend Following strategy would have experienced greater variations in factor weights⁶⁸ and hence higher index turnover. Taking into account the performance drag, the Trend Following strategy generated only 10 basis points of outperformance against the Simple Diversification multi-factor index.

The Fundamentals-Based Approach

The fundamentals-based approach to multi-factor indexing refers to the systematic implementation of fundamental or valuation-based investment strategies following specified rules or algorithms. The core tenet of the approach is that fundamental data contain important signals that can be used to understand the drivers of volatilities and correlations among assets. The concept

⁶⁷ We used the past six months of return data to be consistent both with the MSCI Momentum Index and the rebalancing frequency of the strategy. We performed the same analysis by varying the length of the momentum signal but did not see material differences in their performance characteristics.

⁶⁸ See Appendix C for more detail.

is, in principle, similar to the Systematic Equity Strategies (SES) discussed in recent Barra modeling papers.⁶⁹

For example, valuation plays an important role in any fundamental investment strategy. The same concept can be applied to building multi-factor index strategies: Equity factor indexes can be over- or under-valued versus their historical valuation level. Investing in factors that are cheap compared to their historical mean could provide a better chance of achieving excess returns. In addition, valuation could also serve as an indicator of crowding. A factor index that is trading above its »fair price« may be perceived as a crowded trade and would, therefore, be avoided.

Some investors may prefer to use quality metrics such as profitability. For example, the spread of Return on Equity (ROE) between high quality companies and the broad market could be considered a more appropriate indicator for factors such as Quality and Yield.⁷⁰ The hypothesis is that the market will pay a premium to high ROE and high dividend companies when there are huge differences in terms of the average level of profitability and dividend yield of companies.

While using valuation or a measure of quality to weight each factor index is a rational and sensible approach, we also recognize that each factor premium may be better captured by a different fundamental signal. For instance, the Minimum Volatility Index has historically delivered superior risk-adjusted returns during high volatility regimes. A volatility indicator such as the VIX may provide a better signal to help manage the volatility factor exposure. Thus, we can anchor different factor exposures to relevant signals. We call this the »Blended Factors« approach.

The three main fundamentals-based strategies are summarized in Exhibit 7. Similar to the previous strategies, all fundamentals-based strategies are rebalanced in May and November.

69 Zangari, Peter, Winkelmann, Kurt, Bayraktar, Mehmet and Radchenko, Stan, »Employing Systematic Equity Strategies« (2013), http://www.msci.com/resources/research_papers/research_insight_-_employing_systematic_equity_strategies_-_june_2013.html Available to clients only.

70 The MSCI High Dividend Yield Index explicitly incorporates a quality screen in selecting high yield index constituents.

*Exhibit 7: Examples of Fundamentals-Based Strategies**

Multi-factor Strategy	Investment Belief	Possible Approach	Weighting Scheme
Valuation Based	Factor indexes may become overcrowded and/or expensive which may impair performance	Overweight cheap factor indexes/ underweight expensive ones	Normalized current E/P level**
Quality Based	Factor indexes with higher ROE will outperform ones with lower ROE	Overweight high ROE indexes/ underweight low ROE ones	Normalized current ROE**
Blended Factors	Factor indexes perform well when the underlying signal is strong	Weight each factor index based on the strength of its underlying signal	Normalized E/P spread** (Value) Normalized effective number of stocks** (Size) Normalized ROE spread** (Quality) Normalized D/P spread** (Yield) Normalized VIX (Low Volatility) Normalized past six-month momentum** (Momentum)

*See Appendix B for more detailed descriptions of strategies.

**Compared to its own history.

Return/Risk Profiles Of Fundamentals-Based Strategies

Historically, the use of valuation or other fundamental signals would have improved the performance of multi-factor indexes without significant increases in the total risk, as can be seen in Exhibit 8. We make the following observations:

2. Choosing Right Factors

- Valuation-based and Quality-based multi-factor indexes produced broadly similar risk and return characteristics over the November 1978 to March 2014 period, but the Valuation-based index produced a higher information ratio and a lower maximum active drawdown.
- The Blended Factors multi-factor index provided the strongest return, outpacing the Simple Diversification strategy by 100 basis points without a significant increase in risk (or 80 basis points after the performance drag). This outperformance was accompanied by an information ratio improvement to 0.65 from 0.49, suggesting that an investor might have been able to add value to a multi-factor index by managing factor exposures with the right signals.

Exhibit 8: Performance of Fundamental Signal Strategies

	MSCI World	Simple Diversification	Valuation-Based	Quality-Based	Blended Factors
Total Return*	10.6%	12.4%	13.0%	12.9%	13.4%
Total Risk*	15.1%	13.9%	13.9%	13.8%	14.0%
Return/Risk	0.70	0.90	0.94	0.93	0.96
Maximum Drawdown	-53.7%	-52.0%	-51.9%	-51.5%	-49.7%
Active Return*		1.9%	2.4%	2.3%	2.9%
Performance Drag (bps)**		26.3	39.0	38.5	44.8
Active Return (Net of Performance Drag)		1.6%	2.0%	1.9%	2.4%
Tracking Error*		3.3%	3.9%	4.2%	3.7%
Information Ratio***		0.49	0.52	0.46	0.65
Maximum Active Drawdown		-10.7%	-9.7%	-12.2%	-10.9%
One-way Index Turnover****	3.0				
Separate Mandates		35.4	63.8	64.5	76.1
Combined Mandate		26.3	39.0	38.5	44.8

* Annualized gross return (USD) from 11/30/1978 to 03/31/2014.

** Performance drag calculated based on annualized two-way index turnover for combined mandate assuming a transaction cost of 50bps.

*** Information Ratio is calculated using active return (net of performance drag).

**** Annualized one-way index turnover for the 05/31/1999 to 03/31/2014 period.

Conclusion

There are many possible ways to construct multi-factor indexes. We use nine weighting strategies to proxy different investment approaches and examine the return/risk characteristics over a 36-year period. The results highlight that a Simple Diversification approach to constructing multi-factor indexes has, historically, proven more effective than many of the more complex approaches – pointing to its potential as a way to combine factors, especially in the absence of active investment views and skills.

Dynamic factor allocation strategies, however, have their merits as well. Our study shows that the fundamentals-based approach to factor allocation, especially the Blended Factors strategy index, which weights each factor index in the strategy based on the strength of fundamental signals, would have provided the best overall return/risk profile among the dynamic strategies analyzed. Understanding how to use fundamental signals in constructing multi-factor index strategies while controlling turnover could provide fertile ground for future research.

Clearly, there are tradeoffs between the various approaches. In considering whether to manage a multi-factor index via a simple equal weighting or more dynamic weighting strategies, the decision depends on the investor's investment beliefs and process and – critically – whether the investor is confident of possessing the insight or skills to manage factor exposures dynamically.

Appendixes

Appendix A: Multi-Factor Allocation And Investment Horizon

Investment horizon plays a critical part in factor investing. Due to the cyclicity of factors, a long horizon is essential in ensuring the success of factor index implementation. To investigate the effect of time, we analyzed the frequency of relative outperformance of multi-factor indexes over multiple time horizons. We used rolling windows of different lengths to eliminate possible biases from picking arbitrary start and end dates, running simulations for individual MSCI Factor Indexes and the three best-performing multi-factor strategies.

Exhibit 9 reveals several key findings:

- As the time horizon expanded, the frequency of outperformance against the MSCI World Index increased. The observation is consistent across both single and multi-factor indexes – highlighting the importance of having a long horizon in factor investing.
- By combining multiple factors into a single index, the frequency of outperformance improved compared to single factor strategies. This finding validates the argument that a multi-factor index would have been more effective in cushioning the effects of market cycles.
- For an investor who has a sufficiently long investment horizon (10 years or more), the historical probability of outperforming the market (regardless of which of the four multi-factor strategies was used) was virtually the same.
- The Simple Diversification strategy produced comparable results in terms of frequency of outperformance versus dynamic approaches such as Trend Following and Blended Factors – highlighting that a strategic allocation of factors even when the weights are static has historically produced favorable results.

*Exhibit 9: Historical Frequency of Outperformance of MSCI Factor Indexes versus MSCI World Index**

Rolling Window	Single-Factor Indexes						Multi-Factor Portfolios		
	Equal-Weighted	Minimum Volatility	Value-Weighted	High Div Yield	Momentum	Quality	Simple Diversification	Trend Following	Blended Factors
1 Y	60%	46%	68%	60%	67%	56%	71%	74%	72%
3 Y	70%	55%	67%	70%	75%	59%	86%	88%	90%
5 Y	81%	60%	75%	80%	90%	64%	90%	90%	94%
10 Y	73%	69%	100%	98%	99%	84%	100%	100%	100%
15 Y	82%	75%	100%	98%	100%	94%	100%	100%	100%
20 Y	88%	86%	100%	100%	100%	100%	100%	100%	100%
25 Y	100%	100%	100%	100%	100%	100%	100%	100%	100%

*On a monthly rolling basis for the period of 11/30/1978 to 3/31/2014.

Exhibit 10 repeats the same analysis with respect to index volatility. By construction, single-factor indexes such as Minimum Volatility tended to exhibit lower volatility than the market. Defensive factors such as High Dividend Yield and Quality also tended to display lower volatility compared to the market, especially as the time horizon increased. On the other hand, cyclical factors such as Momentum and Equal-Weighted showed higher volatility on average compared to the market.

The three multi-factor indexes – whether static or dynamically managed – demonstrated very attractive volatility profiles compared to most single-factor indexes. Significantly, the Simple Diversification approach has the highest historical probability of achieving a lower risk than the market.

*Exhibit 10: Frequency of Factor Indexes Experiencing Lower Volatility than MSCI World Index**

Rolling Window	Single-Factor Indexes						Multi-Factor Portfolios		
	Equal-Weighted	Minimum Volatility	Value-Weighted	High Div Yield	Momentum	Quality	Simple Diversification	Trend Following	Blended Factors
1 Y	42%	92%	52%	63%	34%	58%	80%	72%	78%
3 Y	32%	99%	60%	73%	35%	66%	89%	86%	87%
5 Y	28%	100%	66%	72%	28%	67%	100%	95%	96%
10 Y	15%	100%	78%	80%	12%	79%	100%	100%	100%
15 Y	9%	100%	74%	100%	0%	73%	100%	100%	100%
20 Y	1%	100%	69%	100%	0%	81%	100%	100%	100%
25 Y	12%	100%	57%	100%	0%	90%	100%	100%	100%

*On a monthly rolling basis for the period of 11/30/1978 to 3/31/2014.

Appendix B: Description Of Multi-Factor Strategies

Simple Diversification

Simple Diversification is the simplest way of allocating to multiple factor indexes. It does so by giving the same weight to each of the factor indexes. The strategy rebalances every six months back to equal weights, selling the better performing factors and buying underperformers, potentially capturing mean reversion of factors.

Inverse Of Variance

The rationale for this strategy is to allocate to each factor index based on its level of risk, as defined by historical volatility based on trailing 36-month standard deviation. The result will overweight (underweight) factor indexes that have lower (higher) volatility. One expectation for this strategy is to have lower total volatility than the Simple Diversification strategy. This strategy, as well as all other strategies explained in this paper, adhere to a six-month rebalancing frequency.

Risk Parity

The objective of this optimization-based approach is to achieve an equal risk contribution of each factor index. In creating the Risk Parity factor strategy, individual factor index correlations are taken into account.

Inverse Of Tracking Error

This approach is similar to the inverse variance approach, but instead of using the variance of the factor index returns, the strategy employs the squared tracking error (variance) of the active returns, based on a trailing 36-month window. This strategy overweights (underweights) factors that have lower (higher) tracking error to the parent index. By construction, the resulting index is expected to have low tracking error to the parent index.

Tracking Error Optimization

This approach aims to maximize the expected active return while constraining the tracking error using mean-variance optimization. Expected alphas are estimated for this type of optimization. For this simulation, we base expected alphas on actual factor index returns of the past three years and the tracking error constraint is based on the realized 36-month tracking error of an equally weighted multi-factor strategy index.

Trend Following

The Trend Following approach allocates to different factor indexes based on their recent (six-month) performance. The assumption is that momentum exists in factor performance and the factors that have performed well over the recent past will continue to perform well over the next six months.

Valuation-Based

We use valuation to measure how cheap or expensive a factor index is and to adjust the weight of each index in the multi-factor strategy based on its valuation. While the aim here is to avoid expensive indexes or stocks, the strategy may also help to identify crowded strategies. There are several fundamental parameters that can be used individually or in combination to measure value. In this study, earnings yield (E/P) is used as a measure for value. It is recognized that factor indexes have inherent valuation biases. Factor indexes such as Value-Weighted tend to be cheap by construction while Momentum or Quality tend to be more expensive in terms of relative valuation. Therefore, there will be a constant overweight (underweight) for inherently cheap (expensive) factor indexes, which is not desirable in this study.

To avoid systematic valuation biases of factor indexes, we first normalize valuation of a factor index of time t against its own history. To capture as much information as possible, the normalization is performed on an expanded history, i.e., a shorter history is used at the beginning period of the simulation and a full history is used at the ending period of the simulation.

$$Z_i(t) = \text{Normalized } \frac{E}{P}(t) \text{ within } \left\{ \frac{E}{P}(t-1), \frac{E}{P}(t-2), \dots, \frac{E}{P}(t-n) \right\}$$

We repeat the same process for each factor index and use normalized scores at time t to allocate more weight to factor indexes that are cheap compared to their historical long-term average.

Quality-Based

Looking at the cheapness or richness of an index is one way to weight factors. A similar exercise can be made for other fundamental variables such as Return on Equity (ROE), a measure of quality.

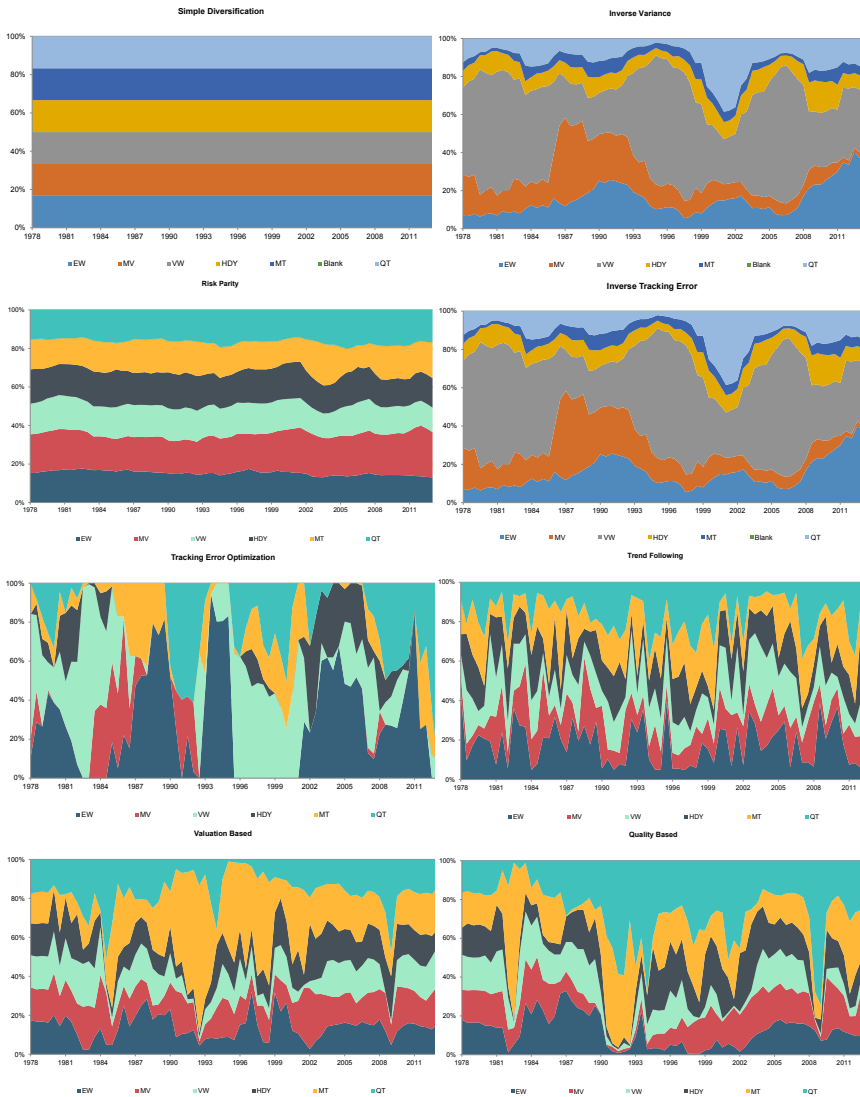
Blended Factors

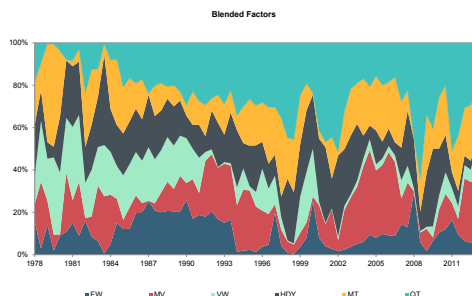
In Blended Factors, we weight each factor index based on the relevant signal strength. This time, we use the normalized earnings yield spread (the difference between the factor index and the parent index) at time t by comparing it to its own history (the long-term historical average) for the Value Index, the Dividend Yield spread for the High Dividend Index and the quality spread for the Quality Index. For the size index, we use the effective number of stocks of the parent index (MSCI World) as the proxy for index concentration at time t and compare it to its own history. For the Minimum Volatility index, we use the normalized volatility score based on the CBOE VIX index at time t and compare it to its own history. For the Momentum Index, we use the recent six-month performance spread against its own history.

Exhibit 11: Summary of Signals Used in the Blended Factors Strategy

Factor Index	Factor Specific Signal	Weighting Scheme
Value-Weighted	Earnings yield spread	Overweight (underweight) Value-Weighted Index when the earnings yield spread against the parent index is high (low) relative to its historical range
Equal-Weighted	Effective number of stocks – a measure of index concentration	Overweight (underweight) Equal-Weighted Index when the effective number of stocks of the parent index (MSCI World) is low (high) relative to its historical range
Momentum	Six-month performance spread	Overweight (underweight) Momentum Index when the six-month return differential with the parent index is high (low) relative to its historical range
Minimum Volatility	VIX	Overweight (underweight) Minimum Volatility Index when VIX is high (low) relative to the historical range
Quality	ROE spread	Overweight (underweight) Quality Index when ROE spread against the parent index is high (low) relative to its historical range
High Dividend Yield	Dividend yield spread	Overweight (underweight) High Dividend Yield Index when the dividend yield spread against the parent index is high (low) relative to its historical range

Appendix C: Weights Of Different Factor Indexes Within Multi-Factor Strategies





List Of Authors

Altat Kassam is Managing Director and Head of Equity Applied Research, Americas and EMEAI for MSCI, with responsibility for research, to support new and existing indexes and risk models including factor and economic exposure indexes, as well as performance and risk attribution. He also serves as an ambassador to asset owners and managers on the role of indexes and factor models in asset allocation and portfolio construction.

Prior to joining MSCI, Mr. Kassam was Global Head of the Equity Market Strategies Group for UBS Investment Bank, which worked on the creation of and studies of custom and 3rd party equity and multi-asset indexes for the Bank's hedge fund, asset manager and private wealth clients, including factor and thematic products, as well as customized hedging products to enable the mitigation of regional, sector and country risks, often optimized to account for short-selling restrictions and liquidity constraints.

Mr. Kassam also spent over seven years at Deutsche Bank in London and Hong Kong, where he was the Head of the Equity Derivatives Strategy Teams for Europe and Asia-Pacific, publishing research to support clients' use of equity derivatives, thematic exposure baskets and regional and country-based optimized hedging products, as well as keeping clients apprised of special situations and relative value opportunities throughout the regions. A key component of his work was the encouragement and support of asset managers, asset owners and consultants, in their use of equity derivatives to reduce risk and customize equity exposure in their portfolios. His teams in Europe and Asia were both multi-year winners of Institutional Investor Magazine's regional equity derivative research awards. Upon returning to London, Mr. Kassam started up, grew,

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Before his time with Deutsche Bank, Mr. Kassam was a member of the Goldman Sachs Equity Derivative Strategy Group, and started his career with Swiss Bank Corporation in Equity Derivatives in 1994.

Mr. Kassam received a first-class joint honors degree in Engineering and Computing Science from St. John's College, Oxford University and an MSc. In Finance from London Business School. He received his CFA charter in 2005. He has published practitioner articles on the relationship between equity volatility and credit spreads, the »optimal« use of protective equity derivative strategies from a risk/return point of view, »how to« guides on call overwriting and analyses of dividend futures markets.

Abhishek Gupta is Senior Associate in the Index Applied Research team at MSCI. He joined MSCI in 2010 and is responsible for conducting research on the existing suite of MSCI Indexes such as MSCI Global and Regional Benchmark Indexes, MSCI Sector, Factor and Economic Exposure Indexes. He regularly contributes to research publications encompassing themes on global investing and factor investments. He also represents the Index Applied Research team to clients within India. In the past he has participated in new product development initiatives as well as index management.

Prior to joining MSCI, Mr. Gupta was a member of the Risk Management team at ICICI Bank (one of the largest private banks in India). In this role he was engaged in portfolio credit risk modelling, derivatives valuation and risk management. Mr. Gupta is a CFA Charterholder and holds a graduate degree in Electronics and a post graduate in Management.

Raghu Suryanarayanan is Executive Director at MSCI and works in Analytics Research. Raghu is leading the macro-economic risk and asset allocation research group. Raghu joined MSCI in 2007 as a member of Applied Research. He then led the firm's research efforts on private asset classes. Prior to joining the company, Raghu worked in the Economics Team at Morgan Stanley.

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As Head of Equity Applied Research for the Americas, Raman Aylur Subramanian conducts research on MSCI product applications, and presents the results in interactive sessions with clients. Raman joined MSCI in 1999, and has been working in a variety of research roles. Raman has established close relationships with clients and gained an in-depth understanding of their needs with respect to the MSCI indexes.

Prior to joining MSCI, Raman worked in the Indian petroleum industry where he worked on sales and product management.

Remy Briand holds an MSc in Computer Sciences from INSA (Lyon) and an MBA from HEC (Paris).

As Global Head of Research at MSCI, Remy Briand is responsible for research across all MSCI products, including indexes, risk models and ESG (environmental, social and governance) ratings. He leads a global team of 150 researchers designing products and services that help institutional investors make more informed investment decisions.

In 2010, Remy led the restructuring of the various ESG businesses acquired through MSCI's purchase of the RiskMetrics Group. MSCI ESG Research Inc. was created with a mission to help investors integrate ESG factors into their investment processes. As Head of Index Research from 2007 to 2010, Remy initiated an innovation programme which led to the creation of the MSCI Frontier Markets Indexes, MSCI Minimum Volatility Indexes and MSCI Factor Indexes.

Prior to joining MSCI in 2001, Remy was an equity portfolio manager and head of research at Credit Lyonnais Asset Management in Paris. He began his career as a private equity analyst at Credit Lyonnais.

Over his 25 year career as an investor, independent thinker and business leader, Remy has gained unique insights on topics such as global investing, emerging markets, sustainable investment and financial innovation. He regularly shares his views at industry conferences and with financial news media.

Dimitris Melas is a Chartered Financial Analyst (CFA) and holds an MSc in Electrical Engineering, an MBA in Finance, and a PhD in Financial Mathematics from the London School of Economics.

As Global Head of New Product Research, Dimitris Melas and his team are responsible for all research efforts to enhance existing MSCI indexes and to develop new index methodologies. The team is also responsible for conducting research, publishing articles, and giving client and conference presentations to promote the use of MSCI indexes in the institutional investment process.

Prior to joining MSCI in 2006, Dimitris worked at HSBC Asset Management where he was Head of Research and Head of Quantitative Strategies, leading a team of analysts and fund managers that provided analytical research to the global asset management business and managed several quantitative equity portfolios. At HSBC, he was also a member of the Global Investment Strategy Group, a committee of CIOs and other senior investment professionals responsible for setting investment strategy and asset allocation policy for all balanced and multi-asset class portfolios managed by HSBC Asset Management.

Dimitris has published several research papers in leading academic and industry journals. His paper »Efficient Replication of Factor Returns« published in the Journal of Portfolio Management was voted »Best Index-Related Research Paper« at the 6th Annual William F. Sharpe Indexing Achievement Awards. Dimitris is frequently invited to speak at investment conferences and has contributed several articles and interviews in leading financial media including the Financial Times, Bloomberg and CNBC.

Mehdi Alighanbari is Vice President in the Equity Applied Research team for EMEAI at MSCI, based in London. He has responsibility for producing research and working with clients to support them to most efficiently use new and existing MSCI indexes, including factor and economic exposure indexes, as well as the full range of capitalization-weighted indexes. The Equity Applied Research team also serve as ambassadors to asset owners and managers to educate them, their teams and their clients on linking MSCI indexes to their investment processes.

Prior to MSCI, Dr. Alighanbari spent five years as an Equity Derivatives Strategist at Deutsche Bank in London. In this role he published several research reports and spoke to clients globally on topics from hedging to optimal benchmark tracking to volatility trading.

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Chin Ping Chia holds a BA (Hons) in Economics from the National University of Singapore and is a CFA Charter holder. His research interests cover a broad range of topics including global investing, asset allocation, quantitative portfolio construction and risk management.

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Chin Ping is responsible for formulating the firm's research and product development agenda as well as delivering clear insights for leading institutional investors, asset managers and sovereign wealth funds in the region. Chin Ping also works closely with regional regulatory bodies to help them to address capital market development issues.

Prior to joining MSCI in 2001, Chin Ping was a buy-side analyst at HSBC Asset Management and Assistant Director of the Department of Statistics, Ministry of Trade & Industry of Singapore.

3. Loading Factor Exposures To Portfolios

Tactical Tilts And Multi Asset Class Risk Premia

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Patrick Zimmermann

UBS Global Asset Management⁷¹

Abstract

Factor-based investing has become popular in recent years. The portfolio construction approach that prevails in the practice is risk-parity whereby contribution to risk is equalized across markets or asset classes. In this article, we depart from this assumption and consider risk-parity as a starting point. We develop a simple framework and show that risk-parity may be improved upon through the adjunction of a slow-moving macroeconomic signal and a counter-cyclical momentum strategy. We discuss the characteristics of each strategy and why the combination of the three, tends to generate more robust portfolio characteristics.

Introduction

Much has been written about the benefits of risk-premia investing for Equities. The literature, academic or practitioner-oriented, tends to focus on risk-based portfolio construction methods. Approaches include volatility-parity with portfolio weights inversely proportional to the standard deviation of

⁷¹ The authors gratefully acknowledge the comments and suggestions from Jay Eisenhardt, Michele Gambera, Art Gresh, Scott Payseur and Andreas Razen.

returns, risk-parity, whereby the marginal contribution to risk is equalized across assets, and maximum diversification⁷².

In this paper, risk-parity constitutes the starting point, instead – in other words, the passive core of the portfolio. We then depart from this setup in two ways. First, we consider a multi asset-class investment universe. Equities are traded via instruments replicating the MSCI equity factor premia. We also include sources of alternative beta from fixed income, foreign exchange and commodities. The portfolio seeks to achieve a balanced risk profile and benefits from diversification between alternative beta sources, as well as complementary asset classes. Next, we show how a risk-parity approach can be enhanced by the addition of tactical tilts that possess counter-cyclical properties. We propose an active trading strategy that relies on simple rules and seeks to avoid over-engineering and hindsight bias. The tactical shifts are composed of two parts. First, we introduce a switching strategy whereby asset tilts depend upon the state of the economy. This low-frequency strategy complements the risk-parity core of the portfolio by shifting the portfolio into a conservative direction when ›things go bad‹. We, then, build on recent asset allocation research suggesting that macro-based tilts tend to be more effective in recessionary rather than expansionary times. To benefit better from style diversification near business cycle peaks, we add a countercyclical momentum strategy and apply it to the same investment universe. We discuss the benefits of this combined approach and propose avenues for further research.

Risk Parity Applied To Risk Premia

We consider a long-only portfolio composed of the following assets, of which there are 13 in total:

- *Equity*: MSCI minimum volatility, momentum, quality, risk-weighted, value-weighted and high dividend yield⁷³ (starting in 1989)

⁷² See Clarke et al. (2013), Qian (2006) and Baltas et al. (2014) for an in depth discussion.

⁷³ Details of the construction and characteristics of the MSCI equity risk premia can be found in Subramanian et al. (2014).

- *Fixed Income*: US 7-10 year government bonds, US investment grade, US high yield (starting in 1989)
- *Alternative Beta*: Commodity carry, FX carry, short rates volatility strategy and equity volatility strategy (starting between 1994 and 2003)

The commodity carry strategy seeks to profit from the observation that commodities in backwardation tend to have a positive return expectation. For simplicity, we have used, as a proxy, the Morgan Stanley Roll Select index that goes long a basket of commodities that are in backwardation and goes short the DJUBS index⁷⁴. The FX carry strategy goes long (conv. short) currencies with the highest (conv. lowest) short term interest rates. Finally, the interest rate and equity strategies⁷⁵ seek to benefit from shifts in volatilities by investing in short dated straddles. All alternative beta strategies are easily accessible.

To construct the »core« of the portfolio, we equalize the marginal contribution to risk of each of the 13 assets, under constraints. For computation of the covariance matrix we consider a rolling 5 year period⁷⁶. The portfolio is rebalanced monthly and targets an ex-ante 5% volatility level. Risk-parity is notorious for its large allocations to assets with low risk and low correlation, and elevated leverage. To mitigate this issue, we constrain the allocations so that 40% of the risk budget is allocated to equities, 50% to fixed income and 10% to alternative beta. We show in Figure 1 the evolution of asset weights through time. We allow for a maximum leverage of up to 150%. We note the decrease in portfolio allocations during the 2008 crisis, consistent with the increase in volatility of risky assets, and their correlations.

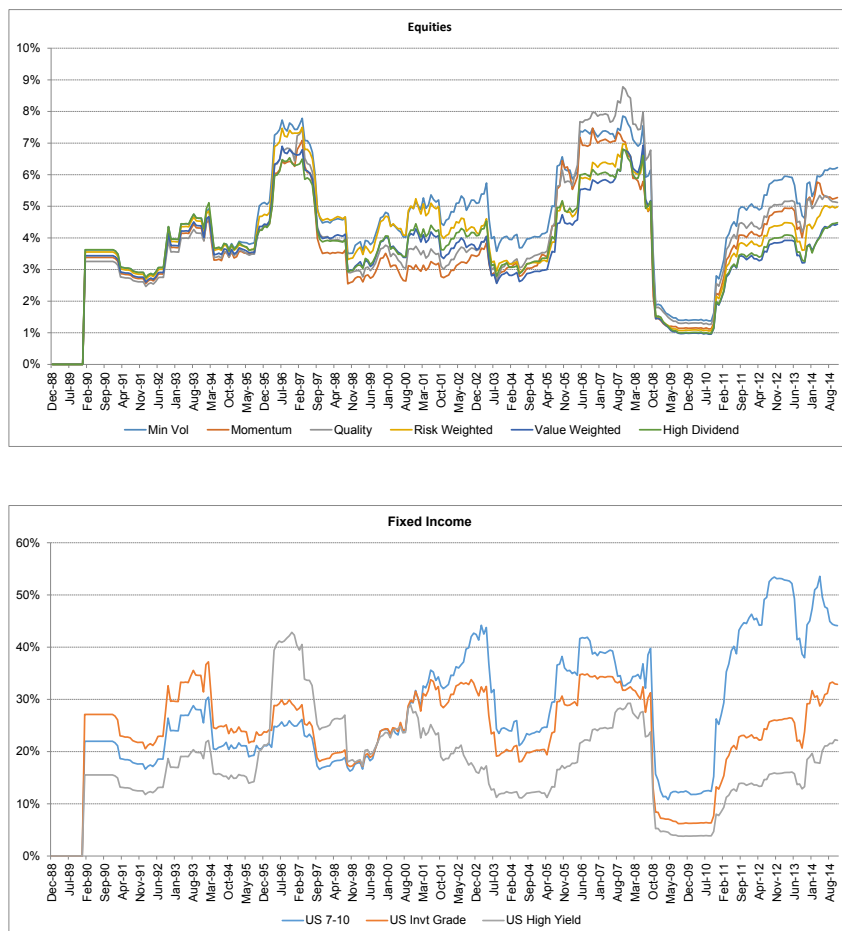
74 A more rigorous implementation of the carry strategy would short commodities that are in contango.

75 Respectively, proxied by the Deutsche Bank Impact Dollar Rates 3M and Goldman Sachs Volatility Carry strategy.

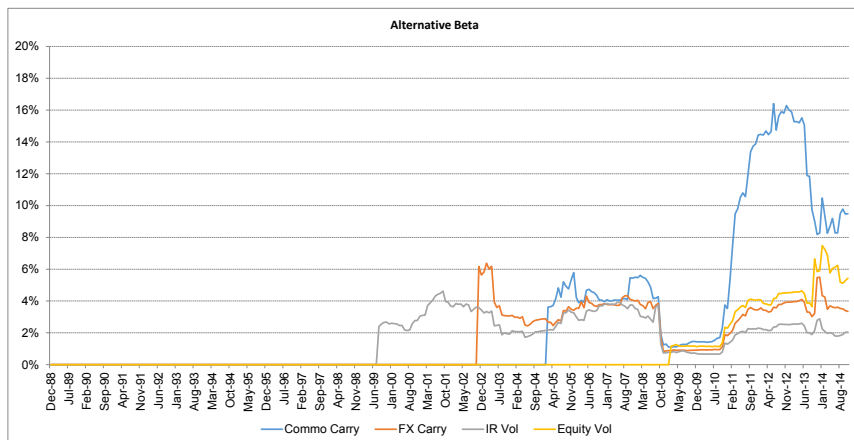
76 Note that the data for alternative beta starts late, their weights are equal to 0 up to 5 years after the data become available.

Figure 1: Risk Parity – Portfolio Weights

We show the evolution of asset weights through time. For computation of the covariance matrix we consider a rolling 5 year period. The portfolio is rebalanced monthly and targets an ex-ante 5% volatility level. We constrain the allocations so that 40% of the risk budget is allocated to equities, 50% to fixed income and 10% to alternative beta. The data for alternative beta start later and their weights are equal to 0 up to 5 years after the series start.



3. Loading Factor Exposures To Portfolios

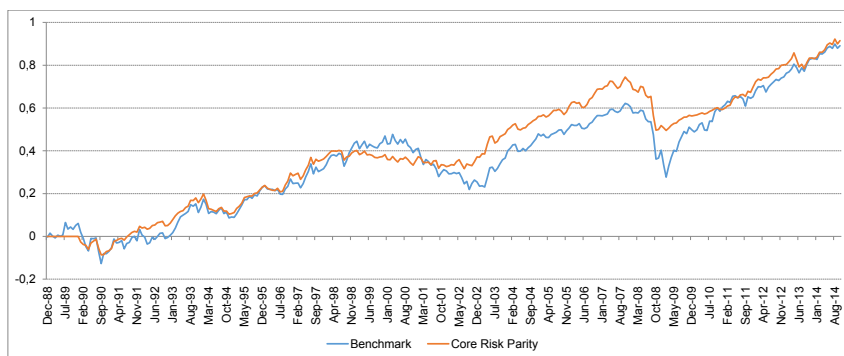


Source: UBS Global Asset Management

In Figure 2 we plot, in the top graph, the cumulative returns (e.g. log wealth) of the risk premia portfolio versus the benchmark, which we assume to be 50% MSCI World and 50% World Global Bond Index (WGBI). Returns are in excess of cash (i.e. US 3 month Libor). We notice that, despite the lower volatility of the risk parity portfolio, both benchmark and risk-parity portfolios have close to identical final wealth. The first source of added value comes from holding a diversified universe of risk factors rather than the benchmark. The second, somewhat trivial source, is that the benchmark allocates 50% to fixed income expressed in terms of percentage weights, whereas the risk parity portfolio allocates 50% in terms of risk. Given the lower volatility of fixed income and its performance during the backtest period, some outperformance relative to the benchmark, mechanically, comes from the overweight to bonds. Given the 40% risk allocation to equities, the portfolio exhibits, as expected, a substantial drawdown in 2008 (-22.1%). Part of the added value seems to relate to the avoidance of the bear market following the Dot Com bubble, reflecting the fact that the equity risk factors may have underlying sector over or under exposures. In the bottom graph, we plot the ratio of risk premia to benchmark wealth. The risk parity outperformed the benchmark for most of the period.

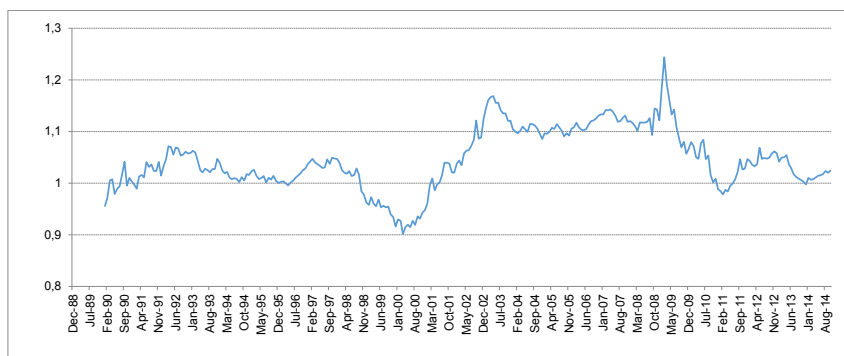
Figure 2: Cumulative Returns and Ratio of Risk Premia to Benchmark Wealth

We plot in the cumulative returns (e.g. log wealth) of the risk premia portfolio versus the benchmark, which we assume to be 50% MSCI World and 50% World Global Bond Index (WGBI). Returns are in excess of cash. In the bottom graph, we plot the ratio of risk premia to benchmark wealth.



Past performance, whether simulated or actual, is not a reliable indicator of future results.

Ratio of Risk Parity to Benchmark



Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: UBS Global Asset Management

In Table 1, we provide the summary statistics of the risk premia portfolio and the benchmark⁷⁷. The risk parity portfolio generates a higher risk-adjusted performance over the entire sample, with less volatility due to diversification across factors.

Table 1: Summary Statistics

The benchmark is assumed to be 50% MSCI World and 50% WGBI. Returns are annualized and in excess of cash (i.e. US 3 month Libor).

	Benchmark	Risk Premia
Sharpe 3yrs	1,51	1,48
5 yrs	1,32	1,58
10 yrs	0,61	0,66
since 1989	0,47	0,67
Avg Return (% ann.)	3,76	3,69
Std Deviation (% ann.)	7,91	5,52
Max Drawdown (%)	(29,17)	(22,11)
Skewness	(0,66)	(1,13)
Kurtosis	4,70	6,95

Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: UBS Global Asset Management

Tactical Tilts: The Macroeconomic Cycle And Momentum

There is considerable evidence that asset returns or risk premia are not constant through time, nor are correlations static. Bear markets are characterized by high correlation across risky asset classes and flight to quality. This gave rise to a number of studies on conditional asset allocation, whereby the composition of a portfolio depends upon the »state« of the economy, or underlying risk

⁷⁷ We assume it to be 50% MSCI World and 50% WGBI. Returns are annualized and in excess of cash (i.e. US 3 month Libor).

regimes⁷⁸. It is beyond the purpose of this study to present a survey of these studies but a few preliminary observations are necessary.

Firstly, the link between asset returns and macroeconomic or financial variables is weaker than most would admit. A time series predictability is low. As Hammerschmid and Lohre (2013) report, »It is well known that not only stock returns but also the strength of return predictability is varying over time and that the predictive power of some instruments seems to be diminishing over time«. A R^2 of 0.5%-1%, which in other contexts would indicate a poor model fit, is generally regarded as high for stock returns. One should, therefore, not expect the added-value of trading rules that are conditional upon macroeconomic or financial variables to be high, and evidence to the contrary might suggest in-sample over-fitting and rule snooping. Secondly, recent research in the area of macro-dependent asset allocation has highlighted poor, out-of-sample forecast performance⁷⁹, model instability⁸⁰ and time-varying performance in expansions and recessions. With regards to this last point, we build upon the observation made by Neely et al. (2012) that macroeconomic and technical indicators both tend to capture counter-cyclical information, but at different points of the cycle: »While technical indicators detect the typical decline in the equity risk premium near business cycle peaks, macroeconomic variables more readily pick up the typical rise in the equity risk premium later in recessions near cyclical troughs«. It follows that combining macro and technical based indicators should improve performance.

The tactical tilts we propose are comprised of two components:

- A momentum strategy, including both cross-sectional and time series momentum, with a signal constructed over the previous 3, 6, 9 and 12 months, and more emphasis on the longer time windows for stability
- A macro based strategy that applies fixed tilts to each asset based upon the state of the economy

⁷⁸ See Ang and Bekaert (2002).

⁷⁹ See Bossaerts and Hillion (1999).

⁸⁰ See Pettenuzzo and Timmermann (2007).

The inclusion of momentum enables capturing shifts that are typically not detected by fundamental variables, as well as time-varying risk-aversion. It also serves as a proxy for well-documented behavioral phenomena. The construction of the momentum strategy is straightforward, the indicator being the moving average of past returns. Time-series momentum consists in going long (conv. short) all assets with positive (conv. negative) momentum. If we consider instead a cross-section of assets, we go long the top half of the sample, and short the bottom half, regardless of the sign or strength of the signal. To avoid rule snooping, we only impose an ex-ante target volatility of 5% per annum on a rolling 2-year⁸¹ basis but refrain from additional constraints, minimum or maximum exposure limits, or stop-losses in order not to distort the signal.

The construction of the macroeconomic trading rule is characterized by simplicity. Kritzman et al. (2012) estimate Markov switching models, explicitly, to assess the probability of expansion or recession, and apply the framework, convincingly, to a number of macro-economic factors. We choose not to follow that route because of the risk of estimation error that may lead to »false positives« (i.e. predicting a recession when there is none). We use only one indicator, the OECD composite leading indicator (CLI), and classify the regime of the economy based upon two dimensions: its 3-month rate of change and its level. Regime classification is done out of sample, using the latest available release each month. This indicator may be viewed as a base-case indicator given that there is a 2-month lag in the release of the latest data. Depending upon whether the economy is growing (e.g. positive 3-month change), and whether it is above average (e.g. level above 100), the regime is classified as being in recession, recovery, expansion or slowdown.

81 Using a 2-year rolling window provides a reasonable middle ground. We avoid excessive portfolio turnover and yet adapt to shifts in market risk. We apply equal weights to monthly observations. Using an exponentially weighted moving average usually yields little difference at this frequency and is highly dependent upon the decay factor.

This is not the only way of using macroeconomic indicators. Gupta et al. (2014) classify regimes using both the CLI and inflation. This leads to a 2x2 regime classification (i.e. goldilocks, stagflation, slow growth and heating-up). This methodology amounts to imposing a classification that is *fully* conditional on 2 variables, a strong assumption in itself. An alternative is to identify regimes switching for each variable separately, so that the resulting bets have more chance of being independent. In addition, practical consideration must be paid to noise and turnover. Since the Great Moderation of the 1980s, the magnitude of macroeconomic cycles has dampened considerably. Identifying a change in the economic regime, depending on whether a given variable exceeds or is lower than 0, could be noise. Small changes may trigger a shift when there was none, thus creating unnecessary changes in portfolio allocations. This observation is confirmed by the low probabilities the authors report of staying in a given regime, implying a duration not exceeding a few months for each, leading to potentially high turnover.

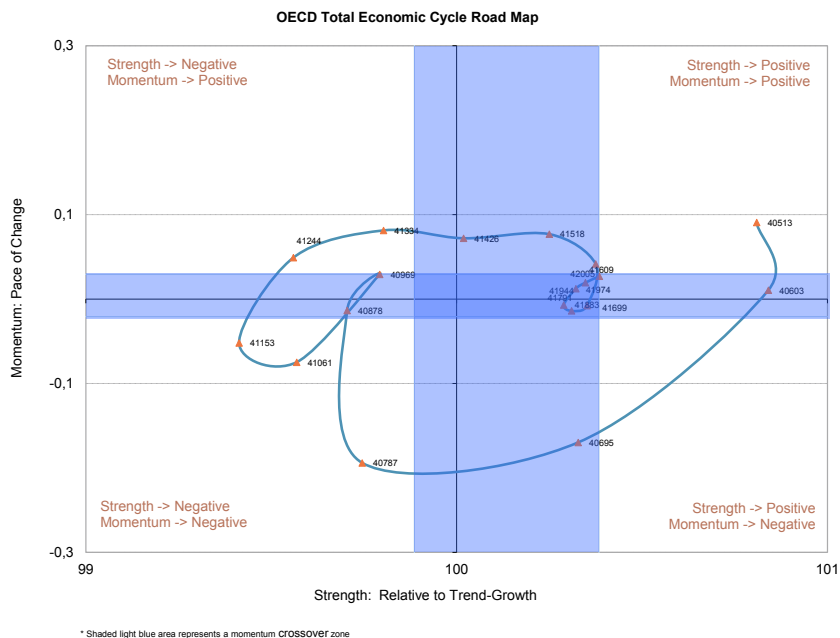
The solution we apply, for the sake of simplicity, is to impose a »no-change« band so that, unless the indicator has crossed the 40% or 60% percentile (computed over past observations), we classify the current regime as being identical to the one the previous month. Figure 3 provides a graphical representation of how we classify regimes⁸²

Figure 3: Regime Classification

We illustrate how we classify regimes. We impose a »no-change« band so that, unless the indicator has crossed the 40% or 60% percentile (computed over past observations), we classify the current regime as being identical to the one the previous month. For ease of representation, the graph is based upon the latest release whereas in the practice each data point pertains to its release date at the time.

⁸² Note that the graph, for ease of representation, is based upon the latest release whereas in the practice each data point pertains to its release date at the time.

3. Loading Factor Exposures To Portfolios



Source: OECD, UBS Global Asset Management

During recessions, portfolio tilts are conservative. We overweight US government bonds and reduce allocation to high yield and investment grade. In equities, we increase allocation to minimum volatility and quality and reduce momentum and allocation to alternative beta. During recoveries, we tilt the portfolio in a more aggressive direction. We acknowledge, in line with previous research, that tilts tend to be more successful in recessionary environments and that it may be difficult to distinguish between recoveries and expansion, or between slowdown and recessions. To simplify, we assume less confidence for slowdown regimes (e.g. tilts are half that during recessions), and are identical during recoveries and expansions. In effect we use three regime-dependent allocations. Table 2 summarizes the portfolio tilts. On average, we have one portfolio tilt a year. The magnitude of the tilts is small and the realized volatility of the resulting portfolio is inferior to 3%. In what follows we rescale the weights obtained to reach a target volatility of 5%

Table 2: Factor Tilts in Economic Regimes

We provide the summary of factor tilts in each of the four regimes as illustrated in Figure 3. On average, there is one portfolio tilt a year.

Asset	Recovery	Expansion	Slowdown	Recession
Minimum Volatility	-8%	-8%	2%	5%
Momentum	10%	10%	-5%	-10%
Quality	-6%	-6%	5%	10%
Risk Weighted	4%	4%	-3%	-5%
Value Weighted	4%	4%	-3%	-5%
High Dividend Yield	4%	4%	3%	5%
US 7-10 yr bonds	-9%	-9%	14%	28%
US Investment Grade	6%	6%	-3%	-5%
US High Yield	-13%	-13%	-4%	-9%
Commodity Carry	2%	2%	0%	0%
FX Carry	2%	2%	-3%	-7%
Interest Rate Vol	2%	2%	-1%	-3%
Equity Vol	2%	2%	-2%	-4%
Sum	0%	0%	0%	0%

Source: UBS Global Asset Management

Combining Risk Parity With Tactical Shifts And Momentum

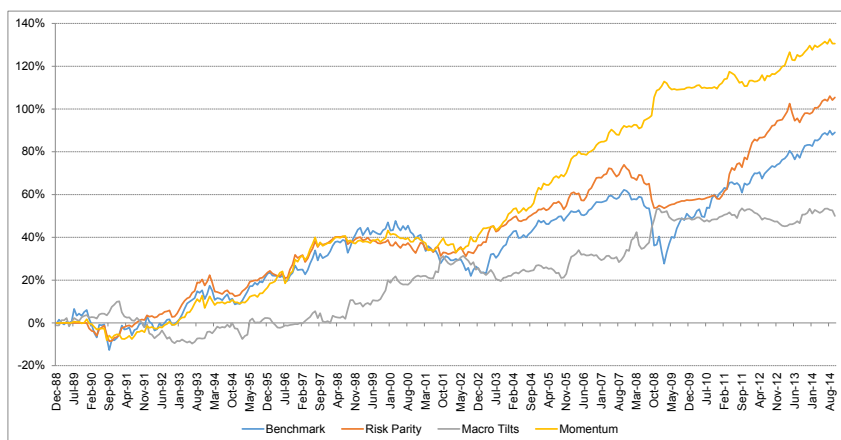
The final portfolio is a combination of 3 components: the risk-premia portfolio, the momentum strategy and the tactical macro tilts.

In Figure 4, we plot the cumulative returns of the 3 components and the balanced benchmark. All 3 components are rescaled in order to achieve an ex-ante rolling 5% target volatility, with no leverage constraint imposed. The momentum strategy tends to outperform during crisis (and volatile) times. It was profitable in 2008, in line with the performance of Commodity Trading Advisors (CTAs), and flat in the following 2 years, consistent with the fact that medium to long term trend followers took a longer time to rebuild positions in risky assets. The macro tilts tend to outperform in recessionary environments

(an observation highlighted in earlier research) and underperforms in quieter economic regimes. Because of their counter-cyclical properties at different points of the business cycle, both complement the risk parity strategy well.

Figure 4: Cumulative Returns of the Strategy Components

We plot the cumulative returns of the three strategy components (risk parity, macro tilts and momentum) and the balanced benchmark. All three components are rescaled in order to achieve an ex-ante rolling 5% target volatility, with no leverage constraint imposed.



Past performance, whether simulated or actual, is not a reliable indicator of future results.
Source: UBS Global Asset Management

Finally, to construct the final portfolio we assign 50% of the weight to the risk-parity portfolio, 25% to macro tilts and 25% to momentum. We now constrain portfolio weights to be positive and gross leverage not to exceed 1. Because of these controls, and the low correlation of the 3 components, the realized volatility of the portfolio falls just under 4%. In Table 3 we provide the summary statistics for three components and their combination in the final portfolio. Cumulative returns are shown in Figure 5.

The strategy exhibits flat returns in 2008, as losses in the core portfolio were offset by the gains in momentum and macro-based strategies. Momentum

and risk-parity strategies achieve the highest risk-adjusted performance over the whole sample. As expected from a low-turnover strategy, the performance of the macroeconomic strategy is much weaker, especially in the last 5 years, given that most of this time period was spent in recovery; a regime where tactical tilts tend to underperform. However, this component has a 0% correlation with risk-parity and 30% with momentum, hence contributing to diversification. The combination exhibits a more stable, risk-return profile through the entire sample and positive skewness⁸³. Fat tails, as measured by kurtosis, are less pronounced. Finally, the ratio of maximum drawdown to standard deviation is more favourable.

Table 3: Summary Statistics

We provide the summary statistics for the three components and the final portfolio (Combination). To construct the final portfolio, we assign 50% of the weight to the risk-parity portfolio, 25% to macro tilts and 25% to momentum. The final portfolio weights are constrained to be positive and the gross leverage not to exceed 1. The individual returns are not adjusted by volatility.

	Risk Premia	Macro Tilts	Momentum	Combination
Sharpe 3yrs	1,50	(0,17)	1,23	1,47
5 yrs	1,53	0,10	0,86	1,46
10 yrs	0,86	0,46	1,33	1,18
since 1989	0,72	0,37	0,92	0,88
Avg Return (%)	4,24	2,09	5,21	3,36
Std Deviation (%)	5,86	5,73	5,63	3,84
Max Drawdown (%)	(18,35)	(17,95)	(9,76)	(5,40)
Skewness	(0,33)	0,64	0,06	0,16
Kurtosis	5,07	6,20	6,44	4,64

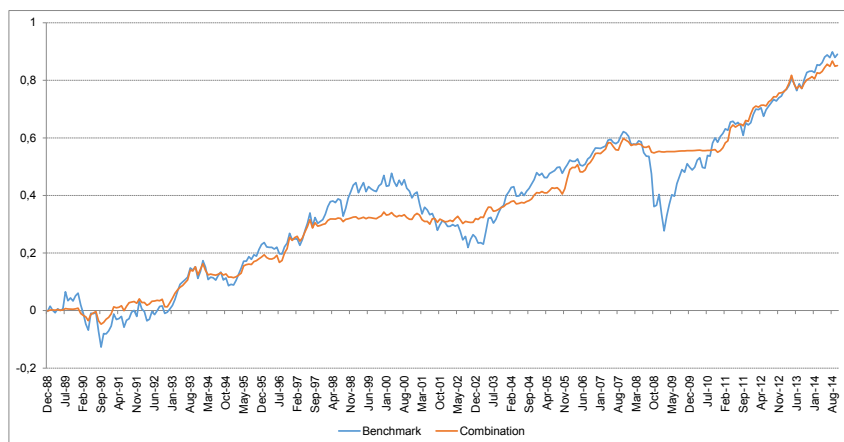
Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: UBS Global Asset Management

⁸³ Please note that the Sharpe ratio and other statistics for the risk premia portfolio differ, slightly, from those in Table 1 because the portfolios in Table 3 have been rescaled to achieve an out-of-sample target volatility of 5%.

Figure 5

We plot the cumulative returns to construct the final portfolio (Combination) and the balanced benchmark. We ascribe 50% of the weight to the risk-parity portfolio, 25% to macro tilts and 25% to momentum. The final portfolio weights are constrained to be positive and the gross leverage not to exceed 1. The individual returns are not adjusted by volatility. The risk of the combination is half that of the benchmark.



Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: UBS Global Asset Management

Conclusions

In this paper, we have outlined a simple strategy that attempts to go beyond risk-based portfolio construction procedures, and applied it to a diversified universe of equity and alternative beta factors. We have sought to avoid hindsight bias and used simple portfolio construction rules. The key observation is that the countercyclical properties of momentum and macro-based strategies at different points of the cycle can be exploited to strengthen a risk-parity framework.

Further research may take different directions. Firstly, we have not used risk premia *per se* for fixed income, but rather for instruments. A direct extension would be to introduce proxies for risk premia in the form of inflation, term and default premia. Secondly, the macroeconomic switching rule could be enhanced, but an effort should be made to keep modelling simple because overfitting is a common pitfall. Finally, the construction of the equity risk premia and portfolio construction can also be improved upon. We describe in companion papers how the equity value and minimum volatility premia can be refined.

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Building Smarter Equity Value Strategies

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Abstract

This paper provides a brief historical overview of value investing and touches on explanations as to why the premium exists. From there we examine issues associated with the current strategies available in the marketplace that allow investors to make a direct bet on the premium. Finally, we take a further dive into the valuation signal construction and propose a more granular, economically relevant methodology that effectively captures the value premium.

Introduction To Value Investing

The golden rule to successful investing is short and simple: »buy low, sell high«. This principle is at the core of value investing which seeks to earn superior returns by preferring cheap assets over expensive ones, according to a predefined measure of cost. The first formal framework to derive the value of a security based on the underlying company's fundamentals was established by Graham

⁸⁴ The authors would like to thank MSCI for graciously providing us with historical constituents and GICS classification of the MSCI World. In addition, the authors gratefully acknowledge the comments and suggestions from Art Gresh, Scott Payseur, Anne-Sophie van Royen and Patrick Zimmermann.

and Dodd⁸⁵ followed shortly by Williams⁸⁶. To this day, their framework constitutes a building block of modern security analysis as it provides the link between equity valuation and financial accounting variables. Since their seminal papers were published more than 80 years ago, practitioners and academics have attempted to harvest this apparent value premium. Vast research on the topic has emerged, offering analyses across markets and asset classes and suggesting competing reasons for the existence and persistence of the value premium.

Both rational and behavioral theories have been proposed in order to explain the value premium. According to the efficient market view, assuming only rational participants in capital markets, the value premium is the compensation for taking on risk. Fama and French (1995) propose this argument based on increased distress risk while later work by Zhang (2005) attributes the premium to a firm's ability to navigate through the economic cycle. From the behavioral perspective, the value premium may be attributable to anchoring or loss aversion biases. Lakonishok et al. (1994) provide a justification based on the sub-optimal behavior of market participants who anchor and extrapolate past performance into the future, leading to the conclusion that betting against these participants will generate superior returns once pricing errors are taken into account. Tversky and Kahneman (1991) offer another behavioral explanation and attribute the premium to loss aversion bias as investors are reluctant to accept losses and will hold on to positions longer than fundamentals warrant.

Current Alternative Beta Value Indices

In the current marketplace, value investors take systematic views towards the premium in one of two broad types of strategies offered by most index providers. The first is the concept of fundamental weighting, proposed by Arnott et al. (2005), which weights the universe by underlying company fundamentals rather than market capitalization. The second set of products incorporates

85 Graham and Dodd (1934) advocate a thorough analysis of financial reports to value securities

86 Williams (1938) proposes to discount future dividends to determine a stock's present value

price-level valuation ratios and security selection into the portfolio construction process for a cleaner, more consistent exposure to the value premium.

The rationale behind fundamental weighting is that over the long term, weighting the universe by fundamental measures, rather than prices, will guard against market bubbles and crashes as deviations between fair price and fundamental value can be long-lived. These types of strategies employ a combination of sales, operating cash flow, uses of cash, and book value to determine the company weight in the portfolio. By construction, fundamental indexing has historically exhibited a tilt towards value versus the capitalization weighted universe. The value tilt is a natural byproduct of the strategy not incorporating the share price into the signal construction process, thereby not overloading on larger names in the capitalization weighted universe.

Fundamental indexing does not explicitly exploit the value premium and the underlying factor exposures of the portfolio will change through time following the market and economic cycle. If the investor's goal is to maintain exposure to the full market while avoiding pitfalls of capitalization weighted indices, then fundamental weighting should be considered an option. However, due to the time varying exposures as well as the market price not being used in the construction process, fundamental indexing should not be used to make a direct bet on the value premium.

The other solutions investors commonly use to exploit the value premium go one level further than fundamental indexing to incorporate specific price-relative value metrics and security selection into the portfolio construction process. The goal of these »purer« value strategies is to maintain a constant exposure to the value premium by aligning the definition of the value score with the definition of the value factor used in the construction of multi-factor risk models. After calculating the value score for each company, the security selection rules will either target a fixed number of securities or use a normalized score cut-off to limit the number of non-value companies in the portfolio. The weights of the portfolio are then determined in part by the market capitalization and a multiplier determined by the value score.

Although these strategies are a cleaner way to gain exposure to value, they still suffer from various methodological issues. First, the inclusion rules of

the portfolio construction process are too broad and lead to portfolio holdings that do not exhibit significant value characteristics. In general, the universe should not be viewed as dichotomous, meaning half the universe should not be viewed as »value« companies. Attention should be given to the tail(s) of the value score distribution to ensure only undervalued (or overvalued if working in the long-short space) companies will be included in the final portfolio to effectively capture the premium. The second issue with the construction process concerns the final company weights within the portfolio. The security weights should be more closely tied to the effective value score rather than using the score as a multiplier of the market capitalization, which leads to a portion of the value bet being negated.

Finally, the greatest concern with these strategies is the way the underlying valuation signals are constructed. The methodology employed should be examined empirically to ensure the signal is capturing undervalued shares within the economic sectors. In general, the standard product shelf of alternative beta value indices applies the same valuation ratios across all industries of the underlying universe, with the exception of financial companies. This naturally leads to an extension where every sector has a separate, more granular definition of value, taking into account that capital needs and profitability cycles differ across the market.

Valuation Signal Construction

The goal of defining our signal is to identify relative value across companies with similar economic exposures. Taking a deeper look at the GICS Consumer Discretionary sector, one would find companies that face distinct challenges during different economic periods. For example, the revenue stream and capital requirements of homebuilders and auto component manufacturers are very different from that of luxury goods retailers or travel companies, yet as we mentioned above, these companies are often scored and normalized within the same basket using the same inputs.

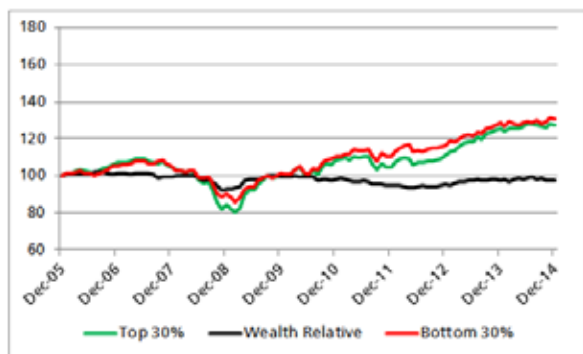
The MSCI value signal construction methodology (MSCI 2014) employed by the MSCI Enhanced Value suite of indices uses an equal weighting of the forward Price to Earnings (P/E), Enterprise Value (EV) to Cash Flow from Operations (CFO), and Price-to-Book value (P/B) to construct the composite value score. While all three components have been shown to exploit different aspects of company value, these signals should not be applied uniformly across the market. For example, the P/B ratio tends to work well when firms have large amounts of fixed assets on their balance sheet, however, the efficacy of this ratio will suffer when companies have large, intangible assets (e.g. brand name, patents, etc.), as do many of the companies included in the luxury goods and retailing industries within the Consumer Discretionary sector.

Using EV/CFO as an input presents drawbacks as well. This ratio is an attempt to approximate how long it would take the company to »buy« itself using only internally generated cash, but relating EV to CFO is inherently flawed. Enterprise value is the total amount required to acquire the firms' debt and equity (minus cash on hand) while CFO is generally regarded as cash available for equity holders (rather than firm) as interest payments are taken out of the total cash flow (rather than being classified as a financing cash flow). The mismatch results in mixing quantities attributable to the firm's various stakeholders in the numerator and denominator of the calculation, leading to worse scores for companies that are more levered than their peers. This runs counter intuitive from including a valuation factor that incorporates total firm value.

Using the scoring methodology described in the MSCI Enhanced Value Indexes Methodology white paper (MSCI 2014), we present a backtest based on equally weighted baskets designed to isolate the information contained in the tails of the value score distribution in order to test for efficacy within the Consumer Discretionary space (stocks classified in the Consumer Discretionary Sector of the MSCI World between December 31, 2005 and December 31, 2014). In Figure 1, we plot the results for the baskets of the top and bottom 30% of the universe value scores as well as the wealth relative line (i.e. the ratio between the levels of the two portfolios) to illustrate the cumulative return differential between the top and bottom baskets.

Figure 1: MSCI Consumer Discretionary Valuation Signal

Monthly time series of returns for the top and bottom 30% of the sorted Consumer Discretionary sector based on MSCI scoring methodology. Portfolios are indexed to 100 as of 31 Dec 2005.



Source: MSCI, UBS Global Asset Management, 31 Dec 2005 - 31 Dec 2014 (Returns in USD)

Upon inspection of the results, relative value is not being captured by the signal for the Consumer Discretionary space. In fact, the bottom (scored as most overvalued) basket of companies outperformed the top (scored as most undervalued) basket in the sample period. However, by tailoring the valuation signals for the baskets of economically similar companies (more granular grouping definitions), we show that these results can be strongly improved upon.

We have attempted to refine what relative value means on an industry-by-industry basis. In order to avoid over fitting and data mining, we have developed industry specific definitions of relative value based strictly on input given by the fundamental investment analysts at UBS Global Asset Management. In their eyes, the core of value investing is free cash flow generation, but should then be tailored to the economic drivers of the respective industries. In the Consumer Discretionary space, we start with the free cash flow yield and P/E as the basis for the value score calculation across all of the industries. From there, the bespoke valuation score is based on underlying industry dynamics.

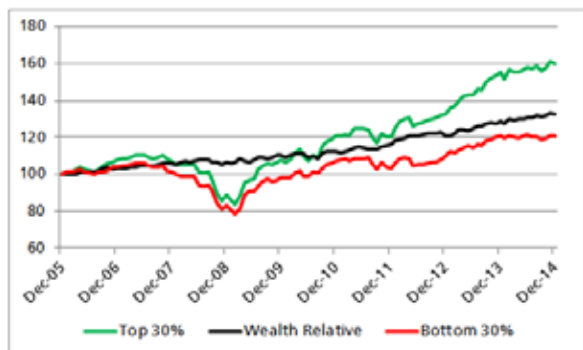
As discussed above, the P/B ratio can be applied to industries without large intangible assets. The P/B ratio is therefore a component of the score for homebuilders as well as auto and auto-component industries, but it is excluded from the others. The economic rationale behind the EV/CFO is valuable in the consumer electronic, household furnishings, household appliance, and educational services industries, but in order to adjust for the drawbacks of the ratio outlined above, we have replaced the CFO component with earnings before interest, depreciation, and amortization (EBITDA) to have a better gauge of value generated by the firm irrespective of financing. In addition to EV/EBITDA in the consumer electronics and home furnishings space, the amount of invested capital (IC) is an important component for cash flow generation so we have included the EV/IC ratio. The final adjustment to the scores is applied to the textile, apparel, and luxury industry group as the price to earnings growth ratio is included to gauge how consumer trends are flowing through to the underlying companies.

After further segmenting the Consumer Discretionary universe to create a more economically relevant way to score the underlying segments, we performed the same equal weighted backtest as presented above, this time using the bespoke signals instead of the original MSCI value definition. Based on Figure 2, the more granular factor definitions are able to partition and capture relative value in the Consumer Discretionary space more effectively than uniformly applying P/B, P/E, and EV/CFO. To compare the signal efficacy, we present a histogram in Figure 3 to show the monthly return spread (top 30% monthly return – bottom 30% monthly return) for both the MSCI and UBS Consumer Discretionary valuation scores. This illustrates the advantage of sorting based on ratios that are better suited to segment the Consumer Discretionary universe, as the resulting distribution is shifted to the right. This indicates that a greater return differential between the top and bottom baskets is captured using the new signal construction methodology. A 99% confidence t-test for mean equality confirms the average UBS spread capture as being greater than that of the MSCI methodology.

3. Loading Factor Exposures To Portfolios

Figure 2: UBS Consumer Discretionary Valuation Signal

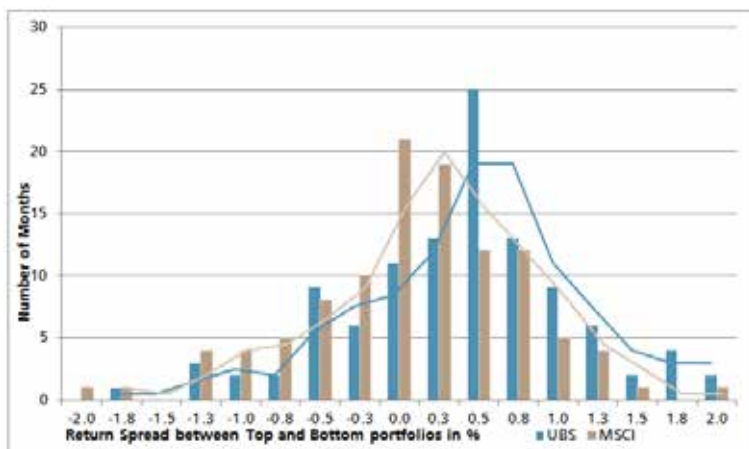
Monthly time series of returns for the top and bottom 30% of the sorted Consumer Discretionary sector based on UBS scoring methodology. Portfolios are indexed to 100 as of 31 Dec 2005.



Source: MSCI, UBS Global Asset Management, 31 Dec 2005 - 31 Dec 2014 (Returns in USD)

Figure 3: Return Spreads for Consumer Discretionary Signals

Distribution of monthly return spreads between the top and bottom 30% of the sorted sector.

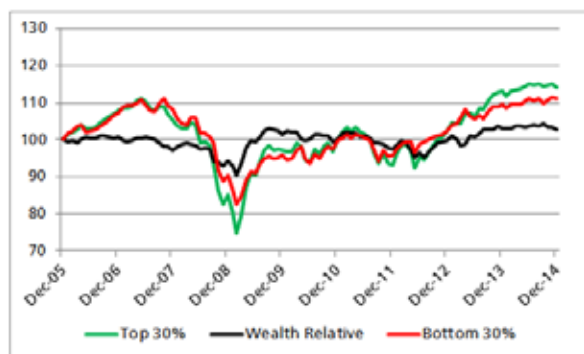


Source: MSCI, UBS Global Asset Management, 31 Dec 2005 - 31 Dec 2014 (Returns in USD)

To further illustrate the benefits of creating tailored definitions for capturing relative value, we also examine the Financials sector. Currently, MSCI uses a different signal construction methodology for this sector compared to the remaining nine: only two of the three ratios mentioned earlier are used to create the valuation score for Financials. EV/CFO is logically excluded as EV is distorted by the bank loan books, thus, leaving only P/E and P/B ratios to distinguish relative value. Using the same equal weighted basket methodology as above, the results for the top and bottom 30% of the sorted universe are presented in Figure 4.

Figure 4: MSCI Financials Valuation Signal

Monthly time series of returns for the top and bottom 30% of the sorted Financials sector based on MSCI scoring methodology. Portfolios are indexed to 100 as of 31 Dec 2005.



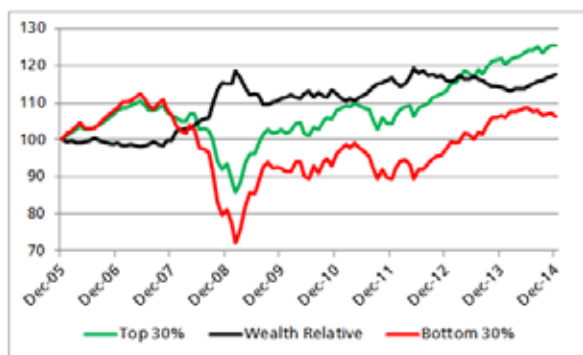
Source: MSCI, UBS Global Asset Management, 31 Dec 2005 - 31 Dec 2014 (Returns in USD)

As can be seen from this illustration, sorting the universe of Financials by P/B and P/E alone does not capture relative value within the sector. This is not a surprise, based on the various financing and cash flow needs, sensitivity to interest rates, and capital return expectations within this sector. For example, yield and payout information are relevant to the insurance and real estate segments of the financial sector, as these industries are highly regulated.

Conversely, income generation for equity holders is important to capital markets, brokerage, and investment banking segments of the sector. After building the economically relevant elements into the score calculation, the signal more effectively segregates the sector to generate dispersion between the top and bottom 30% as seen in Figure 5.

Figure 5: UBS Financials Valuation Signal

Monthly time series of returns for the top and bottom 30% of the sorted Financials sector based on UBS scoring methodology. Portfolios are indexed to 100 as of 31 Dec 2005.

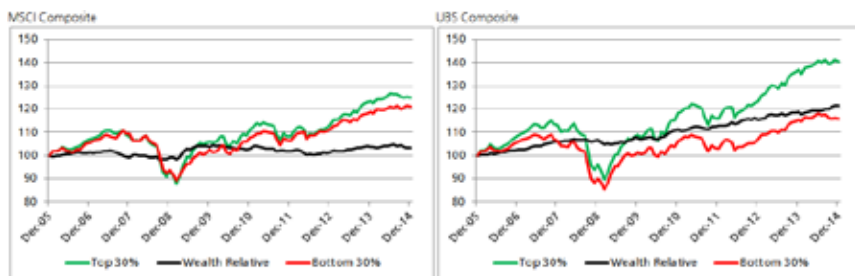


Source: MSCI, UBS Global Asset Management, 31 Dec 2005 - 31 Dec 2014 (Returns in USD)

Creating valuation scores based on the economically relevant data across the remaining sectors leads to improvements in most of the remaining sectors, similar to those we have presented in the Financials and Consumer Discretionary space (see Table 1 and Figure 8 in the Appendix for the full sector summary). To aggregate and summarize the results across sectors, we created composite portfolios based on the top and bottom 30% baskets of each sector. We rebalance the composites on the first business day of each month using the preceding month-end capitalization weight of the corresponding sector in the MSCI World.

Figure 6: Monthly Return Spreads for the Composite Signal

Monthly time series of composite portfolios created with the top and bottom 30% of the individual sectors using the beginning monthly weight of MSCI World for both MSCI and UBS scoring methodologies. Portfolios are indexed to 100 as of 31 Dec 2005.

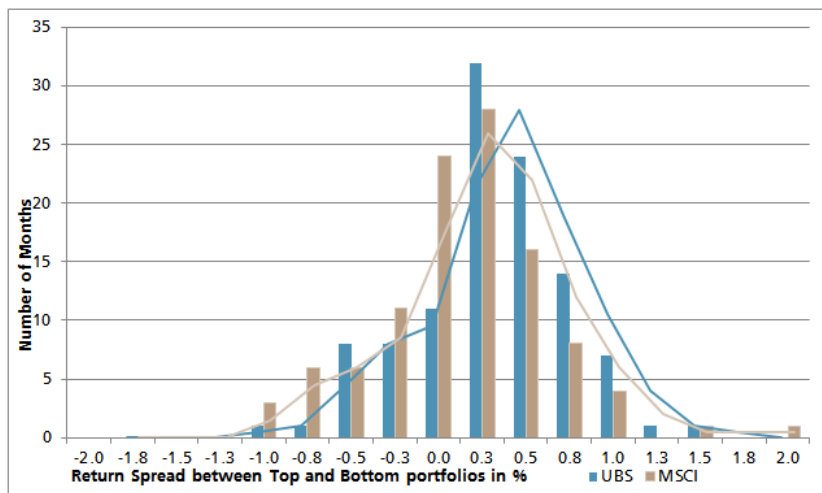


Source: MSCI, UBS Global Asset Management, 31 Dec 2005 - 31 Dec 2014 (Returns in USD)

Based on Figure 6, the MSCI signal construction methodology does not sort the universe efficiently into over and undervalued baskets, while the UBS signal construction methodology shows a clear distinction between the top and bottom baskets, thus better capturing the value premium. As shown in Figure 7, the UBS composite spread distribution is shifted to the right of the MSCI distribution indicating a higher return differential between the top and bottom baskets. A 99% confidence t-test for mean equality confirmed that the average UBS spread capture across the universe was greater than that of the MSCI methodology.

Figure 7: Return Spreads for Composite Signals across the Universe

Distribution of monthly return spreads between composites of the top and bottom 30% baskets of each sorted sector.



Source: MSCI, UBS Global Asset Management, 31 Dec 2005 – 31 Dec 2014 (Returns in USD).

Further Research And Conclusion

The results presented in this paper are based on the first attempt to quantitatively capture relative value on an industry-by-industry basis after discussions with the fundamental equity analysts from UBS Global Asset Management. There are multiple avenues of further research to continue from this point: at the signal construction level, an additional examination of the Energy and Utilities sectors will be undertaken to determine if the investment thesis is incorrect or if we need to develop a more innovative way to capture and score industry drivers. Beyond the signal construction research, various portfolio construction methodologies will be tested to determine the most effective way to capture and translate the valuation signal into the final portfolio.

This study addresses multiple aspects of investing in off-the-shelf products to exploit the value premium ranging from valuation signal construction to the inherent issues embedded in the portfolio construction process. It is imperative to realize that, for strategies attempting to take advantage of the value premium, the input signals need to be defined in a manner that will segment the market effectively as no portfolio construction technique will compensate for inefficient underlying signals. The value premium still exists in the global market, but is not being captured by most off-the-shelf solutions. As the market evolves, smarter signal construction is necessary in order to continue capturing the value premium.

Appendix

Industry Summary

Table 1: Return Spreads Summary

Annualized means of monthly return time series for the baskets created from the top and bottom 30% of the individual sector signals.

	MSCI Valuation Signals			UBS Valuation Signals			Difference in Average Spreads (UBS-MSCI) ¹
	Top 30%	Bottom 30%	Annualized Difference	Top 30%	Bottom 30%	Annualized Difference	
Consumer Discretionary	2.7%	3.0%	00%	5.4%	2.1%	03%	04%***
Consumer Staples	2.9%	3.3%	00%	9.4%	1.6%	08%	08%***
Energy	2.0%	1.1%	01%	0.5%	1.4%	-01%	-02%
Financials	1.5%	1.2%	00%	2.5%	0.7%	02%	02%***
Industrials	3.0%	2.4%	01%	4.8%	2.1%	03%	02%**
Health Care	4.4%	3.9%	00%	4.3%	3.9%	00%	00%***
Information Technology	2.0%	1.3%	01%	3.0%	1.0%	02%	01%***
Materials	2.3%	2.4%	00%	2.7%	0.6%	02%	02%
Telecommunications	3.5%	2.7%	01%	6.2%	2.2%	04%	03%***
Utilities	3.2%	1.7%	01%	2.3%	2.2%	00%	-01%
Composite Portfolio ²	2.5%	2.1%	00%	3.9%	1.6%	02%	02%***

1. Significance level based on P-value from t-Test with hypothesis of UBS mean spread > MSCI mean spread without assumption of equal variance

* - Significant at 10% level, ** - Significant at 5% level, *** - Significant at 1% level .

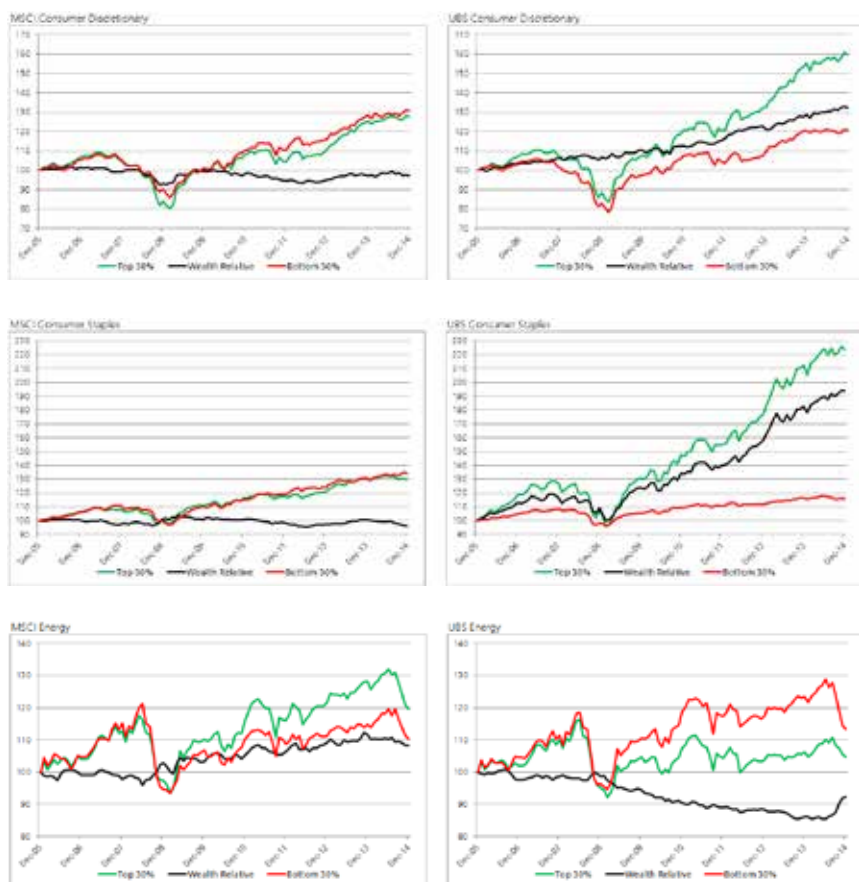
Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: MSCI, UBS Global Asset Management, 31 Dec 2005 – 31 Dec 2014 (Returns in USD).

3. Loading Factor Exposures To Portfolios

Figure 8: Sector Results

Monthly time series of returns for the top and bottom 30% of the sorted sectors based on MSCI and UBS scoring methodologies. Portfolios are indexed to 100 as of 31 Dec 2005.



Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: MSCI, UBS Global Asset Management, 31 Dec 2005 – 31 Dec 2014 (Returns in USD).

Figure 8: Sector Results (continued)

Monthly time series of returns for the top and bottom 30% of the sorted sectors based on MSCI and UBS scoring methodologies. Portfolios are indexed to 100 as of 31 Dec 2005.



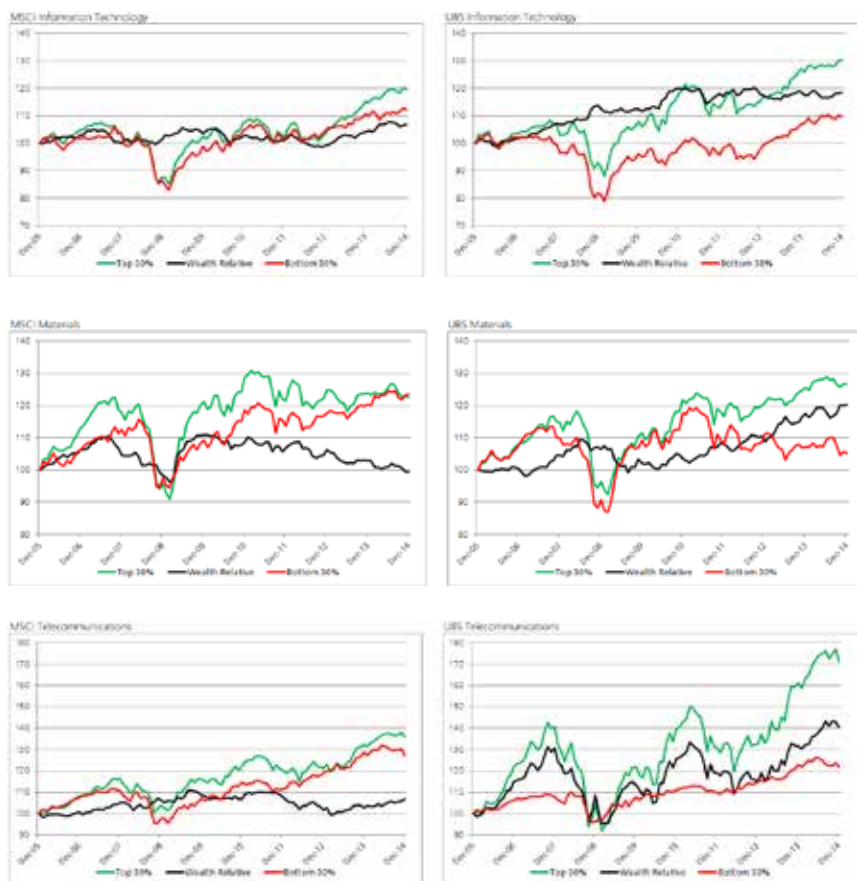
Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: MSCI, UBS Global Asset Management, 31 Dec 2005 – 31 Dec 2014 (Returns in USD).

3. Loading Factor Exposures To Portfolios

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Monthly time series of returns for the top and bottom 30% of the sorted sectors based on MSCI and UBS scoring methodologies. Portfolios are indexed to 100 as of 31 Dec 2005.

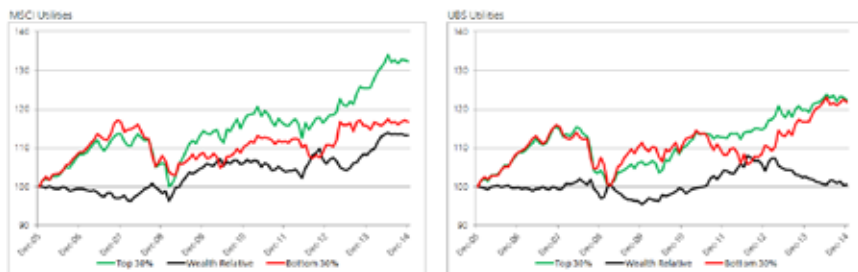


Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: MSCI, UBS Global Asset Management, 31 Dec 2005 – 31 Dec 2014 (Returns in USD).

Figure 8: Sector Results (continued)

Monthly time series of returns for the top and bottom 30% of the sorted sectors based on MSCI and UBS scoring methodologies. Portfolios are indexed to 100 as of 31 Dec 2005.



Past performance, whether simulated or actual, is not a reliable indicator of future results.

Source: MSCI, UBS Global Asset Management, 31 Dec 2005 – 31 Dec 2014 (Returns in USD).

Backtest And Basket Creation Methodology:

The raw data inputs for the valuation signals are sourced from the Thomson Reuters QA Direct database with underlying data stemming from Worldscope, IBES, and IDC.

On a monthly basis, the valuation scores were created and normalized within each economic segment. The normalized scores were then ranked by deciles within each of the ten MSCI GICS Sectors. The stocks that fell within the top and bottom three deciles were equally weighted to create the top and bottom 30% baskets presented in the analysis. In order to gauge the signal efficacy, turnover has been excluded from the analysis.

The equally weighted, basket monthly returns are calculated in USD. The results in Table 1 are in USD and annualized, based on geometric returns.

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Minimum Expected Tail Loss Equity Portfolios

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Abstract

Minimizing Expected Tail Loss (ETL) is an encouraging alternative to minimizing volatility in portfolio construction. The first section of this paper expands on the drawbacks of using minimum volatility for portfolio construction. In the second section, ETL is introduced as a more suitable alternative for portfolio construction. Finally, we present empirical results comparing portfolios created with minimum volatility and ETL as well as discuss the more robust features of the latter.

Introduction

The wealth eroding losses experienced in the financial crisis of 2008 elicited a strong interest in low risk equity strategies including minimum variance, inverse variance weighted, equal-weighted, maximum diversification, and equal risk contribution. The most popular, by far, is minimum volatility portfolio construction. The MSCI suite of Minimum Volatility indices has increased to over 44 billion USD in assets under management (MSCI 2014); in addition, Nomura Research estimates that the percentage of total equity assets invested in minimum volatility strategies has increased 430% over the past five years.

⁸⁷ The author gratefully acknowledges the comments and suggestions from Art Gresh, Jay Eisenhardt, Andreas Razen, Anne-Sophie Vanroyen, Patrick Zimmermann and Teresa Yeo. Thanks to MSCI for providing historical constituents and GICS classification of the MSCI USA.

Minimum volatility strategies reduce the uncertainty of expected returns, decreasing expected loss but also limiting gains. Minimum volatility is also consistent with the behavioural notion of loss aversion bias (Kahneman, 1984), where investors strongly prefer loss avoidance to acquiring gains. The design of a low risk strategy should focus on limiting losses during turbulent markets but must also have the ability to rebound after a drawdown. Minimizing an alternative statistic, ETL, allows for loss reduction without explicitly limiting upside potential and leads to better risk-adjusted returns.

Minimum Volatility

Minimum volatility dates back to 1952 when Harry Markowitz (1952) demonstrated how to construct low risk portfolios using only two statistics: individual asset volatilities and correlations. This is a relatively straightforward task for the practitioner; however, these two statistics do not contain enough information to properly model the required dimensions of risk. Their estimation process does not properly capture asset relationships during crashes, as the average correlation between equities increases sharply during market turmoil and the symmetry imposed on the distribution is undesirable, as there is no distinction between down-side and up-side volatility, leading to a portfolio reducing both negative and positive returns. Additional metrics are needed to capture the fat-tails of equities returns. Given that large negative returns cannot be modeled using only the volatility and correlation statistics, kurtosis, must be included to make use of the information contained in these fat-tails.

The minimum volatility portfolio is set up to have the lowest total portfolio volatility and does not simply hold assets with the lowest individual volatility. As an example, throughout 2014, the MSCI Minimum Volatility Index held the gold mining stock Newmont Mining which was one of the most volatile stocks in the possible universe. The reason this stock was held in the portfolio is that it had low correlation with the other assets in the portfolio providing diversification and a lowering the effect on the expected portfolio risk.

This diversification effect is dependent on robust correlation estimates which are typically estimated using historical data. The problem is that most of the

estimation window corresponds to a relatively calm state of the market. This leads to issues during market turmoil when correlations are known to increase, drastically, just when diversification is urgently needed. Under this minimum volatility methodology, the correlation estimates do not accurately reflect the relationships between assets in this high risk environment and are ill-equipped to provide diversification during extreme conditions.

Volatility is a statistic that describes how observations are dispersed around the average value. Its symmetrical nature means that values below and above the average both contribute equally. Minimizing the volatility of a portfolio limits negative portfolio returns as desired; however, minimizing volatility will also decrease positive returns. A low risk strategy should lower the magnitude of losses but also attempt to make the portfolio recover quickly. For example, after a 10% portfolio drawdown, an 11% gain is required to climb out of the trough, as a result of the lower asset base. The portfolio constructed to minimize volatility, limits both downside and upside potential of the portfolio.

Minimizing volatility assumes asset returns are distributed log normally; when graphed on a histogram they look like a bell curve. Both ends of the bell curve are thin which means that extreme gains or losses do not occur often; in practice, we know this assumption to be untrue. For instance, Xiong (2010) points out that the normal distribution with a monthly standard deviation for the S&P 500, calculated from January 1926 to April 2009, predicts a negative return of 15.5% to happen just over once in 83 years. In reality, the S&P 500 suffered a monthly loss of at least that much 10 times. A Gaussian distribution fails to properly model the tails of stock return data.

In recent years, a number of asset management firms have rationalized the use of minimum volatility strategies with the findings from Ang (2006) that imply low volatility stocks outperform higher volatility stocks. This result is contrary to the basis of modern portfolio theory where investors are compensated for higher risk by higher returns. This »low volatility anomaly« is not a complete rationale to use as an investment thesis, for two reasons: First, the anomaly applies to the volatility of individual assets and completely ignores the diversification benefits of including correlations in the optimization. Second, these findings have been shown to lack robustness if sampled at a different

data frequency (Bali, 2008), liquidity of assets is considered (Bali, 2008) or short-term reversals adjustments are made (Huang, 2010).

Expected Tail Loss

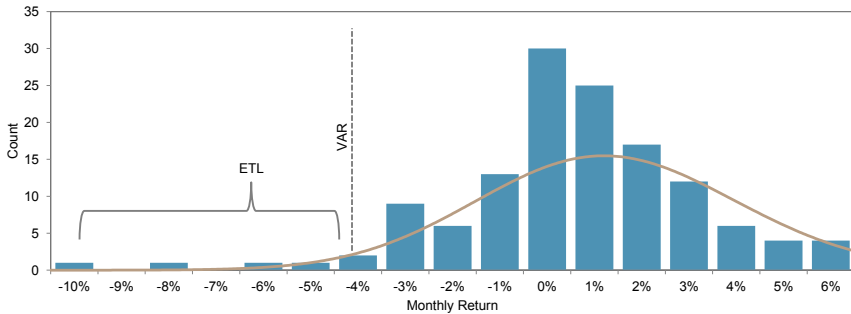
We set out to find a risk measure that can be applied in portfolio construction without facing the issues of minimizing volatility described above. The ideal measure would neither require symmetry, nor assume normality, and should be robust during market downswings in an attempt to prevent drawdown. Expected Tail Loss, which is an extension of the commonly used Value at Risk (VaR) statistic, fits these requirements. VaR (Jorion 1996) is a threshold statistic defined as the minimum amount of portfolio loss at a specified probability and horizon. For example, a particular portfolio might have a 5% VaR value of one million dollars. This means that, 5% of the time, the portfolio will lose one million dollars or more in a specified time horizon.

VaR is used, extensively, in the world of risk management and is included in the stress testing requirements of the Basel accords and the US Volker rule. Its mainstream use is troublesome, due to the fact that investors not only care that they will lose at least one million dollars, they also need to know how much more than one million they expect to lose. However, VaR constitutes the basis for the Expected Tail Loss (ETL) statistic.

ETL is a threshold statistic defined as the expected amount of portfolio loss at a specified probability level. Instead of using VaR, and knowing that the portfolio will lose at least one million dollars 5% of the time, using the ETL measure will tell us that we expect to lose, on average, 1.2 million dollars 5% of the time. Figure 1 shows this, graphically, using a histogram of portfolio returns. The VaR value, represented by the dotted line, is the return value when 5% of the distribution is to the left. The ETL value is the expectation, or average, of the distribution to the left of the VaR. One note about semantics: in the example, we used the cash value of the loss, but this can also be expressed in return space, which we did for Figure 1, and continue to do for the rest of the paper.

Figure 1: Calculating VaR and ETL

Value at Risk represents the minimum amount of loss that will happen $x\%$ of the time. Expected Tail Loss represents the expected amount of loss that will happen $x\%$ of the time.



Source: UBS Global Asset Management.

ETL is calculated by averaging the losses that are beyond a certain threshold of a portfolio return distribution. There are many ways to create the distribution, but the simplest is to use the empirical portfolio returns. The minimum ETL portfolio optimization finds the combination of portfolio weights that result in the lowest ETL value (by summing the weighted individual asset return distributions). Minimum ETL portfolio construction uses every past return to capture co-movement of assets and does not suffer from the issues associated with a parametrized estimate of correlation. Upside reduction, due to symmetry, also does not exist for ETL because the statistic focusses purely on the loss side of the distribution. Additionally, assumptions of normality are not needed with ETL; the optimization process uses every observation of the asset return data within the specified historical window to model the fatter tails in the distribution. Most importantly, the concept of ETL is in line with an investor's true objective: reducing portfolio loss and not just portfolio uncertainty. In simulations, it yields superior risk adjusted returns and lower drawdowns than its minimum volatility counterpart.

ETL Versus Minimum Volatility Simulated Backtest

There is a lot of research demonstrating the advantages of using ETL in the asset allocation space. These studies show encouraging results for a small number of assets, but due to computational complexity they failed to address portfolios with hundreds of potential assets. Rockafellar (2000) created a linear approximation for ETL optimization which decreased complexity and made large dimension ETL portfolio construction fast enough to become feasible for practitioners. Murakami (2010) used this approximation on the Japanese TOPIX universe to show an ETL portfolio outperforms minimum volatility in the Japanese equity space. In particular, the study points out that there is a non-linear relationship in the portfolio backtest returns, with downside being limited more than upside is being penalized. Tokpavi (2012), however, finds that minimum volatility outperforms ETL in their unconstrained US equity market simulations; most of the underperformance coming from transaction costs associated with the excess turnover of the unconstrained backtest.

Results from a 14-year minimum ETL backtest, using the MSCI USA universe of stocks, are presented. The backtest uses MSCI USA Minimum Volatility as the benchmark (MSCI 2012). For comparison purposes, the portfolio is constructed with similar constraints to the MSCI Minimum Volatility Index. ETL is minimized at a 10% threshold, using five years of split-adjusted, daily stock returns; when a stock does not have five years of returns, it is left out of the analysis⁸⁸. The sector constraints were set to +/- 5% relative to the MSCI USA index. The maximum weight for each individual asset was set to the minimum of 1.5% or 20 times the weight in the MSCI USA Index. Exposures are constrained, for all but one of the common USE3L BARRA risk factors, to +/- 0.4 standard deviations from the MSCI USA index; the volatility factor is

88 For MSCI USA this is a fairly small portion of the portfolio: less than 5%.

left unconstrained. The constrained factors are: size, momentum, non-linear size, earnings yield, value and yield (MSCI 2010). In order to control transaction costs, the ETL turnover is limited to 8% (buys + sells / 2) and includes average daily volume constraints on traded and held assets⁸⁹.

The minimum ETL simulation results in Figure 2 include transaction costs⁹⁰. The ETL portfolio is preferable to minimum volatility and the capitalization-weighted benchmark, not only in annualized returns but also across multiple aspects that investors care about: it has lower maximum drawdown, lower ETL, and even lower realized volatility leading to a portfolio with greater annualized risk-adjusted returns. It closely matches the performance of MSCI Minimum Volatility during the down months and outperforms during the up months, leading to greater wealth accumulation.

Conclusion

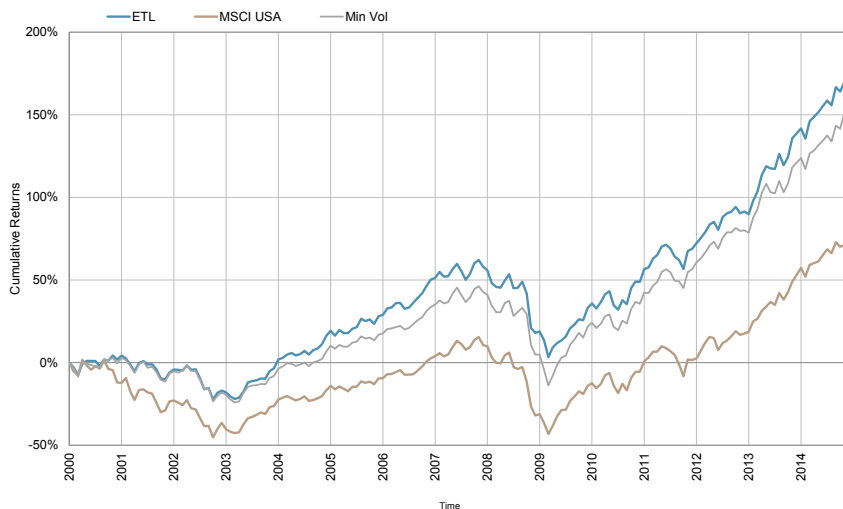
When adding a low risk strategy to an investor's portfolio, it is important that the strategy captures the investors actual risk objective: wealth preservation. Minimizing volatility lowers uncertainty about portfolio outcomes, but for many reasons is not designed to reduce long-term losses. ETL addresses the shortcomings of minimum volatility by not assuming symmetry and focusing on the loss side of the portfolio return distribution. After creating a backtest simulation, with similar constraints to the MSCI USA Minimum Volatility index, the resulting portfolio characteristics are encouraging and demonstrate better risk and return characteristics.

89 These were set to +/- 25% ADV Traded and one day held ADV, for a starting portfolio of 500M USD.

90 Transaction costs as estimated by the ITG ACE model (ITG 2007).

Figure 2: ETL vs MSCI Minimum Volatility

14-year backtest for the Minimum ETL strategy, compared to the MSCI USA Index and MSCI Minimum Volatility Index.



	Minimum ETL	MSCI USA	Minimum Volatility
Annualized Returns	7.1%	3.9%	6.6%
Annualized Volatility	11.1%	15.4%	11.8%
Risk Adjusted Return	0.64	0.26	0.56
Maximum Drawdown	-36.3%	-50.8%	-41.0%
Value at Risk 95%	-5.0%	-7.9%	-5.1%
Expected Tail Loss 95%	-7.4%	-9.9%	-8.0%

Source: UBS Global Asset Management.

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Equity Factors In Action: The Need For Multi-Regional- Currency-Hedged Exposures

Dr. Pawel Janus

UBS Global Asset Management

Introduction

The three-factor model of Fama and French (1993) has proved that inclusion of factors like size and value – in addition to the market beta – can significantly help to explain the cross-section of long-run average stock returns. Carhart (1997) defined a four-factor model which extends the Fama–French three-factor model including a momentum factor and finds additional strong empirical support in the return data. Academic literature and empirical studies have up until today documented more than 300 factors, see Ang (2014), and references therein. Recent years have seen an increasing interest in new forms of *indexation*, referred to as Factor Beta strategies (called also Smart Beta or Alternative Beta strategies). In essence, factor beta indexed strategies provide tilt towards preferred factor (e.g. value, size etc.), with selected securities weighted differently than by their market capitalization. The factor indices can provide tilts towards *size, value, quality, low volatility, yield or momentum*, or possibly some blend of those. In addition, factor beta indices explicitly account for underlying liquidity to be investable, including cost-efficient and transparent solutions like Exchange Traded Funds (ETFs). Factor beta is not a replacement of market beta, but this exposure is used to enhance passive return. With the development of investable factor indices, portfolio return can be represented by market return (standard market cap exposure), factor return (typically a mix of factor exposures) and alpha (value added from active management due to stock picking, timings skill, etc.). In this representation,

the first two return components can be accessed with the use of cost-efficient index replicating portfolios, including market cap ETFs and factor ETFs.

The introduction of factor ETFs necessitates considering a number of important aspects. Firstly, given that most of the existing factor indices are revised annually or semi-annually based on *realized* data, the time-to-market response definitely has a time delay, meaning that current market and corporate *goings-on* would be reflected in the factor indices only with the next rebalancing review (unless some predictive criteria are applied, which is rare), implying higher turnovers. Secondly, the level of aggregation for factor exposure impacts the objectives of asset managers, bearing in mind that higher aggregation (i.e. world factors as opposed to regional ones) implies that region-specific opportunities get *crowded out*. In particular, the world-aggregated factors may be biased towards the most representative sub-universe, e.g. US equities in the case of world equity markets. Thirdly, the investable factor indices are »wrapped« into investable products (e.g. ETFs), with assets valued in a particular base currency, e.g. USD. This creates currency-related risks. Given that global factor indices can be represented by regional building blocks (bottom-up with proper weighting), such building blocks can be valued in local currency or with a currency hedge overlay, as opposed to having broad exposure to world-aggregated factors with multi-currency risk. The remainder of this chapter discusses these empirical considerations, by illustrating some examples. Importantly, this chapter does not provide any recipes on how to best allocate amongst factors, it rather discusses the motivation for providing a more granular factor framework behind UBS factor ETFs, which is more flexible for investors aiming to *enhance passivity*⁹¹.

The Importance Of Time-To-Market Response

Factor indexed portfolios are constructed by means of a *sorting* approach to filter out and rank stocks – in view of pre-defined criteria – from the broad universe which best captures the desired factor exposure. Once the stocks are

⁹¹ The data used in this chapter includes live and back-tested MSCI factor time series. Past performance is not indicative of future results.

sorted, accordingly, the second step involves a choice of a weighting scheme, subject to additional constraints, if any, dealing with a number of targeted securities or a country cap, etc. The data applied to constructing factor exposures can be classified into three groups:

- Accounting data (price-to-book value, earnings-per-share, dividends, return on equity, profitability measures, etc.)
- Market data (liquidity measures, market capitalization, risk domicile, etc.)
- Statistical data (price momentum, covariance structure, volatilities, Sharpe Ratios, etc.)

The accounting data of public companies arrives when quarterly financial results are being published, whilst market data is readily available in *continuous* time (when markets are open). The frequency of computing statistical data depends on the underlying data, and can be done on an intra-daily or quarterly basis. The difference in arrival rates of the data (intra-daily to quarterly), implies that a consistent indexed factor construction process is determined by the lowest frequency. In practice, the selection of securities for factor indices is carried out on an annual or semi-annual basis, to keep index turnover within a reasonable range. For example, the MSCI factor indices are rebalanced, typically, on a semi-annual basis, usually as of the close of the last business day of May and November, coinciding with the May and November Semi-Annual Index Review of the MSCI Global Investable Market Indices, with the *pro forma* indices being announced a few business days before the effective date. The low-frequency framework behind the *indexation* implies that current market and corporate *goings-on* will be reflected in the index only with the next rebalance, i.e. with a time delay. As a consequence, this results in the increased index turnover as compared to market cap indices, as shown in Table 1 (note the entries in Table 1 for factor indices show the turnover multiplier; e.g. *Turnover* for MSCI Quality = $7 \times 2.34\% = 16.4\%$).

Table 1: Index Turnover (MSCI Standard) and Turnover Multiplier (MSCI Factors)

Factor		Quality	Yield	Momen- tum	Size	Value	Low Volatility
Region	MCSI Stan- dard	MCSI Quality	MCSI High Divided Yield	MCSI Momen- tum	MCSI Equal- Weighted	MCSI Value- Weighted	MCSI Risk- Weighted
MCSI World	2.34%	7	4	4	8	5	8
MCSI USA	2.58%	8	5	43	7	4	7
MCSI Europe	2.25%	6	10	45	8	5	7
MCSI Emer- ging Markets	5.95%	4	3	13	5	3	5

Source: MSCI. UBS Global Asset Management. For illustration purposes only.

The indexed factor strategies can become inefficient from a turnover viewpoint, implying higher trading costs, potentially higher tracking risk, and operational risk, etc. For example, momentum strategy features relatively high turnover figures, in particular, at regional level, e.g. 11% for MSCI USA ($43 \times 2.58\%$). MSCI indices can impose turnover constraints to cost-balance the buy-and-sell orders, but this can cause *path-dependency*, i.e. the factor indices as of this rebalance would also have constituents from the previous rebalance, and this could impact the sorting result. This is an undesired feature which actually dilutes the desired *pureness* of factor exposure.

The UBS factor ETFs intend to be cost-efficient, flexible, low tracking-error additions to the traditional market cap toolbox. The implication is fourth-fold. Firstly, to provide cost-efficient passive solutions, the turnover figures eliminate some factor exposures to mitigate tracking error risk and inflated costs. Secondly, investors need cost-efficient factors at hand, which are driven by different sources of return, to mitigate the concentration risk towards one

source of premium. Thirdly, investors need regional factor exposure (i.e. if a more opportunistic approach is desired) to mitigate the response risk. In addition, globally aggregated factor exposures can *systematically* concentrate in some regions, due to disproportional representation in the parent index. Finally, if regional exposure is considered, investors ideally have the possibility to invest in local currency, or with a currency hedge overlay, to mitigate the currency risk.

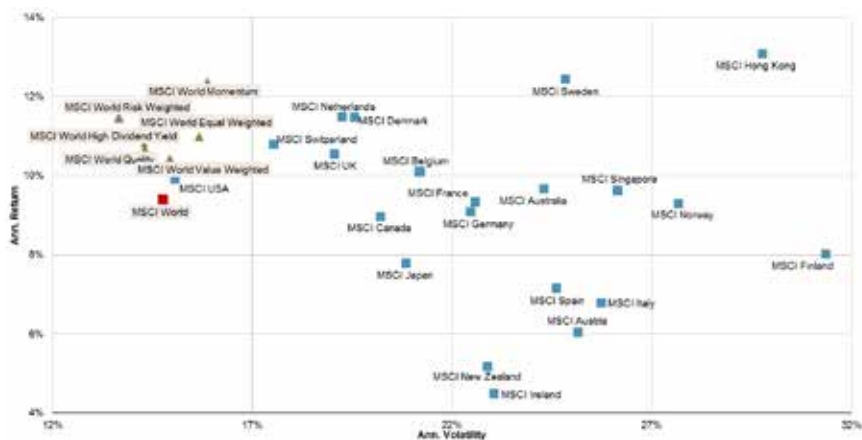
The Need For Multiple Factors

Numerous academic studies provide an ever growing number of possibly relevant factors, see Ang (2014). Many of the newly *discovered* factors are justified by their ability to explain variation in the cross-section of average returns, see Fama and French (1993). In addition, most of the factors have very strong economic and/or behavioural rationale; for example, size factor exposure should result in higher rewards for carrying liquidity risk and non-systematic risk (smaller companies are more vulnerable to idiosyncratic events). Factor beta exposures – as opposed to market cap exposure – increase the risk of concentration towards certain portfolio characteristics (e.g. value, size etc.) and, therefore, ought to result in rewards (premium) for carrying the additional risk of factor concentration. Of course, it is important to understand the risks to which investors are exposed when they adopt alternative benchmarks. From a practitioner viewpoint, it is insightful to learn why there is a need for multiple factors, going beyond the standard arguments of their ability to explain return variation. Simply put, one of the reasons is that risk premia have different levels of persistence, different magnitudes of relative performance to market cap and they all mean-revert (to zero excess return over market cap) with different levels of speed in mean-reversion.

Example 1: This case study is based on the MSCI data for the developed (according to the current classification) countries (Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hong Kong, Ireland, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, United Kingdom and United States). The data is for net total

returns in USD going back to 28 November 1975 until 27 February 2015 (approx. 40 years of data). This study uses six factors: *size* (MSCI World Equal-Weighted), *value* (MSCI World Value-Weighted), *momentum* (MSCI World Momentum), *quality* (MSCI World Quality), *yield* (MSCI World High Dividend Yield), and *low volatility* (MSCI World Risk-Weighted). Figure 1 shows the risk-return space of country exposures, world market cap exposure, and world factor exposures.

Figure 1: Risk-return space of the MSCI developed markets (TR Net in USD, Nov. 1975 – Feb. 2015)



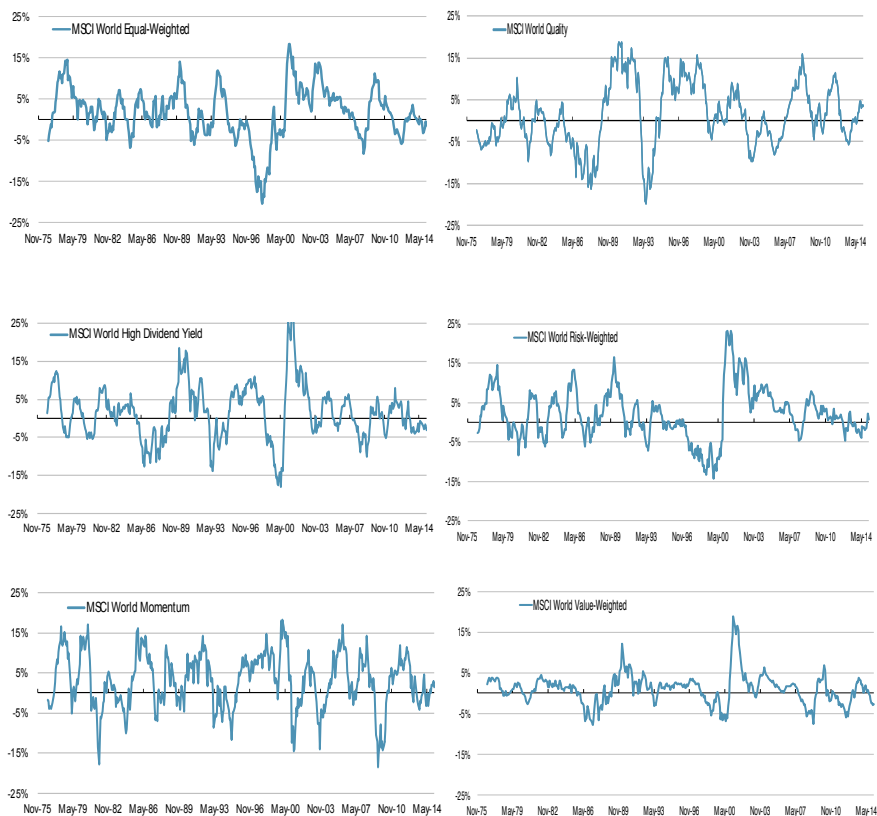
Source: MSCI. UBS Global Asset Management. For illustration purposes only.

The world factors provided better risk-adjusted performance than individual countries and the world market cap.

This result is very appealing and suggests that factor exposures can generate incremental return when added to the portfolio as well as being able to reduce the total risk. It is, however, insightful to see how the premium (excess return over market cap) evolves over time. Figure 2 shows the time-varying premia, each defined as a twelve-month rolling average excess return over the market cap MSCI World.

3. Loading Factor Exposures To Portfolios

Figure 2: Time-varying premia of MSCI World factor exposures (TR Net in USD, Nov. 1975 – Feb. 2015)



Source: MSCI. UBS Global Asset Management. For illustration purposes only.

The sample correlations between premia are shown in Table 2 and they highlight relatively low correlations (on average 0.23) or even negative correlation (e.g. size and momentum factors). The high pair-wise correlation of 0.75 is observed for yield and value factors, which is not surprising, here, as the MSCI Value-Weighted index has a dividend yield as one criterion for defining the value stocks on top of other three fundamental ratios. Similarly, the high correlation of size and low volatility factor is also not surprising, here, as the MSCI

Risk-Weighted framework compresses constituent weights by weighting, due to inverse variance with capped variance estimates which underweight large-caps and overweight mid-caps compared to the market cap index.

Table 2: Sample correlations between factor premia (Nov. 1975 – Feb. 2015)

	MCSI World Equal- Weighted	MCSI World High Dividend	MCSI World Momen- tum	MCSI World Quality	MCSI World Risk- Weighted	MCSI World Value- Weighted
MCSI World Equal- Weighted		0.27	- 0.14	- 0.19	0.79	0.56
MCSI World High Dividend			- 0.05	0.39	0.51	0.75
MCSI World Momentum				0.22	0.01	- 0.33
MCSI World Quality					0.01	0.13
MCSI World Risk- Weighted						0.51
MCSI World Value- Weighted						

Source: MSCI. UBS Global Asset Management.

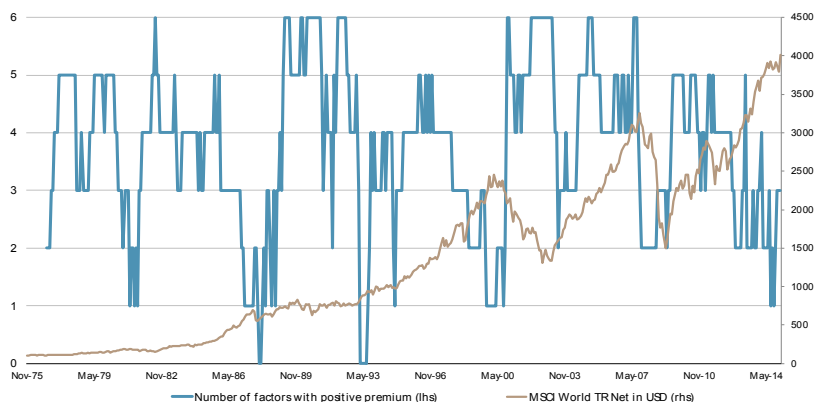
A number of highlights are summarized below:

- The factor beta had positive excess return in the long-run of about 2–3% in annualized terms.
- The premia vary over time with high excess return (reaching even 15–20% excess return in some periods) but also with active drawdowns, i.e. periods of underperformance to the market cap.
- The factor premia are rather weakly correlated (expect some cases with overlapping criteria in the construction of factor indices).

- The factor premia mean-revert to zero excess return over the market.
- The factor premia have different levels of persistence; for example, the momentum factor mean-reverts relatively fast, while the quality factor appears to have longer cycles.
- The magnitude of the premium also differs across factors; for example, the value factor seems to move relatively close to the market (say, -5% to +5% in most of the times), whereas yield factor had substantially larger relative premia (incl. excess return from -15% to 25%).

This set of features justifies and encourages the need for introducing multiple factors, i.e. the possibility to harvest premia along market cycles utilizing different sources of factor return. For example, if investors only had a momentum factor at hand, he or she would need to accept higher risk to enhance return, as seen in Figure 1. However, if combined with a low volatility factor, it is clear from Figure 1 that investors could reduce the risk associated with not yet enhanced return compared to the market. Given that premia vary over time, it is interesting to see how many factors *jointly* have excess return. Figure 3 shows the number of factors which have excess return (a twelve-month rolling average, as in Figure 2) over the market (note that excess return in a downside market is equivalent to a smaller loss than in a broad market).

Figure 3: Number of world factor exposures having positive premium (TR Net in USD, Nov. 1975 – Feb. 2015)



Source: MSCI. UBS Global Asset Management. For illustration purposes only.

The number of factors that deliver positive premium is pretty dynamic, including periods when all factors were over or under the market. Remarkably, there were only two time periods (Dec. 1987 – Jan. 1988 and Mar. 1993 – Aug. 1993) when no factor combination would deliver a return above the market. It is clear, however, that most of the time in this 40-year period, the simple blend of factors could deliver a persistent excess return. Understanding how factor exposures could be combined, as well as the advantages and disadvantages of numerous approaches, is key to making optimal asset allocation decisions. Several other chapters in this book discuss multi-factor investing (including equal weighting, risk parity and trend following, among others), yet an attractive starting point – particularly, in the absence of strong investment opinions – is the equal weighting approach which diversifies risks across all factor exposures.

The Need For Regional Factor Exposures

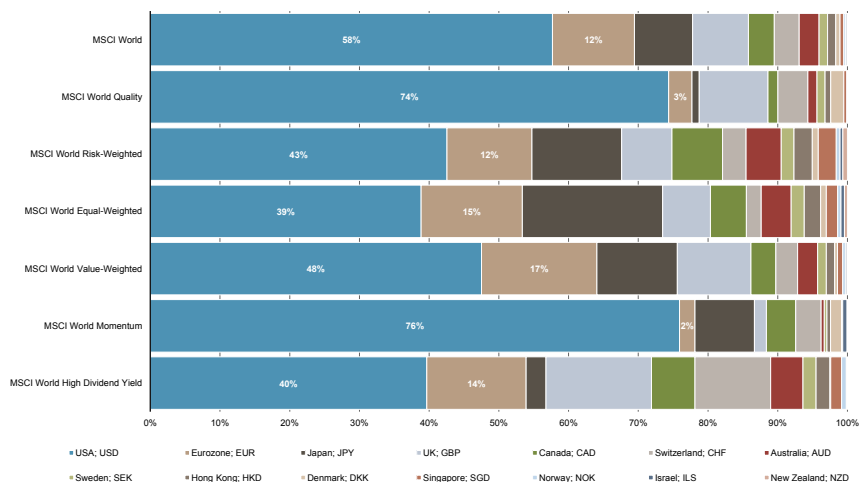
A lot of empirical studies document the *working* features of factor exposure at the global level, i.e. world factors. When on-boarding factor exposure to the portfolio, investors need to decide, not only which factors to add in which proportion, but also at which aggregation level. To illustrate the importance of this concern, Figure 4 shows the exposure to regions – defined by currency blocks – for the six factors and standard world exposure.

In the parent index (MSCI World), the exposure to USD and US equities was approx. 58% at that date, followed by exposure to EUR and Eurozone with a weight of approx. 12%. Given that world factor indices are derived from the equity universe of the world parent index, it is expected that the US equities would have a dominant portion in the world factor indices. This structural *bias* results from the heaviest representation (not only in terms of market capitalization) of the US stocks in the world parent index, with 632 stocks out of 1635 in total as of February 2015. For example, the MSCI World Momentum had excess exposure to US equities of 18%, whilst the MSCI World Equal-Weighted had excess exposure to US equities of –19%. When equally averaging across all factors, the US equities have an average share of 53%, while Eurozone equities have an average share of 11%, which roughly reflects the market capitalization of the world equity markets represented by the parent MSCI

3. Loading Factor Exposures To Portfolios

World index. This, generally, holds true over time and for investors who seek to diversify away from the US equities or investors with a non-USD funding currency, the on-boarding of the world-aggregated factors might turn out to be sub-optimal from a risk perspective.

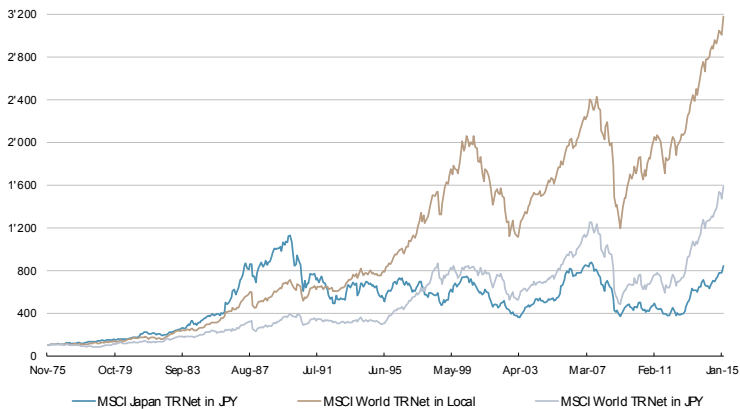
Figure 4: Regional breakdown by market capitalization (Feb. 2015)



Source: MSCI. UBS Global Asset Management.

Example 2: This study is based on the Japanese equity market. Figure 5 shows that the standard market cap MSCI Japan had relatively strong performance in the 70s and 80s, but since the early 90s, this equity market had delivered substantially lower return than the world equity markets. The passive investor holding the MSCI Japan over the last 25 years would have a *flat* return, with massive *opportunity costs* when contrasted against world equity exposure. The beta of the Japanese stocks against the MSCI World is low and on average 0.52. Figure 2 also shows that the MSCI World in JPY (currency unhedged global equity portfolio) also had delivered higher absolute return over that period.

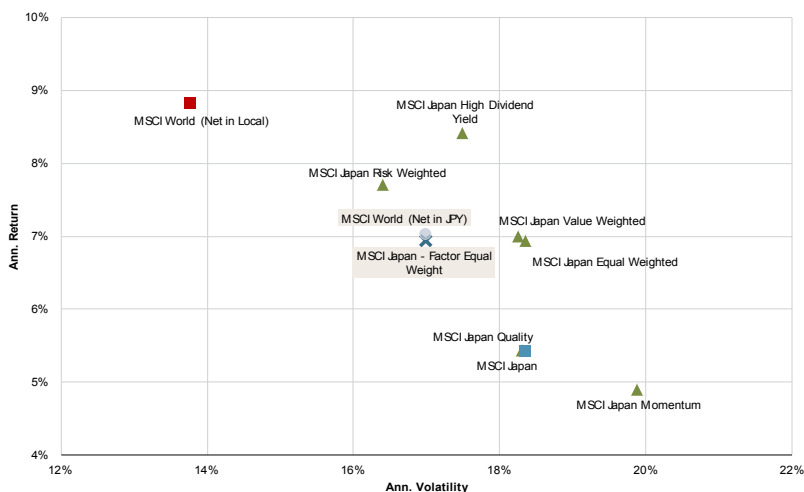
Figure 5: Performance of Japanese equity market vs. world equities (Nov. 1975 – Feb. 2015)



Source: MSCI. UBS Global Asset Management.

From the market cap exposure viewpoint, the outcome shown in Figure 5 appears to discredit Japanese stocks, both large and mid-caps, as captured by the standard MSCI Japan index. Does it mean that there were no equity opportunities – for passive investors – in Japan over that long period? Recall, from Figure 4, that the factor exposure capturing any Japanese source of return is strongly limited (except for the size factor) and currently, on average, 9.5% in factor exposures compared to 8.3% in the MSCI World index, i.e. factor exposure slightly overweights Japanese equities as compared to market cap. The world-aggregation means that any Japanese (or other regional exposure except for the US) is largely averaged out. Suppose an equity investor had a handful of investable factors extracted from the parent MSCI Japan exposure. Figure 6 shows the risk-return space of the MSCI World, MSCI Japan and the six factors, represented by MSCI Japan Equal-Weighted, MSCI Japan Momentum, MSCI Japan Quality, MSCI Japan Risk-Weighted, MSCI Japan High Dividend Yield and MSCI Japan Value-Weighted, respectively. The performance of these indices is measured in total net return terms in JPY, but the MSCI World is also measured in local currencies.

Figure 6: Risk-return space of the MSCI Japan equity market (TR Net in JPY, Nov. 1975 - Feb. 2015)



Source: MSCI. UBS Global Asset Management.

All factors, but momentum, delivered better risk-adjusted return than the market. Strikingly, when combining all six factors with an equal weight (we label it »MSCI Japan – *Factor Equal Weight*«), such a factor blend portfolio would have achieved pretty much this same outcome as world exposure purchased in JPY (MSCI World, Net in JPY) over the period of study. Note that the global portfolio purchased in JPY had underperformed the global equities measured in local currencies due to the appreciation of JPY, in particular, in the 70s–90s. Similar results follow for other regions, like EMU or UK.

Griffin (2002) pointed out in his study that domestic Fama–French factors explain the time series variation in returns and have lower pricing errors than the world Fama–French factors, and the disaggregation improves the explanation of return properties. A possible explanation of that can be the response time discussed above. The *disadvantage* of the consistency behind factor index construction results from the use of a parent universe, implying that in

world-aggregated factor exposures one can expect dominance of the US equities. This complicates the process of diversifying away from the US exposure, but also makes the region-focused factor allocation unfeasible. The *advantage* of consistency behind factor index construction results from the use of the parent universe in local currencies, implying that disaggregation of world factor exposure is feasible, and with the building block approach one can easily reconstruct world factors, while leaving the degree of freedom to fine-tune the weights towards preferred regions, and to avoid a systematic bias towards US equities. In other words, regional building blocks add flexibility, without any loss of diversification when being combined bottom-up.

The Need For Currency Hedged Factor Exposure

Investors may wish to utilize factor ETFs to get exposure to their preferred factor(s), but at the same time they may wish to have geographic diversity in their portfolios. In other words, investors may prefer to on-board regional factor exposures. Within such a *factor-region* investment strategy, the impact of foreign currency exposure and its fluctuation against the funding currency should be carefully considered. MSCI computes the regional indices in the »local currency« and the currency of choice, like USD or CHF, etc. The local currency indices represent the *theoretical* performance of an index without any impact from foreign exchange fluctuations – a *continuously* hedged portfolio – thus indicating a tracking error when contrasted against currency indices.

Example 3: This case study uses the factor data as in Example 1. The indices are expressed in local currency, USD, GBP and CHF. Table 3 tabulates tracking errors – measured by the annualized standard deviation of the monthly return differences between the index calculated in the currency of choice against the local currency index. For example, the tracking error of the MSCI World in USD is 4.5% p.a. In general, the tracking error increases with the decreasing share of the currency in the total currency portfolio. In this example, the indices calculated in CHF have, therefore, the highest risk, as the Swiss equity market plays a relatively minor role in the global context.

Table 3: Tracking error due to currency movements (TR Net, Nov. 1975 – Feb. 2015)

Factor		Size	Yield	Momen- tum	Quality	Low- Volatility	Value
Region	MCSI World Standard	MCSI World Equal- Weighted	MCSI World High Dividend Yield	MCSI World Momen- tum	MCSI World Quality	MCSI World Risk- Weighted	MCSI World Value- Weighted
USD	4.5%	5.5%	4.6%	5.3%	3.2%	5.6%	4.3%
GBP	7.5%	7.1%	6.9%	7.6%	7.7%	7.3%	7.4%
CHF	8.9%	8.2%	9.0%	8.9%	9.7%	8.0%	8.9%

Source: MSCI. UBS Global Asset Management.

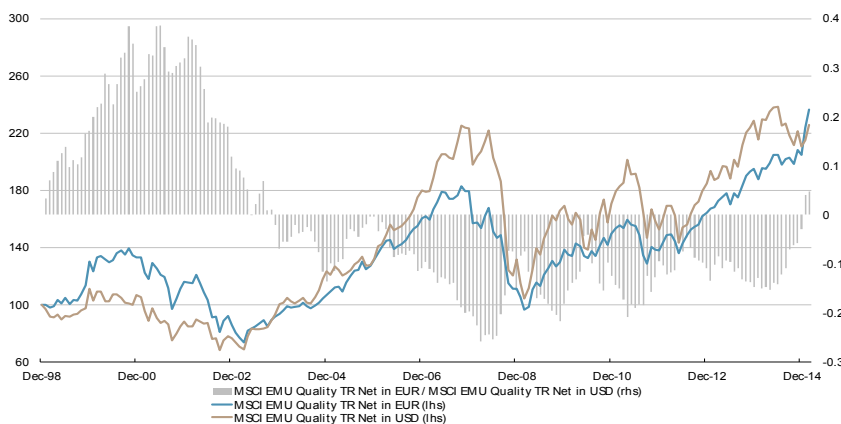
The exchange rate movements are very difficult to predict using economic models, and Meese and Rogoff (1982) showed that a random walk forecasts exchange rates better than many economic models (the Meese and Rogoff puzzle). Various combinations of economic variables and econometric methods have been tested in an attempt to overturn Meese and Rogoff's puzzle, and recent studies document that predictability depends on the choice of predictor, forecast horizon, sample period, economic model, and forecast evaluation method, (see Rossi (2013)). Inarguably, currency movements add volatility to the investment portfolio and can lead to substantial differences of the realized investment returns measured in the funding currency vs. base currency, as seen in Table 3. Recognizing the importance of currency impact on passive investment returns, UBS Global Asset Management has developed a unique framework for passive index replication to offer currency-hedged ETFs which are physically replicated⁹². This framework provides investors with the possibility to control *factor-region-currency* parameters when on-boarding factor exposures to the portfolio. While long-run compensation for taking currency risks is questionable (see Rossi (2013), and references therein), there are periods when short-run compensation is likely due to the »shift« of equilibrium (e.g. regime change in interest rate). To accommodate the possible demand for currency exposure on top of factor exposure, UBS provides factor ETFs,

92 Chapter »Challenges when hedging currency risks in ETFs« discusses in detail the mechanics and specifics of currency hedging in ETFs.

not only in the base currency of the underlying securities, but also currency-hedged to different currencies such as USD, EUR, GBP and CHF.

Example 4: Figure 7 shows index comparison between the MSCI EMU Quality Index in EUR and USD currencies, as well as relative performance between the two exposures.

Figure 7: MSCI EMU Quality (TR Net, Dec. 1998 – Feb. 2015)



Source: MSCI. UBS Global Asset Management.

While targeting MSCI EMU Quality in EUR, investors with USD funding currency would combine quality factor exposure with a *short* position in USD and a *long* position in EUR. The intrinsic currency position implies that investor benefits from EUR appreciation against USD, and vice versa. In this example, MSCI EMU Quality in EUR returns 5.5% p.a. as compared to 4.9% of the MSCI EMU Quality in USD, with the volatility of 16% and 20%, respectively, indicating the absence of compensation for taking currency risk (at least in the long run). Moreover, the tracking error – measured by the annualized standard deviation of the monthly return differences between both indices – equals 10.4%, indicating substantial tracking risk due to currency movements.

Concluding Remarks

This chapter discussed a number of relevant considerations when putting factors into action through investable ETFs. While index providers typically calculate factor indices in an *infinite* number of formats: global vs. regional, single factors vs. blend of factors, local currency vs. currency of choice, uncontrolled turnover vs. controlled turnover, low frequency vs. high frequency of review etc.: the ETF provider has to decide which format is the most versatile for the investors. The decision for a specific range, is basically associated with tradeoff considerations between the two extremes, i.e. highly aggregated factor exposures vs. granular building block factor exposures. The aggregated factor exposures are shown to suffer, here, from slow response, systematic bias towards most representative equity markets and substantial tracking error risk when expressed in a particular currency of choice. Bearing in mind that factor exposures serve as an addition to *enhance passivity*, UBS Global Asset Management has decided to provide investors with a range of non-overlapping factors, covering regional exposures, and with a local currency and currency hedge overlay. UBS factor ETFs are offered as a set of factor exposures for investors aiming to add factor return to their portfolios in a reasonably purified format.

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Using Passive Factor Exposures To Build A Robust Multi-Factor Portfolio

Thomas Merz

UBS Global Asset Management

The conviction that risk factors are the driver behind investment returns has given rise to countless empirical studies over the past 50 years. This approach remains founded on a theoretical model dating back to the 1960s known as the Capital Asset Pricing Model (CAPM). Originally, formulated by Treynor (1961), Sharpe (1964), Linther (1965) and Mossin (1966), and further developed via various multi-factor approaches by Merton (1969, 1973), Ross (1976) and Fama and French (1992, 1993), the model is based on the principle that the factors behind assets should be seen as the driver of what are commonly known as risk premiums. As a result, investors are compensated with a premium that offsets potential losses from holding such assets. Although the CAPM, and with it the rigid assumption that the premium to be obtained is only driven by a single factor (the broad market beta), has been empirically refuted by countless studies since the 1980s, the most significant underlying conclusion of this model retains its universal validity: every investment decision in the portfolio context aims to identify those factors that generate risk premiums with a view to obtaining compensation for the risk of loss that is taken. In the past, the resulting idea of factor-driven premiums in the portfolio context was almost exclusively applied to the active investment process (see Ang, Brandt and Denison (2014)). In this context, attempts were made to obtain access to the factors of value, size and liquidity, as well as momentum and carry strategies, via an active portfolio construction process.

On the basis of an even more differentiated engagement with risk-adjusted returns, there is growing interest amongst institutional investors, in particular, in applying this approach within passive portfolios. The analysis below aims to examine factors from the perspective of a purely passive portfolio. In this context, the goal is to develop a solution that is simple to implement and which can

be applied by both institutional and private investors. The starting point is traditionally formulated asset allocation with two asset classes (bonds and equities) as seen (in a simplified form) in many portfolios throughout Europe. We aim to examine whether the addition of passive factor strategies in a conventionally structured portfolio is capable of generating added value, and to what extent such a portfolio is more robust than the conventional approach based on asset classes.

A European Portfolio As The Benchmark For This Study

The starting point for this study is the conventional breakdown of assets into a bond position with 60% and an equity position of 40%. For institutional investors, this allocation appears to be too heavily weighted in favour of bonds⁹³; however, looking at portfolios that also manage private assets, this seems to be a good approximation for a majority of investors.

As a further requirement, with a view to replicating a real portfolio as closely as possible, the two positions are reset to their strategic weights once a year. As shown in Figure 1, the annual reset of weightings to the initial 60/40 situation produces additional returns of half a percent, which corresponds to the best result in comparison with other frequencies in historical terms. To illustrate the individual implications in isolation, the first step involves reducing the factor selection to equities. This entails the conscious acceptance that potential conclusions will not be equally valid for all factors⁹⁴. For its part, this approach enables a focused engagement and also makes it easier to achieve transparency, as overlapping effects may be minimized. Limiting the equity universe to stocks from the eurozone enables the selection to be narrowed down further, which in turn minimizes overlapping effects from a regional

93 The Global Pension Asset Study 2014 by Towers Watson reports an average equity position for pension funds across the world of 52%, while bond investments amount to 28% (Tower Watson (2014)). The remainder is held as cash holdings (1%) and alternative assets (18% including real estate investments).

94 As well as factor premiums for equities, the literature documents numerous premiums for bond or commodity markets, for example.

perspective. Griffin (2002) also advocates a regional focus. He documents results which support the hypothesis that factor premiums are most visible when regional data are analyzed.

Table/Figure 1: Differences in Returns for Various Weighting Strategies

	Return		Exc. vs static
	full period	p.a.	p.a.
Portfolio	59.50%	3.39%	-
Portfolio rebal. Ann ¹⁾	66.11%	3.69%	0.30%
Portfolio rebal. Semi-ann ²⁾	59.55%	3.39%	0.00%
Portfolio rebal. Quart ³⁾	61.56%	3.49%	0.09%
Portfolio rebal. Feb	65.85%	3.68%	0.29%
Portfolio rebal. Aug	58.57%	3.35%	-0.04%

¹⁾ Rebalanced end of December, ²⁾ Rebalanced end of June and December,

³⁾ Rebalanced end of March, June, September and December.

Source: UBS Global Asset Management, based on a world portfolio with 60% bond and 40% equity exposure, monthly data derived from MSCI index levels (TR Net), log returns from Dec-99 to Dec-14. Past performance, whether simulated or actual, is not a reliable indicator of future results.

Various European Premiums

With regard to selecting factors that reward taking risks with a premium, there are numerous academic studies (Ang, Goetzmann and Schaefer (2011)). The first and most important precondition for the choice of factor portfolios within this study is an in-depth grounding in academic literature. Secondly, there must be clear empirical evidence of a premium, as it is only then worth taking the corresponding active risk versus the benchmark. A third requirement is that the corresponding factors can be replicated by a long-only portfolio. From the perspective of a physical index replicator, this is a very important attribute, as it should be possible to implement the results shown with passive components. This is also based on the requirement of a majority of investors whose legal framework does not (generally) permit instruments with a long-short structure.

Harvey, Liu and Zhu (2014) list a large number of so-called factor premiums. This makes clear – depending on the portfolio structure – that a corresponding selection needs to be made which represents an active decision that is not to be underestimated. Ang, Goetzmann and Schaefer (2009) point out that the switch from a portfolio based on asset classes to a factor-based approach is not without risks. Firstly, there is no consensus with regard to a complete factor universe, while, secondly, there is no clear definition as to how exactly the individual factors should be structured. Melas, Briand and Urwin (2011) find the highest risk premiums for the following factor indices: size (small cap), low volatility (risk wgt or min. vol.), value (val. wgt. or pure val.) and high dividends (yield wgt.). On this basis, the analyses of this study focus on the following four factors: Value (Val), Low Volatility (LVol), Quality (Qual) and Total Shareholder Yield (TSY). In view of the presence of broad-based academic literature, the size factor is excluded, as repeated analysis is likely to yield little in the way of new findings. This does not mean that the study denies the importance of the size factor.

Table/Figure 2: Risk Figures for Standard Market and Factor Betas

Beta Exposures	Return p.a.	Reward-to-Risk ^{*)}	Max. Draw-down
Standard Index	0.84%	0.04	-59%
Val	5.19%	0.29	-49%
LVol	6.46%	0.50	-52%
Qual	3.06%	0.19	-50%
TSY	3.99%	0.23	-54%

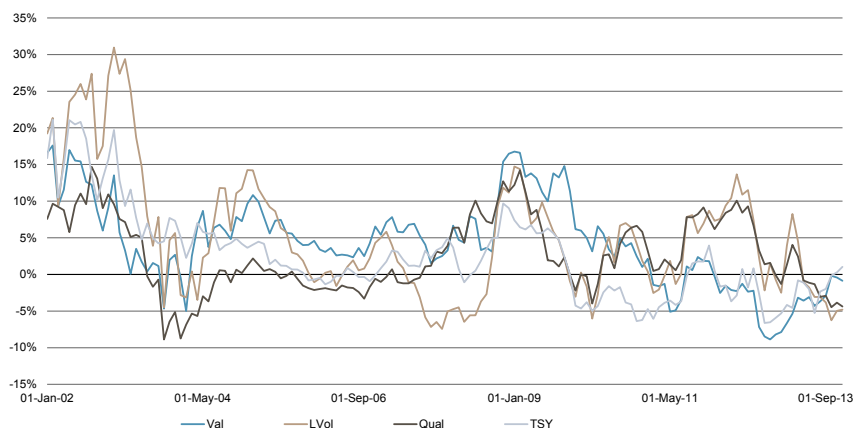
^{*)} Defined as the ratio of the annualized average return to annualized standard deviation of returns.

Source: UBS Global Asset Management, monthly data derived from MSCI index levels (TR Net), log returns from Dec-99 to Dec-14.

Figure 2 shows the extent to which each of the four factors selected was in a position to generate a corresponding premium in excess of the standard index. Although the extent of the premium varies considerably over the period under review (14 years), there is a clear additional positive return on average across the board. Furthermore, the reward-to-risk ratios show that the individual factor indices generate greater returns with lower risk. The highest added value is in low volatility beta. The Val factor also displays a high value. The

picture is similar in relation to the second risk measure, the maximum draw-down, i.e. the maximum loss threshold reached over the entire period. The factor betas achieve better values than the standard index, the MSCI EMU. This makes clear that the partial substitution of standard beta with factor beta, in a portfolio context, should lead to benefits in terms of both risk and return.

Table/Figure 3: Annualized 12-month Rolling Factor Premiums



Source: UBS Global Asset Management, based on log returns of factor and MSCI parent indices (TR Net), monthly data from Dec-99 to Dec-14.

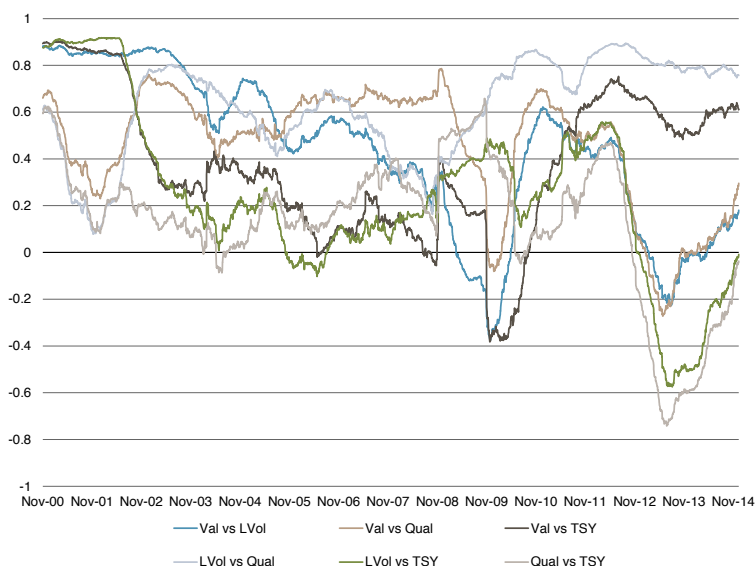
What Does A Combination Of Standard And Factor Beta Achieve?

The results from Figure 3 raise the question regarding the most efficient and profitable combination in which these factor betas can produce their desired effect. On the one hand, the question of how the individual factors behave in relation to one another needs to be clarified. On the other hand, the findings from the study need to withstand testing in a portfolio context with standard betas. Among other aspects, this relates to the question of whether all, or only a selection of, the factors deliver the best results if they are combined with conventional asset allocation. In this respect it would be desirable if the factor

returns selected had an effect on the portfolio as a whole that was as diversifying as possible, with the risk/return profile displaying a corresponding improvement.

An analysis that examines the interdependencies between the individual factor returns is shown in Figure 4. The strong cyclical nature of the correlation between individual factor returns is striking. It is evident that the correlations vary, significantly, at times (from 0.8 to -0.8), while they are considerably closer together in different phases (0.2 and 0.4). The data point to the conclusion that a skillful combination creates added value. In this respect, a combination of the Quality (Qual), Value (Val) and Low Volatility (LVol) factors is particularly interesting.

Table/Figure 4: Intra-Factor Correlation of Value (Val), Low Volatility (LVol), Quality (Qual) and Total Shareholder Yield (TSY)



Source: UBS Global Asset Management, based on 250d rolling correlations of relative returns vs. MSCI standard index TR Net, daily log returns Dec-99 to Dec-14.

In view of this cyclicity, active management of the weights of individual factors appears to be extremely difficult. A detailed look at the correlation between the value and low volatility factor returns, for example, shows how frequently, abruptly and sharply the individual changes of direction take place after one another. For example, their correlation at the start of the financial crisis in February 2008 was 0.4, at end-2009 minus 0.4, then, within two months, back at a value of just under 0.7, while just six months later the value was minus 0.2 before arriving at a value of around 0.2 again, after just four months. On the whole, the range seems to open up somewhat over time, which rather seems to support an implementation of more than one factor pair. In principle, it can be said that a portfolio with a combination of factors will probably be subject to less of a risk of holding the least favorable factor in the portfolio at the wrong time and, therefore, falling well behind the standard beta.

The addition of factor returns to a conventional 60/40 portfolio appears to have a positive effect, overall. It can, generally, be stated that the total return of the portfolio can be improved, while risk is also reduced. In terms of the weighting assigned to the factor returns, a clear pattern can be identified. As the data in Figure 5 shows, the additional return rises steadily, the higher the weighting. From a weight for all factor returns within the equity allocation of 40% upwards, the additional return in the portfolio climbs by more than 0.5% each year. If all the factor returns are deployed, an additional return of more than 1% each year can be achieved from a ratio of 70%. In terms of additional returns, the addition of the Val and LVol factors proved to be the most successful. A restriction to the Val or the equally weighted split across the Val, LVol and Qual factors produced similar results. Within the same period under review, a split across all four factors produced lower returns. In terms of risk, the results are highly comparable. In this respect, focusing on the addition of the Val and LVol factors, within the deployment of the equity ratio, succeeded in lowering the maximum loss threshold (max. draw-down) by 4.37%. This was slightly more successful with the weighted deployment of the Val, LVol and Qual factors, which reduced the maximum loss threshold by 4.83%.

Table/Figure 5: 60/40 Model Portfolio with Factor Returns Added

60/40 Portfolio plus additional factors ^{*)}										
Return full period		Sum Factor Weights								
		20%	30%	40%	50%	60%	70%	80%	90%	100%
Val		55.23%	57.45%	59.67%	61.87%	64.07%	66.25%	68.43%	70.59%	72.75%
Val, LVol		56.19%	58.89%	61.57%	64.24%	66.89%	69.53%	72.15%	74.76%	77.35%
Val, LVol, Qual		55.13%	57.31%	59.47%	61.62%	63.77%	65.90%	68.03%	70.14%	72.24%
Val, LVol, Qual, TSY		54.79%	56.79%	58.79%	60.77%	62.75%	64.72%	66.68%	68.62%	70.57%
Excess Return annualized										
Val		0.32%	0.48%	0.64%	0.79%	0.95%	1.11%	1.26%	1.42%	1.57%
Val, LVol		0.39%	0.58%	0.77%	0.96%	1.15%	1.34%	1.53%	1.71%	1.90%
Val, LVol, Qual		0.31%	0.47%	0.62%	0.78%	0.93%	1.08%	1.23%	1.38%	1.54%
Val, LVol, Qual, TSY		0.29%	0.43%	0.57%	0.72%	0.86%	1.00%	1.14%	1.28%	1.42%
Risk (max draw-down)										
Val		-21.64%	-21.14%	-20.65%	-20.15%	-19.65%	-19.15%	-18.65%	-18.15%	-17.65%
Val, LVol		-21.65%	-21.16%	-20.67%	-20.17%	-19.71%	-19.35%	-18.98%	-18.62%	-18.25%
Val, LVol, Qual		-21.67%	-21.19%	-20.71%	-20.23%	-19.74%	-19.26%	-18.77%	-18.28%	-17.79%
Val, LVol, Qual, TSY		-21.79%	-21.37%	-20.95%	-20.53%	-20.11%	-19.68%	-19.26%	-18.83%	-18.41%
Excess Risk										
Val		-0.99%	-1.48%	-1.98%	-2.47%	-2.97%	-3.47%	-3.97%	-4.47%	-4.97%
Val, LVol		-0.98%	-1.47%	-1.96%	-2.45%	-2.91%	-3.28%	-3.64%	-4.01%	-4.37%
Val, LVol, Qual		-0.95%	-1.43%	-1.91%	-2.40%	-2.88%	-3.36%	-3.85%	-4.34%	-4.83%
Val, LVol, Qual, TSY		-0.83%	-1.25%	-1.67%	-2.10%	-2.52%	-2.94%	-3.37%	-3.79%	-4.22%

**) Equal weights applied to all factor exposures within the multi-factor portfolio. Total weight for all factor exposures corresponds to the »sum factor weights« in each of the simulated multi-factor portfolios and the standard equity portion (MSCI EMU) is reduced, accordingly. Risk refers to max. draw-down. Excess risk is calculated as max. draw-down 60/40 portfolio minus max. draw-down multi-factor portfolio. All portfolios are rebalanced annually.*

Source: UBS Global Asset Management, based on monthly log returns derived from MSCI Net index levels, Dec-99 to Dec-14. Past performance, whether simulated or actual, is not a reliable indicator of future results.

The need for attention to be paid to a sensible combination of individual factors is evident from the fact that they are constantly being added. The data permit the supposition that the skillful modeling of different combinations can lead to the expectation of improvements in terms of both additional returns and risk reductions. We will go on to examine three portfolio variants (addition of two factors, three factors and four factors) with two weighting types in each case (40% and 70% total weight for factors).

Modeling The Factor Combinations In The 60/40 Standard Portfolio

Figure 6 provides a table of the results of all model portfolios. Additional factor returns were assigned to the conventional 60/40 standard portfolio on the equity side, with the weight for the standard market beta (MSCI EMU) being reduced, accordingly. The weight assigned to the factor returns is divided up equally among the factors, as we do not expect any risk-adjusted added value from tactical management due to the high cyclicity of the factor returns. Various studies indicate that naive diversification, i.e. weighting individual sources of returns equally, generally delivers very good results⁹⁵. Then the implementation is straightforward, as there is no need for complicated optimization models and calculations.

⁹⁵ In practice, dynamic allocation models built on optimization mechanisms often find it difficult to beat the very simple but effective equal weighting approach. See DeMiguel, Garlappi and Uppal (2009) for more on this.

As in section 1.3, the portfolio weights of all the positions are reset to the original weighting once a year for this analysis as well. This applies to both the standard beta positions (equities and bonds) and to factor investments. The excess returns of the various factor combination portfolios, with a factor ratio of 40%, range between 0.37 and 0.77% per year. As Figure 6 makes clear, the factor combination of Val and LVol generates the largest excess return at 0.77% per year. The combinations of Val, LVol and TSY or LVol and TSY are similarly successful at 0.66 and 0.67%, respectively, per year.

Table/Figure 6: Annualized Excess Returns for Multi Factor Portfolios (sum of factor weight 40%)

	Multi-Factor Portfolio (sum factor weights of 40%)		
	Factor combinations	Return p.a.	Exc. Return p.a. ^{*)}
1	Val, LVol	4.40%	0.77%
2	Val, Qual	4.10%	0.48%
3	Val, TSY	4.16%	0.53%
4	LVol, Qual	4.24%	0.61%
5	LVol, TSY	4.29%	0.67%
6	Qual, TSY	4.00%	0.37%
7	Val, LVol, Qual	4.25%	0.62%
8	Val, LVol, TSY	4.28%	0.66%
9	LVol, Qual, TSY	4.18%	0.55%
10	Val, LVol, Qual, TSY	4.20%	0.57%

^{*)} Excess returns calculated as multi-factor portfolio return minus the annualized return of the standard 60/40 portfolio (annualized). All portfolios are rebalances, annually.

Source: UBS Global Asset Management, based on monthly log returns derived from MSCI Net index levels, Dec-99 to Dec-14. Past performance, whether simulated or actual, is not a reliable indicator of future results.

Instead of a pure-play equity position in the MSCI EMU, if an investment is also made in the factor returns of Qual and TSY, this only results in additional returns of 0.37% per year. This minimal additional return is, therefore, only slightly higher than that of the portfolio without factor returns. If it is

then assumed that passive factor portfolios have a higher portfolio turnover rate compared to passive equity portfolios (such as MSCI EMU), the reported added value will reduce further, on account of the transaction costs incurred. Accordingly, the addition of factor combinations consisting of Qual and TSY with a factor ratio of just 40% will only be, marginally, worthwhile. However, if the factor ratio is raised to 70%, the excess return of this factor combination comes in at 0.65% on an annualized basis. The additional returns with a factor ratio of 70% for the combinations of Val/LVol, LVol/TSY, Val/LVol/TSY, Val/LVol/Qual and LVol/Qual are well in excess of 1% per year in some cases.

To ensure comparability and to compile a corresponding ranking for the portfolio variations, we calculated the information ratio (IR) for every combination. In this calculation, the annualized excess return is divided by the active risk. Active risk refers to the tracking error versus the standard 60/40 portfolio. The portfolio with the highest IR is, therefore, the preferred combination from a risk-adjusted perspective. As Figure 7 shows, the values for all the portfolio combinations are between 0.134 and 0.165. The portfolio with the highest IR value consists of the standard combination of 60/40, with the equity position of 40% being invested in an equally weighted factor combination of Val/LVol/Qual/TSY. The addition of a dual combination of Val and LVol or a triple combination of Val/LVol/Qual is similarly successful. The calculations conclude that, in the long term, a standard portfolio with the additional combination of Val and TSY will pay out the least in risk-adjusted terms.

If four equally weighted factor positions are also added to the standard portfolio consisting of 60% European bonds and 40% European equities, it is reasonable to assume an annual, additional return of some 60 basis points. This result supports the assumption already made in Figure 4 that the factors of Qual, Val and LVol have an effect when combined. The fact that the combination of all four factors achieves values that are slightly better, still appears to be due to the fact that the four-way combination is in a position to further reduce the cyclicity of the factor returns, thus lowering active risk.

Table/Figure 7: Information Ratios of Multi-Factor Portfolios^{*)}

Sum						
Factor	2 Factor 60/40 Portfolio					
Weight	Val, LVol	Val, Qual	Val, TSY	LVol, Qual	LVol, TSY	Qual, TSY
40%	0.164	0.147	0.134	0.145	0.149	0.140
70%	0.164	0.146	0.134	0.145	0.149	0.139
	3 Factor 60/40 Portfolio					
	Val, LVol, Qual	Val, LVol, TSY	LVol, Qual, TSY			
40%	0.162	0.160	0.157			
70%	0.162	0.160	0.156			
	4 Factor 60/40 Portfolio					
	Val, LVol, Qual, TSY					
40%	0.165					
70%	0.164					

^{*)} Calculated as the ratio of multi-factor portfolio excess return (ann.) divided by the active risk (tracking error vs. standard portfolio). All portfolios are rebalanced, annually.

Source: UBS Global Asset Management, based on monthly log returns derived from MSCI Net index levels, Dec-99 to Dec-14.

Conclusions And Comparison With Other Factor Studies

Over the past ten years, the recognition of the liabilities of individual portfolio components has become far more significant. Indexed investments are now a part of most institutional investors' portfolios. In line with this trend, there has been sharp growth in the scope of investment funds that are invested on the basis of market capitalization. A large majority of passive investments, such as those in the form of Exchange Traded Funds (ETFs), follow an

index basket that specifies the individual securities in line with their market weights. The systematic pursuit of a strategy such as this has repeatedly attracted criticism. For example, Haugen and Baker (1991) or Tabner (2009) as well as Malevergne, Santa-Clara and Sornette (2009) argue that excessively concentrated portfolios often arise on account of the prevailing index composition. Golz and Le Sourd (2011) also document poorer risk-adjusted returns for market capitalized index portfolios.

The conventional model of capital market theory only offers a single risk premium: the equity risk premium. The shift from cash holdings and bond investments to equity investments enables portfolios to obtain a higher return expectation value. Countless empirical studies and refinements of the theoretical foundation now clearly show that price trends are influenced by a wide variety of risk sources. The common denominator in all explanation models, currently prevalent, is that the corresponding premium is greater, the higher the potential for value losses are in challenging markets. It is still unclear whether the premiums depend on rational or irrational investor behaviour.

The study confirms the finding that factor premiums are certainly not stable and, instead, are often subject to large fluctuations (see Ang, Brandt and De-nison (2014)). The data set examined, documents repeated phases in which the factor premiums feature negative values (see Figure 3). The duration of such phases with a relative underperformance ranges from 13 to 22 months, confirming the fact, from earlier studies, that factor premiums are of a cyclical nature. Regarding the level of the premiums documented in Figure 8 as well, the results are relatively consistent with the values listed in literature⁹⁶. This points to the conclusion that factor premiums have the largest effect when they are deployed in combination, and, inherently, also reducing timing risks when implemented.

⁹⁶ Depending on the author and the data set, the premiums fluctuate between 2 and 7% on an annualized basis (for more on this, see Dimson et al. [2010], Baker, Bradley and Wurgler (2011), Amenc (2013)).

Table/Figure 8: Time Varying Factor Premiums

Factor	2000 – Premia p.a.	2000 – 2004	2005 – 2009	2010 – 2014	Dotcom Bubble ¹⁾	Post Dotcom ²⁾	Financial & Sov. Crisis ³⁾
Val	4.36%	7.82%	7.11%	-1.16%	11.69%	4.30%	2.31%
LVol	5.62%	14.61%	1.15%	2.89%	23.60%	1.47%	3.44%
Qual	2.23%	2.43%	2.31%	1.98%	8.59%	-1.09%	2.78%
TSY	3.16%	9.92%	1.71%	-0.81%	15.19%	2.74%	0.02%

¹ 2000 to 2002, ² 2003 to 2007, ³ 2008 to 2014.

Source: UBS Global Asset Management, based on log returns of factor and MSCI parent indices (TR Net), monthly data from Dec-99 to Dec-14. Past performance, whether simulated or actual, is not a reliable indicator of future results.

Overall, it can be stated that the addition of individual factors as part of conventional asset allocation of 60% bonds and 40% equities generates additional returns. The larger the share of factor beta in the equity position, the higher the additional returns for the portfolio as a whole. There are major differences depending on the individual factors added. As well as a two-factor model (Val, LVol), the three-factor model (Val, LVol, TSY) and the four-factor model appear to produce noticeable, excess returns.

As a starting point, our multi-factor model takes into account a conventional asset breakdown into 60% bonds and 40% equities. The addition of a four-factor position at the expense of the equity allocation raises the portfolio value by 0.6% each year. In this regard, the individual factors are weighted equally and substitute 40% of the standard equity beta. At the same time, the portfolio risk reduces from 6.85% for the standard portfolio to 6.45% for the multi-factor portfolio. A higher factor beta ratio, of 70% for example, does generate higher returns, but at the cost of higher risk. The reward to risk ratio is highest for a four-factor model within an equity allocation of 40%.

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4. Indexing A Factor Portfolio

Efficient Implementation Of Alternative Beta Index Portfolios

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Unravelling The Complexity Of Efficient Implementation Of Alternative Beta Index Portfolios

In 1989, to celebrate its 150th anniversary, Swiss watchmaker Patek Philippe created a pocket watch, Caliber 89, which to this day remains the most complicated portable timepiece ever made.

The watch comprises 1,728 parts, including two dials, eight disks, 61 bridges, 129 jewels, 184 wheels, 332 screws, 415 pins, and 429 mechanical components; it took five years of research and development, and another four years to bring it into physical existence.⁹⁷ Managing and efficiently implementing alternative, beta index portfolios is analogous to the level of complexity (but thankfully not the cost!) of Caliber 89: a portfolio benchmarked to a global, alternative, beta index could typically comprise around 3,000 index constituents from around 50 countries and around 25 industries and be exposed to more than 20 equity return factors. Using sophisticated systems and processes, each index constituent has to be analysed carefully and index changes have to be implemented, optimally, as they occur; a process involving timely, detailed, precise and pragmatic consideration of liquidity, turnover, cost, corporate events and valuation metrics. In this note, we discuss our approach to unravelling and managing these complexities and how we, the indexing team of UBS Global Asset Management, efficiently implement alternative, beta index portfolios for our clients.

⁹⁷ Source: Patek Philippe.

Trading For Index Changes: The »Mainspring« In The Index Portfolio Management Process

Our experience over many years has shown that when stocks are added to, or deleted from, major indices, prices can be distorted by market trading at the impact point due to aggregate market demand. Thus, adjusting the timing of purchases and sales by a short period in a risk controlled manner can often add value without significant risk to the performance of the portfolio. These small but incrementally significant gains could be a major source of added value for index equity strategies. Due to the importance of index changes, it is essential that analysis of index changes is initiated at the earliest possible time. Our attention to detail of stock trading patterns has allowed our index clients to capture significant added-value from pricing inefficiencies at the impact point of index changes, with minimal impact on tracking accuracy.

Two factors are specific to trading for index changes in alternative beta indices: size and relative illiquidity. There are significantly fewer assets currently managed against alternative beta indices (approx. USD 330 billion⁹⁸) compared to market capitalization indices (approx. USD 4 trillion⁹⁹), however, alternative beta indices tend to be relatively less liquid in comparison, and their turnover levels higher, hence, the price formation around index changes that we see in market cap indices may also be observed in alternative beta indices. This pattern may increase as the size of assets, managed against non-market cap-weighted indices, grows. Index liquidity and turnover are two key aspects of the equity index portfolio implementation process. Below, we provide colour on their specifics for alternative indices and how we deal with them in practice.

⁹⁸ Source: The Economist, 3 May 2014.

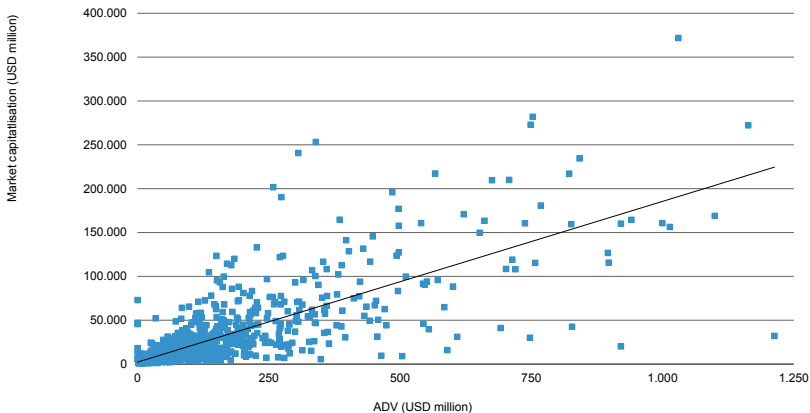
⁹⁹ Source: MSCI, UBS Global Asset Management.

Liquidity

As illustrated in Exhibit 1, a positive correlation between company size, as measured by market capitalization, and stock liquidity, as measured by average daily trading volume (ADV) is typically observed. Therefore, an index with a tilt toward smaller stocks, such as some of the alternative beta indices, would typically be less liquid compared to a market cap-weighted index.

Exhibit 1: Positive relationship between size and liquidity

MSCI World Index



Source: UBS Global Asset Management, MSCI. Data for MSCI World Index (excluding outliers) as at 28 April 2015.

Alternative beta indices tend to be either factor-based (capturing specific equity return factors) or diversification-based (spreading exposure across a number of factors). Factor-based indices, by design, exploit a particular market segment (for example value stocks, quality stocks, low volatility stocks, small caps, etc.), therefore, they could comprise less liquid securities compared to a broad market cap index. Additionally, many factor indices are more concentrated (e.g. comprising around 400–500 stocks) than their parent market cap index (e.g. comprising around 3,000 stocks), consequently they could have

significant overweight positions in certain stocks vs. a broad market cap index. Exposure to particular market segments and large positions in certain stocks could, potentially, create liquidity problems for some non-market cap indices during the quarterly and semi-annual index rebalancing. Additionally, construction methodologies of alternative beta indices differ, resulting in some indices having more breadth and higher capacity, while other indices are more concentrated and the capacity there might become problematic.

We undertake in-depth liquidity analyses of the rebalancing trades for the estimated total global assets currently tracking alternative beta indices against which we manage portfolios. Such analyses indicate areas of potential liquidity concerns and help us develop and determine our optimal trading and implementation strategy at the time of the quarterly and semi-annual index rebalancing.

In addition to liquidity analyses of rebalancing trades for current assets tracking alternative indices, we also undertake liquidity analyses of rebalancing trades for the estimated maximum capacity AUM in these indices. Such analyses indicate whether capacity and liquidity in these strategies could become problematic in the future. If we believe this might be the case, we work with our clients and index providers to explore avenues to ease potential capacity and liquidity constraints. For example, customized rebalancing dates or customized construction processes could be implemented for alternative beta indices – having the index optimized on a customized date could, potentially, alleviate liquidity problems at the time of index rebalancing, while building a customized construction process could, potentially, reduce excessive turnover and reduce trading costs.

Turnover

In the construction and maintenance of alternative beta index portfolios, we consider, simultaneously, three key factors: transaction costs, tracking error and index imbalances, using our in-house proprietary portfolio management system. Portfolios are, therefore, constructed to minimize costs and risks

while closely replicating the index, as we believe that this is the most effective approach to ensuring efficient portfolio management.

Two aspects of turnover are most relevant to alternative indices: turnover of individual indices resulting from quarterly and semi-annual rebalancing and, for multi-factor portfolios, turnover resulting from rebalancing of component indices to specified weights in an index blend.

- Index rebalancing: unlike market cap indices, which by design are largely self-rebalancing and exhibit low turnover (approx. 5% p.a. two-way for a global index), alternative beta indices are characterized by higher turnover, ranging from 20–40% p.a. two-way for global value, quality and low vol indices to over 250% p.a. two-way for global momentum indices, and the associated higher transaction costs. While it could be argued the excess returns achieved by alternative vs. market cap indices could be attributed to a large extent to the rebalancing effect, thereby necessitating higher turnover levels, we question if such high turnover levels at the semi-annual rebalancing are always required. For example, our analysis of some low volatility optimized indices indicates that turnover at the semi-annual index rebalancing could be significantly reduced without impacting the volatility reduction of the index, i.e. the majority of the turnover does not lead to further reduction in the index volatility during the index optimization. Therefore, reduction in the excess turnover could lead to lower transaction costs without impacting the volatility reduction achieved by the index, along with achieving better performance outcomes for investors.
- Multi-factor portfolios: rebalancing of component indices to specified weights in an index blend. One of the key themes in the area of alternative, beta index investing at present is combining indices capturing different factors to form a multi-factor portfolio. There are several approaches to creating multi-factor portfolios, including the building blocks approach

comprising, for example, three portfolios benchmarked to three alternative beta indices, with each index capturing one factor, or one portfolio benchmarked to a custom index capturing the three factors. Although, the latter approach benefits from automatic offsetting positions and turnover savings, we favor the flexibility of index blends constructed via building blocks because such approach allows our clients to select the best of breed indices, capturing the underlying equity factors. Additionally, the building blocks approach still allows potential crossing opportunities between the funds. We are index agnostic and as new alternative beta indices are constantly being launched, we continuously research the space for new benchmarks and aim to select the best index in each category. Applying the building blocks approach (separate funds capturing individual factors) vs. one index capturing several factors allows investors to:

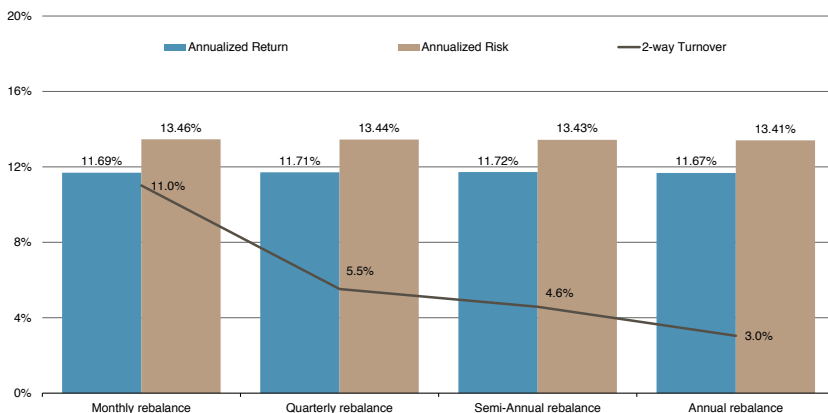
1. *create tailored solutions: invest in one or more funds*
2. *adjust/change allocations to different factors*
3. *select best of breed indices that suit their requirements*

When a building blocks approach is applied, we work with our clients to determine the optimal frequency of rebalancing component indices to the specified weights in the index blend. As illustrated in Exhibit 2, the risk-return profile of an example index blend, comprising three USA alternative beta indices capturing size, volatility and quality factors, is similar for the different rebalancing frequencies, but the turnover is significantly higher when the component indices are rebalanced to the specified weights in the blend monthly (11% p.a. two-way turnover) compared to annually (3% p.a. two-way turnover). We would, therefore, suggest to clients that they consider semi-annual or annual rebalances of component indices in order to reduce the transaction costs associated with more frequent rebalancing.

4. Indexing A Factor Portfolio

Exhibit 2: Tradeoff between rebalancing frequency and cost

MSCI USA alternative beta index blend (1/3rd Equal-Weighted + 1/3rd Min Vol + 1/3rd Quality) annualized return, risk and turnover 31 May 1988 – 31 December 2014



Source: UBS Global Asset Management, MSCI.

Note: Data from 31 May 1988 (earliest date when data is available for all examined indices) to 31 December 2014. TR gross index performance data in USD. Annualized monthly standard deviation of index returns is used as a measure of risk. Data for alternative indices contains live and back-tested data. Turnover data is for rebalancing component indices to specified weights in the alternative beta index blend and does not include the rebalancing of the individual indices. Past performance is not a reliable indicator of future results.

Corporate Actions: The »Wheels« That Make The Index Tick

Specific rules apply to corporate action treatment in alternative beta indices compared to market cap indices. Additionally, rules regarding the treatment of certain corporate events differ amongst the various alternative beta indices. When corporate actions affecting index constituents are announced, we have to decide how to trade and implement them in our clients' portfolios. Due

to the above mentioned nuances and intricacies, often the treatment of the same stock impacted by the same event would differ between our market cap and non-market cap portfolios. We have been managing alternative beta index portfolios for more than five years now and over this time have developed in-depth knowledge and hands-on experience of how to implement corporate events in portfolios benchmarked to market cap and non-market cap indices.

Corporate activity in market cap indices is, largely, self-rebalancing. This might not be the case with alternative beta indices. The main difference in the treatment between market cap and non-market cap indices is in the case of corporate events such as spin-offs, mergers, takeovers and rights issues. While these events require little or no actions in market cap indices, they may result in far more significant trading in alternative beta indices. When events are treated on a »case-by-case« basis, we analyse, carefully, the treatment of the event in question in different indices, and, if necessary, we seek further clarity from the index providers, in order to determine our trading and implementation strategy for our clients' portfolios.

Valuation Metrics: The »Hands« Of The Index »Clock« Measuring The Spirit Of Time

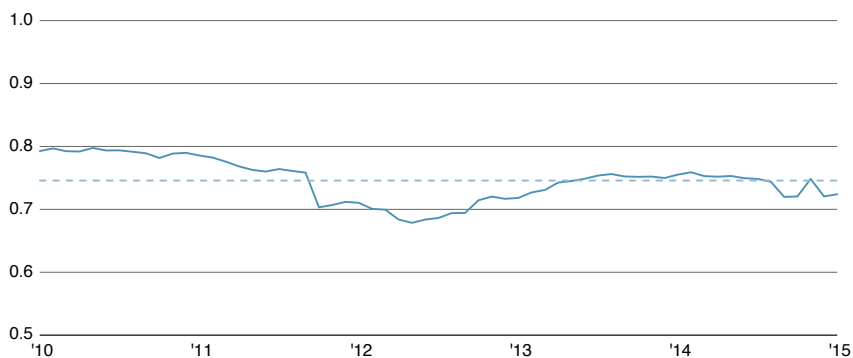
As part of our index analysis, we examine a range of fundamental valuation metrics of alternative beta indices and compare these to the valuation levels of market cap indices and of »pure« equity return factors. Such analysis provides us with an indication of the valuation levels of alternative beta indices, relative to:

- their long-term historic trading levels
- the underlying market cap index
- »pure« equity return factors valuation spreads

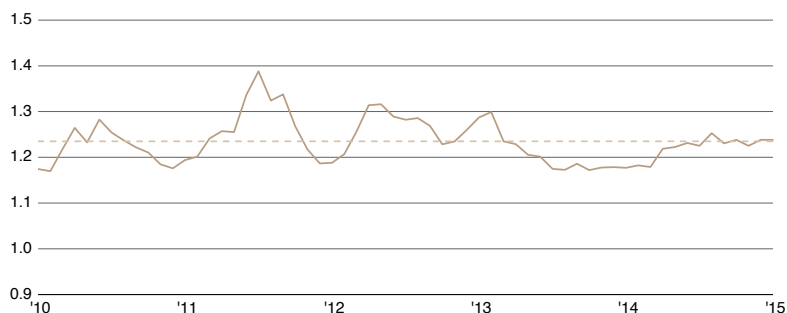
In Exhibit 3 we present the valuation levels of several flagship alternative beta indices exploiting three key equity factors: MSCI World Value-Weighted (value), MSCI World Min Vol USD optimized (volatility) and MSCI World Quality (quality). It is hard to accurately time entry and exit points into a particular alternative beta strategy (and we would not, necessarily, advise clients to use index valuation levels as a tool to »time« their entry and exit points) but we use valuation metrics analysis as a broad indication and a »common sense« check of the trading levels of alternative beta indices. If we observe that certain indices are trading outside of their long-term historic levels or in significant disparity to the respective »pure« equity factors, we investigate the causes of the valuation patterns. For example, if a factor index trades at high multiples, it might be tempting to conclude that stocks capturing this factor are overvalued. But further valuation analysis of the index might reveal a different story: some alternative indices are highly concentrated, consequently, addition/deletion or large weight changes of several major index constituents could impact the overall index valuation metrics. However, this does not mean that all stocks capturing the factor in question are overvalued.

Exhibit 3: Alternative beta indices valuation: value and volatility indices trade close to their historic averages

Relative P/BV: MSCI World Value-Weighted vs. MSCI World



Relative P/BV: MSCI World Min Vol USD opt vs. MSCI World



Relative P/BV: MSCI World Quality vs. MSCI World



Source: UBS Global Asset Management, MSCI.

Data for developed world P/BV ratio and historic average to 31 March 2015. MSCI indices valuation metrics based on past years' monthly data.

Past performance is not a reliable indicator of future results.

Here, we illustrate the relative Price/Book Value multiples of these indices vs. the respective market cap index, but we also conduct valuation analysis based on other metrics including Price/Earnings, Price/Cash Earnings, Price/Sales and Dividend Yield. The alternative beta indices exploiting the value and volatility factors appear to be fairly priced, relative to their historic levels, and, as expected, the value index is cheaper, relative to market cap. The MSCI Min Vol Index appears to be fairly priced, relative to its historic average, but more

expensive, relative to market cap, while the MSCI Quality Index is slightly more expensive than its historic average, and, as expected, it is priced significantly higher than market cap. The recent spike in the valuation of the Quality index is attributable to reduction in the index book value following the November 2014 semi-annual review, where seven out of the top ten largest index constituents recorded book value decrease.

Implementing Alternative Beta Index Portfolios Efficiently: Staying Ahead Of Time

In this note, we discussed how we, the indexing team of UBS Global Asset Management, efficiently implement alternative, beta index portfolios for our clients. Index portfolio implementation in the alternative beta space is an elaborate process requiring skill and dedication, just like Swiss watchmaking. Our approach involves in-depth, timely and precise analysis of index changes, liquidity, turnover, cost, corporate events and valuation metrics, which, we believe, allow us to stay ahead of time in the indexing industry.

Challenges When Hedging Currency Risks In ETFs

Christian Kunth

Markus Goetschi

UBS Global Asset Management

Currency Risk Protection

One of the fundamental investor motivations for utilizing Exchange-Traded Funds (ETF) is adding geographic, sector and/or strategy-based diversity to their portfolios. While this is true for ETFs tracking traditional market capitalization-weighted indices, it is also valid for investors seeking factor beta¹⁰⁰ exposure to further diversify their investments. Besides the underlying investment strategy, the impact of foreign currency exposure and its fluctuation against the home currency of the investor, should be considered in this context. Currency exposure adds an often undesired volatility to the portfolio and, as a consequence, can lead to material changes of the realized investment returns measured in the home currency of the investor.

Currency Impact On Expected Returns

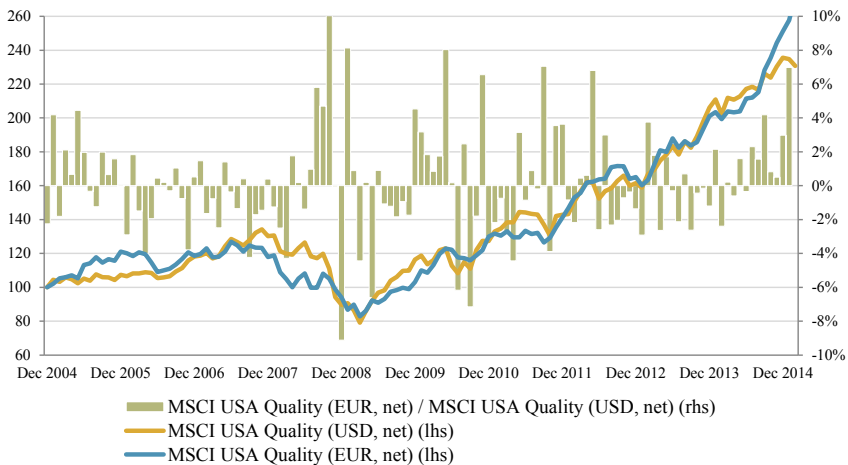
Figure 1 shows a performance comparison between the MSCI USA Quality index in EUR and USD currencies and the monthly relative performance of the two benchmarks. Due to the intrinsic USD currency exposure, a EUR investor faces the risk of diluting the USD equity return. Academic studies¹⁰¹ of foreign

¹⁰⁰ Factor beta is a consistent, rules-based and transparent indexed strategy aiming to create a tilt towards pre-defined fundamental and/or statistical characteristics.

¹⁰¹ MacDonald, R. (1988). Floating Exchange Rates: Theories and Evidence, London.

exchange rates suggest that investors cannot predict currency fluctuations and that the currency risk is not compensated by systematically higher returns. Therefore, currency hedging can protect investors from the movements between foreign and home currency. The tracking error, measured by the volatility of the monthly return differences between both indices, is 10.82% p.a. over the period Nov 2004 to Jan 2015.

Figure 1: Performance comparison and monthly relative performance between the MSCI USA Quality index in EUR and the MSCI USA Quality index in USD



Source: MSCI, end of month data, Nov 2004 to Jan 2015.

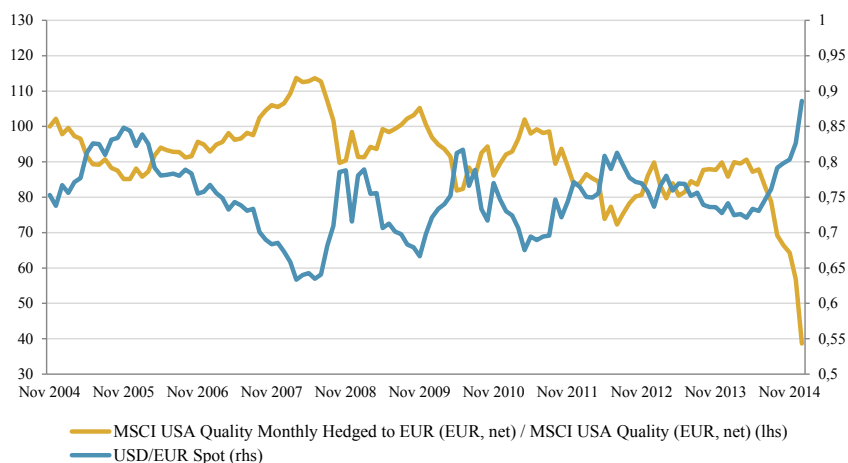
Hedging Currency Risk

With its currency-hedged ETFs of various asset classes, UBS enables investors to continue holding the benchmark stocks while mitigating the intrinsic currency risk. Hence, the investor can still access foreign exposure without taking the downside to unsystematically¹⁰² compensated currency risks.

¹⁰² MacDonald, R. (1988). Floating Exchange Rates: Theories and Evidence, London.

Investors with tightened exchange rate expectations looking for additional return sources can accept the currency risk and switch between currency-hedged and unhedged ETFs. A depreciation (appreciation) of the foreign currency, as shown in Figure 2, leads to an outperformance (underperformance) of the currency-hedged index relative to the unhedged index.

Figure 2: Relative performance of the MSCI USA Quality Monthly Hedged to EUR index to the MSCI USA Quality index in EUR



Source: MSCI, end of month data, Nov 2004 to Jan 2015.

Hedging Model

The quality of the hedge is dependent on the predefined hedging methodology of the underlying index, which should correspond with the ETFs replication method.¹⁰³ With their daily and monthly hedging methodologies, MSCI

¹⁰³ Alexiev, J., Fenty, S., and Moore, J. (2011), »Currency Hedged Benchmark Replication: Challenges and Improvements«, *Journal of Performance Measurement*, Summer 2011, Vol. 15, No. 4, 38–48

offers two concepts which have become the most common hedging models used in the ETF industry.¹⁰⁴

UBS Global Asset Management, as one of the European pioneers in terms of passive index replication, offers a broad range of currency-hedged ETFs which are passively managed and physically replicated. Being directly invested in the underlying components, and with the objective to perform in-line with the index, brings transaction costs in the focus of any currency hedging considerations.

Relation Between Tracking Error And Hedge Frequency

The accuracy of a currency hedge on a foreign currency denominated portfolio is defined by:

1. the frequency a profit or loss resulted if a currency hedge transaction is rolled and
2. the ratio between the Hedge Nominal on t-1 and the Total Net Assets of the foreign currency portfolio on t

$$Hedge\ Ratio_t = \frac{Hedge\ Nominal_{fc,t-1}}{Total\ Net\ Assets_{fc,t}}$$

Hedge Ratio_t: proportion of Hedge Nominal on t-1 and the Total Net Assets on t

Hedge Nominal_{fc,t-1}: the amount in foreign currency sold forward on t-1

Total Net Assets_{fc,t}: the total net assets in foreign currency on t

¹⁰⁴ MSCI, daily and monthly hedging methodology, accessible through their public website (valid hyperlink as of March 2015): http://www.msci.com/eqb/methodology/meth_docs/MSCI_Hedged_FX_Hedge_and_Global_Currency_methodology_book_jul2013.pdf

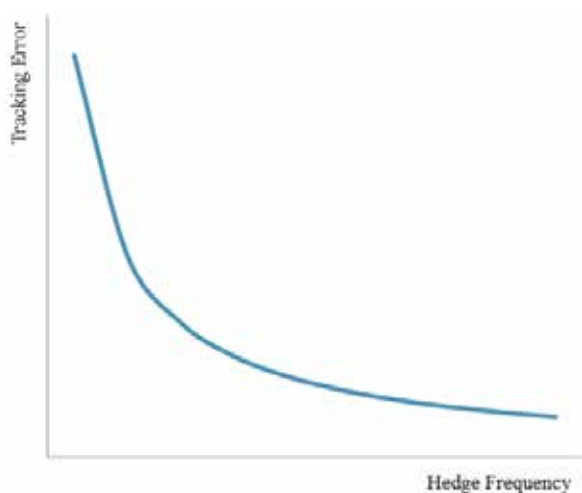
t: the current business day

t-1: last business day of the previous month (hedge frequency = monthly),
previous business day (hedge frequency = daily)

fc: foreign currency is the local currency of the portfolio

The higher the hedge frequency (to the point of continuous hedging) and the closer the hedge ratio to 100% (assuming a fully currency-hedged portfolio strategy), the smaller the volatility of the performance deviation (tracking error) between the currency-hedged portfolio and the parent index in foreign currency (relation shown schematically in Figure 3).

Figure 3: Relation of tracking error and hedge frequency

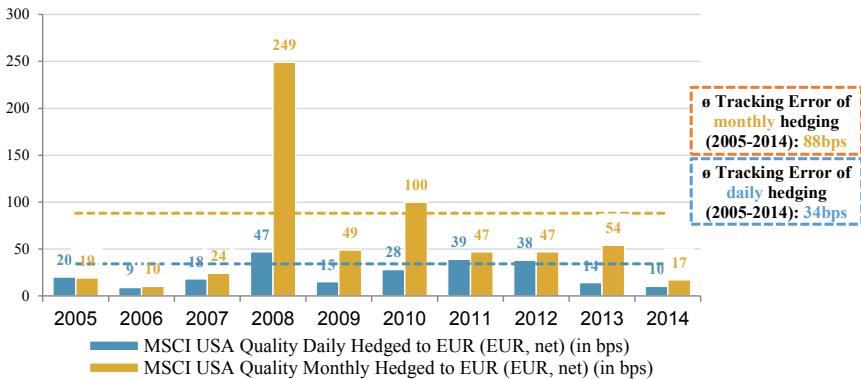


Source: Illustration by the authors.

Figure 4 illustrates the relation between the tracking error and the hedge ratio/hedge frequency using the example of the daily and the monthly EUR hedged MSCI USA Quality index returns. The annualized tracking error,

measured by the volatility of the monthly performance differences over the period December 2004 to January 2015, was 34bps for the daily EUR hedged and 88bps for the monthly EUR hedged MSCI USA Quality index relative to the standard MSCI USA Quality index in USD.

Figure 4: Annualized tracking error in bps of the MSCI USA Quality Daily Hedged to EUR index and the MSCI USA Quality Monthly Hedged to EUR index relative to the MSCI USA Quality index in USD

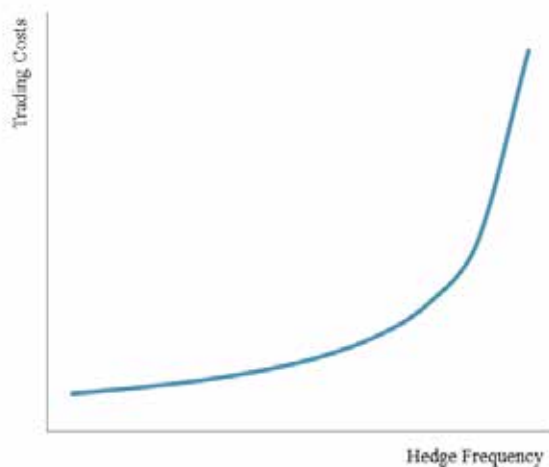


Source: MSCI, end of month data, Jan 2005 to Dec 2014.

Relation Between Transaction Costs And Hedge Frequency

Hedge frequency, and the hedge ratio, highlight the hedging challenges only from an index point of view. Ultimately, the products UBS offers to investors are ETFs which should replicate the referenced index as closely and cost effectively as possible. While the index does not consider transaction costs, ETF trades are, of course, subject to transaction costs and these must be taken into consideration.

Figure 5: Relation of trading costs and hedge frequency



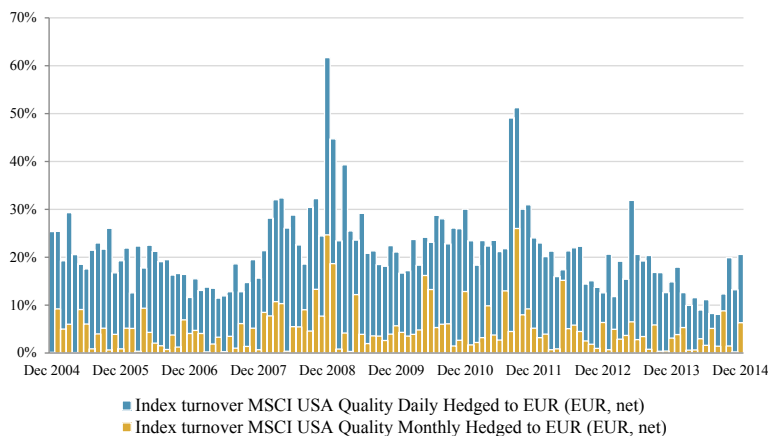
Source: Illustration by the authors.

The higher the hedge frequency, the more frequently currency hedge profits or losses (P/L) have to be crystalized. If a profit is crystalized, the ETF has to invest in portfolio securities to be in line with the index. In the case of a currency loss, the ETF has to divest portfolio securities to compensate for the loss of the currency hedge. As a consequence, and as shown in Figure 5, the higher the hedge frequency, the higher the portfolio turnover rate and hence the higher the portfolio trading costs.¹⁰⁵

Figure 6 presents the monthly index turnover rates as a measurement of trading costs, calculated as the sum of the monthly weighting changes for each security in the underlying MSCI USA Quality Index. The index turnover rate and, hence, the trading costs for an ETF following an index on a daily hedge model, are more than four times higher than for an index based on a monthly hedge methodology over the period December 2004 to January 2015.

¹⁰⁵ Kerzérho, R. (2010), «Currency-Hedged S&P 500 Funds: The Unsuspected Challenges», PWL Capital Inc.

Figure 6: Monthly index turnover in % of the MSCI USA Quality Daily Hedged to EUR index and the MSCI USA Quality Monthly Hedged to EUR index



Source: MSCI, end of month data, Dec 2004 to Jan 2015.

Tradeoff Between A Perfect Currency Hedge And Trading Costs

The investors' goal is a currency-hedged ETF performance which is as close as possible to the performance of targeted index in local currency. But what is achievable on an index level, would be impossible through a physically replicating ETF. MSCI has taken this into account and provides investable indices using both daily and monthly-hedged methodologies.

Providing such an offering, it is left with the ETF provider to decide whether (1) the best practicable currency protection should be achieved (daily hedge) while accepting higher trading costs at the expense of the ETF performance, or (2) whether the best tradeoff between trading costs and currency protection should be achieved (monthly hedge) albeit while accepting a higher tracking error. UBS has decided to structure their suite of currency-hedged ETFs using a monthly hedge approach.

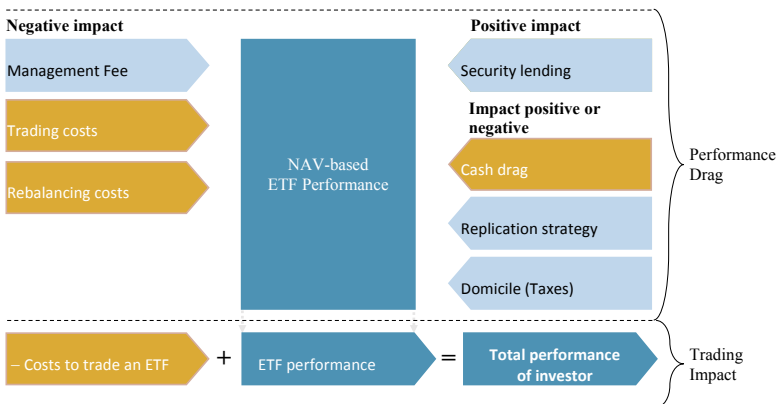
Implementation

With rapid asset growth and increased investor demand, the ETF industry is calling for greater levels of clarity and transparency, along with more robust risk measures. As one of the leading ETF providers in Europe, UBS is concentrating its business model on physical replication wherever this is efficiently achievable. This section focuses on the technical setup which is key to the success of physically replicating ETFs incorporating a monthly hedging feature. The technical product structure, thereby, provides the foundation for an efficient portfolio management setup characterized by high tracking quality at competitive costs. In the following sections, different implementation models are explored whereby master/feeder structures are deliberately ignored. The reasons are:

1. Depending on the master/feeder domicile, both master and feeder fund have to be in different fund umbrellas, which, from an economic standpoint, is quite inefficient.
2. From a tax point of view, the structure is not transparent, as the master fund receives dividends from the feeder instead of from underlying equities.

For physically replicating ETFs, the following major parameters, as shown in Figure 7, determine the total performance of an investor.

Figure 7: Performance drivers for an investor when investing in an ETF



Source: Illustration by the authors.

Focusing on currency-hedged, index investments, the orange arrows in particular are impacted by the technical product structure:

1. *Trading Costs*

The MSCI Hedged Indices are rebalanced monthly on the last trading day of the month when the index will take into account the effect of rolling into new one-month currency forward contracts. The value of a forward is based on the newly determined weight of the currency to be sold for the next month's index calculation. The profits or losses from the currency hedge are realized and evened out by buying (profit) or selling (loss) securities in the portfolio. These monthly recurring FX costs are carried by the fund and reduce the NAV-based ETF performance.

2. *Rebalancing Costs*

During the quarterly, semi-annual and/or annual index reviews, the MSCI indices are rebalanced, e.g. securities are newly included, excluded and their weighting is adjusted. ETFs, as passive index trackers, must adjust their fund holdings in order to stay in line with the underlying benchmark. The costs for these trading activities are carried by the fund and, as a consequence, reduce the NAV-based ETF performance.

3. *Cash Drag*

Cash drag describes a situation in which a portfolio lags behind the index that it is designed to track. Performance drag can occur for a variety of reasons. ETFs keep a small part of their assets in cash, due to round lot constraints. In general, it can be assumed that the higher the total assets of a given portfolio, the smaller the fractional cash and thus the resulting cash drag.

4. *ETF trading costs*

The trading costs are composed of broker fee, bid/ask spread and local tax duties (e.g. Swiss stamp duty). Before proceeding, two important roles in the ETF trading need to be highlighted:

a. *Authorized Participants in the Primary Market*

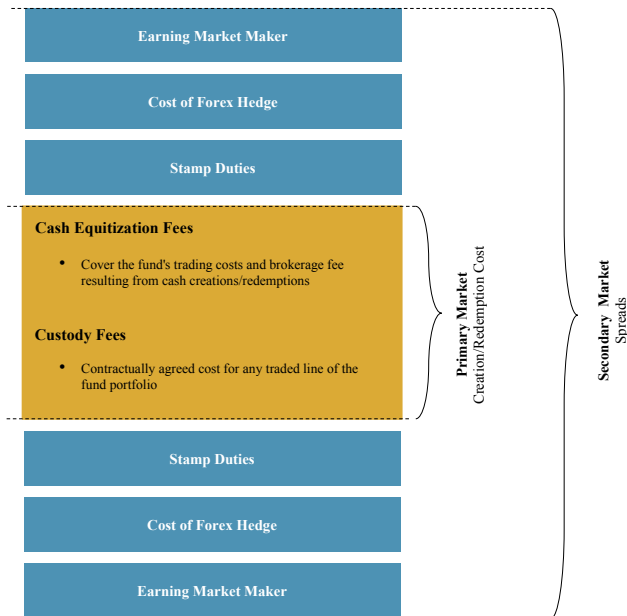
Authorized Participants (APs) are authorized from the ETF sponsor to provide primary market transactions in the ETF, e.g. creating or redeeming ETF shares.

b. *Market Makers in the Secondary Market (Exchange or OTC trading)*

Market makers quote, buy and sell prices for the ETF at a bid/ask spread and, hence, are seen as a liquidity provider to the ETF sponsor. In practice, Market Makers and Authorized Participants are often one and the same legal entity.

As long as the supply and demand of particular ETF shares are balanced, there is no need to create or redeem ETF shares via the primary market. However, if there is an excess demand (supply), shares are created (redeemed). Initially, the Authorized Participant has to pay all the trading costs and fees involved in processing the creation/redemption activity. The recovery of the AP's costs are carried out via the bid/ask spread on the secondary market. Figure 8 illustrates the composition of the bid/ask spread. The AP does not only take into account the creation/redemption costs but also adds a risk premium for the currency risk as well as a profit margin.

Figure 8: Composition of the bid/ask spread of an ETF



Source: Illustration by the authors.

Hedging On Sub-Fund Level

The main quality indicator for ETFs is the tracking accuracy of the indexed investment, represented by the tracking error and the tracking difference. To achieve this, it has been best practice in the industry to structure currency-hedged, physically replicating ETFs as single sub-funds. Using the example of the MSCI USA Quality ETF, as shown in Figure 9 with a sub-fund setup, all the instruments, incl. currency forwards and P/L-related cash, are held in each of the single sub-fund accounts, dependent on the currency to be hedged.

Figure 9: ETF single sub-fund structure



Source: Illustration by the authors.

The main advantage of this setup is that dilutions between the different equity and currency exposure combinations are impossible since the investment structures are completely separated.

But such a setup also incorporates fundamental inefficiencies:

1. *Trading costs*

The trading activities related to FX profits and losses have to be done separately for each of the sub-funds. Netting opportunities between the different hedged currency sub-funds are impossible (one has profit and needs to invest, another one has a loss and needs to disinvest). Custody per line cost has to be paid for each sub-fund, separately.

2. *Rebalancing costs*

The rebalancing has to be performed for every sub-fund, separately. Custody per line cost has to be paid for each sub-fund, separately. The size to be traded is smaller and the conditions in the market are very likely worse for smaller sizes.

3. *Cash drag*

This setup leads to multiple smaller asset exposures as each sub-fund holds fractional cash. Trading in line with the index of smaller sub-funds leads to more fractional cash and, consequently, to a bigger cash drag.

4. *ETF trading cost*

The primary market trading costs drive the secondary market trading costs. Conversions between different (hedged) currency combinations of the same underlying investment exposures is impossible. If the AP wants to convert from (1) MSCI USA Quality hedged to EUR ETF to (2) MSCI USA Quality hedged to CHF ETF, a redemption must be placed for (1), a creation for (2). Both sub-funds require the complete trading activities, that the AP has to recover for both via the bid/ask spread in the secondary market.

5. *Portfolio management efforts*

The Portfolio Manager has to maintain several, separate asset portfolios. This is labor-intensive and error-prone.

Hedging On Share Class Level

Driven by client demand, UBS has also introduced currency-hedged fixed income and equity sub-funds using a share class structure. In general, share class hedging is already well-known and well established in the traditional active funds market. But the setup for an index tracking sub-fund is rather different since the investment objective of the ETF is not to deliver alpha but to deliver beta – a performance as close as possible to the underlying index.

The ETF faces the following challenges:

1. providing exact tracking of the underlying currency-hedged index,
2. handling daily intra-month inflows and outflows (the index does not) and
3. preventing dilutions between the different unhedged and currency-hedged share classes.

The main objective is to bundle the assets of the same market exposure for all the different hedged and unhedged currencies. In order to prevent dilutions, all the currency hedging-related transactions have to be:

1. isolated from the common asset portfolio,
2. performed at share class level,
3. at the cost of the individual share classes.

Separated technical accounts for the individual share classes are introduced, as highlighted in Figure 10 (compared to the sub-fund setup as illustrated in Figure 9).

Figure 10: ETF share class structure



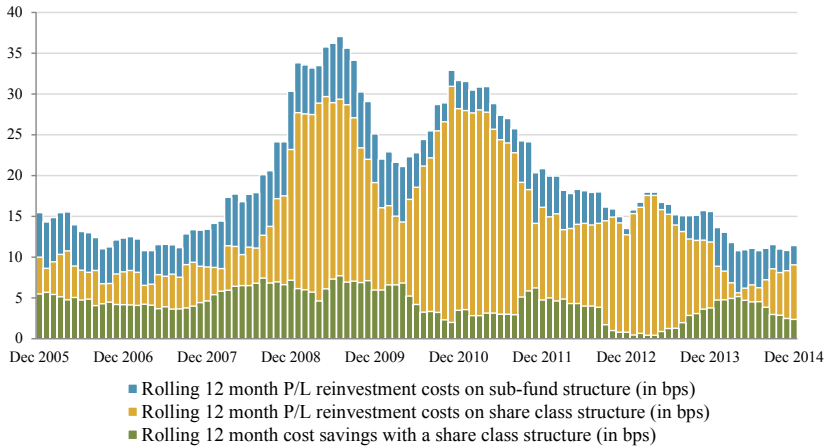
Source: Illustration by the authors.

Compared to the sub-fund setup, this model is considerably more efficient:

1. Trading costs

The trading activities related to currency profits and losses can be netted. The resulting P/L is then invested or divested only once on the common asset portfolio. Custody per line costs also have to be paid only once for the sub-fund. Figure 11 illustrates the rolling 12 month trading costs for the MSCI EMU UCITS ETF due to the monthly P/L reinvestments, assuming five currency-hedged sub-funds and share classes respectively (equal sub-fund/share class asset sizes, currencies USD, CHF, GBP, SGD, CAD and a cash equitization fee of 18bps). For the period Jan 2005 to Dec 2014, the cost savings resulted in over 4bps per year (custody per line cost not included) due to netting opportunities.

Figure 11: Trading cost (monthly P/L reinvestment) comparison between a sub-fund and a share class setup based on the example of the MSCI EMU UCITS ETF



Source: UBS, MSCI, end of month data, Jan 2005 to Dec 2014

2. Rebalancing costs

The rebalancing has to be done only once on the common asset portfolio. As the custody fee is not an asset-based fee but a fee levied per line traded, it also has to be paid only once for the sub-fund, in contrast to the sub-fund structure where the fee applies to each of the different asset portfolios. As a result, the custody fee at the expense of each of the share classes is only $1/n$, where n = no. of share classes in the same sub-fund. In addition, as all share classes contribute to the common asset portfolio, the size to be traded is larger, hence, the probability of better market trading conditions.

3. *Cash drag*

Again, the share class setup leads to one common asset portfolio. Trading in line with the index of a bigger asset portfolio results in less fractional cash and, consequently, in a smaller cash drag.

4. *ETF trading cost*

As previously mentioned, the primary market trading costs essentially drive the secondary market trading costs. Having a share class setup in place, conversions between different currency-hedged share classes of the same sub-fund are supported. Table 1 illustrates this advantage, having an AP who wants to convert in the primary market from:

- a. MSCI USA Quality UCITS ETF (hedged to CHF) A-acc to
- b. MSCI USA Quality UCITS ETF (hedged to EUR) A-acc

Further assumptions:

- a. No. of shares for the conversion out is set to 250,000
- b. Unrealized profit for the MSCI USA Quality UCITS ETF (hedged to CHF) A-acc of 8%
- c. Unrealized loss for the MSCI USA Quality UCITS ETF (hedged to EUR) A-acc of 2%

Table 1: Conversion in the primary market

A	B	D	E	F	G	H	I	J	K	L
Conver- sion	Fund Name / Share Class	NAV in Share Class CCY	No. of shares to convert	Conver- sion amount in Share Class CCY	FX to EUR	Con- version amount in EUR	Unreal- ized FX hedge profit or loss	Invest- ment Ratio	Equity proportion in EUR to be traded	Cash to be traded for a con- version
				D*E		F*G		100%-I	H*J	H-K
Conver- sion out	MSCI USA Quality UCITS ETF (hedged to CHF) A-acc	13,3827	250.000	3,345,675	0.93750	3,136,561	8.00%	92.00%	2,885,636	250.925
Conver- sion in	MSCI USA Quality UCITS ETF (hedged to EUR) A-acc	13,0487	240.373	3,136,555	1	3,136,555	-2.00%	102.00%	3,199,286	-62.731
Total Equity Trading Creation / Redemption (Sum column H)										
Total Equity Trading Conversion (Abs (sum) column L)										313.656

Source: UBS, illustration by the authors.

In the case of a sub-fund setup, the total amount to be traded for both sub-funds would be *EUR 6,084,922*. With a share class setup, only the currency hedging proportion of involved share classes needs to be traded, as both share classes contribute to the same asset portfolio. The amount to be traded in the above example would only be *EUR 313,656*.

Assuming trading cost of 5bps for an MSCI USA Quality portfolio, the costs compared would be:

1. *Sub-fund setup*: 5bps of *EUR 6,084,922 = EUR 3,042*

2. *Share class setup*: 5bps of *EUR 313,656 = EUR 157*

This savings effect depends, on the one hand, on the unrealized currency hedging P/L differential of the involved share classes in proportion to the total flow. The lower the differential, the smaller the trading costs for conversions under a share class setup. On the other hand, the savings depend on the trading costs for the underlying portfolio. The bigger the trading costs of the underlying portfolio, the higher the absolute savings of conversions in a share class setup. Since primary market trading costs are one of the major drivers for the secondary market spreads, the example of share class hedging emphasizes the advantages for authorized participants who can finally offer essential, tighter spreads for hedged share classes compared to currency-hedged ETFs based on a sub-fund structure.

5. *Portfolio management efforts*

The Portfolio Manager has to manage only one common asset portfolio. S/he saves doing a lot of work, and the probability of errors is, essentially, lower.

Conclusion

The mitigation of currency risks as an intrinsic element of a foreign asset portfolio is a major challenge for every passive product team. The decision to go for a specific currency hedge model is, basically, linked to trade-off considerations between the quality and accuracy of the hedge and related trading and rebalancing costs. With regard to technical and regulatory limitations in the implementation of the product structure, there are numerous advantages to setting up the currency hedge on a share class level rather than on a sub-fund

level. Although, this introduces new challenges, such as minimizing the risk of diluting the performance of other share classes in the same sub-fund, in the case of primary market transactions. UBS offers currency-hedged ETFs based on both share class and sub-fund structure for various exposures and asset classes and, thus, will continue to be the provider of choice in this field.

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Prior to his current role he was a member of the passive team. He began his career at UBS Global Asset Management in 1985 as a trainee active European equity fund manager and later became a member of the Quantitative department dealing with indexation and risk measurement.

Ian Ashment is Chairman of FTSE EMEA Advisory Committee, and is also a member of FTSE's Policy Committee and a member of S&P's Global Index Advisory Panel.

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Boriana Iordanova joined UBS Global Asset Management in 2010. Prior to this, she was an equity research analyst at Sanford C. Bernstein covering financials, and an investment banking analyst at Putnam Lovell working on M&A in the asset management sector.

Markus Goetschi started his professional career at Zuercher Kantonalbank as a Business Analyst in the area of electronic sales channels. After completing his studies, he went on to work as Index Analyst at SIX Swiss Exchange in 2004. From 2007 to 2009 he led the index unit and then joined STOXX as part of the transfer of the operative index business, where he focused on the development of new global index concepts. Since 2011, Markus Goetschi has been with UBS Global Asset Management heading the product development team of physically replicated ETFs.

Markus Goetschi completed his studies at the University of Zurich in 2004 with a Master of Science in Computer Science (commercial information technology). In 2008 he also completed a postgraduate training as a financial analyst with the CIIA ® diploma.

Christian Kunth has been a product developer for UBS ETF since August 2009 and is responsible for the development of physically replicated ETFs across all asset classes. In this role, he is in charge of masterminding and implementing substantial parts of product families like the FX hedged UBS ETFs, the UBS precious metal ETFs and the UBS fixed income ETFs.

He joined UBS AG in 1999. In the first years he headed up the development of various IT banking solutions with a focus on internet business.

Before joining UBS AG, he worked as a business information scientist for Mondaine Watch Ltd and Comicro Netsys AG.

Christian Kunth started his carrier as a system engineer at Deutsche Shell AG.

5. Factor Portfolios Put In Practice

A Swiss Pension Fund Case Study

Andreas Toscan

UBS Global Asset Management

The Start

In-depth discussions on the subject of factor investing and smart beta, started in the Swiss pension fund market in the course of the 2008 credit crunch. Beforehand, topics such as smart and alternative beta, as well as their various implementation forms in the Swiss institutional market, were very much on the radar of larger, very sophisticated pensions fund managers, but hardly known by the broad pension fund sector.

The initial client inquiries almost always related to fixed income. The ever increasing sovereign debt levels, particularly in countries in highly developed economic areas and the then prominent PIIGS countries (Portugal, Ireland, Italy, Greece and Spain) prompted Swiss institutional investors to question their standard investment behaviour: as a rule, bond investments in foreign currencies were made in government bonds (less often in corporate bonds), these being indexed or, at least, very close to indexed. The most common indices in this investment segment, however, take a market capitalized approach (for government bonds, it is a debt-weighted approach). Finally, the individual countries are more strongly represented in a benchmark, the more debts they have outstanding.

The consequence of dealing with these systematic challenges and issues, was a growing investment in corporate bonds, a stronger preference for active management over indexing, as well as the increasing consideration of alternative benchmarks, such as GDP-weighted and equally weighted government bond indices.

Gradually, the world of equity investments also caught on to this in the Swiss pension fund landscape. The search for indexing methods, »smarter« than the uncritical implementation of existing market capitalized benchmarks, began. This gradually resulted in the considerable – almost autodidactic – investment

theory training of the Swiss pension fund world. Today, many pension funds and their governing bodies (foundation boards, investment committees, etc.) understand the principles of the smart or alternative beta approaches in the equity area, and inquiries with asset managers in this area are increasing.

UBS Global Asset Management offered complex and tailor-made investment solutions in the area of alternative beta on various occasions in 2014.

A Specific Client Case Study

The following highlights a specific case study of a major Swiss pension fund, which was closely involved with multi-factor investing over the course of last year and considers UBS a preferred partner in this area. The pension fund has assets in excess of 10 billion (in CHF) and insures tens of thousands of active employees and pensioners. The investment strategy provides for investments in global equities (excl. Switzerland) amounting to some 17% of all investments. They have been cost-efficient to date, broadly diversified and predominantly invested on an indexed basis. In 2015, a track record with a multi-factor equity portfolio is to be established for the first time. The initial expected investments of CHF 200 million are to be considered as a starting point, if the – from an academic perspective – attractive properties (back tests, etc.) of the multi-factor portfolio should also manifest in the future.

The mandate specifies the following:

Asset Class:	US equities
Portfolio Size:	CHF 200 million
Benchmark:	MSCI USA (net. div. reinv.)
Implementation:	1) MSCI USA Equal-Weighted 2) MSCI USA Quality 3) MSCI USA Minimum Volatility (Indices are tailor-made and are provided to the clients' specifications by MSCI. All three factors are to be wrapped into one portfolio with half-yearly rebalancing.)
Fund Structure:	Private label fund (wrapping) with an optimal structure for a Swiss pension fund (i.e. exempt from CH stamp tax and relief at source, in terms of US taxes)

The Benefits Of Index Blending

We believe that alternative beta indices, that exploit value, volatility and quality factors, could provide investors with significant benefits. Our detailed proprietary research has shown that, over time, blending alternative beta indices exploiting different equity return factors can provide investors with a more efficient risk/return profile and diversification benefits than simply investing in one. Why? Each factor typically experiences a different performance pattern. For instance, indices with value-based methodology tend to outperform the market in a rising or strong macro environment, whereas indices that exploit quality and low volatility factors tend to be more defensive. We appreciate that each investor's needs are different. This is where we believe our expertise and experience may be beneficial. By working in partnership we aim to deliver a diversified blend of indices to meet the desired objective. This includes not only agreeing the blend, but other factors, such as when to rebalance in order to maintain the agreed target allocation in the most efficient manner possible.

Investment Rationale For The Chosen Three Factors (Favored By The Client)

Equal-weighted:

Small cap stocks outperform large cap stocks: there is a negative relation between stock returns and the company market value. Equally weighting stocks, automatically, gives more attention to smaller-sized companies.

Quality:

High quality stocks outperform low quality stocks: more profitable companies, as measured by accounting data (e.g. profit margin, profit growth), generate higher returns than less profitable companies.

Volatility:

Low volatility stocks outperform high volatility stocks: known as the »low volatility anomaly« because it contradicts the conventional finance wisdom of a tradeoff between risk and return.

What Else?

The Swiss institutional market's (clients and providers) ongoing debate of the subject of factor investing also led to the question of whether it might not also be practical to map factors – i.e. beta sources – in a well-indexed and cost-efficient manner in an equity portfolio, but to actively weight them over time against one another. Studies show that low volatility stocks, for example, are not ideal in all phases of an equity market cycle: in times of markets performing euphorically with a high level of volatility, so-called minimum volatility approaches underperform small-cap or momentum strategies. Therefore, the asset managers who are increasingly in demand are those who are competent in efficiently managing indexed factor components, and who can actively manage their tactical weighting.

Digression: »Straight Alpha« – A Promising Approach To Portfolio Construction

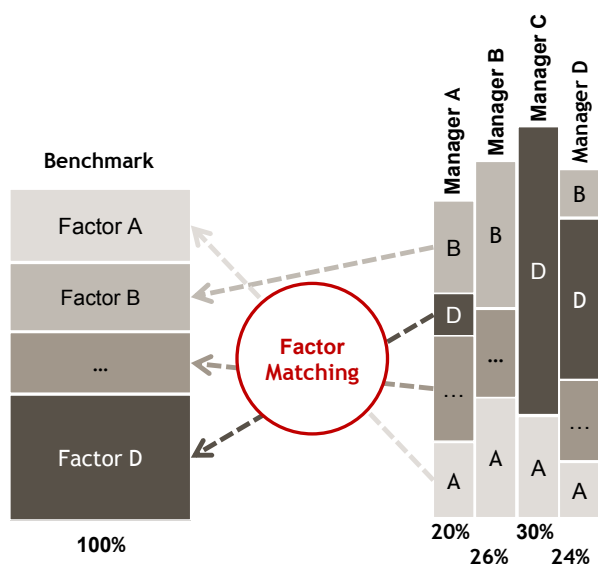
The use of equity factors such as a company's size, the volatility of its shares, its industry or regional exposure, etc., has long been best-practice to create low tracking error index portfolios. However, equity factors can also be of use when constructing customized active portfolios for clients with a specific benchmark in mind.

During the last quarter of 2014, we implemented a portfolio for a large Dutch client using multiple active equity capabilities from various regions (US, Japan, Europe, Emerging Markets) with the objective of outperforming the MSCI AC World (benchmark).

Historically, all managers considered, had a strong track record with high information ratios, outperformance (alpha), established processes and stable teams. Assuming most of that outperformance was skill based, and skill remains constant, the challenge was to combine the managers in a way that would allow us to transfer as much alpha as possible. The solution we found involved using the same equity factors and techniques (factor matching) we

usually use for index funds but this time we tried to reduce the basis risk which we define as the differences in systematic factor exposure between the benchmark and the client portfolio without active bets (see figure 1). A smaller basis risk means more of the outperformance can be directly attributed to manager skill instead of a systematic mismatch between the client's portfolio and its benchmark. We called this approach »straight alpha« and client response has been entirely positive to date.

Figure 1: Factor matching of benchmark and client portfolio



Manager weights get adjusted to minimize a factor mismatch between the benchmark and the client portfolio ignoring active bets.

Source: UBS Global AM. For illustrative purposes only.

An Alternative And More Efficient Approach For Investment Managers To Access Broad Commodities

Andrew Walsh

UBS Global Asset Management

Abstract:

This paper looks at some key considerations for investors when looking to access broad commodities indices via UCITS ETFs. There are some major potential pitfalls to contend with – notably, the varied underlying weightings in different commodities indices as well as the impact of negative roll yield on investment returns. It also highlights the merits of the UBS Bloomberg Constant Maturity Commodity Index as a way to access a more balanced broad commodities exposure and mitigate negative roll yield.

Introduction

Investing in commodities is not as straightforward as with most other asset classes. Accessing commodities poses a number of challenges, most notable is contending with how best to participate in movements in underlying spot prices via the use of futures contracts.

Many investors will be aware of the distinction laid out under the UCITS framework which distinguishes between Exchange Traded Funds (ETFs) and Exchange Traded Commodities (ETCs). That is to say, only a diversified commodities fund with at least five underlying assets being tracked can be

deemed to be an ETF. This means that under UCITS rules, a »gold ETF« is in fact an oxymoron. An exchange traded product tracking a single commodity is in fact merely an Exchange Traded Commodity (ETC) which is, effectively, a debt security issued by an investment bank. This contrasts markedly with a UCITS ETF which has a higher level of security embedded into it. A UCITS ETF's assets are owned and ring-fenced by the fund itself. The fund's assets are not on the balance sheet of the issuer.

Accordingly, this paper will focus on Commodities ETFs and the underlying indices which are tracked by them. Broad commodities indices can have quite varied characteristics and these features can have a major impact on investment returns. Thus, when investing in a commodities ETF, it's very important for investors to look carefully at the underlying constituent weightings and methodologies being employed. Put simply, all broad commodities indices are not created equal.

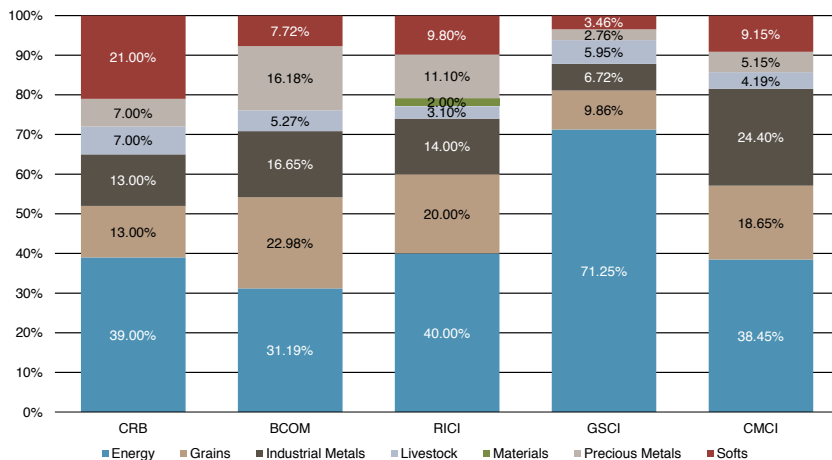
A Look At A Selection Of Index Weights Of Key Commodities Indices

A surprising feature of broad commodities indices is the varied weightings – particularly as regards Energy. Figure 1 shows a selection of five broad commodities indices. The first four are what can be considered »1st generation« indices while the last one, the CMCI, is a »2nd generation« commodities index (touched upon ahead).

Energy makes up over 71% of the weighting of the S&P GSCI while making up just 31% on the Bloomberg Commodities Index. As one can imagine, this can cause the former to behave like a proxy for energy prices in many market conditions. Also of note is the CRB's heavy weighting to Soft Commodities and the CMCI's relatively higher weight of 24% in Industrial Metals.

Figure 1: Index Construction: Target Weights

Commodity Index weights for 2015



Source: Index providers, target weights applicable for H1 2015.

RICI: Rogers International Commodity Index, BCOM: Bloomberg Commodity Index, CMCI: UBS Bloomberg CMCI Composite Index, GSCI: S&P GSCI, CRB: Thomson Reuters/CoreCommodity CRB Index.

These differences occur as a result of index providers' methodologies. If for example worldwide production is being used to determine target weights, this would tend to produce heavy weights towards the energy sector. However, if Consumption, Economic Weights (e.g. CPI) and Liquidity factors are also used, this can produce a rather different and, arguably, a more balanced exposure to broad commodities.

In summary, it is worth investigating what's »under the bonnet« in terms of weightings, as the performance will be dictated to a great extent by this. Clearly, the S&P GSCI's performance in the past year has suffered to a greater extent due to its high energy tilt.

The Fundamental Problem Faced By Commodities Indices – Negative Roll Yield

In addition to the UCITS diversification rules highlighted above, UCITS eligibility rules in the context of commodities investments dictates that direct investment in physical commodities is not allowed. Neither is investing directly in commodities futures. Thus, the only way to access them under UCITS guidelines is via investing in a wrapper (such as an ETF) which tracks a Commodities index through the use of derivatives, such as swaps. The index, itself, will be made up of a basket of commodities futures which are selected, weighted, rolled and traded according to a specific methodology as defined by the index provider.

Investment returns delivered by a commodities index are primarily made up of the price movements of the underlying commodities it holds, as well as the »roll returns« resulting from the ongoing requirement of having to roll the futures contracts to keep the ongoing exposure to each of the respective underlying commodities. The latter is in fact a considerably more important contributing factor to performance than many investors may appreciate. Some investors will recall the extraordinary circumstances in 2009 where commodity prices on average were up about 50%, but many investors in broad commodities indices were only getting returns of about half that. This was of course directly related to a very damaging negative roll yield scenario.

A Simple Explanation Of Roll Yield And Some Solutions To Help To Reduce It

The very nature of the index gaining exposure to these commodities via futures markets means that a major feature of the performance of that index will be dictated by how the rolling of these futures contracts is managed. To maintain exposure to a specific point on the commodities (e.g. copper) futures curve there is ongoing requirement to roll futures exposure as time passes. The rolling process involves selling the soon-to-expire futures contract and buying another later dated futures contract to keep exposure to the commodity. If the later dated contract being bought has a higher price (for simplicity's sake, let's say \$105/ton) than the one which is about to expire (e.g. \$100), this produces

negative roll yield which can, clearly, get quite expensive for investors if this persists. Selling low and buying high was never a very successful investment technique, after all.

Backwardation is the opposite (where the price of the later-dated contract is lower than the current contract which needs to be sold). This scenario would, of course, produce positive roll yield as investors would benefit from being able to sell the expiring copper futures contract at a higher price than the later-dated copper contract they are buying to keep their exposure. Generally, across a wide range of commodities, there will be more constituents in contango than in backwardation. In fact, from 1999 to 2014, of 24 commodities, only four (Brent Crude, Nickel, Copper and Soybeans) had average roll returns which were in positive territory. While the majority of commodities had average roll returns over this period of between 0%–15%, Natural Gas had an average negative roll return of -31%¹⁰⁶ Overall, 1st generation commodities indices have consistently delivered negative Sharpe Ratios.¹⁰⁷

When we talk about a curve being in contango, or backwardation, it's important to understand that in practice, the shape of these curves are not simply linear. Backwardated curves typically reflect a situation where there is a supply shortage or some sort of negative shock to supply which has caused the shorter-dated part of the futures curve to be pushed up higher relative to the longer-dated part. Additionally, it's worth noting that the future curve is not a prediction of where commodity prices will be in the future, but rather what an investor is willing to pay today for delivery of those contracts in future.

2009 exemplified the rather damaging consequences of a broad commodities market which had about 80% of its constituents in contango. This resulted in a situation where spot prices of broad commodities were up + 45%, but the returns of many 1st generation broad commodities indices were only delivering returns of about 20%-25%. Another consideration of this issue is the steepness of futures curves. In addition to the extensive percentage of individual commodities in contango in 2009, there were many which were in steep contango (so, for example, selling the current expiring futures contract

¹⁰⁶ Source: Arnott 2014.

¹⁰⁷ Source: Miffre.

at \$100 and having to buy the next futures contract at, say, \$115). Clearly, such a scenario could easily keep many investors from feeling comfortable with investing in commodities via products which, ultimately, depend on futures contracts to deliver their exposure.¹⁰⁸

The fundamental issue is that many 1st generation commodities indices have a number of shortcomings. They are only participating in the rolling process in the »congested« front month part of the curve (with everyone selling their expiring contracts and trying to buy the next available contract within the same, specific roll period – and there are few market participants on the other side). This exacerbates the contango situation. Miffre noted: »Broadly speaking, 1st generation commodities indices do not pay much attention to the fundamentals of backwardation and contango«.¹⁰⁹

This has led to index providers looking at new methodologies to gain access to commodities via the use of futures. This can include:

- Deferred Rolling: Similar to standard 1st generation, but rolling contracts with longer-dated maturities.
- Optimized Rolling: Where the index provider actively seeks contracts that will improve the roll yield by incurring fewer congestion-related costs.
- Constant Maturity: A rules based approach which rolls smaller amounts on a frequent basis and thus maintains a consistent contract maturity. In addition to reducing negative roll yield, the market impact is considerably lower.

¹⁰⁸ This is, of course, a simplistic explanation of the shape of commodities futures curves. In reality, these can be in various shapes. for example, an upward sloping contango shape in the first three months due to perceived short-term supply shortages of the commodity, but thereafter, a downward sloping backwardation pattern going out to the three-year futures contract.

¹⁰⁹ »DBLCI« is the Deutsche Bank Liquid Commodity Index, »JPMCI« is the JP Morgan Commodities Index, »MLCX« is the Bank of America Merrill Lynch Commodity Alpha fund. The »CSCI« is the Credit Suisse Commodities Index.

The Case For The CMCI Composite Broad Commodities Index And UBS ETF Tracking

The UBS Bloomberg CMCI index («CMCI») is a 2nd generation commodities index which uses a »Constant Maturity« (the CM in CMCI) concept which has been proven to deliver better performance than traditional 1st generation broad commodity indices (such as the S&P GSCI and the Bloomberg Commodities index) yet with lower levels of volatility. The constant maturity concept means that small amounts of each of the various commodities are being rolled, continually, thus, avoiding the »congestion« associated with rolling futures contracts at predetermined days, each month or quarter. The CMCI represents a modern investment benchmark, tracking 26 commodities, which reflects the real nature of commodity markets.

The CMCI methodology is distinct from 1st generation indices in that the underlying futures contracts are bought across the various commodities at five different points in the various curves (3m, 6m 12m, 2yr and 3yr subject to liquidity). This allows for more balanced participation in the curve. Thus, it means accessing the parts of the curve which are in backwardation, not just in contango. As highlighted previously, futures rolling in a market in contango would get very expensive and exacerbate the negative roll yield. Figure 2 shows the key characteristics of the CMCI vs. traditional commodities indices.

Figure 2: Roll Mechanism of Constant Maturity

Example of UBS Bloomberg CMCI

UBS Bloomberg CMCI has been developed in order to deal with this problem of negative roll yield ...

... the innovative methodology involving daily rolling and tenor diversification provides enhanced exposure to the commodity markets.

UBS Bloomberg CMCI

- **Diversified along entire curve**
Advantage: diversified exposure, no congestion, access to a less volatile part of the forward curve
- **Continuous daily roll**
Advantage: lower daily trade volumes, less impact on market, no »announced« transactions
- **5 constant maturities:** 3 months, 6 months, 1 year, 2 years and 3 years



First Generation Indices

- **Front month futures only**
Problem: congestion, most volatile part of the curve, which historically exhibits more contango and less backwardation
- **Punctual roll**
Problem: high volumes, »announced« transactions



Source: UBS AG

What really matters is that this methodology delivers superior results. Since its inception in 2007, the CMCI Commodities index has, healthily, outperformed 1st generation indices, and done so, with lower levels of volatility. For example, the CMCI Composite has outperformed the S&P GSCI by more than 35% over that period. Volatility levels in the CMCI were 17.2% while the S&P GSCI stood at 23.8%.

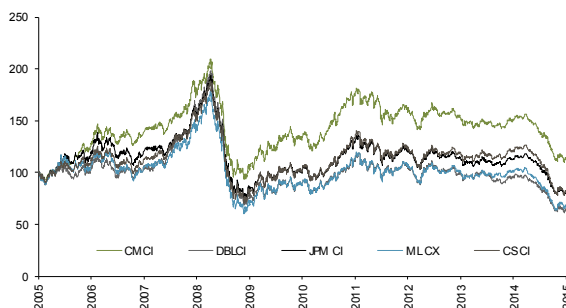
Against the Bloomberg Commodities Index («BCOM»), the CMCI has outperformed by 4.4% on an annualized excess return basis since its inception (Jan 2007). Of this outperformance, approximately 90% has been driven by the daily rolling and constant maturity effects of the CMCI methodology. The remainder is attributed to the difference in weights between the two indices.

Even against other 2nd generation indices such as the DBLCI, JPMCI, MLCX and CS CI 4, the CMCI has delivered better risk-adjusted returns with a Sharpe Ratio of +0.14. Figure 3 summarizes the performance characteristics of the CMCI vs. these peers.

Figure 3: Roll Mechanism of Constant Maturity

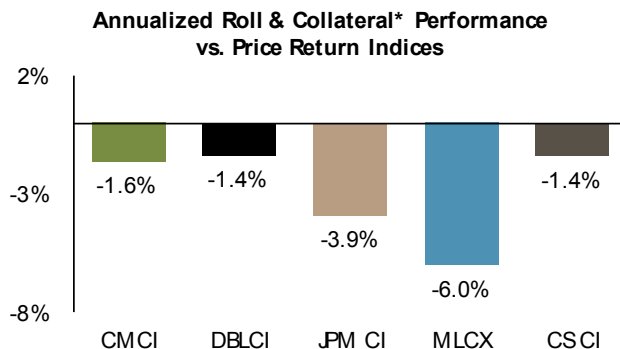
Example of UBS Bloomberg CMCI

	CMCI	DBLCI	JPMCI	MLCX	CS CI
Performance p.a.	1.1%	-4.6%	-2.3%	-4.0%	-2.2%
Volatility p.a.	17.2%	21.7%	19.0%	22.0%	20.4%
Sharpe Ratio	0.06	-0.21	-0.12	-0.18	-0.11



Source: UBS AG

Figure 4: Annualized Roll & Collateral* Performance vs. Price Return Indices



*Collateral refers to 91 day US Treasury Rate earned on the investment.

Conclusion

For investors who are looking to access the broad commodities market via a UCITS ETF wrapper, it's important to look carefully at both the underlying constituent weighting makeup of the index which is being tracked. However, it is equally important to consider how the underlying index manages the way the commodities futures contracts are accessed. The impact of negative roll yield can create a sizeable divergence between the performance of spot prices and that delivered by the futures market. The 2nd generation UBS Bloomberg CMCI Commodities Index and the UBS ETF which tracks it, can help investors access a balanced broad UCITS commodities portfolio in a more efficient manor than ETFs tracking more traditional broad commodities indices.

Italian Portfolio Managers Using Passive Factor Tilts

Simone Rosti

UBS Global Asset Management

The Italian ETFs market has been growing a lot since its launch in October 2002. At the end of February 2015 there were more than 40 bn EUR invested in these products on Borsa Italiana. In 2014 net new flows in ETFs from Italian clients reached almost 8 bn EUR, whilst in the first two months of 2015, more than 3 bn EUR have been attracted by ETFs listed on Borsa Italiana.¹¹⁰

ETFs have becoming more and more popular in Italy but despite a strong interest by private investors, the market is still very much institutional, with funds, funds of funds, discretionary portfolios and insurance products still being the largest ETFs users.

As Italian portfolio managers of the above-mentioned products have been the first users of Exchange Traded Funds, we have asked three of them to highlight the usage of ETFs in their portfolios, mainly focusing how do they utilize alternative beta products and what is their perspective of this »emerging« frontier of ETFs.

The Experience Of Eurizon Capital SGR

*Alberto Luraschi, Portfolio Manager,
Global Strategies and Total Return*

What products do you manage and what is your investment policy and philosophy?

At Eurizon Capital SGR, I manage a number of risk-budget flexible mutual funds in the »Global Strategies & Total Return« sector. The management goal

¹¹⁰ Source: Borsa Italiana, data as of end of February 2015.

is to maximize the portfolio's absolute returns within the confines of the risk budget, recognized as 99% VaR (value at risk) over one month. Market investment is made with no binding benchmark through flexible management designed to smooth out market cycles and reduce drawdown. The possibility of losses remains within pre-set levels, since maximum risk is defined in advance and does not change over time.

Do you use alternative/strategic beta ETFs in your portfolios? How and to what end?

The decisive factor is not the instrument but the diversification of investments within the limit of overall risk. Other elements to keep in mind include market conditions, strategy and dynamic asset allocation, on the basis of which, through an integrated investment process, we determine the asset class in which to invest and what instruments are best suited to our decisions. Alternative/strategic beta ETFs are used less for increasing performance than for improving risk/return (the Sharpe Ratio). Although quantitative models do provide some forecasting ability, strategic/alternative ETFs remain passive instruments that replicate an index with no emotional stake and therefore none of the idiosyncratic risk entailed in active investment.

How do you view the advent of ETFs in factor indexes? Do you consider them useful tools in management?

They increase the choice of instruments available to diversify investment in terms of nominal composition and risk. Although the investment horizon is not short-term, the current low interest rate environment points to greater use of factor tilts to more effectively remunerate risk. Always bear in mind that market betas are »different«: they are not alphas, but they have biases. They could be more useful if we had greater information symmetry, since today many people take the factor/strategic/alternative indexes together as one single »smart beta«, but the risks are neither identical nor smart.

In the future, will they replace core capitalization indexes in portfolios or will they be complementary?

They will likely be complementary, although at the end of 2014 they made up less than one-fifth of ETPs (Exchange Traded Products). Almost certainly ETFs will contribute to the development of these indexes, which have been much more extensively researched for over 20 years (Fama and French in 1993; while the first smart beta ETF was listed in the USA in June 2006). For anyone managing risk-budget multi-asset products, it will be of utmost importance to take advantage of the greater number of possibilities to monitor overall portfolio volatility. Such volatility is influenced in part by invested liquidity and the appeal of the instruments. The deciding factor in using any given instrument should always be dictated by investment optimization and not by what is currently »in fashion«.

The Experience Of Symphonia SGR

Andrea Dolsa, Co-CIO

What products do you manage and what is your investment policy and philosophy?

Direct management in terms of the portfolios at Symphonia SGR concerns mainly two single manager hedge funds and two multi-asset total-return products.

With respect to hedge funds, the Arbitrage fund has been operating for ten years and is one of the first single-manager funds under Italian law. As the name implies, the fund's strategy is to take advantage of price imbalances within the same asset class or among different asset classes. Convertible arbitrage is one of its most prevalent strategies, but over the years the fund has also taken up positions more typical of a macro fund. A substrategy is then implemented through the second hedge fund, called Long/Short Equity.

As regards total return, Symphonia has two products: *Patrimonio Reddito*, which is the company's flagship fund (maximum equity exposure 30%), and *Patrimonio Attivo* (maximum equity exposure 70%). The funds' investment philosophy is to seek out, on the basis of macro-fundamental analyses, one or more asset classes able to guarantee a positive result in a relatively short time period using stop/loss and/or hedging techniques to generate absolute return, while seeking to minimize volatility across the portfolio.

Do you use alternative/strategic beta ETFs in your portfolios? How and for what purpose?

Some macro ideas implemented in the fund in reality have always been mini alternative/strategic beta portfolios. For example, one classic equity investment over the years has been to trade on a basket of stocks sensitive to the euro/dollar exchange rate or shares with higher dividends or related to the internal consumption of a particular geographic region. For some of the funds we manage, stock-picking is not the aim, but we are trying to capture the macro idea behind it. In this sense, the growth in the range of ETFs has enabled us to implement our ideas quickly, ideas that before we had to build up by screening the investable universe and extracting stocks individually. Moreover, the biggest plus we have found about these products is their incredible liquidity and hence the short time necessary to implement or liquidate an investment idea.

How do you view the advent of ETFs in factor indexes? Do you consider them useful tools in management?

In this specific case, factor indexes are extremely useful because they provide a different way of seeing the equity market, a vision that for too long has been limited to replication of the major indexes. More specifically, in some portfolios they are well-suited to implementing short-term ideas, in others as a core holding since they may perfectly embody the spirit of some funds (low volatility ETFs, for example) to which other investments can be added. In our view they are also very useful in portfolio management because they reduce the number of trades and focus the portfolio.

In the future, will they replace core capitalization indexes in portfolios or will they be complementary?

They will be complementary. Indeed, core capitalization indexes, despite many distortions caused by the excessive weight of some blue chip stocks in different national indexes (even Apple on the SPX), at this stage remain unrivaled when it comes to demonstrating stock market performance in numerical terms. Even various corrections that have been proposed over the years, such as limiting weightings, have not had their hoped-for success, in part for historical and reputational reasons (such as the continued use of the Dow Jones Index). For these reasons, and also because they do not include all of the major stocks for a given market and the financial culture is not sufficiently widespread, factor indexes will not likely, within a reasonable amount of time, be able to supplant the major capitalization indexes. But they could be placed alongside such indexes, at least in the specialized press.

The Experience Of Ersel Asset Management

Gianluca Oderda, Head of Quantitative Investments

What products do you manage and what is your investment policy and philosophy?

Since 2012 the Ersel Group decided to complement traditional discretionary approaches to the management of total return products with systematic approaches, where all investment decisions are taken based on algorithms, rules and models.

In November 2012 we launched Leadersel Diversified Strategies, a multi-assets, multi-strategy fund, whose objective is to deliver positive absolute returns, providing resilience in adverse market environments.

The fund has a target volatility of 5% p.a., with a target annualized return objective of 3% above the return of overnight deposits in Euro, parametrized by the EONIA Index. The return objective is relative to an investment horizon, corresponding to a market cycle.

Risk Cluster	Traditional Risk Premium	Alternative Risk Premium	Alpha	
			Systematic	Portable alpha
Rates	Bund US Treasury Note Gilt	G4 Interest Rates Global Carry Premium G4 Interest Rates Value Premium	Selection of strategies based on proprietary models and on proprietary model and on proprietary/ third party-developed trading rules	Selection of portable alpha strategies, both internally and third party-managed
Credit	EM Hard Currency Bonds Corporate HY US Bonds Corporate EUR US Bonds			
Equities	Developed Market Stocks Emerging Market Stocks	»Size« Risk Premium »Value« Risk Premium »M&A Arbitrage« Risk Premium		
Real Assets	Energy, Agricultural Commodities, Industrial Metals, Gold, Real Estate Investment Trusts	Commodity Supply Risk Premium Commodity Lease Rate Premium		
Currencies		G10 Carry Premium G10 Value Premium EM Carry Premium		
Volatility		Equity Volatility Carry Premium		

Figure 1 Investment strategy taxonomy in the Leadersel Diversified Strategies Fund, the innovative multi-assets, multi-strategy fund of the Ersel Group

The fund leverages on three different performance engines, which are synthesized in the matrix shown in Figure 1:

1. a diversified selection of directional risk premiums, covering all liquid asset classes: stocks, government bonds, credit and real assets. Within

each risk premium cluster, we further diversify, either geographically or by sector. The allocation to risk clusters is based on the risk parity principle: the risk budget allocated to the traditional risk premium strategy is evenly split among risk clusters, and then again evenly among financial instruments. A timing engine is applied to each single financial instrument used, providing us with a »risk on«-»risk off« signal: should any given asset enter an adverse market environment, its risk parity weight would be allocated to cash. Our timing engines allow us to avoid the risk of remaining stuck with some of the intrinsic biases of the risk parity allocation approach. For instance, should interest rates suddenly start to rise, our timing engines should timely reduce the allocation to government bonds. In a nutshell, the traditional risk premium strategy can be seen as a dynamic balanced portfolio.

2. a diversified selection of alternative risk premium exposures, extracted from stocks, commodities, currencies, rates and volatility. All alternative risk premium exposures are obtained by going long and short different segments of the same underlying market, without remaining directionally exposed to the market itself. On the equity side, for instance, we expose ourselves to the value¹¹¹, size¹¹² and low volatility¹¹³ risk factors by buying a selection of smart beta indices and by shorting the corresponding cap-weighted indices. For instance, exposure to the value risk factor is obtained by means of a long exposure to a set of quality and high dividend market indices and by a short exposure to the corresponding cap-weighted indices.

On the commodity side we aim to extract supply risk premium by going long a selection of backwardated commodities and by shorting a diversified commodity index, while remaining sector neutral. Commodities

111 See for instance Fama, E. F. , and French, K. R. »The Cross Section of Expected Stock Returns.«, *The Journal of Finance*, 47 (1992), 427–465.

112 See for instance Banz, R. W. »The Relationship between Return and Market Value of Common Stocks« *Journal of Financial Economics*, Vol. 9, No. 1 (1981), 3–18.

113 See for instance Oderda, G., Berrada, T., Messikh, R. J. and Pictet, O. (2015), »Beta-Arbitrage Strategies: When do They Work, and Why?«, *Quantitative Finance* 15, 185 {203; or, alternatively, Frazzini, A. & Pedersen, L. H. (2014), »Betting Against Beta«, *Journal of Financial Economics* 111, 1–25.

with high supply risk tend to be market backwardated: the price of nearby futures tends to be higher than the price of futures farther away on the curve, providing investors with the benefit of a high roll yield.

On the FX side we have both carry and value strategies active on the universe of G10 and emerging market currencies. Carry strategies acquire long exposure to highly remunerated currencies, while funding themselves with low interest rate currencies. Value strategies, on the contrary, compare market exchange rates with fair value levels, based either on a model or on purchasing power parity considerations.

The same kind of carry vs. value paradigm is replicated on the rates side. Carry strategies get a long exposure to the 10Y bond futures corresponding to the steepest yield curves, and a short exposure to the 10Y bond futures corresponding to the flattest or inverted yield curves. Value strategies, on the contrary, aim to select points on different yield curves characterized by the highest roll yield potential over one month.

Finally, on the volatility side, we invest a very limited risk budget into a selection of volatility carry strategies aiming to extract the positive differential between implied and realized volatility. As in the case of traditional risk premium, each alternative risk premium exposure is timed with »risk on«-»risk off« signals, based on a variety of trend following and implied volatility-driven indicators. In fact we are aware of the fact that also alternative risk factors are subject to market cycles. The timing signals are designed to help us avoid being exposed to alternative risk factors which are not remunerated in a given phase of the cycle.

3. a module of alpha strategies, including proprietary systematic strategies based on internal research, and, to a more limited extent, third party systematic strategies (offered mainly by investment banks), and portable alpha strategies, extracting value from both Ersel and third party-managed products.

Allocation to the three performance engines is done based on risk budgeting considerations. Each one of the above three modules is assigned a maximum annualized risk budget, defined as a contribution to annualized volatility.

A maximum risk budget of 2.5% p.a. is allocated to the alternative beta module. The portfolio of alpha strategies receives a maximum risk budget allocation of

4% p.a., whereas the traditional risk premium module has a maximum risk allocation of only 2% p.a., since we prefer to limit the directionality of our fund.

Each strategic risk budget is then divided among the different risk clusters and financial instruments according to the nested risk parity principle explained above. The »risk on«-»risk off« timing signals are systematically applied (weekly or daily) to switch off unwanted risk factor exposures.

Do you use alternative/strategic beta ETFs in your portfolios? How and to what end?

In our portfolios we use some of the smart beta indices available in the market in order to extract equity-related alternative risk factors.

The availability of smart beta ETFs has been instrumental for the design of the alternative risk premium equity strategy which is used in our Leadersel Diversified Strategies fund.

In 2013 we conducted an extensive research study on the indices underlying all the available smart beta ETFs.

Knowing that smart beta indices exploit a combination of alternative risk factors in the equity space (size, value, low risk, etc.)¹¹⁴, and that they need to be sufficiently liquid, in order to be investable by ETFs, we thought that a possible way to extract equity alternative risk factors in a low-cost, liquid fashion was to acquire a long exposure to a selection of smart beta ETFs – or directly to their underlying reference smart beta indices¹¹⁵, shorting the corresponding cap-weighted indices in order to hedge away market risk.

¹¹⁴ See for instance Chow, T., Hsu, J., Kalesnik, V. and Little, B.M., »A survey of alternative equity index strategies«, Financial Analyst Journal, (2011), 67(5), 37–57.

¹¹⁵ In our fund we are subject to the 10% investment limit in other UCITS. Furthermore, we prefer to acquire exposure to smart beta indices without capital absorption. For this reason we always invest in smart beta indices via total return swaps. By doing so, we also reduce cost, since we do not have to pay the management and administrative fees we would incur into, should we invest in smart beta indices via ETFs.

In a nutshell, instead of implementing a portable alpha strategy, whose typical objective is to extract an active manager's alpha generation potential relative to a given benchmark, we aim to extract the combination of alternative risk premiums, which are responsible for the long-term outperformance of smart beta indices relative to their cap-weighted analogs, and to transport it into our fund.

We group smart beta ETFs by underlying index methodology, as shown in Figure 2.

Methodology	Smart Beta Index Principle	Selected Index Families
Diversification	Weight Assets equally	
Risk Parity	Weight assets in such a way that their contributions to total risk are equal	Stoxx
Inverse Volatility/ Variance	Weight assets according to the inverse of volatility (risk parity with equal correlations)	S&P MSCI
Minimum Variance/ Volatility	Lowest risk point on Markowitz's efficient frontier	Ossiam Stoxx MSCI
High Dividend	Weight assets according to dividends	Stoxx (DivDAX, DAXPlus) S&P Dividend Aristocrats Dow Jones Select Dividend Stoxx MSCI
Fundamental/ Quality	Weight assets according to a blend of fundamental measures	RAFI MSCI
Momentum	Weight assets based on six- and twelve-month momentum scaled by volatility	MSCI

We distinguish between:

- high dividend indices: large index families in this space are represented, for instance, by the S&P Dividend Aristocrats, and by the Dow Jones Select Dividend index series. These two index families weigh constituents

by dividend yield, after screening the universe based on dividend sustainability over time, as well as liquidity considerations

- low risk indices: in this space we have both minimum variance indices provided by Ossiam and Stoxx, and minimum/low volatility indices provided by MSCI and S&P Dow Jones. The nature of the indices varies, from simple indices – which weight constituents by the inverse of volatility – to indices employing optimizers from risk model providers in order to define the weighting
- quality and fundamental indices: in this domain the largest index family is the RAFI index series, which uses the audited dollar values of cash flow, book value, total sales and gross dividends to derive each constituent index weight
- diversification indices: in this space we consider a selection of plain vanilla equally weighted indices
- momentum indices: in this domain we consider the MSCI Indices, weighting assets based on six- and twelve-month momentum scaled by volatility

By performing a quantitative assessment of the strategy which is long each given smart beta index and short the corresponding cap-weighted index, we try to isolate indices which might be of interest for the purposes of alternative risk premium extraction. Depending on the smart beta index, the short position in the corresponding cap-weighted index is beta-adjusted, to avoid over or under hedging directional market exposure.

Our aim is to diversify both by underlying index methodology and by regional coverage. In fact the behaviour of any alternative equity risk factor in different regions of the world could be subject to non-synchronous cycles, making regional diversification beneficial.

Our primary criteria for selecting the smart beta indices of interest are related to the robustness of the smart beta index methodology and to the possibility of hedging the corresponding market risk exposure.

This means that we tend to avoid indices whose methodology is not fully transparent, or indices which might be too complex to hedge.

At the moment, in the equity component of our alternative risk premium module we have exposures to high dividend indices, low risk indices and diversification indices, hedged with their cap-weighted analogs.

How do you view the advent of ETFs in factor indexes? Do you consider them useful tools in management?

For each alternative beta index group which we currently use (high dividend indices, low risk indices and diversification indices), by using the Carhart four-factor model, we can identify the sources of outperformance relative to cap-weighted indices as exposures to the value and size factors. The risk-adjusted alpha is hardly different from zero, confirming that the outperformance comes entirely from alternative risk factor exposures. Nonetheless, the smart beta indices hedged with cap-weighted index futures represent for us an efficient and low-cost way to access the value and size premiums, because direct replication of pure style indices can be impractical and costly.

The availability of factor indices in the market might be interesting for us to invest in »pure factors« (for instance, to invest in the pure Fama–French factors), although »pure factors« might be more research rather than investability-oriented, given their potential implementation costs.

»Pure factors« will certainly be interesting from an analytical perspective, to understand factor exposures of any of our strategies going long smart beta indices – defined over universes for which liquid tradable futures are available – and hedging market risk with cap-weighted index futures.

In the future, will they replace core capitalization indexes in portfolios or will they be complementary?

We do not think that smart beta indices will substitute cap-weighted market indices in portfolios.

Smart beta indices, in our opinion, will contribute to make the asset management industry more efficient. Whereas in the past the entire results of active portfolio management were attributed to the skill-related alpha component of performance, it is gradually becoming clear that alternative risk factors are responsible for an important portion of such results.

Smart beta indices and ETFs will increasingly provide investors with low cost exposure to combinations of alternative risk factors and will contribute to lower the level of management fees charged by the active management industry.

They will hardly replace cap-weighted indices as market benchmarks.

In order to be eligible for benchmarking purposes, an index should be representative of a given universe and investable. Smart beta indices are certainly investable, but, due to their small cap size bias, questions might be raised about their representativeness.

A Need For Factor Allocations: A Showcase Of A German Insurance Company

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Michael Plönnes*

UBS Global Asset Management

The Institutional Framework For Insurance Companies

Under the existing regulatory system of the German Insurance Supervisory Law (VAG), in combination with the Investment Directive (AnlV), insurance companies are allowed to invest up to 35% of their assets in stocks and subordinated debt¹¹⁶. Historically, however, the share of assets invested in equities has been significantly lower than that. Over the course of the last ten years, the stock market exposure of insurance companies has declined even further. The main trigger for this was the market turmoil caused by the financial crisis of 2008 and the European debt crisis that reached its first peak in 2011. Since then, there have been no significant reallocations of funds to the stock market, despite a multi-year bull market that is still ongoing at the time of writing. According to the German Association of Insurance Companies (GDV), primary insurers, on average, had invested only 3.4% of their assets in equities at the end of 2013 – not even a tenth of what would have been possible in theory. The biggest part of equity investments, 3.2%, was held through mutual funds. Direct investments in stocks made up only 0.2% of assets.¹¹⁷

¹¹⁶ AnlV § 3 (2) No. 10

¹¹⁷ Gesamtverband der Deutschen Versicherungswirtschaft: Statistisches Taschenbuch der Versicherungswirtschaft 2014, 28

With the advent of Solvency II – the EU directive has to be implemented by January 2016 – things are certainly going to change. It is, however, not yet clear in which direction, exactly. The strictly rule-based system of Solvency I, with the fixed asset-class limits mentioned above, is going to be replaced by a more flexible risk-based approach in order to ensure that economic capital is measured accurately at all times. Or, as the European Commission puts it: »These new solvency requirements will be more risk-sensitive and more sophisticated than in the past, thus enabling a better coverage of the real risks run by any particular insurer. The new requirements move away from a crude ›one-model-fits-all‹ way of estimating capital requirements, to more entity-specific requirements.«¹¹⁸

How the new regulatory framework will affect the asset allocation of insurance companies remains to be seen. It stands to reason, however, that equity allocations will not rise significantly, and for more than one reason. Firstly, Solvency II treats stocks as high-risk instruments, which in turn leads to high solvency capital requirements (SCRs) for most equity investments. Secondly, the main objective of insurers' asset management strategies will remain unchanged: to manage the economic risk of their liabilities. Stocks do not play a significant role in this undertaking. They have their place in the performance seeking part of the portfolio, however.

In order to be as capital efficient as possible, the part of the asset portfolio that is not used to match liabilities should be invested in assets with low correlations with the liability-hedging portfolio. Alternative beta stock investments are an attractive asset class for these surplus investments as they not only have low correlations to the fixed income portfolio, they also usually provide a risk premium. Alternative beta ETFs have the added benefits of being easily investable and highly transparent, thereby helping investors to fulfil the new reporting requirements that also come with Solvency II.

The following sections of this article are going to show how insurance companies can use factor investing and alternative beta ETFs to efficiently manage their performance-seeking portfolios and achieve an attractive risk-return profile.

¹¹⁸ European Commission: »SOLVENCY II«: Frequently Asked Questions (FAQs), http://ec.europa.eu/internal_market/insurance/docs/solvency/solvency2/faq_en.pdf, request from 17th March 2015

Empirical Study: Investments In Factor Indices To Enhance Portfolio Yields And Annualized Returns

The following empirical analysis points out the return characteristics of investments in certain factor indices. The assumption for the following analysis is a global, balanced portfolio of a German life insurance company. The equity portion of the portfolio is being represented by investments in the MSCI EMU and the MSCI USA amongst other regional investments. The assumption for the analysis is that the investments in the MSCI EMU and the MSCI USA will be partly substituted by the factor investments in the MSCI EMU Dynamic Risk-Weighted Index, the MSCI EMU Total Shareholder Yield Index, the MSCI USA Dynamic Risk-Weighted Index and the MSCI USA Total Shareholder Yield Index.

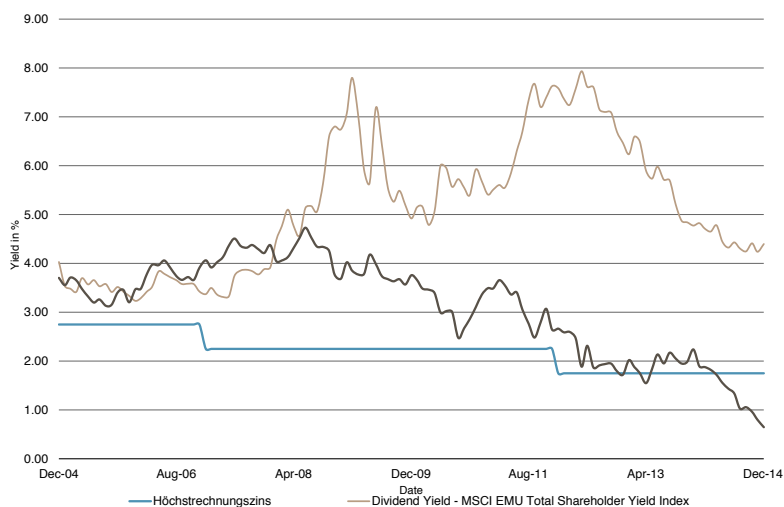
We concentrate the analysis on the Dynamic Risk-Weighted and Total Shareholder Yield factors. We consider them particularly suitable to complement a portfolio of a German life insurance company. The observation period starts on December 31, 2004 and ends on December 31, 2014. Additionally, we simulated mixed portfolios consisting of 50% MSCI EMU Total Shareholder Yield Index and 50% MSCI EMU Dynamic Risk-Weighted Index for the European equity market, and 50% MSCI USA Total Shareholder Yield Index and 50% MSCI USA Dynamic Risk-Weighted Index for the US equity market. Both mixed portfolios were rebalanced on a monthly basis. We find for the European equity market, as well as for the US equity market, that all three-factor strategies compare favourably to the standard market capitalization-weighted index regarding return-risk properties and are, furthermore, particularly attractive for insurance companies due to higher average dividend yields.

The yield environment for German insurance companies

For a German life insurance company a maximum technical interest rate (Höchstrechnungszins) is being applied. This rate is defined as the interest rate that a particular life insurance is allowed to promise its customers as a minimum yield per annum on their insurance premiums. It is also often referred to as a »guaranteed interest rate« (Garantiezins), meaning it is what the owners of the insurance policy can expect as a minimum annual return on their premiums. Since January 2015, the maximum technical interest rate has

been at a new low of 1.25%. The development of the maximum technical interest rate during the observation period is shown in Figure 1. Further, Figure 1 shows the yield of an aggregate basket of AAA-rated, ten-year European Government Bonds and the dividend yield of the MSCI EMU Total Shareholder Yield Index. Between Dec 2004 and Dec 2009, the »Höchstrechnungszins« was on average 1.44% below the yield of the above mentioned government bonds. However, from Dec 2009 until Dec 2014, this average premium was only 0.43%, and it has been consistently negative since the end of April 2014. Hence, it is becoming more and more difficult for insurance to achieve the »Höchstrechnungszins« via investments in top-rated European Sovereign bonds. On the other hand, Figure 1 shows that the dividend yield of the above mentioned factor index surpassed the yield of the aggregate bond index in January 2008 and since then remained on yield levels which are significantly higher than those of the government bonds. Therefore, an investment in yield-oriented factor indices can be considered as an adequate complementary investment to improve the yield of a portfolio.

Figure 1: Yield comparison



Sources: European Central Bank, www.ecb.eu, request from 9th March 2015; Gesamtverband der Deutschen Versicherer e.V., www.gdv.de, request from 9th March 2015; MSCI, monthly data (December 31, 2004 to December 31, 2014).

Alternative Beta vs. Standard Market Capitalization Indices

MSCI EMU vs MSCI EMU Dynamic Risk-Weighted Index and MSCI EMU Total Shareholder Yield Index

Table 1 shows the return characteristics of the standard MSCI EMU index, the above-mentioned factor indices, and a combination of the two-factor indices (mixed portfolio). The numbers reveal that each factor index has superior risk and return characteristics over the observation period, expressed by lower empirical standard deviations and maximum drawdowns as well as higher annualized returns. Furthermore, the MSCI EMU Dynamic Risk-Weighted Index and the mixed portfolio can be characterized as defensive equity investments, which is expressed by a beta of 0.76 and 0.87 respectively.

Table 1: Return characteristics of MSCI EMU Factor Indices¹¹⁹

	MSCI EMU	MSCI EMU Dynamic Risk-Weighted	MSCI EMU Total Shareholder Yield	Mixed Portfolio
Return	3.3%	5.4%	3.7%	4.6%
Volatility	25.8%	20.3%	25.6%	22.6%
Return/Risk	0.13	0.26	0.15	0.20
Beta	1.00	0.76	0.97	0.87
Max. Drawdown	-64.7%	-57.3%	-61.1%	-59.1%
Rel. Drawdown		-15.9%	-18.4%	-13.0%
Dividend Yield	3.6%	3.7%	5.2%	4.4%

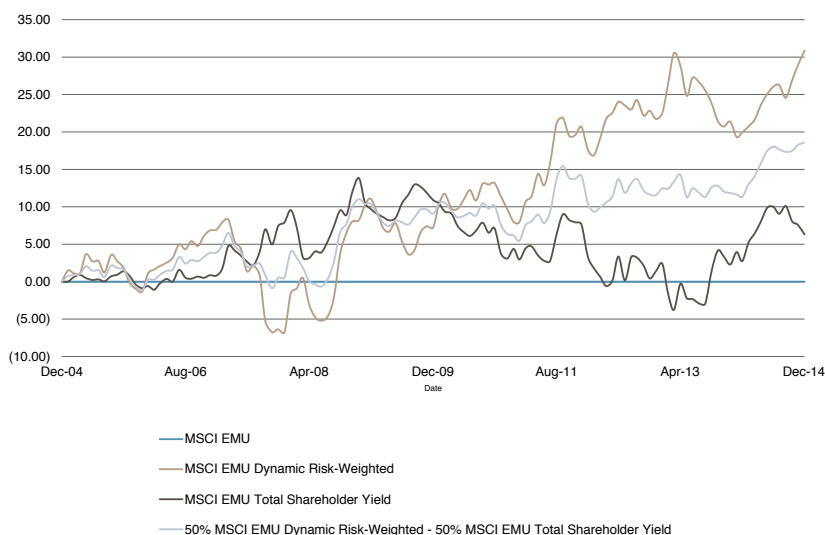
Source: MSCI, daily data for return and risk figures, monthly data for yield figures (December 31, 2004 to December 31, 2014).

The average dividend yield of the MSCI EMU Dynamic Risk-Weighted Index only exceeds the yield of the standard index by an average of 0.1%. However the dividend yield of the MSCI EMU Total Shareholder Yield Index and of the mixed portfolio exceeds that of the standard index considerably – by 1.6% and

¹¹⁹ Figures are based on the performance of the total net return indices; the simulation has been done without taking transaction costs into account.

0.8% respectively. The figures for the relative drawdown show the maximum drawdown of the factor indices relative to the standard index. It is -15.9% for the Dynamic Risk-Weighted version and -18.4% for the Total Shareholder Yield version of the MSCI EMU Index. These numbers represent, to some extent, the cyclicity of individual factors, which can be reduced by combining the factors into the mixed portfolio. The relative drawdown of the monthly rebalanced mixed portfolio is -13.0%, therefore, 2.9% and 5.4% lower, respectively, than that of the single-factor indices. To underline the superior return characteristics of the factor indices and a combination of them in the MSCI EMU universe, Figure 2 shows the relative performance of the respective indices compared to the market capitalization-weighted index.

Figure 2: Relative Performance of MSCI EMU factor indices in %



Source: MSCI, monthly data (December 31, 2004 to December 31, 2014).

MSCI USA vs MSCI USA Dynamic Risk-Weighted Index and MSCI USA Total Shareholder Yield Index.

Table 2 shows the return characteristics of the standard MSCI USA Index, the above-mentioned factor indices, and a combination of the two-factor indices in the mixed portfolio.

Table 2: Return characteristics of MSCI USA Factor Indices¹²⁰

	MSCI USA	MSCI USA Dynamic Risk-Weighted	MSCI USA Total Shareholder Yield	Mixed Portfolio
Return	7.2%	8.1%	7.9%	8.0%
Volatility	20.0%	15.7%	17.9%	16.6%
Return/Risk	0.36	0.52	0.44	0.48
Beta	1.00	0.75	0.88	0.81
Max. Drawdown	-55.4%	-47.6%	-50.8%	-48.9%
Rel. Drawdown		-17.6%	-10.9%	-14.1%
Dividend Yield	2.0%	2.9%	2.6%	2.7%

Source: MSCI, daily data for return and risk figures, monthly data for yield figures (December 31, 2004 to December 31, 2014).

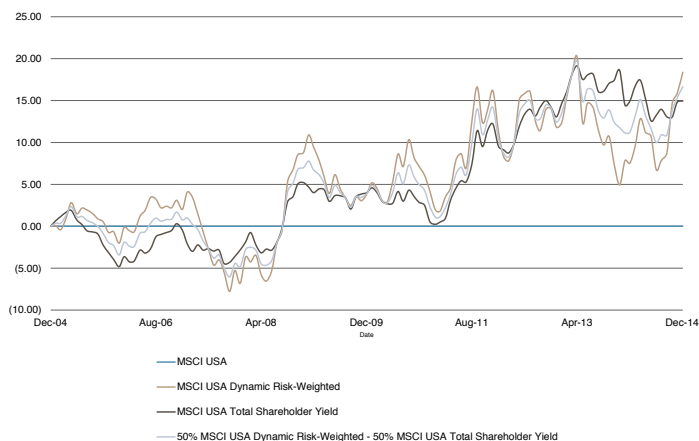
We can see that the factor indices outperform the standard MSCI USA index regarding return and risk properties. Annualized returns are higher with 8.1% for the MSCI USA Dynamic Risk-Weighted Index, 7.9% for the MSCI USA Total Shareholder Yield Index and 8.0% for the mixed portfolio versus 7.2% for the parent index. The dividend yields of the factor indices are higher as well. The yield of the standard index is 2.0%, whereas the yield is 2.9% for the Dynamic Risk-Weighted version, 2.6% for the Total Shareholder Yield version and 2.7% for the mixed portfolio.

In addition to that, both factor indices can be considered defensive equity investments. This is shown by their relatively low beta – 0.75 for the Dynamic Risk-Weighted index and 0.88 for the Total Shareholder Yield index – as well as their lower volatility and maximum drawdowns.

In general, the factor indices seem to be more alike on the US stock market. Consequently, the combination of the two-factor indices does not reduce the cyclicity of the individual factor indices to the same extent as in the MSCI EMU universe. This is shown by the curves in the relative performance chart (Figure 3) and the relative drawdown of the mixed portfolio (-14.1%), whose value exceeds the relative drawdown of the MSCI EMU Total Shareholder Yield Index.

¹²⁰ Figures are based on the performance of the total net return indices; the simulation has been done without taking transaction costs into account.

Figure 3: Relative Performance of MSCI USA factor indices in %



Source: MSCI, monthly data (December 31, 2004 to December 31, 2014).

Conclusion

The search for yield is a major challenge for insurance companies in today's environment. Economic pressure from the extremely low interest rates and increasing regulatory requirements make it ever more difficult for insurers to cover their future liabilities and at the same time achieve an attractive return in the performance-seeking part of their portfolios.

Against this backdrop, ETF's tracking above-mentioned factor indices offer an opportunity to enhance portfolio returns with a defensive equity exposure combining superior risk-return properties and an attractive dividend yield. By deploying alternative beta ETFs, investors can profit from the transparency and low costs traditionally associated with indexing solutions without foregoing the chance to generate above-market returns.

Our analysis above shows that, for a German life insurance company, it can be beneficiary to partly substitute European and US equity exposure (MSCI EMU and MSCI USA) with the factor indices MSCI EMU Dynamic Risk-Weighted, MSCI EMU Total Shareholder Yield, MSCI USA Dynamic Risk-Weighted and MSCI USA Total Shareholder Yield.

Combining Core Satellite Investing With Factor Allocation: A Swiss Private Bank Model

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Introduction: The Core Satellite Approach In Private Banking

The core satellite approach to investing is popular among private banks in Switzerland and Liechtenstein. Generally, portfolios based on this strategy consist of two clearly distinct components. The core consists of a broadly diversified portfolio of basic investments, designed to ensure that investors achieve their long-term investment goals. More and more, these investments are made using passively managed investment products such as Exchange Traded Funds (ETFs). The core portfolio is, figuratively speaking, orbited by satellite investments that are used to exploit short-term opportunities and achieve risk-adjusted excess returns (alpha). Investors typically deploy actively managed instruments in order to achieve these goals. The relative weight of the two portfolio components, as well as the specific asset allocation within the core and the satellite portfolio, typically depend on the investment objectives and risk tolerance of private banks' clients.

Just like other investment strategies, the core-satellite approach has distinct strengths and weaknesses. One of its major strengths is the basic idea behind it: the concept of using a low-cost passive strategy well-founded on science and decades of practice to meet the investor's needs and – having made sure they are taken care of – using the rest of the portfolio to pursue alpha, i.e. above average returns. The concept makes sense, intuitively, and it is not a matter of chance that it became as widely accepted as it is, today. However, there are problems associated with the concept, and it is the most ambitious part of the typical core-satellite portfolio, the satellite component, that causes

most of these problems, in practice. One problem is that the actively managed instruments usually deployed as satellite investments are relatively costly, reducing the chance of actually generating alpha net of costs. Another, is the black box risk that often comes with active instruments; for not only may they change their asset allocation over time, but also their investment strategy as a whole. In addition to these strategic considerations, regulatory changes in Switzerland and other European countries have moved pricing models away from retrocession-based models towards retrocession-free solutions like ETFs. Due to this development, there is a common interest among banks and their customers to find low-cost investment solutions wherever possible.

Hence, we have seen that more and more Swiss private banks have started to employ features of alternative beta in the equity allocation of their clients' portfolios. Against the backdrop mentioned above, one might be tempted to simply replace active strategies with alternative beta in the satellite component. However, in most cases, the alternative beta proposition is more often seen as an attractive complement to actively managed funds. In accordance with this point of view, private banks use alternative beta products as an effective way to access specific market exposures with their most cost-conscious clients.

In this article, we intend to illustrate different ways of combining a set of alternative beta strategies which, taken together, have the potential to improve excess returns with less volatility in the overall portfolio context.

Factor Investing In Theory And Practice

The search for the factors that drive stock market returns has a long and illustrious academic history. The oldest, and probably still best-known, result of this search is the capital asset pricing model (CAPM), which was introduced in the 1960s by Sharpe¹²¹, Treynor, Lintner and Mossin, independently. In the CAPM, two drivers of stock returns are identified: systematic stock market risk (captured by beta) and idiosyncratic risk that can be diversified away and does not carry a risk premium. French and Fama built on the CAPM when they developed their

¹²¹ See Sharpe 1964.

now famous three-factor model in the 1990s.¹²² It introduced two additional factors: size (small vs. large caps) and value (high vs. low book-to-market ratio). Since then, further research has suggested that there is, in fact, an even higher number of factors that drive stock returns. Figure 1 summarizes the factors that are most widely accepted in academic research today, and their characteristics. These factors include value, low size, low volatility, dividend yield, quality and momentum.

Figure 1: Well-known factors from academic research

Systematic Factors	What It Is	Commonly Captured by
Value	> Captures excess returns to stock that have low price relative to their fundamental value	> Book-to-price, earnings to price, book value, sales, earnings, cash earnings, net profit, dividends, cash flow
Low Size (Small Cap)	> Captures excess returns of smaller firms (by market capitalization) relative to their larger counterparts	> Market capitalization (full or free float)
Momentum	> Reflects excess returns to stock with stronger past performance	> Relative returns (3-mth, 6-mth, 12-mth, sometimes with last 1-mth excluded), historical alpha
Low Volatility	> Captures excess returns to stock with lower than average volatility, beta, and/or idiosyncratic risk	> Standard deviation (1yr, 2yrs, 3yrs), Downside standard deviation, standard deviation of idiosyncratic returns, Beta
Dividend Yield	> Captures excess returns to stock that have higher-than-average dividend yield	> Dividend yield
Quality	> Captures excess returns to stock that are characterized by low debt, steable earnings growth, and other »quality« metrics	> ROE, earnings stability, dividend growth stability, strength of balance sheet, financial leverage, accounting policies, strength of management, accruals, cash flows

Source: MSCI Index Research: Foundations of Factor Investing.

¹²² Fama, Eugene F. French, Kenneth R. (1992), »The Cross-Section of Expected Stock Returns«, The Journal of Finance 47 (2), 427–465.

The majority of banks we have spoken to, have a preference for the alternative factors yield, value, quality and low volatility and are open to implementing these factors consistent with their regional allocation approach. Other factors, like momentum, bring up the problem of potentially high turnover costs due to the relatively short-term nature of the momentum effect, itself: portfolios have to be changed, regularly, in order to profit from positive autocorrelations on the stock market. In the following two examples, we are going to show how private banks and their clients can benefit from systematic exposure to these factors, and how factor investing can be implemented using ETFs. The first example explains how investors can use alternative beta strategies to express their macro-economic views as well as be positioned, optimally, during each of the different phases of the economic cycle. In contrast to that, the second example aims at using factor investing, strategically, in order to systematically implement general factor tilts. Furthermore, we will show how currency-hedged factor investing allows for pure exposure to certain factors, regardless of possible differences between the funding currency and the fund currency.

Using Factor Investing To Express Economic Viewpoints

Historically, all of the factors introduced above have earned long-term risk premia. This does not mean, however, that the factors have consistently outperformed the general market. In fact, factor returns have been highly cyclical, in the past, due to their sensitivity to macro-economic developments and other market forces¹²³. Fortunately, for investors, not all factors react to the same drivers in the same way. This opens up interesting diversification and tactical possibilities. To demonstrate these possibilities we have developed an approach that utilizes alternating exposure to the factors low size, value, quality, yield and low volatility through the phases of the economic cycle. This is based on the fact that some factors, like low size and value, have shown pro-cyclical behaviour, while others, like quality, yield and low volatility, have

¹²³ Bender, Jennifer et. al. (2013), »Foundations of Factor Investing«, MSCI Research Insight.

shown defensive qualities in a weakening economic environment.¹²⁴ Figure 2 gives an overview of the most important factors and their cyclicalities.

Figure 2: Factors and their cyclicalities

Factor	Historical Risk	Historical Correlation	Historical Business Cycle
Value	Comparable to market	Low with Momentum and Quality	Pro-cyclical
Momentum	Comparable to market	Low with Value, Yield, and Quality	Pro-cyclical
Low Size	Higher than market	Low with Min Volatility, Yield, and Quality	Pro-cyclical
Quality	Lower than market	Low with Value, Size, Yield, and Quality	Defensive
Low Volatility	Lower than market	Low with Value and Momentum	Defensive
Yield	Lower than market	Low with Size, Quality and Momentum	Defensive

Source: MSCI Index Research: Deploying Multi-Factor Index Allocations in Institutional Portfolios.

It follows that deploying the different factors according to their cyclicalities can lead to excess returns, while at the same time lowering the risk of the portfolio. In practice this means that:

- Portfolios should have systematic exposure to the low size and value factors in the early phase of the economic cycle, characterized by recovery from recession.
- During the phase in the midst of the cycle, typically characterized by slowing and, finally, peaking growth, the quality factor, potentially, offers the greatest rewards.

¹²⁴ Bender, Jennifer et. al. (2013), »Deploying Multi-Factor Index Allocations in Institutional Portfolios«, MSCI Research Insight.

- The defensive qualities of the low volatility, yield and quality factors come into play late in the cycle when the economy starts to slow down.
- During the trough of the cycle, i.e. recession, yield and value stocks start to outperform again, so investors should allocate funds back to value investments.

Investors can gain access to all of these factors using UBS ETFs on the respective MSCI factor indices on Selective Value, Quality, Total Shareholder Yield, Low Volatility and Small Caps.

Implementing General Factor Tilts

A more strategic approach to factor investing includes the implementation of general factor tilts in order to systematically alter the characteristics of the satellite portfolio and/or achieve excess returns. Investors can, for example, use UBS ETFs on low-size and value to follow the French–Fama three-factor model as closely as possible. They should be aware, however, that this approach could lead to higher cyclical nature of the overall portfolio, due to the pro-cyclical nature of both the size and the value factor.

Investors who want to avoid this, or even want to add defensive qualities to their portfolio while keeping their chances of excess returns intact, should add a tilt in a different direction to their satellite investments, favouring the low volatility, quality and yield factors by deploying the respective UBS ETFs.

General factor tilts can further be added to the portfolio by deploying multiple factors at the same time, thereby taking advantage of the low correlations of factors among each other, which can be seen from Figure 2, above. In doing so, investors can take advantage of the risk premia associated with factor investing while avoiding the need to explicitly express an opinion on the future development of the economy or the stock market. Seeking exposure to multiple

factors at the same time allows investors to avoid possible pitfalls of portfolios that are completely based on market capitalization. At the same time, it offers a superior risk-return profile, even in comparison to portfolios with exposure to single factors. This means that investors can potentially reach their target returns with an even lower volatility of returns. As market exposure in regions that have their own base currency adds unwanted currency volatility to the portfolio, which carries no risk premium, so investors might prefer to concentrate on pure factor exposure. Currency-hedged factor ETFs can then free up risk budgets that are better employed elsewhere – namely, for investments in the factor-driven stock market.

Conclusion: High Transparency, Low Cost – The Benefits Of Factor Investing With ETFs

In summary, factor ETFs from UBS open up a whole new universe for investors on the lookout for viable strategies to use in the satellite component of their portfolios. In particular, they offer three distinct advantages: low cost, high transparency by virtue of their rule-based approach to earn excess returns by harvesting factor premia, which are well-founded in academic research.

Of course, finding the best way to implement alternative beta strategies in accordance with the individual investor's aims and risk tolerance is an undertaking that lies beyond the scope of this article. It is, at this point, that the expertise of private banks, with their proficiency in tailoring portfolios that fit their clients' needs, comes into play. Index and ETF providers can only hand advisors the tools they need to do their job properly. Alternative beta ETFs make for great additions to the traditional toolbox. This toolbox is further enlarged by currency-hedged factor ETFs. These innovative products allow investors to concentrate on pure factor exposure, regardless of the regions they invest in, by shutting out unwanted currency volatility.

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Exploring The Liquidity Of A Risk Weighted ETF

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Introduction

This analysis aims to explore the liquidity of an Exchange Traded Fund (ETF) tracking a Risk Weighted Index. In order to gauge the results, a comparison will be made between a Risk Weighted ETF and a comparable Cap Weighted ETF tracking the MSCI All Country World Index (ACWI). Finally, the liquidity measure used in this study will be the ETF bid-ask spread.

A Risk Weighted Index takes exactly the same constituents as a Cap Weighted Index but over weights the constituents with low volatility.¹²⁵ It implies that such an index is less volatile over time. But can we assume that the liquidity, i.e. the bid-ask spread, of a Risk Weighted ETF will be less affected by an increase of volatility in the market than compared to a Cap Weighted ETF? This analysis answers this question by means of a concrete example.

Liquidity

Liquidity defines the degree to which a financial instrument can be bought or sold without affecting its price in the market. It is, generally, measured with different parameters such as the daily trading volume and the bid-ask spread. Securities with high volume are largely preferred among investors

¹²⁵ http://www.msci.com/products/indexes/strategy/factor/risk_weighted/

as it reduces their liquidity risk. Low liquidity leads to wider bid-ask spreads, meaning that trading becomes inherently more costly for investors.

It is worth noting that ETF liquidity is different from a single security. If the ETF does not have a high trading volume, it does not mean that one investor should avoid trading it. The key is in understanding the determinants of the ETF liquidity. ETFs are affected by the so-called primary and secondary market liquidities.

Primary market liquidity is determined by the underlying assets of an ETF. It represents an aggregation of the liquidity of its underlying securities. Indeed, the primary market takes place between the ETF and the Authorized Participant (AP). In order to create ETF shares, the AP first needs to get the underlying assets in the market, where he is directly exposed to the liquidity of each security. This explains why ETFs replicating emerging markets are less liquid than ETFs tracking European benchmarks, for instance.

Secondary market liquidity concerns the ETF's own share in the market, where APs bargain for ETF shares with investors and market makers. This liquidity is a function of the number of ETF shares available, its demand and supply, and other market forces. It is why ETFs tracking the same benchmark can have different levels of liquidity; ETFs with higher turnover usually have better bid-ask spreads.

ETF Bid-Ask Spread

The bid-ask spread is certainly the most popular liquidity measure. As explained previously, the ETF bid-ask spread is first determined by the spreads of its underlying assets and then by the trading volume of its ETF shares. Additionally, the costs from the primary transactions (creation and redemption process) should also be taken into account. Finally, market makers must also cover their own costs like hedging, inventory holding and other fees.

Risk Weighted Index

A Risk Weighted Index has exactly the same constituents as a Market Capitalization Weighted Index. The only difference is that a Risk Weighted benchmark gives higher index weights to stocks with lower volatility.¹²⁶ The graph, below, shows the historical performance of the MSCI ACWI Risk Weighted in comparison to its parents index, the MSCI ACWI.

Figure 1: Historical performance of the MSCI ACWI and MSCI ACWI Risk Weighted

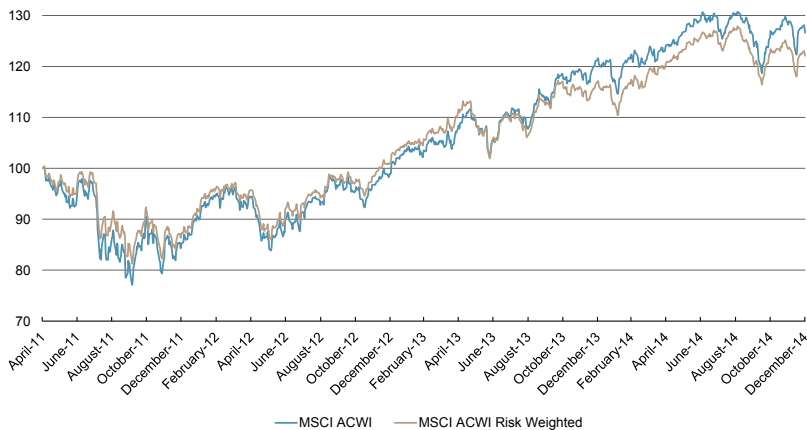


Table 1: Performance and risk of the MSCI ACWI

	Cap Weighted	Risk Weighted
Annualized return	6.35%	5.34%
Annualized standard deviation	13.74%	11.31%

Source: Global Asset Management, Bloomberg. Own computations. Data from 29.04.2011 to 31.12.2014

By emphasizing low volatility stocks in this way, the MSCI ACWI Risk Weighted has, historically, shown lower volatility than compared to the MSCI ACWI.

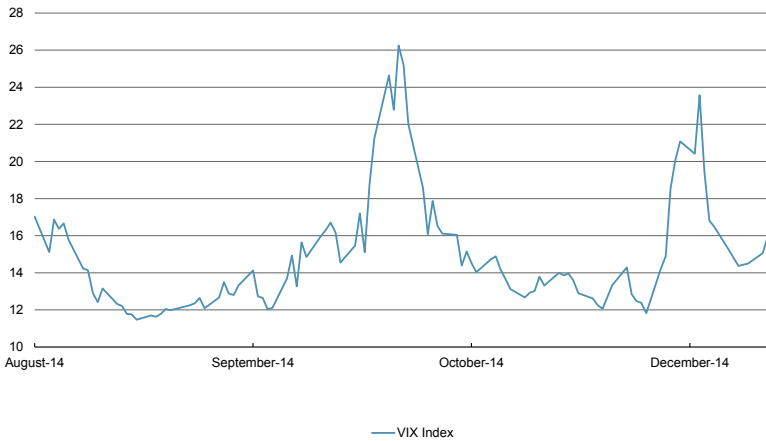
¹²⁶ http://www.msci.com/products/indexes/strategy/factor/risk_weighted/

Annualized volatility for the Risk Weighted and Cap Weighted Indexes is 11.31% and 13.74%, respectively.

Methodology

Simple regressions were run in order to test the statistical relationship between market volatility and ETF liquidity. As most of the constituents of the MSCI ACWI are US-based stocks, the VIX Index was chosen as the volatility measure. Intraday prices were gathered every minute from August 1, 2014 to December 31, 2014.

Figure 2: Historical prices of the VIX Index



Source: UBS Global Asset Management, Bloomberg. Data from August 1, 2014 to December 31, 2014.

Intraday bid and ask prices were sourced for a: Risk-Weighted ETF and a comparable Cap-Weighted ETF tracking the MSCI All Country World Index (ACWI). From this data, bid-ask spreads were calculated according the following formula.

$$Spread_t = \frac{Ask_t - Bid_t}{\frac{1}{2}(Ask_t + Bid_t)}$$

Where Ask_t and Bid_t are, respectively, the ask and bid prices at time t .

A simple regression is used to test if the liquidity of an ETF, i.e. the bid-ask spread, can be explained by the market volatility, i.e. the VIX Index. This relation can be expressed thanks to the following formula:

$$y_t = a + bx_t + \varepsilon_t$$

Where y_t , x_t and ε_t are respectively the ETF bid-ask spread at time t , the VIX Index at time t and the residuals at time t : a is the constant of the model and b the coefficient of the explanatory variable.

Results And Conclusion

According to the regression results, in the table below, we can observe that a statistical relationship exists between the volatility and the liquidity of an ETF. Both regressions show that the ETF bid-ask spread is positively linked to the VIX Index (coefficient b in the formula above). Indeed, an increase of volatility leads to higher bid-ask spreads, i.e. a loss of liquidity for the investors. This outcome was expected as we observe it in practice.

Table 2: Simple regression results

MSCI ACWI ETF		
	Coefficients	t Stat
Constant (a)	16.6589	3.9636
Coefficient (b)	2.8863	10.5849
MSCI ACWI Risk-Weighted ETF		
	Coefficients	t Stat
Constant (a)	35.4007	69.0928
Coefficient (b)	1.9723	59.5499

Source: UBS Global Asset Management, Bloomberg. Own computations.
Data from August 1, 2014 to December 31, 2014.

The most interesting result comes from the fact that the coefficient b is lower for the Risk-Weighted ETF than for the Cap-Weighted ETF. The results are 1.97 and 2.88, respectively, showing that an increase in volatility will have roughly 1.5 x more impact on the bid-ask spread of the Cap-Weighted ETF. In other words, and by assuming an equivalent level of liquidity for the two ETFs, investors can expect better liquidity in the Risk-Weighted ETF when the volatility increases.

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Marco Vario joined UBS Global Asset Management in November 2013 as ETF Capital Markets Specialist for Europe. In this role, he gained in-depth experience of ETFs and their execution.

Prior to joining UBS, Marco began his career at Credit Suisse in March 2012 where he did a 6-month internship as ETF Product Specialist for Switzerland & Lichtenstein. There, he learned a lot, about the ETF industry, and wrote his Master Thesis on a topic directly related to ETFs and Trading. Marco holds a Master of Science (MSc) in Finance from the University of Lausanne, with a specialization in Financial Engineering and Risk Management.

About The Contributors

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Index-tracking investment solutions have been a core competence of UBS Global Asset Management for over 30 years. As a leading fund house in Europe, UBS launched its first ETF in Europe in 2001 and is currently both one of Europe's foremost providers as well as a dominant force in currency hedged ETFs in the region. The range comprises more than 160 ETFs and offers investors a transparent and flexible opportunity to diversify their investments across key markets and all asset classes, including equities, bonds, real estate, commodities and alternative investments. UBS ETFs are listed on the following stock exchanges around the world – the SIX Swiss Exchange, Borsa Italiana, the London Stock Exchange, Deutsche Börse XETRA, the Australian Securities Exchange ASX, KRX Korea Exchange and the Tokyo Stock Exchange.

For more information on UBS ETFs: www.ubs.com/etf

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